

E.C. JORDAN CO.

ENGINEERS &
SCIENTISTS

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CADILLAC AREA GROUNDWATER CONTAMINATION
FEASIBILITY STUDY
CADILLAC, MICHIGAN

FINAL DRAFT
FEASIBILITY STUDY

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CADILLAC AREA GROUNDWATER CONTAMINATION
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FEASIBILITY STUDY

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MICHIGAN DEPARTMENT OF NATURAL RESOURCES

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1.0 INTRODUCTION

Under contract to the Michigan Department of Natural Resources (MDNR), E.C. Jordan Co. (Jordan) conducted a Remedial Investigation/Feasibility Study (RI/FS) of the contamination (primarily in groundwater) beneath the Cadillac Industrial Park in Cadillac, Michigan. Because the Northernnaire Plating Company (Northernnaire) and Kysor of Cadillac, Inc. (Kysor) Superfund sites are located within the study area, the study is being conducted under joint MDNR/U.S. Environmental Protection Agency (USEPA) guidance, and in accordance with the Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (CERCLA), as amended by the Superfund Amendments and Reauthorization Act of 1986 (SARA), and the National Oil and Hazardous Substances Pollution Contingency Plan (NCP 1985 - Final Rule, Section 300.68).

The Northernnaire site and associated metals contamination were the subjects of an RI and a Focused FS (FFS) for contaminated site soils, also performed by Jordan for MDNR and USEPA. Because the Northernnaire plume was known to mingle with plumes from other sources, it was decided that the Northernnaire operable unit for groundwater would be performed in conjunction with the Cadillac FS, which also includes assessments of volatile organic compound (VOC) contamination from other sources in the Cadillac Industrial Park identified in the RI and other previous studies in the area.

Investigation and remediation of site contamination for the Cadillac site was divided into two parts: soil and groundwater. The RI presented the distribution of VOC and chromium contamination in groundwater beneath the Cadillac Industrial Park and adjacent areas of Haring Township, Michigan, and identified some residual contaminants in soils at suspected source locations. Information sources for the RI included studies performed by Jordan, MDNR, and USEPA, and by consultants for industries within the industrial park. A baseline risk assessment (RA) was then prepared based on findings of the RI. This FS identifies and analyzes alternatives for groundwater remediation of the contaminated plumes and soil remediation of the identified source locations.

1.1 PURPOSE AND SCOPE

The purpose of the FS is to identify remedial action objectives, and to develop and identify alternatives that will achieve those objectives. The development of alternatives includes (1) identifying applicable technologies, and (2) screening out those that would not meet response objectives or that are technically infeasible due to specific site characteristics, waste characteristics, or other physical constraints.

Technologies that pass the screening are then combined to form a range of medium-specific alternatives. SARA emphasizes the use of treatment technologies that reduce toxicity, mobility, or volume of contaminants rather than technologies that prevent exposure. Therefore, the range includes (to the extent possible) alternatives from each of the following categories:

- o no-action
- o containment
- o treatment

Treatment alternatives range from those that eliminate the need for long-term site management to those that reduce toxicity, mobility, or volume of contaminants. The treatment alternatives, once developed, are screened based on specific criteria pertaining to effectiveness, implementability, and cost. Alternatives retained for both groundwater and soil remediation are summarized in Sections 4.4 and 5.4, respectively.

A detailed analysis of each alternative is then conducted. In this phase, alternatives are described and analyzed for several factors, including the following:

- o compliance with Applicable or Relevant and Appropriate Requirements (ARARs)
- o reduction of the toxicity, mobility, or volume of contaminants
- o short-term effectiveness
- o long-term effectiveness and permanence
- o overall protection of public health and the environment
- o implementation
- o community and state acceptance
- o costs and economic analysis

Following detailed analysis, the results are summarized and compared, considering the following factors:

- o protection of public health and the environment;
- o attainment of federal and state public health and environmental requirements identified for the site;
- o cost effectiveness; and
- o use of permanent solutions and alternative treatment technologies or resource recovery technologies, to the maximum extent practicable.

The FS will serve as the basis for the Record of Decision (ROD), which documents remedy selection, and for remedial design and construction.

1.2 SITE DESCRIPTION

The Cadillac Industrial Park includes nearly 40 industrial facilities. Past uncontrolled releases of contaminants to the ground by several of these facilities have contaminated groundwater in aquifers underlying the park. The contaminants of concern are VOCs (i.e., chlorinated and aromatic hydrocarbons such as trichloroethylene [TCE] and xylene) and hexavalent chromium (derived from chromium-plating wastes). The presence of contaminants within the upper aquifers beneath the site has created concern for potential impact on a lower aquifer. This lower aquifer supplies potable water to the city from a well-field located near the center of the industrial park.

The RI identified six contaminant sources and seven contaminant plumes in groundwater in the study area. The six probable or known source locations are (1) Northernnaire, (2) Kysor, (3) Rexair, (4) Four Winns, (5) Four Star/Joynts (former), and (6) Ingraham property.

In the following descriptions, the precise units for concentration, milligrams per liter (mg/l), have been replaced by parts per million (ppm), which are conceptually more easily appreciated. At the low concentrations detected, the concentrations are numerically equal for both sets of units.

The seven plumes are categorized under four major plumes, as described in the following paragraphs.

Northernnaire Plumes. The following two plumes comprise the Northernnaire plumes:

- o A hexavalent chromium plume in the intermediate aquifer situated just above the lower clay aquitard. Northernnaire was the source.
- o A small hexavalent-chromium plume of relatively low concentration in the shallow aquifer, interpreted as a small slug released from Northernnaire. The slug, which is moving rapidly through the upper aquifer and being diluted as it migrates, should soon merge with the intermediate aquifer.

These plumes are depicted in Figure 1-1.

Shallow VOC Plume. The shallow VOC plume is a large, concentrated VOC plume (up to 115 ppm) in the shallow aquifer, caused primarily by past solvent dumping behind Kysor and at the Ingraham property. This plume has merged with other VOC releases from Four Star/Joynts (former) to the northeast and Four Winns to the northwest. The shallow VOC plume is depicted in Figure 1-2.

Intermediate VOC Plume. The intermediate VOC plume is a large VOC plume (up to 12 ppm) in the intermediate aquifer resulting from Kysor- and Ingraham-related VOCs migrating from the shallow aquifer through the upper aquitard. The intermediate VOC plume is depicted in Figure 1-3.

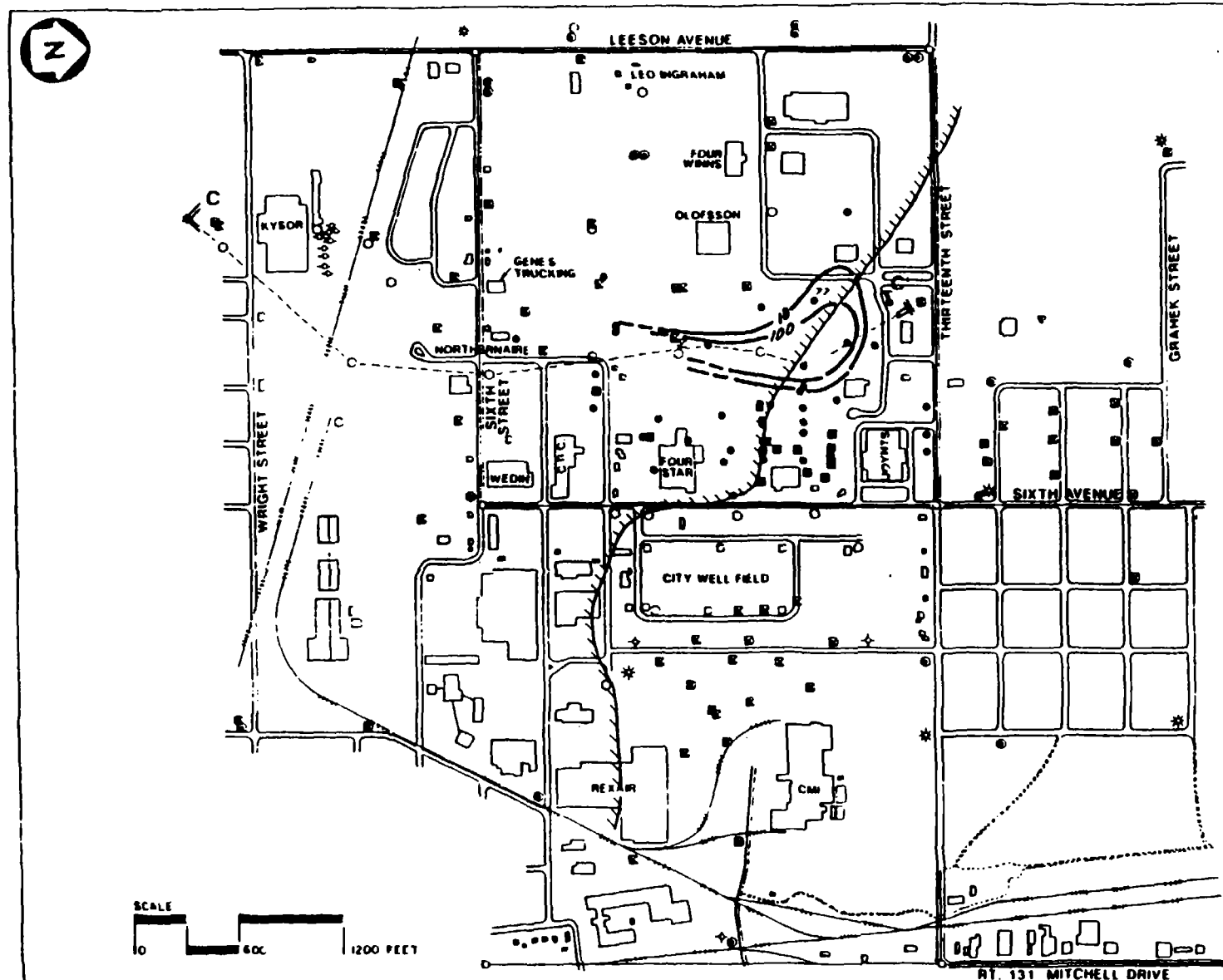
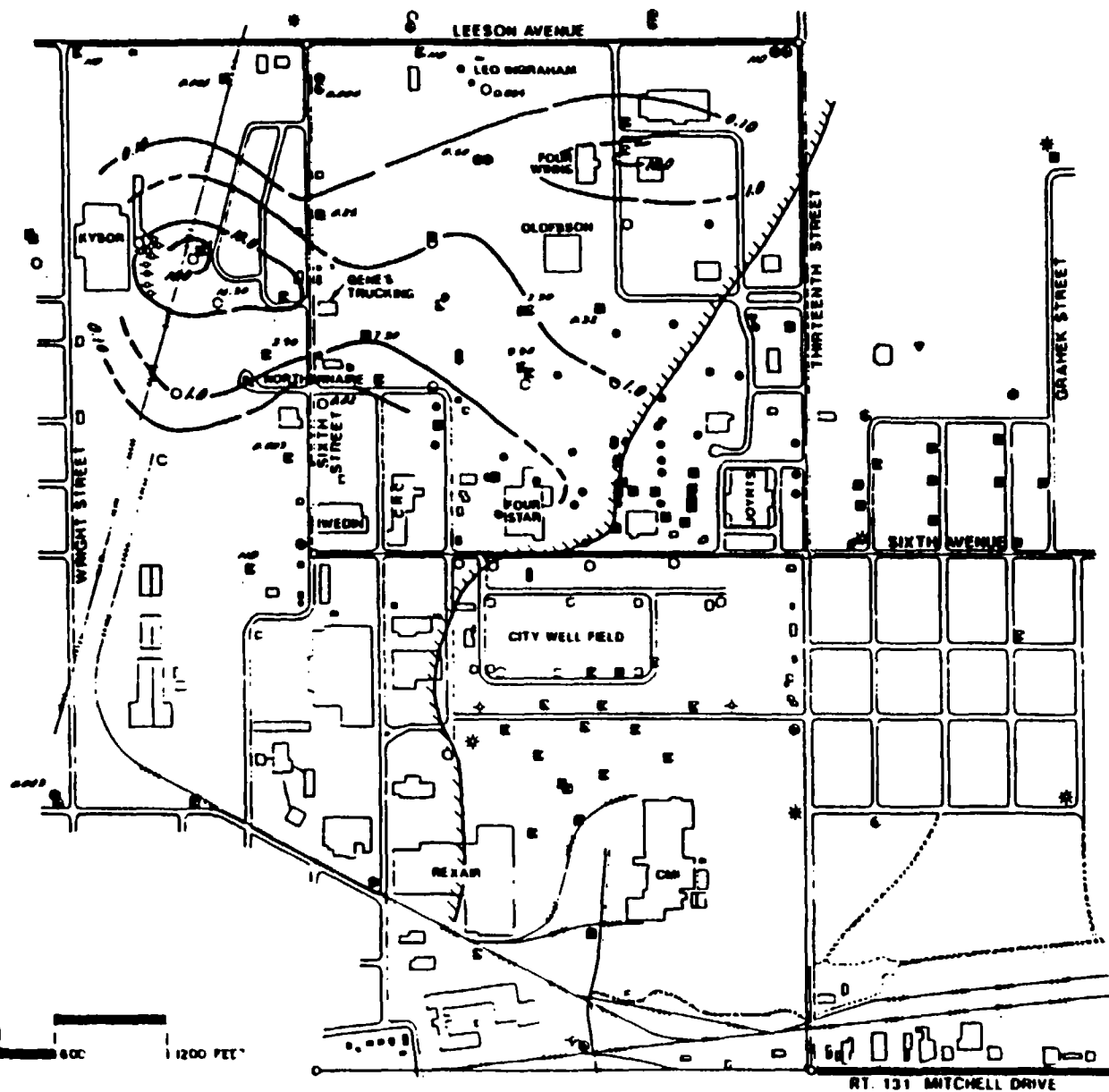


FIGURE 1-1
NORTHERNAIRE PLUMES AND
HEX-CHROME CONCENTRATIONS
MAY/JUNE 1987
CADILLAC AREA FS
CADILLAC, MICHIGAN

EC.JORDANCO

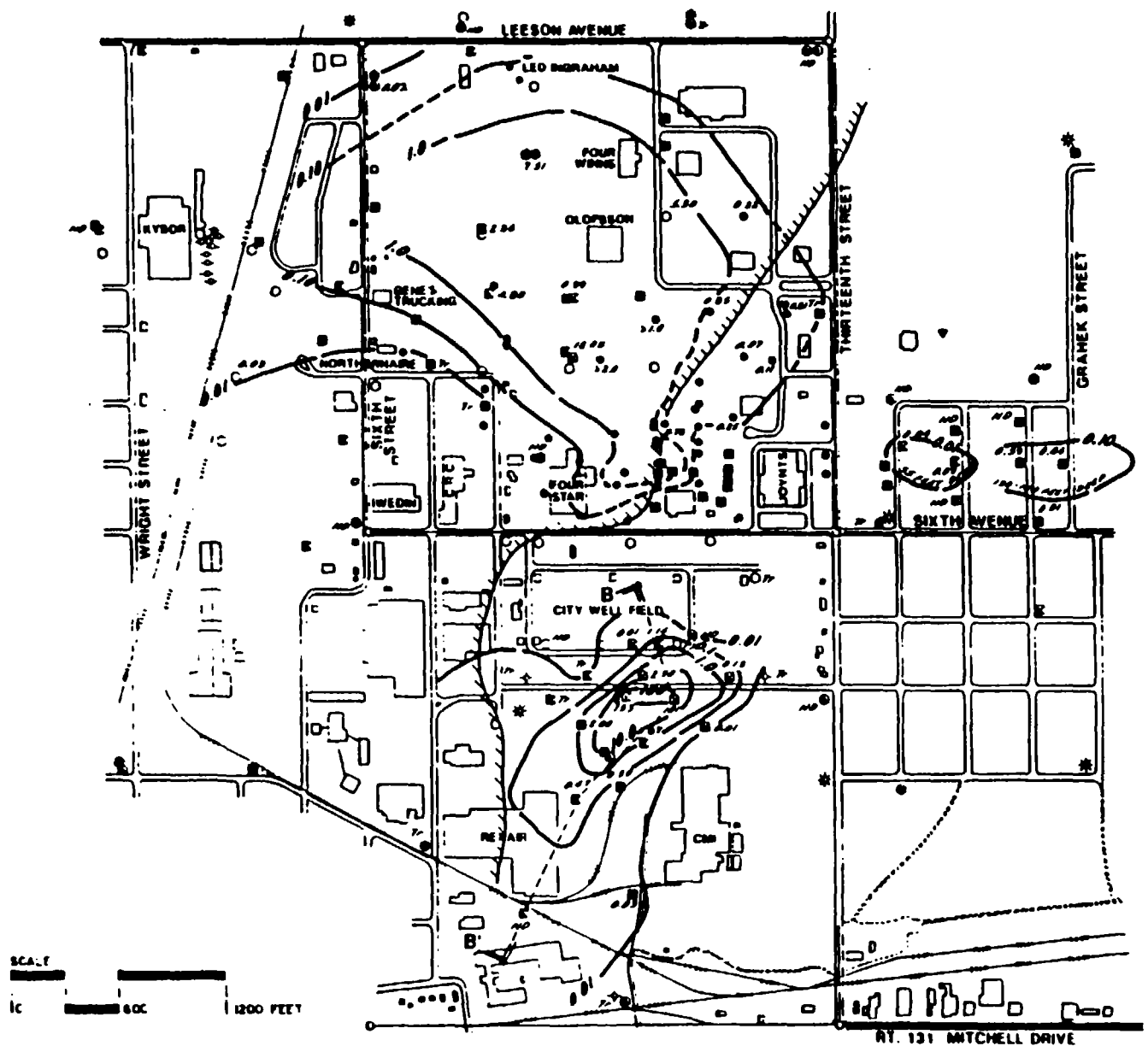


- LEGEND**
- FOUR STAR WELLS
 - D.M.R. WELLS
 - KYSON WELLS
 - M.W. WELLS (NORTHERNAIRE)
 - CITY WELL HOUSES
 - SOURCE LOCATION WELLS
 - GENERAL WELLS
 - ▼ PIEZOMETERS
 - * MUD BORINGS
 - KYSON SOIL BORINGS
 - ✦ MUD ROTARY WELLS
 - - - 1.0 VOC CONCENTRATION CONTOUR IN ppm
 - ____ 2.50 VOC CONCENTRATION IN ppm
 - |||| NORTHERN EXTENT OF UPPER AQUIFARD

NOTE: DATA USED FOR PLUME INTERPRETATIONS
ARE CLP ANALYSES

FIGURE 1-2
SHALLOW AQUIFER AND TOTAL
VOLATILE ORGANIC COMPOUND
CONCENTRATIONS: JUNE 1987
CADILLAC AREA FS
CADILLAC, MICHIGAN
E.C. JORDAN CO.

2



- LEGEND**
- FOUR STAR WELLS
 - D.A.R. WELLS
 - KYSOR WELLS
 - M.W. WELLS (NORTHERNAIRE)
 - CITY WELL HOUSES
 - SOURCE LOCATION WELLS
 - GENERAL WELLS
 - ▼ PIEZOMETERS
 - * MUD BORINGS
 - ◆ KYSOR SOIL BORINGS
 - ◆ MUD ROTARY WELLS
 - 1.0 VOC CONCENTRATION CONTOUR IN ppm
 - ||||| NORTHERN EXTENT OF UPPER AQUIFARD
 - 10.0 VOC CONCENTRATION IN ppm

NOTES. SLW-53, SLW-210, MRW-1, MRW-2, AND MRW-3 ANALYSES ARE FROM AUGUST 1987 SAMPLING EVENT, CLP ANALYSES.

MW WELLS INCLUDE NORTHERNAIRE VERTICAL SCREENING DATA SINCE WELL SCREEN IS NOT IN HIGHEST ZONE OF VOC CONCENTRATION

FIGURE 1-3
INTERMEDIATE AQUIFER AND TOTAL
VOLATILE ORGANIC COMPOUND
CONCENTRATIONS: MAY/JUNE 1987
CADILLAC AREA FS
CADILLAC, MICHIGAN
EC JORDANCO

Rexair Plume. The Rexair plume is a concentrated VOC plume (up to 153 ppm) at depths to 140 feet, located in the intermediate aquifer between Rexair and the city wellfield. The Rexair plume is depicted in Figure 1-3.

Other Plumes. The RI also identified two small VOC plumes of relatively low concentration (up to 0.10 ppm) in the North Park Subdivision (see Figure 1-3). These plumes are located horizontally downgradient of the city wellfield and appear to pose a less immediate threat (if any) to the city water supply. Municipal water was extended and connections were made to all homes within the entire North Park Subdivision. Therefore, these plumes were not assessed as a potential public health risk in the baseline RA. However, 85 wells within this subdivision were not abandoned; therefore, remediation of these two plumes is included in the FS as a possible additional component of each groundwater alternative.

Soils. The former solvent disposal area behind Kysor was positively identified in the RI as having significant residual VOC contamination. This area appears to be a continuing source of groundwater contamination in the shallow and intermediate aquifers southwest of the city wellfield. Figure 1-4 depicts the interpretive extent of soil contamination at Kysor. In addition, soils at Four Winns are assumed to be a potential continuing source of groundwater contamination. However, because Four Winns, Inc., elected to participate as a potentially responsible party in the study with its own consultant, Jordan was not able to completely assess soil contamination at Four Winns as a source of groundwater contamination. Jordan's review of soil data collected by Four Winns' consultant concluded that the sampling and analysis was not extensive and, therefore, some uncertainty remains as to the degree to which the soils below Four Winns are contaminated and may serve as a continuing source of contamination to groundwater. Based on 1985 data from MDNR, Jordan has concluded that concentration levels in Four Winns soils could require remediation. Figure 1-5 depicts the interpretive extent of soil contamination at Four Winns.

Another area of soils contamination that contributed to the groundwater contamination through leaching is the Northernnaire site. An RI and FFS that evaluated remedial alternatives for Northernnaire soils was conducted. Remediation of the Northernnaire soils is currently scheduled. Other sources of soils contamination could be present at the Ingraham property and at Rexair. Studies further defining plumes and sources at Rexair are being conducted by a consultant engaged by Rexair. Other explorations at Ingraham will be performed by MDNR under another contract to MDNR.

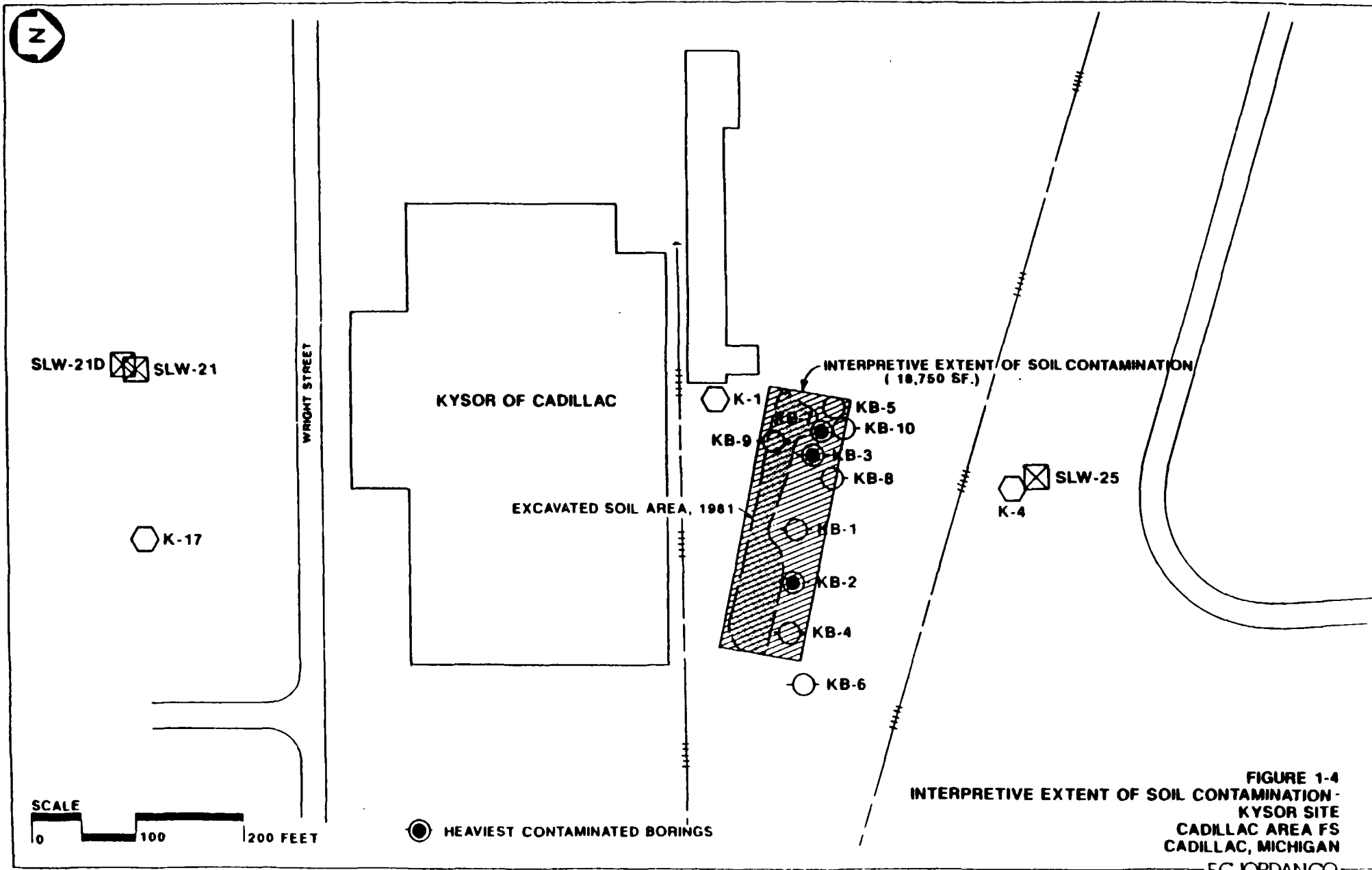
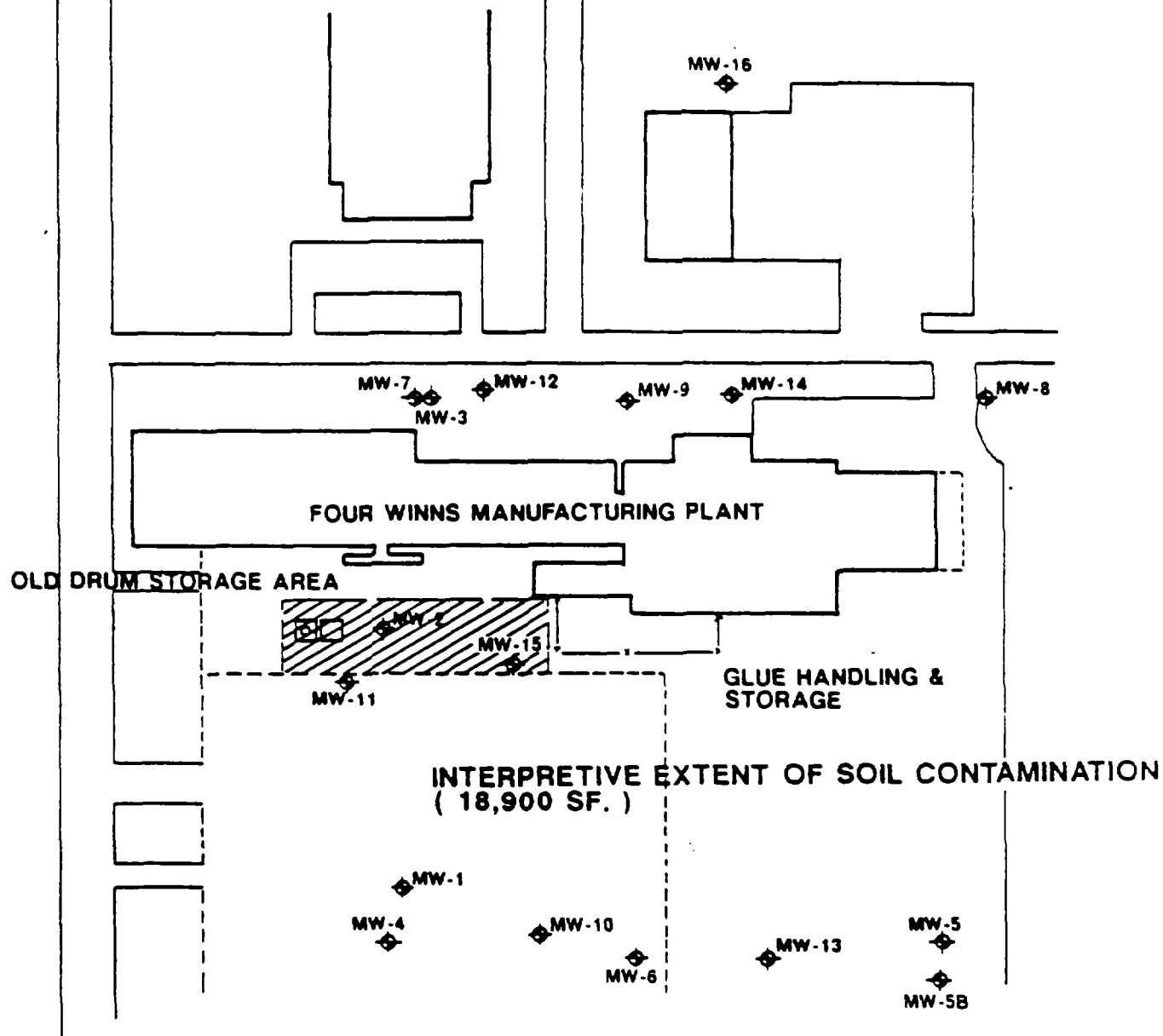


FIGURE 1-4
INTERPRETIVE EXTENT OF SOIL CONTAMINATION -
KYSOR SITE
CADILLAC AREA FS
CADILLAC, MICHIGAN

ECJORDANCO



MW-17



SCALE
0 100 200 FEET

FIGURE 1-5
INTERPRETIVE EXTENT OF SOIL CONTAMINATION -
FOUR WINNS SITE
CADILLAC AREA FEASIBILITY STUDY
CADILLAC, MICHIGAN

EC 1000-1000

2.0 IDENTIFICATION OF REMEDIAL ACTION OBJECTIVES

Remedial action objectives and site-specific considerations must be identified as a basis for selecting the appropriate remedial technologies and formulating remedial alternatives. Remedial action objectives are statements of the proposed level of cleanup associated with the specific chemical contamination problems in the various media at a site. Remedial action objectives are aimed at protecting public health and the environment, and are based on the contaminant(s) of concern, exposure route(s), and receptor(s); they are acceptable contaminant levels or ranges of levels for each exposure route. ARARs that establish clean-up standards are also used to determine remedial action objectives.

Section 2.1 identifies the general response objectives based on results of the RA and other considerations in the NCP 40 CFR 300.68(e)(2). Chemical-specific ARARs, previously identified in the RI, are summarized in Section 2.2. These ARARs are used in conjunction with the general objectives to formulate specific target levels for groundwater and soil remediation. These target levels are presented in Sections 2.3 and 2.4. Discharge requirements for potential discharges of treated groundwater to surface water, publicly owned treatment works (POTWs), or recharge to groundwater are presented in Section 2.5; the requirements are used to determine the appropriate level of treatment before discharge.

2.1 GENERAL RESPONSE OBJECTIVES

In the RA, the only exposure pathway identified as presenting a potential threat to public health is the domestic use of contaminated groundwater, including drinking, showering, bathing, and cooking. Because each contaminated groundwater plume identified and evaluated in the RA is considered an existing or potential source of drinking water, the following separate remedial action objectives are proposed for each plume (i.e., the shallow plume, intermediate plume, Northernnaire plumes, and Rexair plume).

Shallow VOC Plume: Reduce public health risks posed by potential use of the shallow groundwater plume as a domestic water supply (including drinking). The shallow plume is contaminated with 1,1,1-trichloroethane (1,1,1-TCA); 1,1-dichloroethylene (1,1-DCE); TCE; 1,2-dichloroethane (1,2-DCA); and tetrachloroethylene (PCE).

Intermediate VOC Plume: Reduce public health risks posed by the potential use of the intermediate groundwater plume (which is contaminated with trans-1,2-dichloroethylene [t-1,2-DCE]; TCE; 1,1-DCE; and PCE) as a domestic drinking water supply.

Northernnaire Plumes: Reduce public health risks posed by the potential use of the hexavalent chromium-contaminated groundwater in the Northernnaire plumes as a domestic drinking water supply.

Rexair Plume: Reduce public health risks posed by the potential use of the Rexair groundwater plume, primarily contaminated with TCE and methylene chloride (MC), as a domestic drinking water supply.

As discussed in the RA, the residual soil contamination at Kysor does not pose a direct contact threat because the contaminated soil is subsurface. However, the RA did assess residual soil contamination at Kysor as a continuing source of groundwater contamination, based on concentrations of the contaminants and their ability to leach to groundwater. Adequate data were not available to assess existing soil contamination at Four Winns as a source of groundwater contamination. Therefore, it is presently unknown whether a direct contact threat exists at Four Winns. However, based on 1985 MDNR soil data and 1987 soil data collected by Four Winns' consultant, it is assumed that concentration levels at Four Winns are high enough to warrant remediation. Based on this assessment, the following remedial action objectives are proposed.

Kysor: Reduce potential public health risks posed by the continued leaching of TCE, toluene, and 1,1,1-TCA from subsurface soils at Kysor to a potential drinking water source. Reduce potential public health risks posed by xylene leaching from subsurface soils at Kysor and potentially contaminating drinking water source.

Four Winns: Reduce potential public health risks posed by the potential for continued leaching of 1,1,1-TCA, toluene, TCE, and MC from subsurface soils at Four Winns to a potential drinking water source.

2.2 SUMMARY OF CHEMICAL-SPECIFIC APPLICABLE OR RELEVANT AND APPROPRIATE REQUIREMENTS

As discussed in the Cadillac RI, ARARs are federal and state public health and environmental requirements and guidelines that apply to hazardous waste site cleanup. Section 121 of CERCLA, amended by SARA, requires that hazardous waste remedial actions comply with federal ARARs and state ARARs that are more stringent, legally enforceable, and consistently enforced statewide. ARARs are used to determine the appropriate extent of site cleanup, to scope and formulate remedial action alternatives, and to govern the implementation and operation of the selected action.

Applicable requirements and relevant and appropriate requirements are defined as follows:

Applicable Requirements are those clean-up standards, standards of control, or other substantive environmental requirements, criteria, or limitations promulgated under federal or state law that specifically address a hazardous substance at a CERCLA site. An example of an applicable requirement is the use of Maximum Contaminant Levels (MCLs), promulgated under the Safe Drinking Water Act (SDWA), for a site where groundwater contamination is entering a public water supply.

Relevant and Appropriate Requirements are federal- and state-promulgated clean-up standards, standards of control, or other substantive environmental protection requirements, criteria, or limitations that, while not legally applicable, can be applied if, in the decision-maker's best professional judgment, site circumstances are similar to jurisdictionally covered situations, and use of the requirements makes good sense. The term "relevant" is included so that requirements initially screened out as nonapplicable because of jurisdictional restrictions can be reconsidered. For example, MCLs would be relevant and appropriate requirements for use at a site where groundwater contamination could affect a potential, as opposed to an actual, drinking water source. Relevant and appropriate requirements should be given the same weight for consideration as applicable requirements.

To Be Considered Materials (TBCs) are federal and state nonpromulgated advisories or guidance documents that do not have ARAR status. However, if there are no specific ARARs for a chemical or situation, or if existing ARARs are not deemed sufficiently protective, then guidance or advisory criteria should be identified and used to better ensure public health and environmental protection.

As discussed in the RI, ARARs may be characterized as chemical-, location-, or action-specific. Chemical-specific requirements are used to determine the remedial action objectives because they set health- or risk-based concentration limits or discharge limitations in various environmental media for specific hazardous substances, pollutants, and contaminants. Chemical-specific requirements govern the extent of site remediation by providing either actual clean-up levels or a basis for calculating such levels. If a chemical has more than one requirement that is an ARAR, the most stringent generally should be attained. If no ARAR exists, or if the ARAR for a substance is insufficient to be protective, then the federal or state TBC should be used in conjunction with the RA to determine the appropriate clean-up level.

Chemical-specific ARARs for the Cadillac site were identified and discussed in the RI. ARARs that set concentration limits for groundwater contaminants include SDWA MCLs, SDWA Maximum Contaminant Level Goals (MCLGs), and federal Ambient Water Quality Criteria (AWQC). In addition, Michigan's Act 245 and Rule 323, Part 22 Ground Water Quality Rules, provide for the non-degradation of groundwater in usable aquifers. MDNR requires the use of Secondary MCLs when available in order to reflect the "usable" standard of Michigan's Act 245 and Part 22 rules. TBCs include USEPA Health Advisories, USEPA Risk Reference Doses, and USEPA Carcinogen Assessment Group (CAG) Potency Factors. No federal or state regulatory requirements currently govern the extent of cleanup for contaminants in soil. Chemical-specific ARARs that set limitations for discharges to surface water and POTWs were also discussed in the RI and are addressed in Section 2.5. Chemical-specific ARARs and TBCs for the Cadillac site and a synopsis of the requirements are listed in Appendix A.

Actual required concentration levels for the groundwater, which are used to set groundwater remedial action objectives, are listed in Table 2-1, and are compared to the maximum concentrations of groundwater chemicals identified as

TABLE 2-1
CADILLAC AREA GROUNDWATER STUDY: GROUNDWATER
CHEMICAL-SPECIFIC ARARS COMPARISON

CADILLAC AREA FEASIBILITY STUDY

COMPOUND	ARARS (mg/l)			EXISTING MAXIMUM CONCENTRATIONS (mg/l)			
	MCL	MCLG	AWQC*	REXAIR	UPPER AQUIFER	INTERMEDIATE AQUIFER	NORTHERNAIRE PLUME ¹
1,1,1-Trichloroethane	0.2	0.2	18.4	0.017	32.0	1.2	0.44
Trans-1,2-dichloroethylene	--	0.07**	--	0.048	--	4.6	0.04
1,1-Dichloroethylene	0.007	0.007	0.00033	--	1.0	0.099	0.026
1,2-Dichloroethane	0.005	0	0.00094	--	2.0	--	--
Methylene chloride	--	--	0.00019	0.078	0.69	0.24	--
Tetrachloroethylene	--	0**	0.0008	--	0.72	0.17	0.003
Trichloroethylene	0.005	0	0.0028	153.0	78.0	7.5	2.5
Chromium (hexavalent)	0.05	0.12**	0.05	--	--	--	0.773
Xylene	--	0.44**	--	--	1.7	--	--
Toluene	--	2.0**	14.3	0.042	3.2	0.001	--

* AWQC are adjusted for drinking water exposure only (USEPA, 1986) or calculated for water and fish ingestion if drinking water exposure value does not exist.

** Proposed levels.

¹ The VOCs present within the boundaries of the Northernnaire plume are not derived from Northernnaire; the chromium from Northernnaire and VOCs from other sources have mixed in the aquifer.

NOTES: AWQC = Ambient Water Quality Criteria; MCL = Maximum Contaminant Level; MCLG = Maximum Contaminant Level Goal

contributing the most to total risk in each plume. Because contaminants in the soil are either currently leaching or have the future potential to leach to groundwater, concentrations of chemicals that are currently not a problem in groundwater, but which are found at high concentrations in soil, are also compared to groundwater ARARs.

2.3 TARGET LEVELS FOR GROUNDWATER REMEDIATION

Groundwater target clean-up levels (TCLs) are the chemical concentrations in each plume that would be protective of public health if long-term exposures were to occur. The baseline RA for the Cadillac site (Section 8.0 of the RI) provides a basis for determining which chemicals of concern should be targeted for cleanup in each plume. It is not necessary to calculate target levels for every chemical found in the plumes. Cleanup should focus on those chemicals that have been shown in the RA to pose "significant" risks to public health (i.e., carcinogenic risks within or above the target range of 10^{-4} to 10^{-7} , or noncarcinogenic risks greater than 1). This section describes the approach to development of groundwater target levels and presents the target levels for each plume.

2.3.1 Approach

The first step in the development of target levels for groundwater remediation at Cadillac is to identify the chemicals in each plume that present "significant" risks in the baseline RA (as defined previously, carcinogenic risks within or above the target range or noncarcinogenic risks greater than 1). For the Cadillac area groundwater contamination, these chemicals are as follows:

- o Shallow VOC Plume

- 1,1,1-TCA

- 1,1-DCE

- TCE

- 1,2-DCA

- PCE

- o Intermediate VOC Plume

- t-1,2-DCE

- TCE

- 1,1-DCE

- PCE

o Rexair Plume

TCE

MC

o Northernnaire Plume

Hexavalent Chromium (Cr[+6])

Preliminary target levels are calculated for carcinogens on the basis of public health risk calculations. Preliminary target levels for noncarcinogens are defined on the basis of health-based ARARs or, if ARARs do not exist, public health risk calculations. It should be noted that two VOCs included in the RI as chemicals presenting "significant" risks in the Northernnaire plume (TCE and 1,1-DCE) are not included in the FS in the same manner. These VOCs are believed to be intermingling from the larger generalized intermediate plume and will be addressed as part of the intermediate plume.

For noncarcinogens that do not have ARARs, target levels are calculated so that the resultant risk ratio is less than 1. The following equations are used:

$$\text{Target Risk for Individual Compound} = \frac{1}{\text{Total Number of Chemicals}}$$

$$\text{Target Chronic Daily Intake (CDI) (mg/kg/day)} = \text{Target Risk} \times \text{RfD (mg/kg/day)}$$

$$\text{Target Level (mg/l)} = \frac{\text{Target CDI (mg/kg/day)}}{\text{Human Intake Factor (l/kg/day)}}$$

WHERE:

$$\begin{aligned} \text{Human Intake Factor} &= \frac{\text{Amount of Water Consumed per Day}}{\text{Average Adult Body Weight}} \\ &= \frac{2 \text{ l/day}}{70 \text{ kg}} = 0.029 \end{aligned}$$

Target levels for carcinogens are developed to fall within a total risk range of 10^{-4} to 10^{-7} because, as stated in the Superfund Public Health Evaluation Manual, site remedies should reduce ambient chemical concentrations to levels within that range when possible (USEPA, 1986). The formulas used to calculate preliminary target levels for carcinogens are as follows:

$$\begin{aligned} \text{Target Risk for Individual Compound} &= \frac{\text{Target Risk Level } (10^{-4} \text{ to } 10^{-7})}{\text{Total Number of Chemicals}} \end{aligned}$$

(The 10^{-6} target risk level will be used to calculate preliminary target levels in accordance with MDNR policy.)

$$\text{Target Chronic Daily Intake (CDI, mg/kg/day)} = \frac{\text{Target Risk}}{\text{CAG Potency Factor (mg/kg/day)}^T}$$

$$\text{Target Level (mg/l)} = \frac{\text{Target CDI (mg/kg/day)}}{\text{Human Intake Factor (l/kg/day)}}$$

(Human intake factor is the same as for carcinogens.)

In addition to the levels calculated based on risk, health-based ARARs need to be considered in the development of target levels for carcinogens. Several options are presented for each groundwater plume that incorporate risk levels, ARARs, or a combination of the two.

Preliminary TCLs provide options that are used in the FS to assist in the selection and screening of remedial technologies, and in the analysis of risks for remedial alternatives. The preliminary TCLs are revised in Section 2.3.3 in response to engineering, cost, policy, and/or environmental considerations. Final TCLs, therefore, will achieve a specified total target risk, but may also reflect public health, environmental, engineering, policy, and cost considerations. This approach, which integrates several considerations, allows for the most efficient reduction of risk.

2.3.2 Development of Preliminary Groundwater Target Clean-up Levels

The following subsections summarize the identification of preliminary TCLs for groundwater contaminants in each plume.

2.3.2.1 Shallow VOC Plume. Six chemicals of concern, including two non-carcinogens and four carcinogens, in the shallow VOC plume were identified in the RA as presenting potentially significant risks to public health. Both noncarcinogens have available health-based ARARs. The preliminary TCLs (based on the 10^{-6} risk level for carcinogens and ARARs for noncarcinogens) for the shallow plume contaminants are in Table 2-2.

The TCLs for carcinogens that would be associated with a total risk of 1×10^{-6} were calculated and are presented in Table 2-3 (Option 1). These TCLs are extremely low (ranging from 0.015 ug/l for 1,1-DCE to 0.8 ug/l for TCE), and are below the detection limits for these chemicals. Health-based ARARs were also identified for the four carcinogens. All have ARARs available. However, PCE only has an AWQC (0.0008 mg/l), which is below detection limits. Therefore, the TCL corresponding to the 10^{-6} risk level for PCE alone was used. The residual risks associated with these levels for the four carcinogens in the shallow plume are calculated and presented in Table 2-3 (Option 2). The total risk associated with the plume at health-based ARARs is 1.32×10^{-4} , most of which is associated with exposure to 1,1-DCE. This is quite high for a potential drinking water source. Therefore, a third option was considered, where TCE, 1,2-DCA, and PCE levels remain as in Option 2, but the TCL for 1,1-DCE is adjusted to the contract required detection limit (CRDL). The total residual risk associated with Option 3 is 9×10^{-5} ; this is within the target risk range. The residual risks associated with each of these options are also presented

TABLE 2-2
 CADILLAC AREA GROUNDWATER CONTAMINATION
 PRELIMINARY TARGET LEVELS
 SHALLOW PLUME

CADILLAC AREA FEASIBILITY STUDY

CHEMICAL	(mg/l)	SOURCE
1,1,1-TCA	0.2	MCL
Toluene	0.04	SMCL
1,1-DCE	0.00001	Calculated
TCE	0.0008	Calculated
1,2-DCA	0.0001	Calculated
PCE	0.0002	Calculated

NOTES:

MCL = Maximum Contaminant Level

MCLG = Maximum Contaminant Level Goal

SMCL = Secondary Maximum Contaminant Level

TABLE 2-3
RESIDUAL RISK ASSOCIATED WITH TARGET CLEAN-UP LEVELS
GROUNDWATER INGESTION OF SHALLOW PLUME

CADILLAC AREA FEASIBILITY STUDY

SHALLOW PLUME: OPTION 1

Calculation of Preliminary Target Levels at 10^{-6} Risk Level

Compound	MF	Target Risk	Potency Factor ($\mu\text{g/kg/day}$)	Target CDI ($\mu\text{g/kg/day}$)	Amount Contacted (l/day)	Amount Ingested (l/day)	Dermal Absorption Fraction	Ingestion Absorption Fraction	Frequency of Contact (days)	Fraction of Year Exposed	Years of Exposure	Fraction of Lifetime Exposed	Body Weight (kg)	Dermal Intake (l/kg/day)	Ingestion Intake (l/kg/day)	Target Level ($\mu\text{g/l}$)
1,1-DCE	2.50E-01	2.50E-07	5.00E-01	4.31E-07	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	17.00E+01	0.00E+00	2.04E-02	11.51E-02
TCE	2.50E-01	2.50E-07	1.10E-02	2.27E-05	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	17.00E+01	0.00E+00	2.04E-02	7.95E-01
1,2-DCA	2.50E-01	2.50E-07	9.10E-02	7.75E-06	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	17.00E+01	0.00E+00	2.04E-02	9.62E-02
PCE	2.50E-01	2.50E-07	5.20E-02	4.81E-06	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	17.00E+01	0.00E+00	2.04E-02	1.68E-01

SHALLOW PLUME: OPTION 2

Residual Risks Associated with 1,1-DCE, TCE, 1,2-DCA at MCLs; PCE at 10^{-6} Risk Level

Compound	Concentration ($\mu\text{g/l}$)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure	Fraction of Lifetime Exposed	Body Dose ($\mu\text{g/kg/day}$)	Micrograms per milligram	CAB Potency Estimate ($\mu\text{g/kg/day}$) ⁻¹	Risk Estimate
1,1-DCE	7.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.00E-01	1.00E+03	5.00E-01	1.16E-04
TCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	1.10E-02	1.57E-06
1,2-DCA	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	9.10E-02	1.30E-05
PCE	1.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.86E-02	1.00E+03	5.20E-02	1.49E-06
											TOTAL RISK	1.52E-04

SHALLOW PLUME: OPTION 3

Residual Risks Associated with: TCE and 1,2-DCA at MCLs; 1,1-DCE at CML, and PCE at 10^{-6} Risk Level

Compound	Concentration ($\mu\text{g/l}$)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure	Fraction of Lifetime Exposed	Body Dose ($\mu\text{g/kg/day}$)	Micrograms per milligram	CAB Potency Estimate ($\mu\text{g/kg/day}$) ⁻¹	Risk Estimate
1,1-DCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	5.00E-01	8.29E-05
TCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	1.10E-02	1.57E-06
1,2-DCA	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	9.10E-02	1.30E-05
PCE	1.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.86E-02	1.00E+03	5.20E-02	1.49E-06
											TOTAL RISK	9.89E-06

graphically in Figure 2-1. Final TCLs for the shallow plume are identified in Section 2.3.3.

2.3.2.2 Intermediate VOC Plume. Four chemicals of concern in the intermediate VOC plume were identified in the RA as presenting potentially significant risks to public health. These include one noncarcinogen and three carcinogens. Preliminary TCLs for contaminants in the intermediate plume corresponding to health-based ARARs for the noncarcinogen and a 10^{-6} risk level for carcinogens are presented in Table 2-4.

Residual risks associated with preliminary TCLs were also calculated for the intermediate plume in a manner similar to the shallow plume. The risks associated with Options 1, 2, and 3 are presented in Table 2-5 and depicted graphically in Figure 2-2. As with the shallow plume, most of the residual risk is posed by 1,1-DCE; therefore, its TCL was adjusted to the CRDL in Option 3. Final TCLs for the intermediate VOC plume are identified in Section 2.3.3.

2.3.2.3 Rexair Plume. Although most of the risk posed by the Rexair plume is associated with TCE, MC also presented a risk within or above the target range for several exposure scenarios in the RA. Therefore, preliminary TCLs corresponding to the 10^{-6} risk level were identified for both chemicals and are presented in Table 2-6.

Health-based ARARs were also identified for these two chemicals. TCE has an applicable ARAR, a final MCL of 0.005 mg/ℓ. However, MC only has an AWQC (0.00019), which is far below practical detection limits. Therefore, the concentration that corresponds to the 10^{-6} risk level for MC alone was calculated.

The residual risks associated with the 10^{-6} risk level (Option 1) or health-based ARARs (Option 2) for TCE and MC in the Rexair plume are calculated in Table 2-7 and presented graphically in Figure 2-3. The total residual risk for Option 2 is 2.64×10^{-6} ; this is well within the target risk range. Therefore, the preliminary TCL analyses for the Rexair plume will not be taken further.

2.3.2.4 Northernnaire Plume. The chemical of concern that poses a potentially significant risk in the Northernnaire plume is hexavalent chromium (Cr(+6)). This chemical is not carcinogenic via the ingestion route, and it has an applicable health-based ARAR, a final MCL of 0.05 mg/ℓ. Therefore, the preliminary TCL for Cr(+6) is set at its MCL.

2.3.3 Final Groundwater Target Clean-up Levels

The preliminary groundwater TCLs were revised as part of the FS process, in response to MDNR precedent. The final TCLs achieve a specified total risk range and reflect public health, environmental, engineering, and policy considerations. In general, MDNR precedent requires that TCLs for carcinogens are set as close to the 10^{-6} risk level as is technically possible to achieve. However, in many cases, a 10^{-6} risk level is below practical detection limits. Therefore, the final groundwater TCLs for the Cadillac area groundwater contamination were set as close to the 10^{-6} risk level as technically possible, using a combination of ARARs, risk-based levels, and/or the CRDLs. USEPA Region V

FIGURE 2-1. SHALLOW PLUME CONTAMINATION

Residual Risk Associated w/ TCL Options

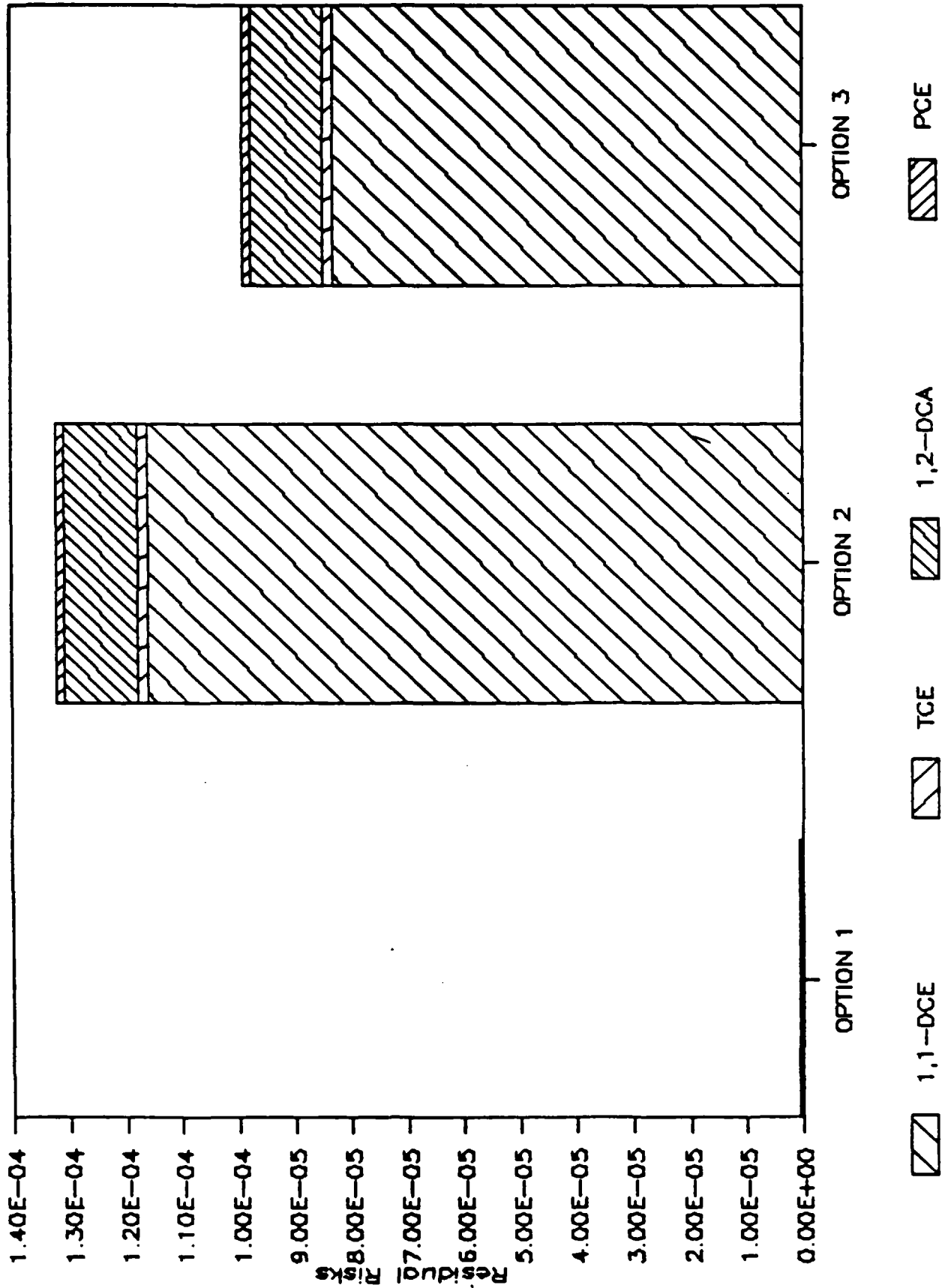


TABLE 2-4
CADILLAC AREA GROUNDWATER CONTAMINATION
PRELIMINARY TARGET LEVELS
INTERMEDIATE PLUME

CADILLAC AREA FEASIBILITY STUDY

CHEMICAL	(mg/l)	SOURCE
T-1,2-DCE	0.07	MCLG
TCE	0.001	Calculated
1,1-DCE	0.00002	Calculated
PCE	0.0002	Calculated

NOTE:

MCLG = Maximum Contaminant Level Goal

CADILLAC AREA FEASIBILITY STUDY

Calculation of Preliminary Target Levels at 10⁻⁶ Risk Level

[illegible]Residual Risks Associated with 1,1-DCE, TCE at HCLs; PCE at 10⁻⁶ Risk Level[illegible]

Residual Risks Associated with: ICE at MEL, 1,1-DCE at CROU, and PCE at 10-6 Risk Level

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure (years)	Fraction of Lifetime Exposed	Body Dose (ug/kg/day)	Micrograms per milligram	CAS Potency Estimate (ug/kg/day)-1	Risk Estimate
1,1-DCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	5.00E-01	8.29E-05
PCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	1.10E-02	1.57E-06
PCE	1.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.86E-02	1.00E+03	5.20E-02	1.49E-06
											TOTAL RISK	9.59E-05

FIGURE 2-2. INTER. PLUME CONTAMINATION

Residual Risk Associated w/ TCL Options

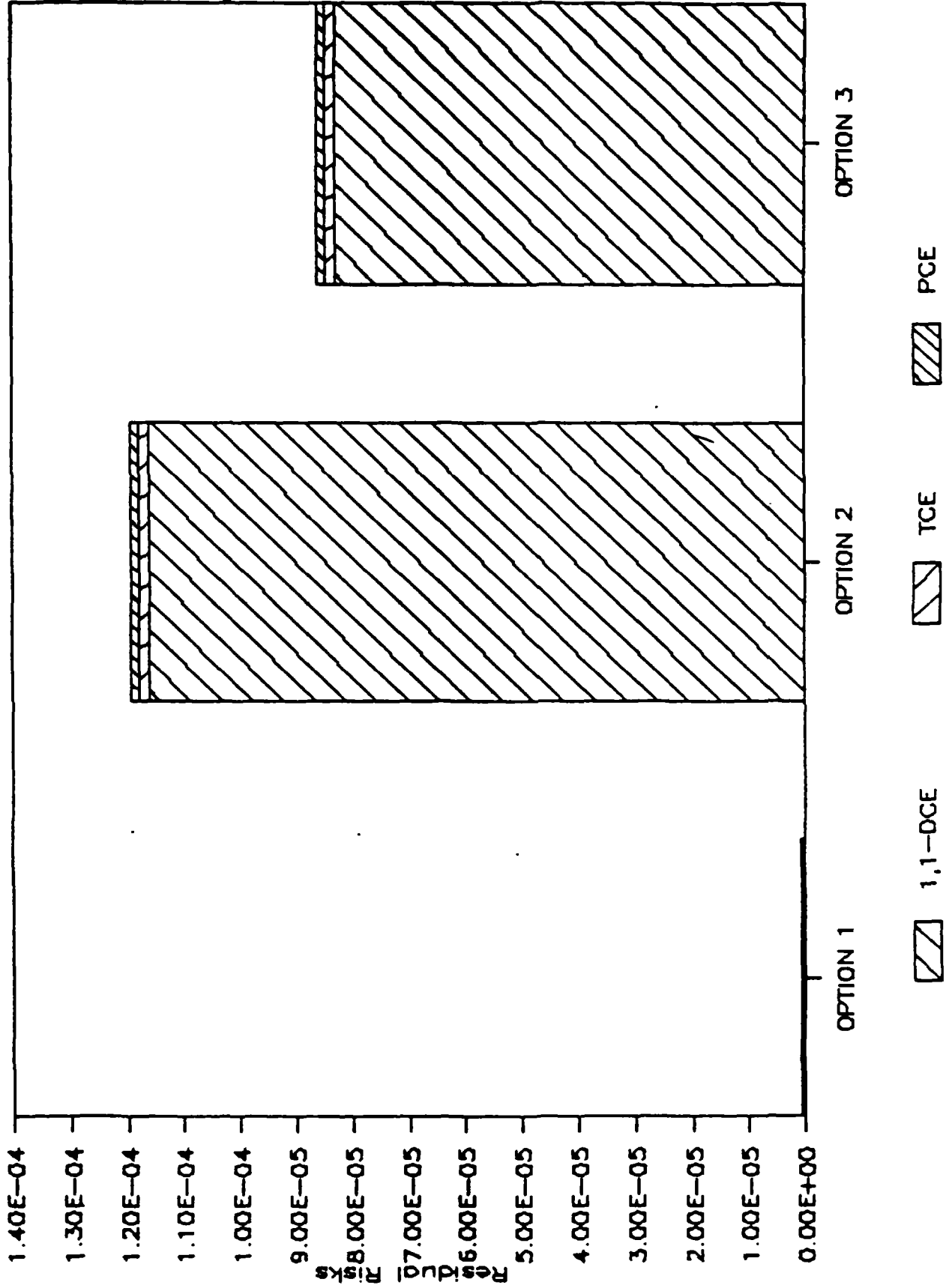


TABLE 2-6
CADILLAC AREA GROUNDWATER CONTAMINATION
PRELIMINARY TARGET LEVELS
REXAIR PLUME

CADILLAC AREA FEASIBILITY STUDY

CHEMICAL	(mg/l)	SOURCE
TCE	0.002	Calculated
MC	0.002	Calculated

CADILLAC AREA FEASIBILITY STUDY

Calculation of Preliminary Target Levels at 10⁻⁶ Risk Level

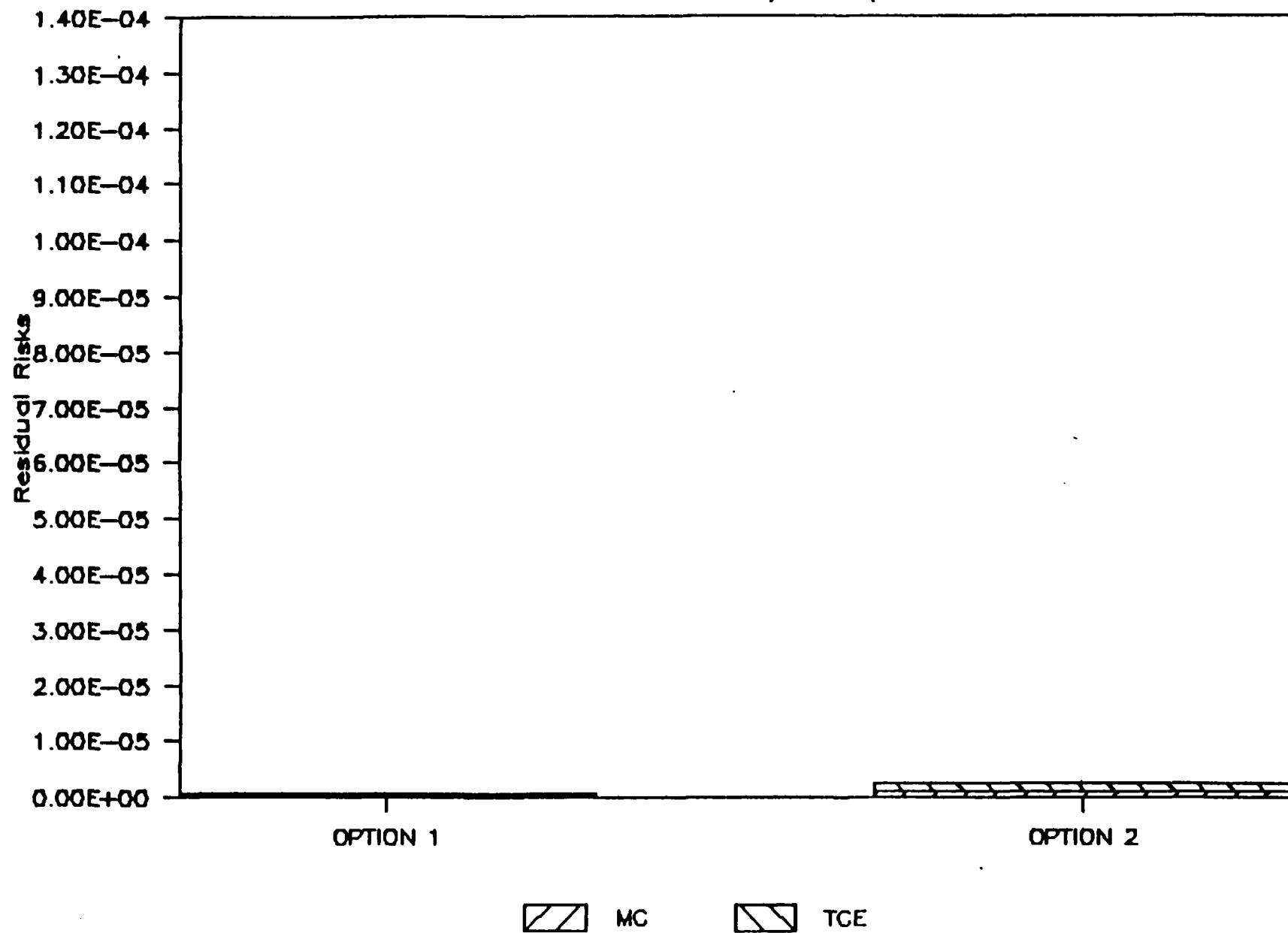
[illegible]

Residual Risks Associated with TCE at MCL, NC at 10^{-6} Risk Level

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure (years)	Fraction of Lifetime Exposed	Body Dose (ug/kg/day)	Micrograms per milligram	CAS Potency Estimate (ug/kg/day)-1	Risk Estimate
HC	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	7.50E-03	1.07E-04
TCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	1.10E-02	1.57E-04
											TOTAL RISK	2.64E-04

FIGURE 2-3. REXAIR PLUME CONTAMINATION

Residual Risk Associated w/ TCL Options



was consulted, and concurred with this decision. Option 3 was used to define final TCLs for the shallow and intermediate groundwater plumes (see Tables 2-3 and 2-5), and Option 2 was used to define final TCLs for the Rexair plume (see Table 2-7).

Final TCLs for groundwater remediation at the Cadillac site are listed in Table 2-8.

2.4 TARGET CLEAN-UP LEVELS FOR SOIL REMEDIATION

The general response objective for remediation of soils at the Cadillac Industrial Park is to reduce residual contamination to levels that will no longer significantly contribute to the groundwater contamination. This section describes the approach to development of soil TCLs, and presents the TCLs for soils in both the Kysor and the Four Winns areas.

2.4.1 Approach

The first step in the development of soil TCLs is to identify the chemicals in each area that are or have the potential to significantly impact groundwater. This was accomplished for Kysor soils by identifying the chemicals of concern that (1) are present in the soils at high concentration, and (2) are posing "significant" risks (or are at levels above ARARs) in the shallow plume. Using these criteria, TCE, toluene, and 1,1,1-TCA are included in this TCL evaluation. Xylene was also included because it was found at the highest concentrations in the Kysor soils. Adequate current data are not available to assess Four Winns soils. However, soil TCLs are developed for those chemicals that may be in the soils at significant concentrations, based on 1985 and limited 1987 data. These chemicals include 1,1,1-TCA, toluene, TCE, and MC in the Four Winns soils.

There are currently no health-based ARARs for these chemicals in soils. Therefore, all preliminary soil TCLs were calculated on the basis of risk. This was accomplished by using the final groundwater TCLs identified for these chemicals in Section 2.3.3, and back-calculating to a "safe" concentration in the soil using the defining equation for the USEPA Organic Leaching Model (OLM). This equation is as follows:

$$C_w = 0.00221 \times C_s^{0.678} \times S^{0.373}$$

WHERE:

C_w = TCL in water

C_s = TCL in soil

S = Contaminant's water solubility at ambient temperature

TABLE 2-8
CADILLAC AREA GROUNDWATER CONTAMINATION
FINAL GROUNDWATER TARGET LEVELS

CADILLAC AREA FEASIBILITY STUDY

PLUME	CHEMICAL	TARGET LEVEL (mg/l)	REFERENCE
Shallow	1,1,1-Trichloroethane	0.2	MCL
	Toluene	0.04	SMCL
	1,1-Dichloroethylene	0.005	CRDL
	Trichloroethylene	0.005	MCL
	1,2-Dichloroethane	0.005	MCL
	Tetrachloroethylene	0.001	Calculated
Intermediate	Trans-1,2-dichloroethylene	0.07	MCLG
	1,1-Dichloroethylene	0.005	CRDL
	Trichloroethylene	0.005	MCL
	Tetrachloroethylene	0.001	Calculated
Rexair	Methylene chloride	0.005	Calculated
	Trichloroethylene	0.005	MCL
Northernaire	Chromium (hexavalent)	0.05	MCL

NOTES:

CRDL = Contract Required Detection Level
MCL = Maximum Contaminant Level
MCLG = Maximum Contaminant Level Goal
SMCL = Secondary Maximum Contaminant Level

2.4.2 Development of Soil Target Clean-up Levels

The following subsections summarize the identification of TCLs for soils in the Cadillac Industrial Park.

2.4.2.1 Kysor. The parameters necessary to calculate TCLs for chemicals in the Kysor soils, and the TCLs themselves, are summarized in Table 2-9. Xylene did not have a preliminary groundwater TCL, as it was not identified as posing a significant risk in the baseline RA. Therefore, a health-based groundwater TCL was identified for xylene, based on its proposed MCLG of 0.44 mg/l. TCLs for Kysor soils range from 0.07 ppm for TCE to 724 ppm for toluene.

2.4.2.2 Four Winns. Table 2-10 summarizes the parameters and preliminary TCLs for chemicals in the Four Winns soils. TCLs for Four Winns soils range from 0.02 ppm for MC to 724 ppm for toluene.

2.5 TREATED GROUNDWATER DISCHARGE REQUIREMENTS

Potential options for the disposal of treated groundwater include discharge to groundwater, surface water, or a POTW. Discharge requirements for these options are identified in Table 2-4, using the chemicals identified in the RA as contributing the most to the total risk in each plume. Although other contaminants were detected in groundwater in past sampling episodes, remediation of the specified compounds identified from the RA to discharge limits should remove other compounds that may be present to acceptable levels.

While the SDWA MCLs and MCLGs and the Clean Water Act (CWA) AWQC all establish TCLs for public drinking water systems, the Michigan Water Resources Commission has established groundwater quality rules for discharges to groundwater, under Michigan Groundwater Quality Rules (MGWQ), Part 22. Generally, these discharge rules are more stringent than MCLs and AWQC, and maintain that the quality of groundwater in all usable aquifers will not be degraded from local background groundwater quality as the result of a discharge. Compounds at concentrations that exceed MCLs will not be discharged into groundwater in usable aquifers, even in those cases where the local background groundwater levels for these compounds exceed the MCLs.

Generally, Michigan's nondegradation policy for groundwater discharges prohibits discharge of organic contaminants at levels above detection. Variances may be granted by the commission for exceptional circumstances where it is determined that (1) strict conformity with the rule is not economically or technically feasible; (2) a prudent alternative does not exist; and (3) such variances are consistent with the promotion of public health, safety, and welfare. Unless a variance is granted, requirements for discharges to groundwater (see Table 2-4) will be based on nondetectable levels.

For surface water discharges, the Michigan Water Quality Standards are applicable. Under these standards, toxic substances will not be present in the waters of the state at levels that are or may become injurious to the public health, safety, or welfare; plant and animal life; or the designated uses of those waters. Allowable levels of toxic substances in discharges to surface water

TABLE 2-9
 CADILLAC AREA GROUNDWATER CONTAMINATION
 TARGET LEVELS
 KYSOR SOILS

CADILLAC AREA FEASIBILITY STUDY

CHEMICAL	GROUNDWATER TARGET LEVEL (mg/l)	SOLUBILITY (l, mg/l)	SOIL TARGET LEVEL (mg/kg)
Trichloroethylene	0.005	1,100	0.07
Toluene	2.0	535	724.0
Xylene	0.44	180	141.0
1,1,1-Trichloroethane	0.2	4,400	7.6

TABLE 2-10
 CADILLAC AREA GROUNDWATER CONTAMINATION
 TARGET LEVELS
 FOUR WINNS SOILS

CADILLAC AREA FEASIBILITY STUDY

CHEMICAL	GROUNDWATER TARGET LEVEL (mg/l)	SOLUBILITY (l, mg/l)	SOIL TARGET LEVEL (mg/kg)
Trichloroethylene	0.005	1,100	0.07
1,1,1-Trichloroethane	0.2	4,400	7.6
Toluene	2.0	535	724.0
Methylene chloride	0.005	16,700	0.02

will be determined by MDNR using technology-based standards, developed by MDNR pursuant to R323.2137 and R323.2140 and imposed under Section 402(a)(1) of the CWA. Water-quality-based effluent limits will also be used, but only when they are more restrictive than technology-based limits.

Technology-based standards include, for toxic and nonconventional pollutants, the best available technology (BAT) that is economically achievable. Technology-based requirements are determined through effluent limitations for specific categories of industries. For CERCLA sites, BAT-based standards are determined on a case-by-case basis using best professional judgment (BPJ). During a BPJ evaluation, treatment technologies are reviewed to ensure the use of technologies that have been proven effective to treat the pollutants present at the site. Once this BPJ determination is accomplished, the National Pollutant Discharge Elimination System (NPDES) permitting authority derives numerical effluent discharge limits that should be achievable by applying the treatment technology.

Based on the contaminants of concern at the Cadillac site, MDNR's Surface Water Quality Division developed effluent limitations for discharge to the Clam River. These limitations are a combination of technology-based and water-quality-based standards, and are presented in Table 2-11.

If groundwater is discharged to a POTW, it must meet pretreatment standards set by the POTW. The City of Cadillac POTW does have an approved pretreatment program with specific limits for the contaminants listed in Table 2-11. If the Cadillac POTW is utilized for disposal, the groundwater will be pretreated to meet appropriate standards.

TABLE 2-11
TREATED GROUNDWATER DISCHARGE REQUIREMENTS

CADILLAC AREA FEASIBILITY STUDY

COMPOUND	MAXIMUM CONCENTRATION DETECTED (mg/l)	DISCHARGE TO GROUNDWATER ¹ (mg/l)	REFERENCE	DISCHARGE TO SURFACE WATER ² (mg/l)	REFERENCE ⁶	DISCHARGE TO CADILLAC POTW ⁷ (mg/l)
1,1,1-Trichloroethane	32.0	≤ 0.005	CRDL	0.01	MDNR	0.1
Trans-1,2-dichloroethylene	4.6	≤ 0.005	CRDL	N/A ⁴	MDNR	0.1
1,1-Dichloroethylene	1.0	≤ 0.005	CRDL	N/A ⁴	MDNR	0.1
1,2-Dichloroethane	2.0	≤ 0.005	CRDL	N/A ⁴	MDNR	0.1
Methylene chloride	0.69	≤ 0.005	CRDL	N/A ⁴	MDNR	0.1
Tetrachloroethylene	0.72	≤ 0.005	CRDL	N/A ⁴	MDNR	0.1
Trichloroethylene	153.0	≤ 0.005	CRDL	0.02	MDNR	0.1
Toluene	3.2	≤ 0.005	CRDL	N/A ⁵	MDNR	
BTX				0.02 ⁵		
Chromium (hexavalent)	0.773	≤ 0.010	CRDL	0.006 ³	MDNR	0.355
Total chromium				0.052 ³		(total chromium)

TABLE 2-11
(continued)
TREATED GROUNDWATER DISCHARGE REQUIREMENTS
CADILLAC AREA FEASIBILITY STUDY

NOTES:

- 1 A nondetectable level is required under MGWQ, Part 22.
 - 2 MDNR Daily Maximum Limitation
 - 3 Monthly average
 - 4 MDNR's effluent limitations assume that if TCE and 1,1,1-TCA are treated to the required levels, the other organic parameters would not exist in the effluent at levels of concern and, therefore, would not require long-term limitations.
 - 5 The effluent limitation is based on benzene plus toluene plus xylene in combination.
 - 6 Letter from William Shaw, MDNR; to Rebecca White, E.C. Jordan; June 20, 1988.
 - 7 Reference - Cadillac POTW Pretreatment Program Requirements
- MCL = Maximum Contaminant Level
CRDL = Contract Required Detection Level

3.0 IDENTIFICATION AND SCREENING OF TECHNOLOGIES

Based on the response objectives and current understanding of the groundwater and soil contamination, treatment technologies suitable to conditions at the Cadillac Industrial Park site were identified. In the identification process, the demonstrated performance of each technology is considered. Those technologies that are only conceptual in development are not identified. However, to the extent possible, innovative technologies are identified (i.e., technologies that have not been proven in hazardous waste application or have been proven only on a pilot scale).

After technologies are identified, they are screened based on engineering feasibility and implementability. The screening consists of evaluating technologies to ensure that they are suitable for the location and conditions at the Cadillac site, and applicable to the contaminants and environmental media. Therefore, the assessment considers both waste and site characteristics.

Waste characteristics include physical properties of the waste (e.g., volatility, solubility, chemical concentrations, and other properties of specific chemical constituents that affect the performance of a technology). Site characteristic information gathered during the RI (e.g., site geology and hydrogeology) are reviewed to identify conditions that may limit or favor the use of certain remedial technologies. Site characteristics also include the availability of space and resources necessary to implement the technology. Those technologies whose use is clearly precluded by waste or site characteristics were eliminated from further consideration.

3.1 TECHNOLOGIES FOR GROUNDWATER REMEDIATION

The groundwater remedial technologies primarily address the reduction of public health risks posed by the potential use of groundwater contaminated with VOCs and hexavalent chromium as a domestic drinking water supply. As required by SARA, a range of technologies was identified. These technologies fall into the following seven response categories:

- o no-action
- o institutional controls
- o containment on-site
- o collection
- o VOC treatment
- o hexavalent chromium treatment
- o discharge

As the list indicates, a set of separate technologies was identified for the VOCs and hexavalent chromium. The identification of groundwater remedial technologies, including a description of each technology, is shown in Table 3-1. The results of screening, including any site- or waste-limiting characteristics of a technology, are presented in Table 3-2. The remaining technologies are summarized in Table 3-3.

TABLE 3-1
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
No Action	Groundwater Monitoring	Perform water quality analyses to monitor contaminant migration and assess future environmental impacts.
Institutional Controls	Alternate Water Supply	Close existing municipal well system and identify new wells to serve city residents.
Containment On-site	Subsurface Slurry Wall	Emplacement of a low-permeability barrier wall to restrict groundwater migration. Should include a cover system to reduce infiltration.
Collection	Barrier Drain System	A series of shallow trenches constructed to intercept contaminated groundwater.
	Extraction Wells	Installation of several strategically located pumping wells to collect contaminated groundwater for treatment.
VOC Treatment	UV/Oxidation	Oxidize organics in extracted groundwater through simultaneous application of UV light with ozone and/or hydrogen peroxide.
	Air Stripping	Reduce concentrations of VOCs through intimate contact of extracted groundwater with air. Water descends a packed column while air is forced up the column to promote mass transfer of organics from aqueous to gaseous phase. Gaseous phase would require further treatment to meet air regulations.
	Steam Stripping	Remove contaminants through intimate contact of extracted groundwater and steam. Similar to air stripping, but steam is utilized to elevate temperatures and enhance removal of VOCs.

TABLE 3-1
(continued)
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
VOC Treatment (continued)		
	Carbon Adsorption	Reduce concentrations of aqueous or gaseous phase organics through adsorption onto granular activated carbon. May be used as a polishing step for treatments such as air stripping to further reduce organic contaminant concentrations.
	Wet Air Oxidation	Destroy organic contaminants in an aqueous solution by inducing oxidation and hydrolysis reactions at high temperature and pressure. Oxygen, at elevated temperatures, enhances oxidation of organic compounds to carbon dioxide and water.
	Thin Film Evaporation	Remove contaminants from extracted groundwater by vaporizing VOCs from water. Process produces a concentrated wastestream requiring further treatment.
	Reverse Osmosis	Remove organics from extracted groundwater using membrane processes. Process will remove organics with >200 molecular weight. Recent studies indicate success in treating organic chemicals with molecular weight >120. At high pressures, membrane allows water to pass while organics are rejected. Process produces a concentrated wastestream requiring further treatment.
	Biological Treatment	Destroy organics through biodegradation, acclimation/ degradation, or chemical conversion of the organic wastes by introducing the extracted groundwater to either an aerobic or anaerobic biological treatment process. Microorganisms and nutrients (if needed) are added to induce one or more of the responses.

TABLE 3-1
(continued)
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
VOC Treatment (continued)	Supercritical Water Oxidation	Destruction of organic compounds in an aqueous solution at high temperature and pressure. Supercritical water oxidation is a high-temperature and pressure version of wet air oxidation.
Hexavalent Chromium Treatment	Oxidation-reduction	Chemical reagents are added to contaminated water, changing the oxidation state of the chromium to a more treatable form. This technology would be used prior to precipitation or ion exchange to reduce hexavalent chromium to a less toxic, precipitable trivalent state before treatment for removal of the chromium.
	Ion Exchange	Metal ions are removed from solution by exchanging ions, electrostatically attached to a solid resin material, for the dissolved chromium ions in solution. Regeneration of the exhausted resin would produce a concentrated wastestream requiring further treatment.
	Precipitation	Chemical precipitation removes dissolved chromium from solution by chemical reaction. The process can be used to complete either chemical oxidation-reduction or electrochemical reduction. Hexavalent chromium would require reduction to trivalent chromium and precipitate as relatively insoluble hydroxide. Process requires close control of pH. Process would generate chromium hydroxide sludge requiring disposal.

TABLE 3-1
(continued)
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
Hexavalent Chromium Treatment (continued)	Reverse Osmosis	Remove hexavalent chromium from extracted groundwater using membrane processes. Process will remove metals or organics with >200 molecular weight. At high pressures, membrane allows water to pass while organics and metals are rejected. Process produces a concentrated wastestream requiring further treatment. However, molecular weight of chromium is only about 52.
	Evaporative Recovery	Vaporize water to concentrate chromium and other metals present in extracted groundwater (e.g., magnesium). Process produces a concentrated wastestream requiring further treatment.
	Electrochemical Reduction	Reduce hexavalent chromium by an imposed DC electrical potential. Consumable iron electrodes are employed, allowing ferrous ions to be released into solution from the electrodes by the electric current. The ferrous ions reduce the Cr(+6) to Cr(+3). The process produces iron sludge with the precipitated trivalent chromium.
	Activated Carbon Adsorption	Reduce concentrations of hexavalent chromium through adsorption onto available granular activated carbon sites. Process produces spent carbon containing chromium which would require disposal.
Discharge	Off-site Water Treatment Facility	Off-site disposal of extracted groundwater to the City of Cadillac POTW. Groundwater would require transport by means of a force main and/or gravity sewer to the POTW.

TABLE 3-1
(continued)
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
Discharge (continued)	Recharge to Groundwater	Reinject treated groundwater using a series of wells and pumps. Can be used to enhance plume removal and accelerate remediation.
	Discharge to Clam River	Discharge treated groundwater directly to nearby surface water body. Transport groundwater by means of a force main.
	Off-site RCRA Disposal Facility	Dispose of extracted groundwater, organic or metals sludge/wastestreams at an off-site RCRA facility. These sludge/wastestreams would be waste residuals from groundwater treatment.

NOTES:

UV = ultraviolet
VOC = volatile organic compound
POTW = publicly owned treatment works

TABLE 3-2
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>No-Action</u>			
Groundwater Monitoring	This action does not contribute to site remediation.	None.	Retained for further consideration, as required by CERCLA, as amended.
<u>Institutional Controls</u>			
Alternate Water Supply	This action does not contribute to site remediation. Would need to locate a new wellfield.	None.	Retained for further consideration. This institutional control would be used in conjunction with a treatment technology.
<u>Containment</u>			
Subsurface Slurry Wall	A low-permeability formation is too deep to seat a slurry wall.	None.	Not retained.
Barrier Drain System	Groundwater contamination is too deep and extensive to make this technology effective.	None.	Not retained.
<u>Collection</u>			
Extraction On-site	None.	None.	Retained for further consideration.

TABLE 3-2
(continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>VOC Treatment</u>			
UV/Oxidation	Electrical generation source required for the production of ozone.	Pretreatment for removal of inorganics (e.g., calcium and manganese) present in the groundwater may be required to prevent fouling of oxidation system. Treatability studies should be performed prior to full-scale design to determine operating parameters and pretreatment requirements necessary to optimize operating efficiency. The contaminant that would control the oxidation retention time and oxidant dose is 1,1,1-trichloroethane.	Retained.
Air Stripping	None.	Pretreatment for inorganics (e.g., calcium and magnesium) present in the groundwater may be required to prevent fouling of air-stripping unit. The contaminant that would control the system design is methylene chloride. Liquid-phase carbon adsorption would be required following air stripping to meet discharge requirements. Off-gases would require treatment by carbon adsorption to meet state air regulations.	Retained.

(continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>VOC Treatment (continued)</u>			
Steam Stripping	Electricity, natural gas, or oil source required for steam generation.	Trichloroethylene would be the design compound due to its its high concentration. May have to pretreat for magnesium or acidify the feed stream to prevent fouling by the production of compounds such as calcium carbonate on the packing material. Treatability studies should be performed prior to full-scale installation to determine actual system requirements.	Retained.
Carbon Adsorption	None.	Methylene chloride is the least adsorbable of the VOC contaminants of concern and would not be removed to the discharge requirement during aqueous-phase application. Aqueous-phase application would be implemented as polishing step for air stripping to meet discharge requirements. Applicable for vapor-phase application as polishing step for air stripping to meet state air regulations.	Retained for vapor- and liquid-phase application following air stripping.

TABLE 3-2
(continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>VOC Treatment (continued)</u>			
Wet Air Oxidation	On-site mobile treatment systems not available. On-site system would require complete building, resulting in high capital costs. Electricity, natural gas, or oil source required for heat due to low BTU content of groundwater.	This technology is not economical for the dilute (<1 percent) VOC concentrations present in the groundwater due to high treatment costs. Generally achieves about 80-percent oxidation of organic constituents. Therefore, discharge requirements would not be met. This high temperature and pressure system would require extensive monitoring.	Not retained.
Thin Film Evaporation	Electricity, natural gas, or oil source required to produce heat.	Difficult to separate organics with boiling points close to water. Produces a concentrated waste-stream requiring further treatment and/or disposal. Not demonstrated at the full-scale level.	Not retained. Process not demonstrated for organics removal.
Reverse Osmosis	None.	Generally applied for dissolved solids and metals removal. Process not proven for lower molecular-weight organics present at the site.	Not retained.

TABLE 3-2
(continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>VOC Treatment (continued)</u>			
Supercritical Water Oxidation	Electricity, natural gas, or oil source required for heat concen- tration due to low BTU content of VOCs present in groundwater. Full-scale systems not available.	Not economically feasible for dilute concentrations (<1 percent) of VOCs present in the ground- water due to high treatment costs. This high-temperature and pressure system would require extensive monitoring.	Not retained.
Biological	None.	Chlorinated organics present in the groundwater are difficult to treat biologically. The VOCs of higher concentrations (e.g., trichloroethylene) would most likely be toxic to the system.	Not retained.
<u>Hexavalent Chromium Treatment</u>			
Oxidation-Reduction	None.	A reducing agent such as sodium metabisulfite would be introduced to the influent to reduce Cr+6 to Cr+3 prior to ion exchange. pH adjustment of wastestream required.	Retained.
Ion Exchange	None.	Hexavalent chromium (Cr+6) to trivalent chromium (Cr+3) prior to ion exchange. pH adjustment of wastestream required.	Retained.

TABLE 3-2
(continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Ion Exchange (continued)			
		Regeneration would produce concentrated wastestream requiring further treatment or disposal.	
Precipitation	None.	Would not be able to achieve discharge requirements as a main treatment.	Retain as possible treatment for back-washing effluent produced during ion exchange process.
		Process would generate chromium hydroxide sludge requiring disposal.	
Reverse Osmosis	None.	Difficult to control leakage rate of chromium through membrane. Therefore, may not be able to meet discharge requirements.	Not retained.
		Would not reduce toxic hexavalent chromium to less-toxic trivalent state. A separation process, not a treatment process. Produces concentrated stream requiring treatment or disposal.	
Evaporative Recovery	Energy source required for evaporation process.	Applicable for process rinse streams to recover chromium for reuse.	Not retained.

TABLE 3-2
(continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Evaporation Recovery (continued)			
		Not cost-effective for chromium levels present in groundwater (<1 ppm).	
		VOCs (i.e., 1,1,1-trichloroethane and trichloroethylene) would be released to atmosphere during evaporation process without treatment. May exceed air emissions standards.	
Electrochemical Reduction	Electrical source required to release ferrous ions into groundwater process stream.	Ferrous ions produce excess (fourfold increase reported) requiring further treatment and/or disposal. The sludge is difficult to settle and dewater due to the presence of ferric hydroxide.	Not retained.
Carbon Adsorption	None.	Treatability testing should be performed to determine actual carbon usage requirements for treatment of hexavalent chromium.	Retained.
		Adsorption system should be able to remove VOCs also present in groundwater (trichloroethylene and 1,1,1-trichloroethane). This would provide single-stage treatment for the hexavalent chromium plume.	
		Pretreatment for removal of metals (e.g., calcium and magnesium) present in groundwater may be required.	

TABLE 3-2
(continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Carbon Adsorption (continued)		Process produces waste residual containing spent carbon, chromium and adsorbed VOCs requiring disposal.	
<u>Disposal</u>			
Off-site Water Treatment Facility	Gravity sewer or force main would be required to transport extracted groundwater to POTW.	Capacity of Cadillac POTW would not permit the discharge of an estimated 1.5 mgd flow.	Not retained.
Recharge to Groundwater	Due to the relatively high hydraulic conductivity and transmissivity of the aquifer, recharge, in conjunction with extraction, would not result in a significant reduction of clean-up time but could potentially cause clean water to be pumped and treated unnecessarily. Also, recharge through infiltration in selected areas might affect groundwater flow within capture zones established for extraction of other plumes. (See Appendix B for further details.)	VOCs and hexavalent chromium in the groundwater must be treated to nondetect levels to meet applicable discharge requirements for recharge to groundwater.	Not retained.
Discharge to the Clam River	Force main or gravity sewer required to transport treated groundwater to discharge point along the Clam River.	Discharged water must meet applicable discharge standards for VOCs and hexavalent chromium, developed by MDNR.	Retained.

TABLE 3-2
(continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Off-site RCRA Disposal Facility	Trucking of extracted groundwater residuals to RCRA facility would be required.	Michigan RCRA facilities accepting metals and organic wastes have been identified. Trucking 1.5 mgd of extracted groundwater is not feasible. Capacity and availability of the RCRA facilities to accept potential waste residuals must be determined.	Retained for disposal of waste residuals. Not retained for disposal of extracted groundwater.

NOTES:

CERCLA = Comprehensive Environmental Response, Liability, and Compensation Act
MDNR = Michigan Department of Natural Resources
mgd = million gallons per day
POTW = publicly owned treatment works
RCRA = Resource Conservation and Recovery Act
UV = ultraviolet
VOC = volatile organic compound

TABLE 3-3
SUMMARY OF SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

TECHNOLOGY	RETAINED	
	YES	NO
<u>NO-ACTION</u>		
Groundwater Monitoring	X	
<u>INSTITUTIONAL CONTROLS</u>		
Alternate Water Supply	X	
<u>CONTAINMENT</u>		
Subsurface Slurry Wall		X
Barrier Drain System		X
<u>COLLECTION</u>		
Extraction On-site	X	
<u>DOC TREATMENT</u>		
UV/Oxidation	X	
Air Stripping	X	
Steam Stripping	X	
Carbon Adsorption	X	
Wet Air Oxidation		X
Thin Film Evaporation		X
Reverse Osmosis		X
Supercritical Water Oxidation		X
Biological		X
<u>HEXAVALENT CHROMIUM TREATMENT</u>		
Oxidation-Reduction	X	
Ion Exchange	X	
Precipitation	X	
Reverse Osmosis		X
Evaporative Recovery		X
Electrochemical Reduction		X
Carbon Adsorption	X	
<u>DISPOSAL</u>		
Off-site Water Treatment Facility		X
Recharge To Groundwater		X
Discharge to the Clam River	X	
Off-site RCRA Disposal Facility	X	

3.2 TECHNOLOGIES FOR SOIL REMEDIATION

The purpose of the soil remediation is to prevent continued leaching of VOCs from subsurface soils at Kysor and Four Winns to a potential drinking water source. Soil technologies were categorized into the following five primary response categories:

- o no-action
- o containment
- o on-site disposal
- o off-site disposal
- o treatment

Because the contaminants and site characteristics are similar at Kysor and Four Winns, treatment technologies applicable to contaminants in both areas were identified and are presented in Table 3-4. Treatment technologies were also screened based on general applicability to both sites. The results of screening are shown in Table 3-5, and summarized in Table 3-6. As soil technologies are combined into remedial alternatives and the alternatives are refined (see Section 5.0), the applicability of an alternative to each site will be discussed.

TABLE 3-4
IDENTIFICATION OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
No-Action	Monitoring/Land Use Restrictions	Environmental monitoring and/or deed restrictions.
Containment	Surficial Capping	Diversion of surface water to reduce infiltration of precipitation and subsequent leaching of contaminants into groundwater.
	Subsurface Contain- ment Wall	A vertical subsurface barrier is constructed to control groundwater, and/or leachate movement, in permeable soils. The barrier is constructed of low-permeability material and is usually keyed into a soil confining layer.
Disposal On-site	On-site RCRA Landfill	Disposal of excavated soils in a RCRA-approved landfill constructed on-site.
Disposal Off-site	Off-site RCRA Landfill	Transport and disposal of excavated soils in an off-site RCRA-approved landfill.
Treatment		
(Biological)	In-situ Biodegrada- tion	Microbes are introduced into the soil, or existing soil microbes are stimulated, to degrade organic contaminants. The process may require nutrient or oxygen additions to the subsurface environment.
	Biological Treatment	Excavated soil is slurried, then introduced into a biological reactor for degradation of organic contaminants.

TABLE 3-4
(continued)
IDENTIFICATION OF SOIL REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
(Physical)	Composting	Excavated soils are mixed with organic matter providing a nutrient source for microbes, which can then adapt to degrade the organic contaminants.
	Land Spreading	Excavated soil is spread out on the ground surface, and the organic contaminants are reduced through microbial degradation or volatilization.
	Low-temperature Thermal Stripping	Excavated soil is heated (350 to 700°F) and mixed with air in a process reactor to volatilize organic contaminants.
	Soil Flushing	In-situ soils are flushed with water or other solvent to remove contaminants. The contaminated extract is recovered, then treated with an appropriate water treatment technology, or disposed of.
	Solvent Extraction	Excavated soils are mixed with a solvent or water in a batch reactor to extract the contaminant. The extracted solvent is then regenerated or disposed.

TABLE 3-4
(continued)
IDENTIFICATION OF SOIL REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
	Steam Stripping	Excavated soils are placed in a reactor where steam is added to remove the more water soluble, less volatile contaminants from the soils.
	Vacuum Extraction	Air is circulated through soils in-situ to remove VOCs. The process gas is recovered and treated to destroy or capture the removed organic contaminants.
	Vitrification	Soil is heated in-situ to its melting point using an electrical current to volatilize or combust the organic contaminants.
(Chemical)	Hydrolysis	Excavated soils are slurried with the addition of acid or base to adjust the pH and degrade organic compounds into simpler compounds through a hydrolysis reaction.
	Oxidation - Hydrogen Peroxide	Hydrogen peroxide is injected into soils in-situ to oxidize organic contaminants.
(Thermal)	Off-site RCRA Incinerator	Excavated soils are transported to an off-site incineration facility for treatment and disposal.

TABLE 3-4
(continued)
IDENTIFICATION OF SOIL REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
	Fluidized Bed Reactors	Excavated soils are fed into a bed of granular material suspended in hot gases to combust or volatilize organic contaminants.
	In-situ Thermal Destruction	Radio frequency electrodes are placed along the ground surface to heat the shallow subsurface. Organics are destroyed or removed in the super-heated steam resulting from groundwater.
	Infrared Incineration	Excavated soils are fed along a moving conveyor under electric heating elements. Organic contaminants are volatilized from the soil matrix.
	Multiple Hearth Incineration	Excavated soils are fed into the top of a vertical combustion chamber with a series of stacked hearths where the soil goes through drying, combustion, and cooling.
	Pyrolysis Process	Organic constituents are heated in the absence of air to thermally degrade organics into a volatile gaseous portion. Typical pyrolysis equipment is designed to convert organic wastes into boiler fuel.
	Rotary Kiln Incineration	Excavated soils are fed into a heated rotating combustion chamber which tumbles the waste. Contaminants removed from the soil matrix are fed into a secondary chamber for additional combustion.

TABLE 3-5
SCREENING OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>No Action</u>			
Monitoring/Land Use Restrictions	None.	None.	Retain for further consideration.
<u>Containment</u>			
Surficial Capping	None.	None.	Retain for further consideration.
Surface Containment Wall	Confining layer located 60 to 90 feet below grade.	High concentrations of organics can affect the integrity of the containment wall.	Retain for further consideration.
<u>Disposal</u>			
On-site RCRA Landfill	None.	Land disposal regulations.	Retain for further consideration.
Off-site RCRA Landfill	Suitable facility would need to be located.	Land disposal regulations.	Retain for further consideration.
<u>Treatment</u>			
In situ Biodegradation	None.	Specific organic compounds may be marginally biodegradable.	Retain for further consideration.
Biological Treatment	None.	Specific organic compounds may be marginally biodegradable. The slurring of sandy soils is not considered practical.	Eliminate from further consideration.
Composting	None.	Specific organic compounds may be marginally biodegradable. The relatively large soil volume limits practicability.	Eliminate from further consideration.

TABLE 3-5
(continued)
SCREENING OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Land Spreading	Limited on-site space.	Specific organic compounds may be marginally biodegradable.	Eliminate from further consideration.
Low-temperature Thermal Stripping	None.	None.	Retain for further consideration.
Flushing	None.	None.	Retain for further consideration.
Solvent Extraction	None.	None.	Retain for further consideration.
Steam Stripping	None.	Technology not sufficiently developed for field implementation.	Eliminate from further consideration.
Vacuum Extraction	None.	None.	Retain for further consideration.
Nitrification	Proximity of treatment system to existing building structures and underground utilities restricts use.	None.	Eliminate from further consideration.
Hydrolysis	None.	Low concentrations of organic contaminants. This process may mobilize naturally occurring heavy metals.	Eliminate from further consideration.

TABLE 3-5
(continued)
SCREENING OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Oxidation - Hydrogen Peroxide	None.	Low concentrations of organic contaminants.	Eliminate from further consideration.
Off-site RCRA Incineration	Suitable facility would need to be located.	None.	Retain for further consideration.
Fluidized Bed Reactors	None.	Suitable for on-site incineration.	Retain for further consideration.
On-site Thermal Destruction	Proximity of treatment system to existing building structures and underground utilities may restrict use.	Conceptual in nature.	Eliminate from further consideration.
Infrared Incineration	None.	Suitable for on-site incineration.	Retain for further consideration.
Multiple Hearth Incineration	None.	Has not been extensively used for hazardous waste destruction.	Eliminate from further consideration.
Pyrolysis	None.	Low concentrations of organic contaminants in soil makes the use of this technology infeasible.	Eliminate from further consideration.
Rotary Kiln Incineration	None.	Suitable for on-site incineration.	Retain for further consideration.

TABLE 3-6
SUMMARY OF SCREENING OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

TECHNOLOGY	RETAINED	
	YES	NO
<u>NO-ACTION</u>		
Monitoring/Land Use Restrictions	X	
<u>CONTAINMENT</u>		
Surficial Capping	X	
Subsurface Containment Wall	X	
<u>DISPOSAL</u>		
On-site RCRA Landfill	X	
Off-site RCRA Landfill	X	
<u>TREATMENT</u>		
In-situ Biodegradation	X	
Biological Treatment		X
Composting		X
Land Spreading		X
Infrared Incineration	X	
Multiple Hearth Incineration		X
Pyrolysis		X
Rotary Kiln Incineration	X	
Low-temperature Thermal Stripping	X	
Flushing	X	
Solvent Extraction	X	
Steam Stripping		X
Vacuum Extraction	X	
Solidification		X
Hydrolysis		X
Oxidation-Hydrogen Peroxide		X
Off-site RCRA Incineration	X	
Fluidized Bed Reactors	X	
On-site Thermal Destruction		X

4.0 DEVELOPMENT AND SCREENING OF GROUNDWATER ALTERNATIVES

This section presents the development and screening of groundwater remedial alternatives. Potential action-specific ARARs, which help to shape remedial alternatives, are presented in Section 4.1. In Section 4.2, these ARARs and the groundwater technologies are used to develop alternatives. In Section 4.3, the alternatives are screened; those retained after screening are presented in Section 4.4.

4.1 POTENTIAL ACTION-SPECIFIC ARARS FOR GROUNDWATER REMEDIAL ALTERNATIVES

Potential action-specific ARARs were identified for the Cadillac site in Section 4.3 of the RI. This section expands on those action-specific ARARs for the development of remedial alternatives. Action-specific requirements generally set performance or design standards, controls, or restrictions on particular types of activities. To develop technically feasible alternatives, it is important that applicable performance or design standards be considered during the development of all remedial alternatives. Furthermore, selected remedial alternatives are required to comply with all ARARs, unless USEPA waives a requirement under one of six conditions identified in Section 4.3 of the RI.

The following paragraphs identify ARARs that will apply to, and must be incorporated in, the proposed remedial alternatives. Based on the technologies that passed initial screening, the proposed alternatives will involve the following activities: no-action, institutional controls, extraction, treatment, and discharges to the Clam River or groundwater.

Resource Conservation and Recovery Act (RCRA) (40 CFR 264) and Michigan (R 299.9600) regulations for owners and operators of hazardous facilities contain general facility requirements that should be undertaken during all remedial activities, including general waste analysis, security measures (i.e., fencing and posting notices), safety equipment and spill control, emergency planning, and closure and post-closure care. Closure includes monitoring, deed notification, and proper decontamination. Post-closure includes groundwater monitoring and maintenance for 30 years. Depending on the effectiveness of the remedial alternative in reducing the volume, toxicity, or mobility of or exposure to the contamination, the post-closure care period, including monitoring, may be reduced.

Occupational Safety and Health Administration (OSHA) standards must be recognized and complied with throughout all on-site activities. Standards for hazardous waste site operations are contained in 29 CFR 1910. Appropriate safety equipment must be located on-site. Allowable levels of contaminants in the atmosphere must be defined and monitored for. If allowable levels are exceeded, proper respiratory equipment must be worn.

The construction and operation of on-site treatment facilities must meet design and operation requirements set forth in RCRA and MDNR Hazardous Waste Management and Air Pollution Control regulations. Permits or other administrative requirements for on-site treatment facilities are not required by CERCLA, as

amended. However, because portions of the study area are not considered Superfund/National Priority List (NPL) sites, MDNR may require licenses and permits for the installation and operation of on-site treatment facilities in those areas.

Residues from on-site treatment operations must be analyzed and properly handled. If a residue (e.g., spent carbon from carbon adsorption) is determined to be a hazardous waste, contingencies must be made for additional treatment and/or removal and proper disposal. The discharge of treated groundwater, resulting from on-site treatment operations, to the Clam River will require state approval and an NPDES permit pursuant to the Michigan Water Resources Commission rules (R 323, Parts 1 and 21). NPDES surface water discharge limits will be developed for the Clam River in accordance with the Michigan Water Quality Standards (R 323.1041). The injection of treated wastewater into the ground must have state approval and be in compliance with RCRA underground injection control regulations at 40 CFR 265 (Subpart R) and 40 CFR 144, and the proposed MGQR (R 323, Part 22).

Off-site treatment and/or disposal must be conducted in a permitted facility in compliance with appropriate federal and state hazardous waste facility regulations. Off-site transport of hazardous waste or substances must meet MDNR hazardous waste transportation requirements (R 299.9401). The transporter and vehicle must be licensed with the state.

Table 4-1 identifies the action-specific ARARs for groundwater remediation and details specific requirements.

4.2 DEVELOPMENT OF GROUNDWATER ALTERNATIVES

Groundwater alternatives were developed by assembling the technologies and discharge options that passed the initial screening, and assessing any restrictions imposed by action-specific ARARs. As required by SARA, a range of alternatives was developed, to the extent possible.

The groundwater alternatives developed for the Cadillac site include the no-action alternative and three treatment alternatives that combine extraction, treatment, and discharge. A containment alternative was not developed, because no containment technologies were retained for further consideration. Off-site treatment was also not retained in technology screening, except for treatment of residues.

To develop alternatives, the Northernnaire, Shallow, Intermediate, and Rexair plumes were considered one area to be remediated by one alternative. Remediation of the two small VOC plumes in the North Park Subdivision is also being evaluated, including no-action. A separate extraction system would be developed for each of these plumes, but the same treatment alternatives would be utilized.

TABLE 4-1
ACTION-SPECIFIC ARARS
FOR GROUNDWATER
REMEDIAL ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS

REQUIREMENT SYNOPSIS

Federal ARARS

- | | |
|---|--|
| 1. CWA NPDES Program
(40 CFR 122, 125) | Point-source discharges from on-site treatment operations must meet NPDES permitting requirements, which include compliance with applicable water quality standards, establishing a discharge monitoring system, and routine completion of discharge monitoring records. |
| 2. RCRA General Facility and Location Standards (40 CFR Subparts A through E) | <p>These requirements outline general facility standards, which include general waste analysis, security 264, measures, inspections, personnel training, and location standards; preparedness and prevention; contingency plan and emergency procedures; and manifest system, recordkeeping, and reporting. The substantive portions of these requirements should be incorporated into the remedial actions involving treatment and disposal. For on-site actions, administrative requirements (e.g., recordkeeping and reporting) will not be considered ARARS.</p> <p>For alternatives incorporating an on-site treatment system, the treatment system must be constructed to withstand a 100-year flood and may not be located within 200 feet of a fault that had displacement in Holocene time.</p> |
| RCRA Groundwater Monitoring Requirements (40 CFR 264, Subpart F) | Groundwater monitoring at CERCLA sites involves the monitoring of hazardous constituents in groundwater during and following remedial actions. Concentration limits for the hazardous constituents are specified at 40 CFR 264.94. Groundwater monitoring will generally follow a corrective action monitoring program at 40 CFR 264.100. The corrective action program specifies that groundwater must be treated or removed where hazardous waste constituents exceed the established concentration limits. |

TABLE 4-1
(continued)
ACTION-SPECIFIC ARARS
FOR GROUNDWATER
REMEDIAL ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>Federal ARARS (continued)</u>	
1. RCRA Closure and Post-Closure Requirements (40 CFR 264, Subpart G)	These requirements apply to alternatives that remove hazardous waste. To protect public health and the environment, closure must minimize the need for further maintenance, and control, minimize, or eliminate post-closure releases of hazardous waste, hazardous constituents, leachate, and contaminated runoff to all environmental media. All equipment used for treatment or disposal must be removed and decontaminated. Post-closure care must incorporate groundwater monitoring and long-term maintenance of the unit and deed notification (if hazardous waste or constituents remain). Post-closure care is usually 30 years.
. RCRA Regulations for Generators of Hazardous Waste (40 CFR 262)	Alternatives involving the movement or removal of hazardous waste will trigger RCRA generator requirements. These requirements specify hazardous waste determination, manifest, pretransport (packaging, labeling, marking), recordkeeping, and reporting, exporting, and importing standards which must be followed following hazardous waste generation. For remedial actions taken solely on-site, manifest, pretransport, recordkeeping, and reporting requirements will not be considered ARARS.
. RCRA Transportation Regulations (40 CFR 262) and DOT Standards (49 CFR, Parts 171 through 179)	If residue from on-site treatment operations is considered a hazardous waste after treatment, the waste must be transported as a manifested hazardous waste to a permitted RCRA facility. The DOT standards specify packaging, labeling, manifesting, and transporting procedures.
.. RCRA Subtitle D Solid Waste Regulations (40 CFR 241 and 257)	Treatment residues determined to be nonhazardous may be disposed of in a RCRA Subtitle D facility.

TABLE 4-1
(continued)
ACTION-SPECIFIC ARARS
FOR GROUNDWATER
REMEDIAL ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS

REQUIREMENT SYNOPSIS

Federal ARARS (continued)

- | | |
|---|---|
| <p>CAA - NAAQS for
Particulate Matter
(40 CFR 50.6)</p> | <p>Site remediation activities, including treatment operations, must comply with the NAAQS. The most relevant pollutant standard is for particulate matter less than 10 microns in size (PM10). The PM10 standard for a 24-hour period is 150 µg/m³ of air not to be exceeded more than once per year. The PM10 standard is based on the detrimental effects of such particles to the lungs.</p> |
| <p>9. RCRA Standards for
Environmental Performance
of Miscellaneous Units
(40 CFR 264, Subpart X)</p> | <p>Treatment technologies, to the extent practical, must follow these general requirements. 40 CFR 264, Subpart X, governing Environmental Performance of Miscellaneous Treatment Units, states that these operations must be protective of public health and the environment, and must prevent releases and migration of contaminants to environmental media.</p> |

State ARARS

- | | |
|---|---|
| <p>10. Michigan Wastewater
Discharge Permits (R 323,
Part 21)</p> | <p>Discharge to surface waters must be permitted (NPDES) by the state and must comply with monitoring, recordkeeping, and reporting requirements. Discharges may not degrade the quality of the water body. MDNR established an NPDES technology-based level for total VOCs discharged to the Clam River at 20 ppb. MDNR established a preliminary standard for hexavalent chromium at 6 ppb and total chromium at 73 ppb. Wastewater discharge to a POTW must meet its pretreatment program.</p> |
| <p>11. Michigan Groundwater
Quality (R 232, Part 22)</p> | <p>Part 22 establishes a nondegradation policy and regulates groundwater discharges. A hydrogeologic report, groundwater monitoring, and a discharge permit are required for all discharges.</p> |

TABLE 4-1
(continued)
ACTION-SPECIFIC ARARS
FOR GROUNDWATER
REMEDIAL ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS

REQUIREMENT SYNOPSIS

State ARARS (continued)

- | | |
|---|--|
| 2. Michigan Liquid Industrial Waste Disposal Act (MCL c 323) | This act requires that detailed records be maintained and a state-licensed transport vehicle be used for for all required industrial waste removals from a facility. Trip records must be maintained, tanks must be tightly closed, and vehicles must be clean. Industrial liquid waste may only be disposed of in a manner and location approved by the commission. |
| 3. Michigan Hazardous Waste Management Rules (MHWMR), Environmental and Groundwater Monitoring (R 299, Rules 611 and 612, respectively) | Environmental and groundwater monitoring (including air and soil monitoring) are required to be performed at all hazardous waste treatment, storage, and disposal facilities. Groundwater rules, including corrective action monitoring, cross reference the federal requirements at 40 CFR 264, Subpart F. Although groundwater monitoring requirements are not considered ARARS for all proposed remedial alternatives, it is best engineering judgment to include groundwater monitoring in all alternatives. |
| 4. MHWMR, Closure and Post-closure (R 299, Rule 613) | Hazardous waste treatment, storage, and disposal facilities must be closed and post-closure care implemented in accordance to 40 CFR 264, Subpart G, RCRA Closure and Post-closure Care Regulation. Closure and post-closure care requirements become ARARS when contaminated waste, soils, or debris are moved or brought to the ground surface. |
| 5. MHWMR, Generators of Hazardous Waste | When hazardous waste or soil and debris contaminated with hazardous waste is removed, generator requirements become ARARS, and the waste or contaminated soils and debris must be managed as a hazardous waste. The Michigan generator regulations cross-reference the RCRA generator regulations at 40 CFR 262. In addition, the Michigan regulations require that the generator use the MDNR-approved manifest forms and a Michigan-licensed transporter or a generator-owned, licensed vehicle for transport of hazardous waste off-site. |

TABLE 4-1
(continued)
ACTION-SPECIFIC ARARS
FOR GROUNDWATER
REMEDIALATION ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS

REQUIREMENT SYNOPSIS

State ARARS (continued)

- | | |
|--|---|
| 6. Michigan Hazardous Waste Management Regulations (MHWMR), Facility Location Standards (R 299.9603), and Facility Design and Operating Standards (R 299.9604) | <p>Hazardous waste treatment, storage, or disposal facilities may not be located within 61 meters of a fault that had displacement in Holocene time, in a floodway or floodplain, over a sole-source aquifer, or in a wetland or coastal high-risk area. Active portions of landfills may not be located within 150 meters of commercial, residential, or recreational property lines.</p> <p>Facilities must have run-on controls that prevent flow from a peak discharge of a 24-hour, 25-year storm. Runoff management must control water volume from a 24-hour, 100-year storm.</p> |
| 7. MHWMR, Construction Permits and Operating Licenses (R 299, Part 5)* | <p>A permit to construct and a license to operate a hazardous waste treatment, storage, or disposal facility are required. Application contents, fact sheet requirements, and a permit fee schedule are specified.</p> |
| 8. MHWMR, Transporters of Hazardous Waste (R 299, Part 4) | <p>Requirements go beyond the scope of RCRA transport requirements. A USEPA ID number and Michigan manifest form must be used. Vehicles and drivers are required to have licenses with the state, as well as liability insurance. Requirements also specify procedures for proper packaging, identification, and handling of hazardous waste.</p> |
| 9. Michigan Air Pollution Control Regulations (R 336, Parts 2, 3, 7, and 9) | <p>Process or control equipment that may be a source of air contamination is required to have permits to install and to operate under Part 2. Fugitive dust from trucks, roads, and stockpiles is regulated under Part 3. Part 7 requires that the maximum allowable emission rate from new sources of VOCs be based on the application of BACT. Furthermore, Part 9 prohibits the emission of air contaminants in quantities that will cause "injurious effects to human health or safety, animal life, plant life of significant economic value, or property" or "unreasonable interference with the comfortable enjoyment of life and property."</p> |

TABLE 4-1
(continued)
ACTION-SPECIFIC ARARS
FOR GROUNDWATER
REMEDIALATION ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS

REQUIREMENT SYNOPSIS

State ARARS (continued)

- | | |
|---|---|
| 20. OSHA Standards for
Hazardous Waste Site
Operations (29 CFR 1910). | OSHA standards include use of proper respiratory and safety equipment and compliance with safety procedures during hazardous waste site operations. Recordkeeping and reporting requirements are also specified. These standards and requirements apply to all site contractors and subcontractors and must be followed during all site work. The degree of safety equipment and procedures required will be determined based on the hazards (present and potential) posed by the site. |
|---|---|

NOTES:

- .ACT = best available control technology
- .AA = Clean Air Act
- CWA = Clean Water Act
- OT = Department of Transportation
- AAQS = National Ambient Air Quality Standards
- NPDES = National Pollutant Discharge Elimination System
- OSHA = Occupational Safety and Health Administration
- = Resource Conservation and Recovery Act

4.2.1 No-action Alternative

The no-action alternative consists solely of groundwater monitoring for changes in contaminant concentrations with time along the pathway of migration. No action may be chosen for the entire Cadillac site, or may only be chosen for the two plumes in the North Park Subdivision, while the other four plumes would be treated.

4.2.2 Pump, Treat, and Discharge Alternatives

The treatment alternatives combine extraction, hexavalent chromium treatment, VOC treatment, and a discharge option. One extraction system has been designed that will capture the four plumes (i.e., Northernnaire, Shallow, Intermediate, and Rexair plumes). If it is determined that remediation of the VOC plumes in the North Park Subdivision is required, a separate extraction system will be utilized to capture those plumes. The same extraction system will be utilized for Alternatives GW-2 through GW-4. In addition, Alternatives GW-2 through GW-4 are all combined with the same discharge technology; that is, discharge to the Clam River.

Hexavalent chromium treatment is associated with each groundwater treatment alternative. Either of the two hexavalent chromium treatment technologies that passed the screening of technologies may be a component of Alternatives GW-2 through GW-4. These two technologies will be compared during detailed analysis for inclusion in the overall alternative.

4.2.3 Additional Component - Alternate Water Supply

Groundwater extracted from the Northernnaire Plume will first be treated by one of the hexavalent chromium technologies, and then discharged to a VOC facility identified in Alternatives GW-2 through GW-4 for further treatment.

An alternate water supply, which is considered an institutional control, may be an additional component of Alternatives GW-2 through GW-4. While providing an alternate water supply does not remediate the groundwater contamination, it does eliminate the exposure to contaminated and potentially contaminated drinking water supplies. This institutional control is being considered as an additional component to the groundwater alternatives because total remediation of site groundwater is uncertain and may take many years.

Site conditions and past waste disposal practices at the Cadillac site suggest that dense nonaqueous-phase liquid (DNAPL) chemicals may be present in the subsurface in the form of residual concentrations retained in the soil column, or as pools of pure product at the aquifer base. Based on the experiments and observations of Dr. Friedrich Schwiller on the mechanisms and conditions under which DNAPLs may migrate as free product into and through an aquifer, it appears that because the sand and soils at the Cadillac site are of sufficiently high conductivity, they could permit extensive intrusion of the DNAPLs into the aquifer.

If DNAPL chemicals are present at the Cadillac site, groundwater contamination cannot be efficiently eliminated without removal of the DNAPL source. In most

situations, removal of the deep subsurface DNAPL source(s) will not be possible because there are no highly effective methods currently available for location and removal. The implication is for possible pumping and treatment for DNAPLs in groundwater for an extremely long period of time, as the residual sources continue to dissolve into the groundwater.

Remedial activities at the Cadillac site will need to be continually assessed to determine any presence of residual DNAPL chemicals. In accordance with Michigan Act 245, P.A. 1929, the groundwater must be returned to a usable condition. However, providing an alternate water supply may be included with any of the groundwater remedial alternatives being considered for the Cadillac site, if it is determined that pumping and treating the groundwater will not remediate the site within a reasonable time frame.

The groundwater alternatives developed for the Cadillac site and a description of the key components are presented in Table 4-2.

4.3 INITIAL SCREENING OF GROUNDWATER ALTERNATIVES

In this section, the four groundwater alternatives and the three additional components to the alternatives are screened against effectiveness, implementability, and cost factors. Criteria considered for each factor during screening are presented in Table 4-3. In addition, preferences stipulated by SARA with regard to site remediation alternatives are considered when evaluating remedial alternatives. These preferences include the following:

- o Remedial actions that involve treatment to permanently and significantly reduce the volume, toxicity, or mobility of hazardous substances are preferred over remedial actions not involving such treatment.
- o Off-site transport and disposal of hazardous substances without treatment is considered the least-favored alternative when practical treatment technologies are available.
- o Remedial actions using permanent solutions, alternative treatment technologies, or resource recovery technologies are preferred.

Innovative technologies should be carried through screening if there is reason to believe that they offer potential for better treatment performance or implementability, fewer or lesser adverse impacts than other available approaches, or lower costs than demonstrated technologies. Results of the initial screening are presented in the following subsections.

TABLE 4-2
DEVELOPMENT OF GROUNDWATER REMEDIAL ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
GW-1: No-Action	- Groundwater monitoring for indicators of contaminant migration.
GW-2: Extraction, Hexavalent Chromium Treatment, UV/Oxidation, Recharge	- Extract contaminated groundwater using pumping well systems. - Hexavalent chromium treatment. - UV/Oxidation for VOC removal. - Discharge treated water back to groundwater system. - Groundwater monitoring for indicators of contaminant migration. - Five-year site review.
GW-3: Extraction, Hexavalent Chromium Treatment, Air Stripping, Discharge	- Extract contaminated groundwater using pumping well systems. - Hexavalent chromium treatment. - Air strip for VOC removal. - Vapor-phase carbon adsorption to meet state air requirements. - Liquid-phase carbon adsorption to meet discharge requirements. - Dispose of spent carbon at off-site RCRA facility. - Discharge treated groundwater to Clam River. - Groundwater monitoring for indicators of contaminant migration. - Five-year site review.
GW-4: Extraction, Hexavalent Chromium Treatment, Steam Stripping, Discharge	- Extract contaminated groundwater using pumping well systems. - Hexavalent chromium treatment. - Steam strip for VOC removal. - Dispose of concentrated wastestream at off-site RCRA facility. - Discharge treated groundwater to Clam River. - Groundwater monitoring for indicators of contaminant migration. - Five-year site review.

TABLE 4-2
(continued)
DEVELOPMENT OF GROUNDWATER REMEDIAL ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
<u>Hexavalent Chromium Treatment Options</u>	
Ion Exchange	- Reduction/Ion Exchange for hexavalent chromium removal. - Dispose of regeneration effluent at off-site RCRA facility. - Discharge effluent to VOC facility for further treatment (GW-2, GW-3, or GW-4).
Carbon Adsorption	- Carbon adsorption for hexavalent chromium removal. - Dispose of spent carbon at off-site RCRA facility. - Discharge effluent to VOC facility for further treatment (GW-2, GW-3, or GW-4).
<u>Additional Components</u>	
..C-1: Alternate Water Supply	- Restrict access to existing public water supply. - Work with the City of Cadillac to locate and test alternate sources. - Michigan Department of Public Health and citizen approval of new water supply source. - Design and construct new wellfield and service connections. - Environmental monitoring of water supply. - Conduct educational programs. - Supply bottled water in the interim (if necessary).

TABLE 4-3
CRITERIA FOR INITIAL SCREENING OF ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

EFFECTIVENESS	IMPLEMENTATION	COST
Compliance with ARARs	Technical Feasibility	Equipment/Construction
Reduction in Mobility, Toxicity, or Volume of Contaminants	Demonstrated Performance	Operation and Maintenance
Protection of Public Health and the Environment	Availability of Equipment, Capacity, and Services	
Adverse Effects Caused by Implementation		

3.1 Alternative GW-1: No-action

This alternative consists solely of groundwater monitoring.

<u>EFFECTIVENESS</u>	<u>IMPLEMENTATION</u>	<u>COST</u>
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
o Site would be monitored for contaminant migration.	o No remedial actions would be implemented.	o Cost includes groundwater monitoring, maintenance, and 5-year review only.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
Would not protect public health from risks due to potential use of VOC- and hexavalent-chromium-contaminated groundwater.	o Long-term monitoring and maintenance would be required.	o None.
o Would not reduce the toxicity, mobility, or volume of VOCs and hexavalent chromium in the groundwater.		

Conclusion. The no-action alternative is retained as a baseline for the remaining alternatives.

4.3.2 Pump, Treat, and Discharge Alternatives

The following subsections present the initial screening of pump, treat, and discharge alternatives. These alternatives apply to the Northernair, Rexair, Shallow, and Intermediate plumes and the two small VOC plumes in the North Park Subdivision, if it is determined that remediation is required in that area. The initial screening of the two options for hexavalent chromium treatment is presented first, followed by the overall pump, treat, and discharge alternatives screening.

4.3.2.1 Hexavalent Chromium Treatment Option 1 - Ion Exchange. This option consists of ion exchange for hexavalent chromium removal, disposal of regeneration effluent and spent resin at an off-site RCRA facility, and discharging effluent to a VOC facility for further treatment (Alternatives GW-2, GW-3, or GW-4).

<u>EFFECTIVENESS</u>	<u>IMPLEMENTATION</u>	<u>COST</u>
<u>Advantages</u> - o Would reduce existing risk to public health due to potential use of hexavalent-chromium-contaminated groundwater. - o Would reduce toxicity of groundwater by removing contaminants.	<u>Advantages</u> - o Installation of an ion exchange system at the site is technically feasible. - o Equipment is readily available and requires minimal time for full-scale installation.	<u>Advantages</u> - o Relatively low capital cost for ion exchange system (approximately \$55,000). - o Precipitation may be added to reduce the quantity of regeneration effluent requiring disposal.
<u>Disadvantages</u> o Would require disposal of regeneration effluent and spent resin at off-site RCRA facility.	<u>Disadvantages</u> o Long-term monitoring and maintenance of treatment system would be required.	<u>Disadvantages</u> - o Long-term costs associated with operation and maintenance of the system. - o Long-term costs associated with disposal of regeneration effluent.

Conclusion. This option is retained for detailed analysis because it would reduce existing public health risks due to potential use of contaminated groundwater, it would be easy to implement, and it has a low capital cost.

3.2.2 Hexavalent Chromium Treatment Option 2 - Carbon Adsorption. This option consists of carbon adsorption for hexavalent chromium removal, disposal of spent carbon at an off-site RCRA facility, and discharging effluent to the Clam River.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u> - o Would reduce existing risk to public health due to potential use of hexavalent-chromium-contaminated groundwater. - o Could reduce existing risk to public health due to potential use of contaminated groundwater by also treating the VOCs contained in this plume. <u>Disadvantages</u> - o Would require disposal of spent carbon at an off-site RCRA facility. Treatability studies should be performed to determine system requirements.	<u>Advantages</u> - o Installation of a carbon adsorption system at the site is technically feasible. - o Equipment is readily available and relatively easy to implement. <u>Disadvantages</u> - o Long-term monitoring and maintenance of the system would be required.	<u>Advantages</u> - o Capital costs for carbon adsorption system estimated at \$60,000. - o Single-stage treatment could reduce overall costs by reducing flow to main VOC treatment system. <u>Disadvantages</u> - o Long-term costs associated with operation and maintenance of the system. - o Long-term costs associated with disposal of spent carbon.

Conclusion. This option is retained for detailed analysis because it would reduce existing public health risks due to potential use of contaminated groundwater, could provide single-stage treatment of the Northernnaire plume, and has a similar capital cost as the ion exchange process.

3.2.3 Alternative GW-2: Extraction, Hexavalent Chromium Treatment, UV/Oxidation, and Discharge. This alternative consists of groundwater extraction, hexavalent chromium treatment, UV/oxidation for VOC removal, recharging treated water to groundwater system, long-term groundwater monitoring, and a five-year site review.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u> Would reduce existing risk to public health due to potential use of VOC-contaminated groundwater. o Would reduce existing risk to public health due to potential use of hexavalent-chromium-contaminated groundwater. Would reduce toxicity of groundwater by removing contaminants. o UV/Oxidation would not produce secondary wastewater requiring disposal. <u>Disadvantages</u> o Either hexavalent chromium treatment (ion exchange or carbon dsorption) would produce secondary waste requiring off-site disposal. o Treatability studies should be performed to determine system requirements.	<u>Advantages</u> o UV/Oxidation to treat the VOC-contaminated groundwater is technically feasible. o Equipment and personnel are readily available for on-site implementation. <u>Disadvantages</u> o Long-term monitoring and maintenance of treatment system would be required. o Electrical energy source required for production of ozone.	<u>Advantages</u> o Carbon adsorption could be used as a polishing step to reduce capital and operating costs. <u>Disadvantages</u> o Capital costs for UV/oxidation system estimated to be \$2.2 million. o Annual operation and maintenance costs estimated at \$1.75 to \$2.00 per 1,000 gallons (approximately \$958,000 to \$1 million).

Conclusion. This alternative is retained for detailed analysis because it would reduce existing public health risks due to potential use of contaminated groundwater; it is an innovative technology; and it would not produce a secondary wastestream.

3.2.4 Alternative GW-3: Extraction, Hexavalent Chromium Treatment, Air Stripping, and Discharge. This alternative consists of groundwater extraction, hexavalent chromium treatment, air stripping for VOC removal, vapor-phase carbon adsorption to meet state air regulations, liquid-phase carbon adsorption to meet discharge requirements, disposal of spent carbon at an off-site RCRA facility, discharging treated groundwater to the Clam River, groundwater monitoring, and a five-year site review.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u> - o Would reduce existing risk to public health due to potential use of VOC-contaminated groundwater. - o Would reduce existing risk to public health due to potential use of hexavalent-chromium-contaminated groundwater. Would reduce toxicity of groundwater by removing contaminants.	<u>Advantages</u> - o Installation of air stripping and carbon adsorption system at the site is technically feasible. - o Full-scale on-site systems are readily available.	<u>Advantages</u> - o Capital costs for the treatment system are estimated at \$730,000. - o Long-term operation cost would be reduced through on-site regeneration of vapor-phase carbon.
<u>Disadvantages</u> - o Would require off-gas vapor-phase carbon treatment to meet state air regulations. - o Would require liquid-phase carbon treatment to meet discharge requirements. - o Would require off-site disposal of spent carbon. - o Hexavalent chromium treatment would produce secondary waste requiring off-site disposal. Pilot-scale studies should be performed prior to full-scale design.	<u>Disadvantages</u> o Long-term monitoring and maintenance of treatment system would be required.	<u>Disadvantages</u> - o Annual operation and maintenance costs are estimated at \$567,000. - o Significant annual cost would also be incurred for transport and disposal of spent carbon from liquid-phase application.

Conclusion. This alternative is retained for detailed analysis because it would reduce existing public health risks due to potential use of contaminated groundwater, and would have a similar capital cost as the steam stripping alternative.

3.2.5 Alternative GW-4: Extraction, Hexavalent Chromium Treatment, Steam Stripping, and Discharge. This alternative consists of groundwater extraction, hexavalent chromium treatment, steam stripping for VOC removal, disposal of concentrated wastestream at an off-site RCRA facility, discharging treated groundwater to the Clam River, groundwater monitoring, and a site review every five years.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
- o Would reduce existing risk to public health due to potential use of VOC-contaminated groundwater. - o Would reduce existing risk to public health due to potential use of hexavalent-chromium-contaminated groundwater. - o Would reduce toxicity of groundwater contaminants. - o Would not produce air emissions.	- o Installation of a steam stripping system is technically feasible for the groundwater VOC contaminants at the site. - o Steam stripping has been demonstrated for groundwater treatment. - o Process equipment is readily available.	o Capital costs (for steam stripping system equipment including reboiler for vapor recompression) are estimated at \$700,000.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
- o Would require disposal of condensate at off-site RCRA facility. - o Hexavalent chromium treatment would produce secondary waste requiring off-site disposal. - o Treatability studies should be performed to determine system requirements.	- o Not as readily implementable as air stripping system because of greater equipment requirements and time required to tie into electrical energy source. - o Long-term monitoring and maintenance would be required.	- o Long-term operating costs associated with generation of process steam could be reduced through use of vapor recompression in the system, versus a straight steam process (approximately \$185,000 versus \$500,000 per year). - o Long-term annual costs associated with disposal of waste condensate.

Conclusion. This alternative is retained for detailed analysis because it would reduce existing public health risks due to potential use of contaminated groundwater; would not produce air emissions; and has a capital cost similar to the air-stripping system.

3.2.6 Alternative GW-5: Extraction, Hexavalent Chromium Treatment, Carbon Adsorption, and Discharge. This alternative consists of groundwater extraction, hexavalent chromium treatment, carbon adsorption for VOC removal, disposal of spent carbon at an off-site RCRA facility, discharging treated groundwater to the Clam River, groundwater monitoring, and a five-year site review.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u> - Would reduce existing risk to public health due to potential use of VOC-contaminated groundwater. - Would reduce existing risk to public health due to potential use of hexavalent-chromium-contaminated groundwater. - Would reduce toxicity of groundwater by removing contaminants. <u>Disadvantages</u> - Would require disposal of spent carbon at an off-site RCRA facility; approximately 3 million pounds per year. - Hexavalent chromium treatment would produce secondary waste requiring off-site disposal. - Treatability studies should be performed to determine system requirements.	<u>Advantages</u> - Installation of a carbon adsorption system at the site is technically feasible. - Carbon adsorption has been demonstrated for groundwater treatment. - Equipment is readily available and relatively easy to implement. <u>Disadvantages</u> - Long-term monitoring and maintenance of the system would be required.	<u>Advantages</u> - Capital costs for the system are estimated at \$380,000. <u>Disadvantages</u> - Annual operation costs estimated at \$3.25 million per year (for carbon usage only). - Long-term costs associated with disposal of spent carbon.

Conclusion. Because this alternative would have very high annual operation costs and would require the disposal of approximately 3 million pounds of carbon annually, it is not retained for detailed analysis.

4.3.3 Additional Component AC-1: Alternate Water Supply. This section presents the initial screening of the alternate water supply which may be added to the groundwater alternatives screened in the previous subsection. This additional component involves relocating the wellfield to provide a new source of public drinking water.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
o Would eliminate exposure of contaminants in drinking water by utilizing a different water supply source.	o The City of Cadillac was advised by the Michigan Department of Public Health (MDPH) to locate an alternate source of drinking water, and has located one potential new source. The technology and construction will be similar to the existing wellfield and water supply facility.	o The long-term costs associated with operation and maintenance will be the same as for the existing wellfield.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Would not remediate existing groundwater contamination. Therefore, this alternative, used alone, would not meet ground-water quality and safe drinking water WRARs.	o The process of locating and constructing an "acceptable" new water supply may take several months to several years, and will require extensive geologic/hydrogeologic studies.	o High capital costs associated with closing the existing wellfield and constructing a new wellfield. Implementation will be labor- and equipment-intensive.

Conclusion. According to conversations with MDPH Division of Water Supply, the City of Cadillac was advised by MDPH to develop a contingency plan for the Cadillac wellfield (i.e., identify another wellfield location). It is the policy of MDPH to pursue elimination of any trace of VOC contamination from public water supplies through constructing new wells or alternate water supply sources. Therefore, the institutional control of moving the wellfield will be retained as a potential additional component of another groundwater alternative.

4.4 SUMMARY OF SCREENING OF REMEDIAL ALTERNATIVES FOR GROUNDWATER

The groundwater remedial alternatives retained after the screening process for detailed analysis are as follows:

- o GW-1: No-action
- o GW-2: Extraction, Hexavalent Chromium Treatment, Ultraviolet (UV)/Oxidation, Discharge
- o GW-3: Extraction, Hexavalent Chromium Treatment, Air Stripping, Carbon Adsorption, Discharge
- o GW-4: Extraction, Hexavalent Chromium Treatment, Steam Stripping, Discharge
- o Hexavalent-Chromium Treatment (Option A) - Ion Exchange
- o Hexavalent-Chromium Treatment (Option B) - Carbon Adsorption
- o AC-1: Alternative Water Supply

5.0 DEVELOPMENT AND SCREENING OF SOIL REMEDIAL ALTERNATIVES

Alternatives applicable to the soil contamination at the Kysor and Four Winns sites are developed and screened in this section. Potential action-specific ARARs that aid in the development of soil alternatives are presented in Section 5.1. The approach and results of the development of soil alternatives are discussed in Section 5.2. In Section 5.3, the alternatives are screened; those retained after screening are summarized in Section 5.4.

5.1 POTENTIAL ACTION-SPECIFIC ARARS FOR SOIL REMEDIAL ALTERNATIVES

Alternatives initially considered for soil remediation include the following activities: no-action, excavation, treatment on-site, treatment off-site, in-situ treatment, and capping. Remedial alternatives involving excavation and on-site treatment operations must comply with federal and state air regulations. The federal Clean Air Act regulations contain National Ambient Air Quality Standards (NAAQS) for Particulate Matter and National Emission Standards for Hazardous Air Pollutants (NESHAPs), which may become applicable. The Michigan Administrative Rules for Air Pollution Control (R 336.1101) contain emissions limitations and prohibitions for emissions that are deemed to have injurious effects on public health and welfare and the environment. Best available control technology (BACT) requirements may also apply to new treatment facilities.

All remedial alternatives must comply with RCRA closure requirements. For remedial alternatives attempting to clean close, all waste residues, contaminated containment and treatment system components, and contaminated subsoils must be removed and managed as hazardous waste. After a site is certified as clean closed, no further monitoring or post-closure care is required (see 40 CFR 264.113, .228, and .258). For alternatives that leave wastes or contaminated soils on-site, the site would need to be capped with a final cover designed and constructed to provide long-term minimization of the migration of leachate. This type of closure requires post-closure care for a minimum of 30 years, at the discretion of the Regional Administrator.

The construction and operation of on-site treatment facilities (e.g., an incinerator) must meet the design and operation requirements set forth in RCRA and MDNR Hazardous Waste Management and Air Pollution Control regulations. Permits, licenses, or other administrative requirements for on-site treatment facilities are not required by CERCLA, as amended. However, because only the Northernaire site is currently a federal Superfund/NPL site (the Kysor site is currently nominated to be placed on the NPL), it is unclear whether MDNR will require permits for the installation and operation of on-site treatment facilities.

Residues, scrubber water, ash, or atmospheric emissions from any on-site treatment activities must be anticipated and properly handled. For example, ash or other residues must be analyzed to determine if they are considered hazardous waste. If so, contingencies must be made for either the treatment of

and/or removal and proper disposal of the waste (e.g., in a permitted facility).

If an alternative involves land disposal, the RCRA Land Disposal Restrictions ("land ban") may become applicable. The land ban prohibits the land disposal of certain hazardous wastes and residuals not meeting specified treatment standards. The Hazardous and Solid Waste Amendments (HSWA) and the RCRA land ban regulations (40 CFR 268) establish categories of hazardous wastes and specify schedules to ban these wastes from land disposal without treatment. Treatment standards are to be achieved by the best demonstrated available treatment technologies (BDATs). A BDAT may either be a concentration- or technology-based standard.

Certain solvent-contaminated wastes, defined as F001-F005 spent solvents, were banned from land disposal if specified treatment standards were not achieved on November 8, 1986. Solvent-contaminated soil and debris resulting from a response action taken under Section 104 or 106 of CERCLA have been granted an extension from the land ban until November 8, 1990. The BDAT technologies for F001-F005 wastes include batch distillation, thin film evaporation, fractionation, incineration, steam stripping, biological treatment, carbon adsorption, air stripping, and wet air oxidation. The treatment standards that must be achieved prior to land disposal in a RCRA facility are listed in 40 CFR 268.41, Table CCWE.

Solvent constituents listed in Table CCWE in 40 CFR 268.41, when used to solubilize (dissolve) or mobilize other constituents, are considered spent solvents under the land ban. A solvent is considered spent when it has been used and is no longer fit for use without being regenerated, reclaimed, or otherwise reprocessed. If it is determined that the contaminated soils at Kysor and Four Winns were the result of RCRA-listed solvent waste spills (i.e., F001-F005), land disposal of these soils untreated will not be feasible after November 8, 1990, unless USEPA has:

- o approved a petition demonstrating that hazardous constituents will not migrate from the land disposal unit for as long as the wastes remain hazardous;
- o already got a capacity variance from November 1988 until November 1990; or
- o granted a variance to the treatment standard because the waste is significantly different from that which is regulated under the land ban.

If there is insufficient evidence to determine that the contaminated soil is the result of a RCRA waste, then the RCRA land ban is not considered applicable. Under current USEPA policy, if the waste is not considered a RCRA waste, then the RCRA ban is neither applicable nor relevant and appropriate. Current USEPA policy also states that to determine that the contaminated soil is a RCRA hazardous waste, a "reasonable level of effort" has to be put forth, and from the information generated it has to be reasonably ascertainable that the

contaminated soil is a RCRA waste. Michigan has adopted the RCRA land ban regulations in R 299.9627.

Atmospheric emissions must be controlled and must comply with the Michigan Administrative Rules for Air Pollution Control. For remedial alternatives incorporating off-site transport of hazardous substances, the transporter and vehicle must be licensed by the state and must comply with Department of Transportation manifest and other MDNR transporter requirements (R 299.9401). Any off-site treatment or disposal facilities must be permitted and in compliance with appropriate federal and state hazardous waste facility regulations.

Table 5-1 identifies the action-specific ARARs for soil remediation, and provides detail on specific requirements.

5.2 DEVELOPMENT OF SOIL ALTERNATIVES

Based on the applicable technologies and potential action-specific ARARs identified in the preceding sections, 11 soil remedial alternatives were developed for the Cadillac site. These alternatives were developed based on their general applicability to both the Kysor and Four Winns contaminated soils. As required by SARA, the range of alternatives includes no-action, containment, and treatment. The alternatives developed for the Cadillac site are presented in Table 5-2 and are briefly described in the following subsections.

5.2.1 No-action Alternative

The no-action alternative would involve groundwater monitoring to indicate any migration of contaminants from the affected soils. Restrictions would be placed on deeds to notify future owners of the presence of contaminated soils. Under the no-action alternative, the site must be reviewed every five years by USEPA.

5.2.2 Containment, Treatment, and Disposal Alternatives

Three containment alternatives were developed for the Cadillac site. These alternatives address applicable closure and post-closure requirements. The nine remaining alternatives incorporate treatment as a principal element, but vary in the degree to which they minimize long-term management and reduce toxicity, mobility, or volume of contaminants. In addition, one alternative involves off-site treatment.

5.3 INITIAL SCREENING OF SOIL ALTERNATIVES

Alternatives were screened for applicability to both the Kysor and Four Winns sites. Two options were considered suitable in formatting the presentation of the screening: (1) screen the developed alternatives considering each site individually, thereby requiring 22 alternatives to be addressed in the initial screening; or (2) screen each alternative considering its applicability to the sites in general terms, noting any site-specific differences or requirements.

TABLE 5-1
ACTION-SPECIFIC ARARS FOR SOIL
REMEDIAL ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>Federal ARARS</u>	
RCRA Regulations, 40 CFR 264 Landfill Regulations (40 CFR 264 [Subpart N])	For capping in-place alternatives, a final cover must be designed, constructed at closure to prevent migration of liquids, have minimal maintenance requirements, promote drainage, minimize erosion, and accommodate settling. The cover integrity and the leachate collection, groundwater monitoring, and runoff/run-on systems must be maintained throughout the post-closure care period.
RCRA Location Standards (40 CFR 264.18)	For alternatives incorporating an on-site cap or treatment system, the facility or cap must be constructed to withstand a 100-year flood, and may not be located within 200 feet of a fault that had displacement in Holocene time. The Cadillac site is not located within a 100-year floodplain or within 200 feet of a fault displaced in Holocene time. Alternatives requiring treatment and/or disposal of contaminated materials outside the site area must consider RCRA location standards.
RCRA Incinerator Standards (40 CFR 264 [Subpart O])	For alternatives involving on-site incineration, the mobile incinerator must be permitted or approved by USEPA. Waste analyses and a trial burn is required to demonstrate removal efficiency. The on-site incinerator must achieve a destruction and removal efficiency of 99.99 percent of the Principal Organic Hazardous Constituent (POHC). Hydrogen chloride (HCl) stack emissions will be controlled to no greater than the larger of 1.8 kg/hr or 1 percent of the HCl in the stack gas prior to pollution control equipment.
RCRA Storage Regulations (40 CFR 264 [Subpart I])	For alternatives specifying incineration, wastes may have to be stored on-site following excavation while awaiting incineration. Wastes must be stored in a manner that precludes leaks and withstands flooding.

TABLE 5-1
(continued)
ACTION-SPECIFIC ARARS FOR SOIL
REMEDIALATION ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>Federal ARARS (continued)</u>	
RCRA Transportation Regulations (40 CFR 262) and DOT Standards (40 CFR 107 and 171)	If incinerator ash residue is considered a hazardous waste after treatment, the waste must be transported as a manifested hazardous waste to a permitted RCRA facility. The DOT standards specify packaging, labeling, manifesting, and transporting procedures.
RCRA Land Disposal Regulations (40 CFR 268)	Restricts the land disposal of certain wastes without treatment and specifies treatment standards for these wastes. Solvent wastes generated at CERCLA sites are banned from land disposal as of November 8, 1988. Any waste meeting this definition at the Cadillac site will need to be treated prior to land disposal.
RCRA Subtitle D Solid Waste Regulations (40 CFR 241 and 257)	Incinerator residues determined to be nonhazardous may be disposed of in a RCRA Subtitle D facility.
RCRA Standards for Environmental Performance of Miscellaneous Units (40 CFR 264, Subpart X)	Treatment technologies, to the extent practical, must follow these general requirements. 40 CFR 264, Subpart X, governing Environmental Performance of Miscellaneous Treatment Units, states that these operations must be protective of public health and the environment and must prevent releases and migration of contaminants to environmental media.
CAA - NAAQS for Particulate Matter (40 CFR 50.6)	Site remediation activities must comply with the National Ambient Air Quality Standards (NAAQS). The most relevant pollutant standard is for particulate matter less than 10 microns in size (PM10). The PM10 standard for a 24-hour period is 150 µg/m ³ of air not to be exceeded more than once per year. The PM10 standard is based on the detrimental effects of such particles to the lungs.

TABLE 5-1
(continued)
ACTION-SPECIFIC ARARS FOR SOIL
REMEDIAL ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>tate ARARs</u>	
0. Michigan Hazardous Waste Management Regulations (MHWMR) Facility Location Standards (R 299.9603), and Facility Design and Operating Standards (R 299.9604)	Hazardous waste treatment, storage, or disposal facilities may not be located within 200 feet of a fault that had displacement in Holocene time, in a floodway or floodplain, over a sole-source aquifer, or in a wetland or coastal high-risk area. Active portions of landfills may not be located within 492 feet of commercial, residential, or recreational property lines. Run-on controls must prevent flow from a peak discharge of a 24-hour, 25-year storm. Runoff management must control water volume from a 24-hour, 100-year storm.
1. MHWMR, Monitoring (R 299.611 - 299.612)	Environmental and groundwater monitoring (including air and soil monitoring) are required at all hazardous waste treatment, storage, and disposal facilities. Groundwater monitoring, including corrective action monitoring rules, cross-reference federal requirements at 40 CFR 264, Subpart F. Although groundwater monitoring requirements are not considered ARARs for all the proposed remedial alternatives, it is best engineering judgment to include groundwater monitoring in all alternatives.
2. Closure/Post-closure, (R 299.613)	Hazardous waste treatment storage and disposal facilities must be closed and post-closure care implemented in accordance to 40 CFR 264, Subpart G, RCRA Closure and Post-closure Care Regulations. Closure and post-closure care requirements become ARARs when contaminated waste, soils, or debris are moved or brought to the ground surface.
3. MHWMR, Use and Management of Containers (R 299.9614)	Refers to RCRA regulations at 40 CFR 264 (Subpart I).

TABLE 5-1
(continued)
ACTION-SPECIFIC ARARS FOR SOIL
REMEDIAL ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>tate ARARS (continued)</u>	
14. MHWMR, Landfills (R 299.9619)	Refers to 40 CFR 264, Subpart N (Landfills). Additional requirements are (1) the final cover must be of compacted clay which meets the liner requirements (a unified soil classification system with over 25 percent of the soil particles less than 5 microns in size); (2) not less than 24 inches of additional material is required over the final cover to prevent animal burrowing into the synthetic cover component; (3) establish shallow-rooted grass to stabilize cap and prevent erosion at the earliest possible point; and (4) establish a venting system to prevent accumulation of gases.
15. MHWMR, Incinerators (R 299.9623 through .9626)	These standards reference portions of the federal RCRA regulations (40 CFR 264, Subpart O), including management of POHCs, incinerator operation standards, and trial burns. Additional requirements specify design criteria including temperature and retention time standards; monitoring of oxygen content; and additional trial operations.
16. MHWMR, Construction Permits and Operating Licenses (R 299, Part 5)	A permit to construct and a license to operate a hazardous waste treatment, storage, or disposal facility are required. Application contents, fact sheet requirements, and a permit fee schedule are specified.
17. MHWMR, Generators of Hazardous Waste (R 299, Part 3)	When hazardous waste or soil and debris contaminated with hazardous waste is removed, generator requirements become ARARs, and the waste or contaminated soils and debris must be managed as a hazardous waste. The Michigan generator regulations cross-reference the RCRA generator regulations at 40 CFR 262. In addition, the Michigan regulations require that the generator use the MDNR-approved manifest forms and use a Michigan-licensed transporter or a generator-owned, licensed vehicle for transport of hazardous waste off-site.

TABLE 5-1
(continued)
ACTION-SPECIFIC ARARS FOR SOIL
REMEDIAL ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>State ARARS (continued)</u>	
18. MHWMR, Transporters of Hazardous Waste (R 299, Part 4)	Requirements go beyond the scope of RCRA transport requirements. A USEPA ID number and Michigan manifest form must be used. Vehicles and drivers are required to have licenses with the state, as well as liability insurance. Requirements also specify procedures for proper packaging, identification, and handling of hazardous waste.
9. Michigan Air Pollution Control Regulations (R 336, Parts 2, 3, 7, and 9)	Process or control equipment that may be a source of air contamination is required to have a permit to install and a permit to operate under Part 2. Fugitive dust from trucks, roads, and stockpiles is regulated under Part 3. Incinerators with a capacity of 100 pounds or less per hour have a maximum allowable emission of 0.65 pounds of particulate per 1,000 pounds of gas. Incinerators with a capacity over 100 pounds/hour have a maximum allowable emission limit of 0.30. Part 7 requires that the maximum allowable emission rate from new sources of VOCs be based on the application of BACT. Furthermore, Part 9 prohibits the emission of air contaminants in quantities that will cause "injurious effects to human health or safety, animal life, plant life of significant economic value, or property" or "unreasonable interference with the comfortable enjoyment of life and property."
20. Michigan Groundwater Quality (R 232, Part 22)	Part 22 establishes a nondegradation policy and regulates groundwater discharges. A hydrogeologic report, groundwater monitoring, and a discharge permit are required for all discharges.

TABLE 5-1
(continued)
ACTION-SPECIFIC ARARS FOR SOIL
REMEDIALATION ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>State ARARS (continued)</u>	
1. OSHA Standards for Hazardous Waste Site Operations (29 CFR 1910)	OSHA standards include use of proper respiratory and safety equipment and compliance with safety procedures during hazardous waste site operations. Recordkeeping and reporting requirements are also specified. These standards and requirements apply to all site contractors and subcontractors and must be followed during all site work. The degree of safety equipment and procedures required will be determined based on the hazards (present and potential) posed by the site.

NOTES:

ACT = best available control technology
 CA = Clean Air Act
 CWA = Clean Water Act
 DOT = Department of Transportation
 NAAQS = National Ambient Air Quality Standards
 NPDES = National Pollutant Discharge Elimination System
 OSHA = Occupational Safety and Health Administration
 POCs = Principal Organic Hazardous Constituents
 RCA = Resource Conservation and Recovery Act

TABLE 5-2
DEVELOPMENT OF SOURCE CONTROL ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
C-1: No-action	- Groundwater monitoring to indicate any migration of contaminants from the affected soils. - Deed restrictions to notify future owners of the presence of contaminated soils. - Long-term monitoring and maintenance of the site.
C-2: Surficial Capping/ Subsurface Containment Wall	- Post-closure plan to monitor and maintain site. - Composite cap system to minimize surface water infiltration. - Bentonite slurry wall to prevent groundwater migration. - Groundwater monitoring to indicate any migration of contaminants from the affected soils. - Surface water management to minimize erosion of final cover. - Vegetative establishment to minimize erosion of final cover. - Periodic five-year review of remedial actions.
C-3: Excavation, Off-site RCRA Landfill	- Excavation of contaminated soil to the water table. - Transport of excavated soil to a RCRA-permitted landfill for disposal.
C-4: Excavation, On-site RCRA Landfill	- Excavation of contaminated soil to the water table. - Construction of an on-site RCRA landfill to contain excavated soils. - Installation of a groundwater monitoring system meeting RCRA regulations. - Groundwater monitoring to indicate any migration of contaminants from the affected soils. - Long-term monitoring and maintenance of the site. - Periodic five-year review of remedial actions.
C-5: In-situ Biodegradation	- Introduction or enhancement of waste-degrading microbes in the affected soils. - A nutrient and oxygen source is provided to sustain the microbial activity. - Post-treatment soil sampling for verification of process effectiveness.

TABLE 5-2
(continued)
DEVELOPMENT OF SOURCE CONTROL ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
C-6: Excavation, Low-temperature Thermal Stripping	- Excavation of contaminated soil to the water table. - Excavated soil is processed in a low-temperature reactor to desorb organic contaminants. - Removed organics are vented to the atmosphere to volatilize, captured in a vapor-phase carbon canister, or thermally destroyed in an afterburner.
C-7: In-situ Soil Flushing	- A source of water (flushing agent) is required. - Construction of infiltration basin typically consisting of aggregate base and polyethylene distribution pipe. - Subsurface system to collect flushate for subsequent treatment (extraction wells). - Flushate treatment system (groundwater treatment alternative). - Groundwater monitoring during process implementation to monitor system effectiveness and reliability. - Post-treatment soil sampling for verification of process effectiveness.
C-8: Solvent Extraction	- Excavation of contaminated soil to the water table. - Process reactor to slurry and mix excavated soils with solvent. - Dewatering device to remove solvent following processing. - Solvent collection for treatment or disposal. - Solvent treatment system if solvent reuse is anticipated.
C-9: In-situ Vacuum Extraction	- Installation of wells in unsaturated zone to circulate subsurface vapor. - Blower mechanism to induce vapor circulation. - Volatilized organics are vented to the atmosphere to be captured in a vapor-phase carbon canister or thermally destroyed in an afterburner. - Post-treatment soil sampling for verification of process effectiveness.

TABLE 5-2
(continued)
DEVELOPMENT OF SOURCE CONTROL ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
SC-10: Excavation, Off-site Incineration	- Excavation of contaminated soil to the water table. - Transport of excavated soil to a RCRA-permitted off-site incinerator. - Residual process ash must be disposed of off-site.
SC-11: Excavation, On-site Incineration	- Excavation of contaminated soil to the water table. - Processing of excavated soil in an on-site mobile incinerator. - Installation of air pollution control equipment to control emissions from process equipment. - Air monitoring during process operation. - Residual process ash disposed of on-site.

To determine the best option, the specific characteristics of each site were first analyzed to highlight similarities and differences; Table 5-3 presents results of this analysis.

Comparing the specific characteristics of each site reveals that the sites are very similar. The estimated volumes of soils to be remediated are relatively close, contaminants of concern at each site can all be categorized as VOCs, and the geology and depth to groundwater at each site are nearly identical. The major differences between the individual sites are the specific VOCs that prompt site remediation, and their identified concentrations (i.e., chemicals in soil at the Kysor site appear in relatively higher concentrations).

Considering the similarities between the two sites, it was decided that it would be most appropriate to screen the alternatives concurrently for both sites. Any differences or requirements specific to each individual site (Kysor versus Four Winns) are noted. Results of the initial screening are presented in the following subsections.

TABLE 5-3
PRELIMINARY SITE EVALUATION
AREAS OF SOIL REMEDIATION
CADILLAC AREA FEASIBILITY STUDY

SITE CHARACTERISTIC	KYSOR SITE	FOUR WINNS SITE
Estimated Surface Area of Contamination	250 by 75 feet = 18,750 sq. ft.	270 by 70 feet = 18,900 sq. ft.
Estimated Depth to Water Table	25 feet	23 feet
Zone of Contaminated Soil depth below grade)	6 to 25 feet	zero to 23 feet
Estimated Volume of Soil to be Remediated	13,200 cy	16,100 cy
Chemicals Present that Exceed TCLs	TCE, Xylene, 1,1,1-TCA	MC*
TCLs (ppm)	TCE: 0.07 Xylene: 141 1,1,1-TCA: 7.6	MC: 0.02
Maximum Concentration (ppm)	TCE: 74.00 Xylene: 520 1,1,1-TCA: 24	MC: 4.70
Soil Type (unsaturated zone)	medium to fine sand	medium to fine sand
Site Accessibility	very good	good some gravel/pavement

NOTES:

TCL = Target Clean-up Level
cy = cubic yards
MC = methylene chloride
1,1,1 TCA = 1,1,1-trichloroethane
TCE = trichloroethylene

* Sparse soil data available for Four Winns are not believed to be sufficient for complete characterization. A groundwater plume containing 1,1,1-TCA was identified as related to Four Winns. A residual concentration of 1,1,1-TCA would also be expected in soils; however, remediation of soils to the TCL for MC may control soil remediation volumes, and a conservative distribution of MC was assumed for the site.

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0030.0.0

.3.1 Alternative SC-1: No-Action

The no-action alternative consists of groundwater monitoring of the site and deed restrictions to protect public health.

<u>EFFECTIVENESS</u>	<u>IMPLEMENTATION</u>	<u>COST</u>
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
Site would be monitored for contaminant migration.	o The installation of additional monitoring wells is technically feasible and easily implementable.	o Cost includes groundwater monitoring, maintenance, and 5-year review only.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
Would not reduce the residual soil contamination currently contributing to the groundwater contamination because chemicals would continue to leach from soil into the groundwater.	o A periodic 5-year review of the site would be required.	o None.
Would not reduce the mobility, toxicity, or volume of residual soil contaminants.	o Long-term monitoring would be required.	
o Would not remediate a source of groundwater contamination.		
o The effectiveness of deed restrictions is considered marginal.		

Conclusion. The no-action alternative is retained for detailed analysis as a baseline for the remaining alternatives.

1.3.2 Containment, Treatment, and Disposal Alternatives

The following subsections present the initial screening of containment, and treatment and disposal alternatives for the Kysor and Four Winns sites. Unless specifically noted, comments are applicable to each site.

1.3.2.1 Alternative SC-2: Surficial Capping/Containment Wall. This alternative consists of surficial capping of the site, coupled with the use of a subsurface containment wall implementing surface water management, groundwater monitoring, and post-closure plan development.

<u>EFFECTIVENESS</u>	<u>IMPLEMENTATION</u>	<u>COST</u>
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
Designed to reduce the infiltration of precipitation through contaminated soils, thereby reducing the mobility of contaminants.	- o Installation of a cap system can be easily implemented and is considered technically feasible. - o Does not require excavation of large volumes of contaminated soil.	o None.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
Would not reduce the volume or toxicity of the residual soil contamination chemicals of concern. Contaminants would remain in on-site soils.	- o Would require long-term monitoring and maintenance. - o Confining layer beneath the site is relatively deep (60 to 90 feet), which limits the desirability of installing a subsurface containment wall. - o Would require a periodic five-year review of the site.	o Installation of a deep containment wall would increase cost.

Conclusion. RI/FS guidance suggests that a containment alternative be carried through the alternative screening for subsequent detailed evaluation. While this alternative may have some constraints associated with the installation of a subsurface containment wall, it is retained for detailed analysis for use at both the Kysor and Four Winns sites. If during the detailed assessment the subsurface containment wall is proven infeasible for implementation at the sites, the alternative will be evaluated based solely on the merits of the cap system.

3.2.2 Alternative SC-3: Off-site RCRA Landfill. This alternative consists of the excavation of all soils that exhibit chemical concentrations exceeding the established TCLs, and transport of contaminated soils to a RCRA-permitted landfill for disposal.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u> - o Would reduce risks associated with residual soil contamination by removing soil. <u>Disadvantages</u> Off-site transport and disposal is the least preferred alternative per SARA. - o Potential exposure risk associated with the transport of contaminated soil. Toxicity, mobility, and volume of contaminants would be reduced by removing soil from the site, but would be transferred to another site.	<u>Advantages</u> - o Eliminates the need to construct and operate process equipment at the site. - o Likely to be the most time-effective alternative. <u>Disadvantages</u> - o Soil from the Kysor site would be subject to land ban restrictions affecting CERCLA response actions. - o Soil from the Four Winns site may be subject to land ban restrictions. - o Requires extensive soil excavation.	<u>Advantages</u> - o Capital short-term costs would be less than on-site landfill alternative. <u>Disadvantages</u> - o None.

Conclusion. Because the soil from the Kysor site would be subject to land ban restrictions, soils from the Four Winns site could be subject to land ban restrictions if they are classified as a RCRA waste and/or if they fail the Toxicity Characteristic Leaching Procedure (TCLP). For this reason, and because disposal without treatment is least-preferred under SARA, this alternative is not retained for detailed analysis.

3.2.3 Alternative SC-4: On-site RCRA Landfill. This alternative consists of the construction of a RCRA landfill cell at each of the sites to contain excavated soils that exhibit chemical concentrations exceeding the TCLs, or construction of a RCRA landfill cell at one area to consolidate materials from both sites.

<u>EFFECTIVENESS</u>	<u>IMPLEMENTATION</u>	<u>COST</u>
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
Affected soils are contained in an engineered landfill cell, thereby reducing the potential for leaching mobility.	o Kysor site would be exempt from obtaining RCRA permit but would have to meet the intent of the regulations.	o None.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Toxicity and volume of contaminants would not be reduced.	- o Four Winns site would require a RCRA Part B permit to dispose of wastes on-site. - o Long-term monitoring and maintenance would be required. - o An on-site RCRA landfill would be subject to the same land ban restriction as an off-site facility. - o Would require a periodic five-year review of the site.	- o Considerable cost and manpower are required to construct, maintain, and monitor an on-site facility. - o Long-term costs associated with operation and maintenance of landfill cover system.

Conclusion. As with Alternative SC-3, the soil from the Kysor site would be subject to land ban restrictions; soils from the Four Winns site could be subject to land ban restrictions if they are classified as a RCRA waste and/or fail the TCLP; and disposal without treatment is least preferred under SARA. In addition, Four Winns soils would require a RCRA Part B permit to dispose of wastes on-site. Therefore, this alternative is not retained for detailed analysis.

3.2.4 Alternative SC-5: In-situ Biodegradation. This alternative consists of in-situ biodegradation of organic chemicals in soils to levels below the established TCLs.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
o Process degrades organic contaminants, thereby reducing their toxicity, volume, and mobility. Permanent treatment remedy.	o Process is performed in situ, eliminating the need for extensive soil excavation.	o None.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
- o Treatability testing would be required to assess site-specific effectiveness. - o Water, microbes, and nutrients would be circulated through subsurface soils and resulting elutriate would be collected for treatment at the groundwater treatment facility.	- o Because contaminants extend to relatively deep levels (25 feet), implementation would require a subsurface circulation system (similar to soil flushing) to introduce and maintain microbes, nutrients, and oxygen supply. - o Typically would be coupled with biological treatment of groundwater.	- o The requirements for biological treatment will make alternative comparably more expensive than soil flushing. Both are theoretically equivalent in terms of effectiveness. - o Actual costs can not be determined prior to site-specific treatability tests.

Conclusion. Biological treatment of deep subsurface soils is most effective when coupled with the biological treatment of groundwater. Because biological treatment of groundwater has not been retained for detailed analysis, this alternative is also eliminated from detailed analysis.

5.3.2.5 Alternative SC-6: Low-temperature Thermal Stripping. This alternative consists of treating excavated soils contaminated with chemicals above TCLs, using a low-temperature thermal reactor to strip or desorb VOCs from the soil. Treated soils would be backfilled on-site.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
- o Process can be controlled to maximize removal efficiencies. - o Would result in a clean soil which can be re-deposited on-site. - o Sandy soils present are suitable for this process. - o Would substantially reduce the toxicity and volume of organic chemicals in soil. - o Highly effective on the chemicals of concern at both the Kysor and Four Winns sites. - o Remedy is permanent.	- o Openly accessible area is readily available at both the Kysor and Four Winns sites for set-up of process equipment. - o Equipment easily transportable to move between the Kysor and Four Winns sites.	o Lower capital costs than other forms of thermal treatment (on-site or off-site incineration).
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Process produces air emissions that would require vapor phase carbon adsorption. Spent carbon would either be regenerated off-site or disposed of at a secured landfill.	- o Process air emissions would require adequate control. - o Requires the excavation of affected soils. - o Damp or wet soils would require greater processing energies to desorb chemicals of concern.	o Additional costs would be associated with controlling air emissions.

Conclusion. Because the low-temperature thermal stripping alternative is technically suitable for both the Kysor and Four Winns sites; would reduce the toxicity and volume of organic contaminants present in the soil; and has the lowest capital cost of the thermal treatments, it is retained for detailed analysis.

3.2.6 Alternative SC-7: In-situ Soil Flushing. This alternative consists of in-situ soil flushing to remove the contaminants of concern from soil to concentrations below TCLs, collecting flushate using groundwater extraction wells, and treating the flushate using one of the groundwater treatment alternatives.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
Effective for removal of chemicals from the zone of contamination by mimicking the natural infiltration.	o Process is performed in situ, eliminating the need for soil excavation.	o This alternative would be cost-effective if implemented in conjunction with a groundwater extraction/treatment alternative.
Most effective on water-soluble, water-mobile chemicals.	o Sandy soils present at both Kysor and Four Winns is well-suited to soil flushing.	o Remedy is considered permanent.
Removes chemicals from soil, thereby reducing the volume and mobility of contaminants in soil.		
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o May require the use of surfactants or additives to improve effectiveness.	o Must be implemented in conjunction with a groundwater extraction/treatment alternative.	o None.
Potential for leaving contaminants in subsurface soils or groundwater if the collection system is not adequately designed.		
o Flushate would need to be treated by one of the groundwater alternatives.		

Conclusion. Soil flushing for treatment of soils at both the Kysor and Four Winns sites could be cost-effective if implemented with a groundwater treatment alternative. Soil flushing is technically feasible and is retained for detailed analysis.

3.2.7 Alternative SC-8: Excavation/Solvent Extraction. This alternative consists of using a process reactor to mix excavated soils with a solvent (possibly water) to physically remove organic chemicals from excavated soils. Treated soils will be backfilled on-site.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
Would reduce the volume and mobility of contaminants in soil by removing chemicals from soil.	o Process equipment is readily available.	o None.
Would result in a clean soil that can be re-deposited on-site.		
Remedy is considered permanent.		
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
May require the use of surfactants or additives to improve effectiveness.	o Would require excavation of affected soils.	o While treatment element is conceptually similar to soil flushing, this alternative would result in higher costs due to excavation, dewatering, and material handling.
o Potential for leaving contaminants or chemical by-products in soil if the solvent is not thoroughly removed.		
o Process would produce solvents that would need to be treated either on-site at the groundwater treatment facility or off-site at a RCRA facility.		

Conclusion. Because solvent extraction would require extensive excavation and has higher capital costs than the in-situ soil flushing alternative (SC-7) while achieving the same level of effectiveness, it is not retained for detailed analysis. If during the detailed valuation of the soil flushing alternative it becomes apparent that soils must be excavated, the solvent extraction alternative will be reconsidered.

3.2.8 Alternative SC-9: In-situ Vacuum Extraction. This alternative consists of in-situ vacuum extraction to physically remove organic chemicals in soil from the unsaturated zone.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u> - Most effective on VOCs with high vapor pressures (TCE, 111 TCA, and MC). May require longer treatment for xylene, which has a lower vapor pressure. Results in a reduction of volume of contaminants in soil. <u>Disadvantages</u> - May require pilot-scale testing to determine site-specific effectiveness. - Process produces air emissions that would require vapor-phase carbon adsorption. Spent carbon would either be regenerated off-site or disposed of at a secured landfill.	<u>Advantages</u> - Treatment process is performed in situ, eliminating soil excavation. - Sandy soils at both the Kysor and Four Winns sites are well-suited to vacuum extraction. <u>Disadvantages</u> - May require capping to effectively remove near-surface contaminants. - Process air emissions may require treatment, typically using activated carbon.	<u>Advantages</u> - In-situ process is cost-effective. - Remedy is considered permanent. <u>Disadvantages</u> - None.

Conclusion: Because in-situ vacuum extraction would reduce the volume of contaminants in the soil at the Kysor and Four Winns sites; is an in-situ process; and is cost-effective, it is retained for detailed analysis.

3.2.9 Alternative SC-10: Off-site Incineration. This alternative consists of the excavation of soils with contaminants exceeding established TCLs, containerization, and transport to a commercial off-site RCRA-permitted incinerator for treatment and disposal.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u> - Would reduce toxicity, volume, and mobility of chemicals in soil through thermal destruction. - Pilot-scale tests have demonstrated removal efficiencies of 99.9999 percent for various organic compounds. - Would remove the contaminated soil from the site. - Remedy is considered permanent.	<u>Advantages</u> - Would eliminate the need to construct and operate process equipment at the site. - Currently, commercial RCRA-permitted incineration facilities are not available in Michigan. Commercial, RCRA-permitted incineration facilities exist near Chicago, Illinois, and St. Louis, Missouri.	<u>Advantages</u> Long-term monitoring would not be required.
<u>Disadvantages</u> Transport of contaminated soil would increase the potential risk of exposure by the public.	<u>Disadvantages</u> - Currently, the availability of an off-site facility with the capacity to handle large quantities of soil is very limited. - Excavated soils must be transferred to small containers (i.e., 55-gallon drums) prior to acceptance by an off-site facility.	<u>Disadvantages</u> - Relatively high cost for a treatment alternative. - Transport of soil to out-of-state facilities will substantially increase costs.

Conclusion. Because of the potential for a large volume of material to be treated at each site, associated transportation requirements, limited facility capacity, facility acceptance of only containerized materials, and high costs associated with off-site incineration, this alternative is not retained for detailed analysis.

3.2.10 Alternative SC-11: On-site Incineration. This alternative consists of excavation of contaminated soils for destruction of organic contaminants in an on-site incinerator, utilizing either a rotary-kiln, infrared, or fluidized bed incinerator. Ash passing the EP Toxicity test will be backfilled on-site.

EFFECTIVENESS	IMPLEMENTATION	COST
<u>Advantages</u> Would reduce toxicity, volume, and mobility of chemicals in soil through thermal destruction. Pilot-scale tests have demonstrated removal efficiencies of 99.9999 percent for various organic compounds. Remedy is considered permanent.	<u>Advantages</u> - o Considered practical for waste volumes greater than 10,000 cy. - o Open accessible area is readily available at both the Kysor and Four Winns sites for setup of an on-site incinerator.	<u>Advantages</u> o Long-term monitoring is not required.
<u>Disadvantages</u> A trial burn is required to demonstrate compliance with emission limits. If ash does not pass the EP Toxicity test, it would have to be solidified. Scrubber effluent would be disposed of at an off-site RCRA facility.	<u>Disadvantages</u> - o The incinerator would have to be set up at both the Kysor and Four Winns sites. - o Ash must be tested for EP Toxicity prior to on-site disposal. - o Air emissions must be controlled.	<u>Disadvantages</u> - o Relatively high cost for a treatment alternative. - o If solidification of ash is required due to residual metals content, costs would increase significantly.

Conclusion. Because on-site incineration is technically feasible; would reduce toxicity, volume, and mobility of contaminants in the soil; and is practical for the volume of waste to be treated, it is retained for detailed analysis.

5.4 SUMMARY OF SCREENING OF REMEDIAL ALTERNATIVES FOR SOIL

The range of remedial alternatives for soil retained after screening for detailed evaluation in the FS is presented as follows. Each alternative will be considered and evaluated for both the Kysor and Four Winns sites.

- o Alternative SC-1: No-action
- o Alternative SC-2: Surficial Capping/Subsurface Containment Wall
- o Alternative SC-6: Low-temperature Thermal Stripping
- o Alternative SC-7: In-situ Soil Flushing
- o Alternative SC-8: In-situ Vacuum Extraction
- o Alternative SC-11: On-site Incineration

6.0 APPROACH TO THE CADILLAC DETAILED ANALYSIS

6.1 GENERAL APPROACH

The detailed analysis of alternatives is intended to provide decision-makers with sufficient information concerning a range of proposed remedial actions in order to select a single remedy that meets the following CERCLA requirements:

- o protective of public health and the environment;
- o attains ARARs (or provides grounds for invoking a waiver);
- o cost-effective;
- o permanent solution that uses alternative treatment technologies or resource recovery technologies to the maximum extent practicable; and
- o preference for treatment that reduces toxicity, mobility, or volume as a principal element.

This section presents the approach to the detailed analysis of the groundwater and soil alternatives that passed through initial screening (see Sections 4.0 and 5.0).

Each alternative evaluation includes a detailed description emphasizing the technology used, specific components, and proposed design specifications. Anticipated work activities are summarized and graphics are included to depict process flows and site layouts of equipment. The description is followed by an assessment of the alternative based on the following nine evaluation criteria:

- o compliance with ARARs
- o protection of public health and the environment
- o reduction of toxicity, mobility, or volume of contaminants
- o short-term effectiveness
- o long-term effectiveness and permanence
- o implementation
- o community acceptance
- o state acceptance
- o cost

Table 6-1 outlines the specific factors considered for each criterion in accordance with "Additional Interim Guidance for FY87 Records of Decision" (USEPA OSWER Directive No. 355.0-19 Supplements); "Guidance for Conducting Remedial Investigations and Feasibility Studies Under CERCLA" (USEPA OERR/OWPE, March 1988); and other CERCLA and SARA guidance.

The only criteria that cannot be fully analyzed are community acceptance and state acceptance. A detailed discussion of these criteria will be presented in USEPA's ROD after the state and public have thoroughly reviewed this document. Community and State acceptance will be addressed herein, based on information

available when the Draft FS report was prepared, and will apply to all groundwater and soil alternatives.

Community Acceptance. To date, the community has not been informed of proposed remedial actions at the site. In accordance with CERCLA requirements, the community will be given a 30-day public comment period, following release of the final FS, to make formal comments. Comments received at that time will be incorporated into the ROD.

State Acceptance. With multiple and overlapping contaminated groundwater plumes from the Superfund and proposed Superfund sites (i.e., Northernair and Kysor) and other sources within the industrial park, MDNR and USEPA agreed it would be more appropriate to include assessments of the Superfund-related and other VOC contamination in a comprehensive groundwater study of the area. Therefore, the FS is being conducted under joint MDNR/USEPA guidance. The state has been actively involved in the review process. An interim deliverable, presenting the remedial action objectives, was submitted to MDNR and USEPA in April 1988. Based on MDNR and USEPA comments on the deliverable, the remedial action objectives were revised and finalized, and a second interim deliverable (the Alternatives Array Document) was submitted to MDNR and USEPA in June 1988. The Alternatives Array Document presented the development and initial screening of alternatives. MDNR and USEPA concurred with the remedial alternatives that passed initial screening and that are evaluated in the detailed analysis.

After each alternative analysis, results of the evaluation will be summarized in the form of tables.

6.2 APPROACH TO COST ANALYSIS FOR GROUNDWATER ALTERNATIVES

In this section, the approach for estimating cost for each of the groundwater alternatives is discussed. The specific costs for each alternative are presented in the detailed analysis for that alternative.

6.2.1 Capital Costs

Capital costs include direct and indirect costs for each alternative. Direct costs consist of construction, equipment, mobilization, and demobilization costs. Because each alternative has not been designed in detail, only major pieces of equipment were costed separately. Other equipment and materials are included as one cost category: undeveloped design detail.

Indirect costs considered for each alternative include engineering and contingency costs. Engineering costs, which consist of design and general engineering costs, were estimated to be 10 percent of the total estimated direct costs. This estimate is based on general engineering practices. Contingency costs cover unpredictable events (e.g., strikes, storms, floods, and price variations). For the Cadillac site, a contingency of 25 percent was applied. This estimate was selected based on the accuracy of the overall estimate (i.e., -30 to +50 percent) and conceptual design, and the variability in costs associated with construction at hazardous waste sites.

For the Cadillac site, capital costs were estimated for each component of the groundwater treatment alternatives. Component costs are presented in Appendix D.

6.2.2 Annual Costs

Annual costs are operation and maintenance (O&M) costs for post-construction activities necessary to provide continued effectiveness of the remedial action. For Cadillac groundwater treatment alternatives, these costs include the following items:

- o five-year site review costs
- o groundwater monitoring costs
- o power and heating for the treatment building
- o maintenance and monitoring for the discharge system
- o hexavalent chromium and VOC treatment system O&M costs
- o maintenance and electricity for the extraction system

The annual costs associated with the five-year site review, groundwater monitoring, treatment building, and discharge system will be assumed for the entire duration of the remedial action, which is conservatively estimated at 64 years. Because different plumes will be remediated in varying time frames, the flow of contaminated groundwater to be treated will be reduced, as the various plumes reach TCLs. The hexavalent-chromium in the Northern Plume is expected to be remediated to TCLs within a four-year period. Therefore, the annual costs for the hexavalent-chromium treatment were estimated for only four years.

The time frames for remediation of the VOC plumes range from 6.5 to 64 years. The model developed for the groundwater extraction system indicates that, as the various plumes are remediated, the flow of contaminated groundwater to be treated will be reduced by 60 percent after 38 years. Because 60 percent is a substantial reduction, annual costs for components of the treatment system that are sensitive to flow (e.g., electricity and chemicals) have been reduced accordingly after 38 years. Therefore, annual costs for VOC treatment were calculated for Years 1 through 38, and a different set of annual costs were calculated for Years 39 through 64. Annual costs associated with maintenance and electricity for the extraction system are also for varying time frames, because each plume will be extracted within a different number of years. These costs are presented in Appendix D and summarized at the end of each alternative analysis.

6.2.3 Present Worth Analysis

The present worth analysis evaluates costs that occur over different time periods by discounting future costs to the present. Present worth values were calculated assuming an interest rate of 5 percent, in accordance with "Guidance for Conducting Remedial Investigations and Feasibility Studies Under CERCLA" (USEPA, 1988). The performance period depended on duration of the activity.

TABLE 6-1
CRITERIA FOR DETAILED ANALYSIS OF ALTERNATIVES^{1,2,3,4}

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	FACTORS TO CONSIDER
1. Compliance with ARARs	o Extent to which alternative attains federal or state, if more stringent, chemical-, location-, and action-specific ARARs.
2. Protection of Public Health and the Environment	- o Existing and potential risks adequately eliminated, reduced, or controlled through treatment, engineering controls and/or institutional controls. - o Extent to which alternative attains federal or state, if more stringent, chemical-specific ARARs governing health and the environment.
3. Reduction of Toxicity, Mobility, or Volume	- o The treatment processes: innovative/alternative or resource technologies. - o The amount of waste treated and the degree of expected reduction in toxicity, mobility, or volume of contaminants. - o The degree to which treatment is irreversible. - o Type and quantity of residuals resulting from treatment process.
4. Short-term Effectiveness	- o Potential short-term impacts on community, workers, and the environment during remedial action implementation (e.g., extraction construction). - o Time until full protection is achieved.
5. Long-term Effectiveness	- o Magnitude of total residual risk (untreated waste and treatment residuals). - o Type and degree of long-term management required, including monitoring and operation and maintenance. - o Adequacy and suitability of engineering and institutional controls used to manage untreated waste and treatment residuals. - o Reliability of controls over time, including potential need for replacement.

TABLE 6-1
(continued)
CRITERIA FOR DETAILED ANALYSIS OF ALTERNATIVES^{1,2,3,4}

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	FACTORS TO CONSIDER
	- o Length of time that level of effectiveness can be maintained. - o Reversible/irreversible effects to environmentally sensitive areas and resources.
6. Implementability	<u>Technical Feasibility</u> - o Expected operational reliability of technologies, considering site- and waste-related characteristics. <u>Demonstrated Performance</u> - o Bench- and/or pilot-scale results - o Previous use in the field - o Failure/downtime estimate <u>Availability</u> - o Equipment, site areas, land - o Location and capacity of needed treatment, storage, and containment services. <u>Installation</u> - o Implementation as a function of studies, design, construction, weather, unknown site and safety precautions. - o Degree of difficulty associated with construction considering site conditions. - o Time to implement. <u>Permitting and Legal Constraints</u> - o Basic requirements and projected length of time to obtain permits.

TABLE 6-1
(continued)
CRITERIA FOR DETAILED ANALYSIS OF ALTERNATIVES^{1,2,3,4}

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	FACTORS TO CONSIDER
	o Siting issues (i.e., land right-of-ways, land acquisition, zoning).
	<u>Impacts on Historical and Critical Issues</u>
7. Community and State Acceptance	- o Potential impacts to recorded/unrecorded archeological/architectural resources. - o Components of alternatives that community and state supports or opposes.
8. Costs	<u>Capital Costs</u> - o Direct costs for site preparation, construction (materials/labor), remedial equipment, buildings and services, and disposal costs. - o Indirect costs for engineering, legal/administrative, and contingencies. - o Accuracy -30% to +50%. <u>Annual Operating and Maintenance Costs</u> - o Costs for labor, maintenance, and auxiliary materials, energy, services and administration, insurance, taxes, and licensing. <u>Present-worth Analysis</u> - o Interest rate = 5% - o Performance period: up to 64 years <u>Sensitivity Analysis</u> - o Effect of variations in efficiency of extraction system on the present worth.

TABLE 6-1
(continued)
CRITERIA FOR DETAILED ANALYSIS OF ALTERNATIVES^{1,2,3,4}

CADILLAC AREA FEASIBILITY STUDY

NOTES:

- ¹ "Guidance for Conducting Remedial Investigations and Feasibility Studies Under CERCLA." USEPA OERR/OWPE; March 1988. (Interim Final to be released by January 15, 1989.)
- ² Superfund Amendments and Reauthorization Act of 1986 (SARA).
- ³ "Interim Guidance on Superfund Selection of Remedy." USEPA OSWER Directive No. 9355.0-19; December 24, 1986.
- ⁴ "Additional Interim Guidance for FY87 Records of Decision." USEPA OSWER Directive No. 9355.0-19 Supplements; July 24, 1987.

6.2.4 Sensitivity of Costs

A sensitivity analysis assesses the effect that variations in specific assumptions have on the estimated cost of an alternative. Because it is difficult to estimate the efficiency of the extraction system and, therefore, the time frame needed to reach TCLs in each plume, the sensitivity of each groundwater treatment alternative's present worth to the efficiency of the extraction system and the treatment duration was analyzed. The sensitivity analyses of the groundwater alternatives are presented in Section 9.1.

6.3 APPROACH TO COST ANALYSIS FOR SOIL ALTERNATIVES

The approach to cost estimating for the soil alternatives is presented in this section. Specific costs for each alternative are presented in the detailed analysis for that alternative.

6.3.1 Capital Costs

The approach to estimating capital costs for soil alternatives is identical to the approach for groundwater alternatives. Capital costs include direct and indirect costs. For the Cadillac site, capital costs were estimated for each component of the containment and treatment alternatives. Component costs are presented in Appendix D.

6.3.2 Annual Costs

For Cadillac soil alternatives, O&M costs include the following items:

- o groundwater monitoring costs
- o post-closure maintenance and inspection costs
- o five-year site review costs
- o O&M of equipment, when applicable

These costs are also presented in Appendix D.

6.3.3 Present Worth Analysis

As previously discussed, the present worth analysis evaluates costs that occur over different time periods by discounting future costs to the present. Present worth values for the soil alternatives were calculated assuming an interest rate of 5 percent. The performance period varied per alternative, depending on the length of time required to achieve protection.

7.0 DETAILED ANALYSIS OF GROUNDWATER ALTERNATIVES

7.1 PRESENTATION OF INFORMATION

Groundwater alternatives that passed the initial screening and that will undergo the detailed analysis include a no-action alternative and three treatment alternatives which combine extraction, hexavalent-chromium treatment, VOC treatment, and a discharge option. The No-action Alternative is analyzed in Section 7.2. Because the three treatment alternatives include the identical extraction and discharge systems, the descriptions and detailed analyses of the extraction and discharge technologies are presented once before the treatment alternative analyses in Sections 7.3 and 7.4, respectively.

Because either of the two hexavalent-chromium treatment technologies may be a component of any of the treatment alternatives (i.e., GW-2, GW-3, or GW-4), these technologies are compared, and also presented before the treatment alternative analyses in Section 7.5. The alternate water supply, which may be an additional component of no-action or any of the treatment alternatives, is evaluated separately from the treatment alternatives in Section 7.9. Figure 7-1 illustrates groundwater treatment alternative scenarios available for the site.

Each detailed analysis section compares the alternative to the nine criteria discussed in Section 6.0, excluding state and community acceptance (see Section 6.1), and addresses remediation of both the industrial park and the North Park Subdivision. If an alternative will impact the North Park Subdivision differently than it will the industrial park, the differences are noted. Information regarding effectiveness, implementability, and costs of the treatment alternatives was obtained primarily from vendors. Costs for each alternative are calculated for cleanup of the entire site, including the Northernnaire, Shallow, Intermediate, Rexair, and North Park Subdivision plumes.

7.2 DETAILED ANALYSIS FOR ALTERNATIVE GW-1: NO-ACTION

7.2.1 Description

The No-action Alternative consists solely of groundwater monitoring and a five-year review for the site. The following sections describe a long-term groundwater monitoring program and five-year review for the site.

Groundwater Monitoring. The No-action alternative would allow the groundwater plumes at the Cadillac Industrial Park to continue to migrate by natural mechanisms. Under this alternative, a groundwater monitoring plan would be necessary to periodically assess the fate and transport of the contaminant plumes (i.e., Northernnaire, Intermediate, Rexair, North Park, and Shallow, which is divided into Kysor Shallow (A), Kysor Shallow (B), and Four Winns Shallow). Table 7-1 lists monitoring wells associated with each of the seven plumes to reach this objective if no-action is selected, and outlines the sampling program for each plume area. Appendix C illustrates the monitoring well locations. Groundwater sampling and analysis episodes should be conducted

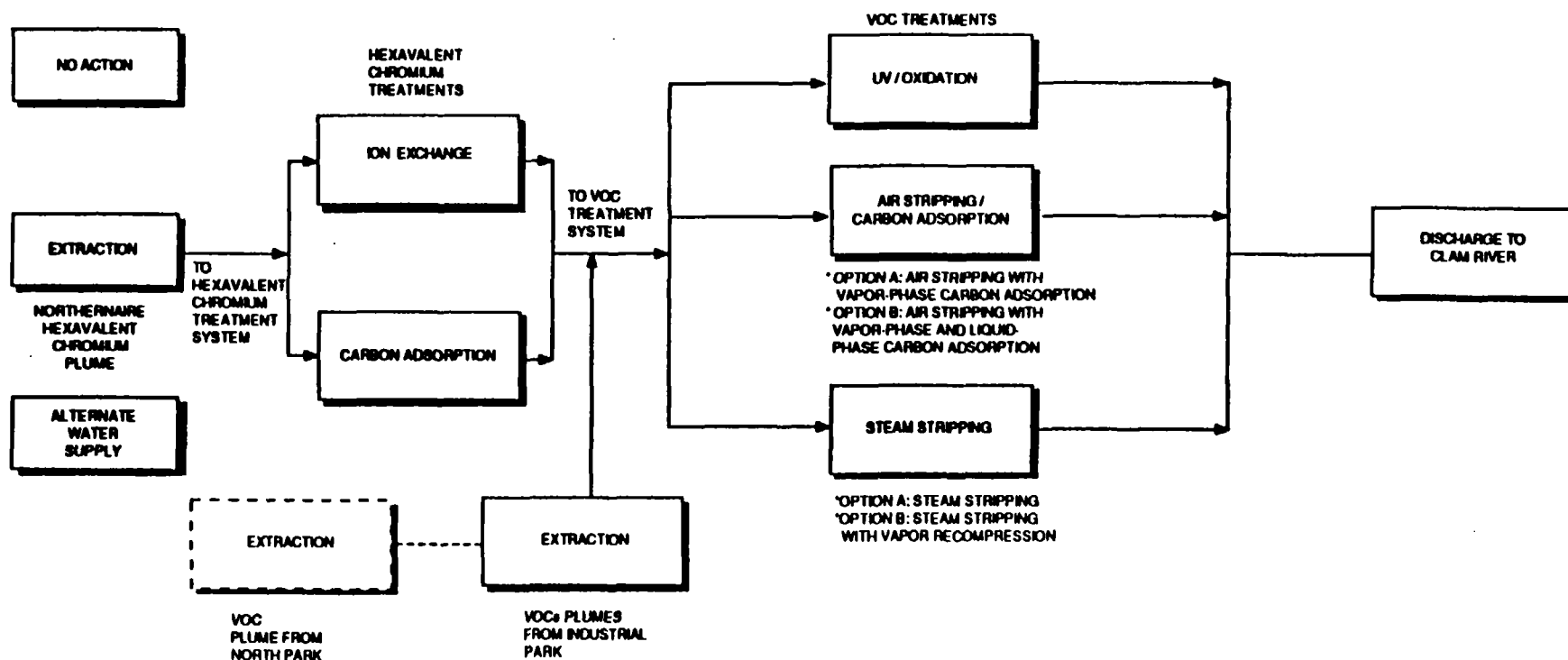


FIGURE 7-1
GROUNDWATER TREATMENT ALTERNATIVE SCENARIOS
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN
E.C.JORDAN

TABLE 7-1
GROUNDWATER SAMPLING AND ANALYSIS PROGRAM
FOR NO-ACTION ALTERNATIVE

CADILLAC AREA FEASIBILITY STUDY

EXTRACTION WELL SYSTEM	NO-ACTION ALTERNATIVE WELL SAMPLING	ANALYSIS
Rexair	SLW-41, 45, 51 MRW-1, 2 CW-6	VOA By #8010
Kysor Shallow (A)	SLW-1, 7, 10S, 16, 25 GW-6S, 9S K-6S FS-4-0	VOA By #8010 Acetone By #524.2
Kysor Shallow (B)	Same As Kysor (A)	VOA By #8010 Acetone By #524.2
Four Winns	SLW-38 GW-6S, 14S FWMW-8, 17	VOA By #8010 Acetone By #524.2
Kysor Intermediate	SLW-10D GW-6D, 8, 9D, 10D, 11, 14D	VOA By #8010
Hexavalent Chromium	MW-10, 15, 16, 18 K-15	Hexavalent Chromium By #7197
North Park	SLW-4, 13, 20 GW-7, 16	VOA By #8010

quarterly for the first year, after which the sampling will be reduced to annual frequency. When the plumes migrate beyond Thirteenth Street, as the interpretation of the groundwater flow system indicates, it may also be necessary to install additional monitoring wells to collect data farther downgradient than the present locations proposed. In addition, it may be possible to drop wells from the program when data indicate that the plume has passed. At some point, the plume may appear to stabilize as the concentrations in the leading edge become too low to detect. It is not possible to accurately predict how many additional wells and samples might have to be included in a future program.

The chemical analysis program will involve laboratory testing of the parameters specific to the contaminant plumes. For six of the seven plumes, the chlorinated fraction of the volatile organic analytes (VOAs) will be tested by Method 8010. The analysis program should not require the additional expense of Contract Laboratory Program (CLP) protocols. Method 8010 should provide necessary gas chromatograph (GC) quantification to meet the sampling program objectives. For three of the VOC plumes (i.e., Kysor Shallow (A), Kysor Shallow (B), and Four Winns), an analysis for acetone by Method 524.2 TCL should also be performed. For compound identification and confirmation of the purgeable VOAs, approximately 10 percent of the samples should be analyzed by Method 524.2.

Hexavalent chromium will be analyzed in the monitoring wells listed in Table 7-1 associated with the extraction system of this plume; Method 7197 will be used. Because a high proportion of the chromium in the plume is known to be in the hexavalent form, monitoring for total chromium by standard atomic absorption methods is also possible, with occasional spot checks for hexavalent-chromium by the wet chemistry method (the hold time for hexavalent chromium samples is relatively short).

Five-year Site Review. Under SARA 121C, a remedial action must be reviewed at least every five years, whenever that action results in any hazardous substance, pollutants, or contaminants remaining on-site. During this review, USEPA must determine whether public health and the environment are protected. If appropriate, additional action may be initiated as a result of this review. In addition to consideration of public health and the environment, treatment technology improvements may also be reviewed.

For the Cadillac Groundwater No-action Alternative, wastes will remain on-site; therefore, a five-year review must be conducted. The five-year review of Cadillac would consist of a review of the monitoring data generated as a result of groundwater monitoring. If the review indicates that the environment or public health is not sufficiently protected, further action may be recommended. A review of any new technologies and improvements may be conducted and appropriate new technologies may be implemented.

7.2.2 Compliance with ARARs

A No-action Alternative at the Cadillac site would not be consistent with the goal and intent of CERCLA or USEPA's Groundwater Protection Strategy (GWPS). The Superfund Program's goal is to restore groundwater to its beneficial uses,

based in large part on its vulnerability to contamination, use, and value. USEPA's GWPS and draft "Classification Guidelines" (Office of Groundwater Protection) focus on the classification and protection of existing and future groundwater resources. Although the GWPS and Classification Guidelines are not promulgated requirements and, therefore, not ARARs, they serve as useful guidance for establishing groundwater TCLs and should be considered during selection of a groundwater remediation alternative.

The No-action Alternative will incorporate a USEPA/state-approved post-closure groundwater/corrective action monitoring program at the site (40 CFR 264.100). Installation of monitoring wells must be performed consistent with OSHA Standards for Hazardous Waste Site Operations. Under RCRA closure regulations (40 CFR 264.119), deeds to all contaminated properties must include a notice stating the type and location of hazardous waste contained on the property and that the property's use is restricted under 40 CFR 264, Subpart G.

A No-action Alternative is also inconsistent with federal SDWA regulations and the Michigan Safe Drinking Water Act and Water Quality Rules. These regulations and rules protect present and future public water supplies and are considered applicable to groundwater at the Cadillac site.

Appendix E identifies the action-specific ARARs associated with this alternative. These ARARs are detailed in Section 4.1 and summarized in Table 4-1.

7.2.3 Protection of Public Health and the Environment

In the case of the No-action Alternative, groundwater contaminants would continue to exceed chemical-specific ARARs. The ARARs relevant to the public health assessment include MCLs, MCLGs, AWQC, and any duly promulgated and consistently applied state standards. As shown in the RI, each of the four plumes analyzed in the industrial park (i.e., the Rexair, shallow, intermediate, and Northernnaire plumes) contains several selected toxic chemicals of concern at concentrations exceeding these relevant health-based chemical-specific ARARs. Under the No-action Alternative, the existing and potential risks posed by using these sources of water for domestic use would not be eliminated.

In the North Park Subdivision, TCE was found in the groundwater at concentrations up to 0.6 ppm at depths of 130 to 150 feet; this exceeds the MCL of 0.005 ppm. Therefore, in the case of no groundwater remediation, a potential public health risk would remain, due to the possibility of potable wells being installed in the area of a contaminated plume.

7.2.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

Because no remedial actions would be implemented, the No-action Alternative would not reduce the toxicity, mobility, or volume of groundwater contaminants.

7.2.5 Short-term Effectiveness

The groundwater monitoring program and five-year site review under the No-action Alternative would not adversely impact the public health of the individuals in the surrounding community or the workers involved with

monitoring activities. In addition, the activities associated with the No-action Alternative would not negatively affect the environment.

Because remedial activities are not associated with the No-action Alternative, full protection of public health and the environment from groundwater contaminants would not be achieved.

7.2.6 Long-term Effectiveness

The No-action Alternative for groundwater at the Cadillac site would not reduce a potential risk to public health. Because there are no controls identified for aquifer use, future potential water users could be exposed to the toxic chemicals through ingestion of water from the shallow or intermediate contaminated aquifers. Therefore, domestic wells could be inadvertently placed in a contaminated area. In addition, the Rexair Plume or the Northernnaire Plume could contaminate the municipal supply wells in the City of Cadillac in the future, if not remediated.

As described in the baseline RA, exposure to contaminants in each of the four plumes investigated poses an unacceptable risk to public health (i.e., exceeds the USEPA acceptable target range of 10^{-4} to 10^{-7}). Risks were quantified based on current VOC or hexavalent-chromium concentrations measured in on-site wells (present potential risk) and future concentrations at selected receptor wells (future potential risks). The potential risks reported in the baseline risk assessment would remain under the No-action Alternative. The following paragraphs summarize these potential risk values for each of the four plumes evaluated.

Shallow Aquifer Plume. Current potential risks to the public from exposure to the contaminants in the shallow aquifer plume, assuming it was used as a drinking water supply, are determined to be significant. For the noncarcinogen 1,1,1-TCA, the hazard index ranges from 1.03 to 10.6. According to USEPA, a hazard index greater than 1.0 indicates the potential for adverse health effects. Exposure to carcinogens in the shallow aquifer presents potential risks at, or exceeding, USEPA's target risk range of 10^{-4} to 10^{-7} for total individual carcinogenic risks. The estimated total incremental carcinogenic risks range in value from 4×10^{-4} to 5×10^{-2} for the generalized VOC shallow plume.

In the baseline RA, future potential risks associated with the generalized VOC shallow aquifer plume are split by source into two plumes in order to model concentrations at a hypothetical downgradient receptor well. In the Kysor plume, three carcinogens (i.e., TCE, 1,1-DCE, and PCE) pose potential future risks at or above USEPA's target risk range. In the Four Winns plume, four carcinogens (i.e., 1,1-DCE, 1,2-DCE, PCE, and TCE) pose potential future risks at or above USEPA's target range.

Rexair Plume. The Rexair Plume potentially poses the greatest threat to public health in the Cadillac area, because it is the only plume determined to be directly upgradient of the city municipal wellfield. None of the three noncarcinogens at the levels measured in the Rexair Plume would be expected to adversely affect public health if ingested over a lifetime. However, the three

carcinogens in the Rexair Plume (i.e., TCE, methylene chloride, and chloroform) present a significant potential risk, with total incremental cancer risks calculated to range from 5×10^{-4} to 5×10^{-2} .

Future potential risk to exposed populations is estimated at two receptor locations. At the hypothetical receptor location on Thirteenth Street, the total future potential incremental carcinogenic risk associated with the Rexair Plume ranged from 1×10^{-3} to 2×10^{-2} . Risk due to TCE accounted for more than 99 percent of this total. At the hypothetical receptor location in the municipal wellfield, the future potential cancer risk is 3×10^{-5} due to lifetime ingestion of TCE. These values are well above USEPA's target risk range.

Intermediate Aquifer VOC Plume and Northernnaire Plume. The intermediate aquifer on the western side of the site includes two plumes: the generalized intermediate aquifer VOC plume and hexavalent-chromium plume from Northernnaire. Each was assessed separately in the baseline RA. All nine chemicals of concern in the generalized intermediate VOC plume pose a potential significant risk at present concentration levels. Hazard indices were estimated to range from 6.06 to 66.5, and total incremental carcinogenic risks ranged from 1×10^{-4} to 4×10^{-3} . The chemicals TCE and 1,1-DCE accounted for over 97 percent of the total estimated risk.

Similarly, the deeper intermediate aquifer plume near Northernnaire poses a high potential risk to the public if the groundwater were ingested, again due mostly to TCE and 1,1-DCE.

Risk calculations for potential future risks under the No-action Alternative showed chemicals in the Intermediate Plume, originating at both Kysor and Four Star, pose a significant risk. The hexavalent-chromium plume in this aquifer also poses a potential risk to public health in the future, if used for drinking water.

North Park Subdivision Plume. Under the No-action Alternative, a risk remains at the North Park Subdivision because the carcinogen, TCE, was measured in the groundwater at levels exceeding its MCL for drinking water.

The baseline environmental exposure assessment was limited to terrestrial organisms because there are no surface water or wetlands areas on the Cadillac site. No potential routes of exposure for terrestrial organisms were identified, because soil contamination was below the ground surface. In the following discussions of groundwater alternatives, there is no further mention of ecological risk, because any potential adverse effects to organisms other than humans at the Cadillac site were predicted at minimal to nonexistent.

The No-action Alternative requires long-term groundwater monitoring for 64 years to monitor the migration of VOCs and hexavalent chromium in the groundwater. Annual maintenance would be required to clean and restore the monitoring wells. Because the monitoring program would extend for 64 years, it is expected that some monitoring wells may require replacement due to weathering. However, information regarding design life of monitoring wells is not available.

7.2.7 Implementability

The No-action Alternative is readily implementable at the Cadillac site. The wells identified for the groundwater monitoring program were installed at the site. Monitoring wells have been well-demonstrated for monitoring the migration of chemicals in groundwater. Well installation equipment and drilling contractors are readily available if replacement of one or more wells is required in the future.

7.2.8 Costs

In this section, the capital and annual costs, as well as the present worth analysis, are presented for the No-action Alternative. These costs, presented in Table 7-2, are accurate to -30 to +50 percent.

Capital Costs. Because monitoring wells already exist at the site, no capital costs are associated with the No-action Alternative.

Annual Costs. Annual costs are O&M costs associated with post-construction activities, necessary to provide continued effectiveness of the remedial action. For the No-action Alternative, annual O&M costs would include costs for implementing the groundwater monitoring program and for conducting the five-year site review.

Groundwater monitoring costs include those costs required for collection and analysis of samples from 37 monitoring wells. Collection costs are based on costs from previous sampling events at the site, and analytical costs are based on current CLP prices. The costs presented do not include potential replacement of the wells. A detailed estimate of groundwater monitoring costs is presented in Appendix D.

Costs for the five-year site review include costs for an environmental consultant to review monitoring data and technology improvements, and to generate a review report. These costs are also detailed in Appendix D.

The annual cost associated with the No-action Alternative is \$178,000.

Present Worth Analysis. The present worth analysis evaluates costs that occur over different time periods by discounting future costs to the present. The net present worth (in 1988 dollars) of the No-action Alternative is \$824,000. This value was calculated assuming an interest rate of 5 percent and a performance period of 64 years.

7.2.9 Summary

The detailed analysis for the No-action Alternative is summarized in Table 7-3.

7.3 GROUNDWATER EXTRACTION SYSTEM

Extraction by pumping has been identified as the remedial technology that will most effectively remove contaminated groundwater from the seven plumes at the

TABLE 7-2
 GW-1 NO-ACTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL
CAPITAL COSTS	
FIVE YEAR SITE REVIEW	\$
MONITORING	
TOTAL CAPITAL COSTS	\$
ANNUAL COSTS	
FIVE YEAR SITE REVIEW	\$3,00
MONITORING - YEAR 1	140,00
MONITORING - YEAR 2 - 64	35,00
PRESENT WORTH FOR 64 YEARS @ 5%	\$824,00

TABLE 7-3
SUMMARY OF DETAILED ANALYSIS
GW-1: NO-ACTION GROUNDWATER ALTERNATIVE
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o Compliance with chemical-specific ARARs would not be attained since contaminants in groundwater will remain above drinking water standards. - o RCRA groundwater monitoring requirements would be met. There are no other action-specific ARARs applicable to the No-Action Alternative.
2. Protection of Public Health and the Environment	o No reduction of existing risks to public health associated with drinking contaminated groundwater.
3. Reduction of Toxicity, Mobility, or Volume	o No reduction in toxicity, mobility, or volume.
4. Short-term Effectiveness	- o There are no significant risks to community, workers, or the environment associated with implementation. - o Full protection from potential risks associated with drinking contaminated groundwater would be achieved only by dilution. Time of dilution cannot be accurately estimated.
5. Long-term Effectiveness	- o Long-term carcinogenic risks to humans from drinking/exposure to contaminated groundwater remain. - o Long-term groundwater monitoring and management of property required. - o Institutional controls would not adequately manage or control migration of untreated waste. - o Monitoring wells may require replacement at some future time.
6. Implementability	- o Groundwater monitoring equipment is readily available. - o No construction would be implemented; monitoring wells have already been installed.

TABLE 7-3
(continued)
SUMMARY OF DETAILED ANALYSIS
GW-1: NO-ACTION GROUNDWATER ALTERNATIVE
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
7. Costs	- o Capital Costs = \$0 - o Annual = \$178,000 - o Present Worth = \$824,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21; "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

A = The long-term uncertainties associated with land disposal.

B = The goals, objectives, and requirements of the SWDA.

C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.

D = Short- and long-term potential for adverse health effects from human exposure.

E = Long-term monitoring costs.

= The potential for future remedial action costs if the alternative remedial action in question were to fail.

G = The potential threat to public health and the environment associated with excavation, transportation, and redisposal, or containment.

Cadillac Industrial Park, based on its suitability to the existing geologic and hydrogeologic conditions. Because significant contamination was detected at depths of 180 feet below ground, only groundwater extraction by deep recovery wells strategically located to intercept contaminated groundwater is practical. Several pumping well systems have been conceptually designed and evaluated to meet response objectives and TCLs for the seven plumes at the Cadillac site.

The extraction wells would be interconnected by an underground piping collection system to transport the extracted groundwater to the VOC and hexavalent-chromium treatment facility. The collection piping provides the means for installing a single treatment facility.

7.3.1 Description

Design of a pumping system to extract contaminated groundwater requires consideration of several hydrogeologic factors of the aquifer (e.g., porosity, hydraulic conductivity, vertical and horizontal gradients, contaminant thickness and extent, and location of confining layers). For purposes of preliminary design of an extraction system, the aquifer response to pumping may be described by analytical functions which may be applied to determine the radius of influence a pumping well will have when placed in a selected area of the contaminated aquifer. These analytical expressions may be generalized to produce type curves (Javandel and Tsang, 1986), which can then be applied to determine the optimal number and placement of the pumping well(s) for contaminated groundwater extraction. With a location selected, a pumping rate in gallons per minute (gpm) can be estimated, which would create a capture zone of appropriate size for the associated plume.

Once a pumping rate is determined, a time estimate for removing one pore volume of water from the aquifer can be estimated based on plume length. Then, using an expected efficiency of contaminant removal with each pore volume, the number of pore volumes required to flush the aquifer and reach specified contaminant TCLs can be estimated. Finally, a time estimate for cleanup can be determined based on pumping rate and the total number of pore volumes that require pumping. Appendix F describes the application of the analytical expressions and type curves to each identified contaminant plume in the Cadillac Industrial Park for preliminary design of an appropriate extraction system.

Certain assumptions concerning the hydrogeologic conditions at the Cadillac Industrial Park and, most importantly, the behavior of contaminant migration in the groundwater, were made for the design of the extraction systems.

The first basic assumption is that the hydrogeologic conditions derived from the drilling explorations and monitoring well sampling and discussed in the RI are representative of the entire study area. Data from several studies were incorporated in the assessment of the geology and hydrology of the industrial park so that the data base is as complete as possible. However, as with any natural setting, local variations not identified by the drilling explorations, or known aberrations in the geology which could not be practically explored (i.e., the dense gravel-silt layer between Rexair and the city wellfield), may be significant factors controlling groundwater and contaminant movement. For the assessment of groundwater extraction of specific plumes, aquifer parameters

(e.g., hydraulic conductivities and horizontal gradients) were estimated using only those borings and monitoring wells located in that general area. If actual conditions vary, the response of the contaminated groundwater to pumping may not be exactly as predicted.

A second assumption is that the solubilized plume in groundwater is the primary target for cleanup by the pumping extraction wells. The possibility exists, however, that concentrated pools of free product exist in both the unsaturated and saturated zones, which will greatly affect clean-up efforts. Contaminant concentration levels, particularly in the Rexair and Kysor plumes, suggest that free product of TCE may be present in the aquifer. TCE, a DNAPL, will tend to sink in water due to gravity. According to recently published literature and based on laboratory experiments, DNAPL compounds will sink to the first impermeable layer, and then may continue to be moved by gravity along this layer's sloping surface unaffected by hydraulic gradients.

With current technologies, it is not cost-effective to attempt to locate pools of free product that can accumulate in small depressions in the surface of the impermeable layer and remain undetected by borings drilled on a relatively large grid system. The presence of these pools will be determined by the observed effectiveness of the extraction system after several years of pumping. Clean-up times are estimated only for the solubilized portion of the plume. The effect of free product in the aquifer, if present, would be to significantly increase the estimates of clean-up periods. If concentrated pools exist, contaminant concentrations in the groundwater will increase shortly after the pumping system has been shut down; obtaining TCLs may not be possible.

The extraction system is not expected to be 100-percent efficient. Incomplete flushing of pore spaces, presence of lower permeable zones with contaminated groundwater, dispersivity, collection of clean water, and retardation will all tend to lower the apparent efficiency of removal. Also, while the plume pore volume may decrease through time, it may also become successively more difficult to recover residual contaminated groundwater. Because the accumulated effect of all these factors cannot be directly determined, ranges of contaminant removal efficiency with each pore volume extracted were used to estimate time to attain clean-up objectives. These ranges also have associated limitations; therefore, a relatively wide range of from 70- to 90-percent removal efficiency was used for this study.

It is anticipated that the final system design will allow for sufficient flexibility in operating parameter ranges so that the system can be adjusted to obtain optimal efficiencies. To develop treatment alternatives and cost estimates, and to provide a starting point for initiating aquifer restoration, a 90-percent efficiency was assumed. However, the sensitivity analyses developed for each treatment alternative (see Section 9.1) assesses the effect that variations in the extraction system's removal efficiency will have on the estimated present worth of the alternative.

Using periodic groundwater monitoring and sampling, the effectiveness of the designed system can be evaluated and pumping conditions changed as required. This periodic sampling is a necessary phase of the remedial alternative to evaluate progress toward reaching clean-up objectives. If the solubilized

plumes as defined by the monitoring wells installed for this and other studies are the major portion of existing groundwater contamination, the time estimates should be reasonable, with some future modification. While greater numbers of wells might be installed within plumes to ostensibly speed up cleanup, it is likely that removal efficiencies will drop and capital costs will increase sharply. More evaluation during detailed design may further optimize costs and clean-up times; however, to evaluate the alternatives, the selected well configurations and assumptions of efficiency are believed appropriate. Lastly, if concentrated pools of TCE and other DNAPL compounds exist, clean-up times may be extended beyond practical objectives.

7.3.1.1 Cadillac Extraction System. A total of seven extraction well series was designed to pump the contaminated groundwater at the Cadillac Industrial Park, including the following:

- o a single-well system for the Rexair Plume
- o a two-well system for the concentrated portion of the VOC (Kysor-related) shallow aquifer plume
- o a two-well system for the VOC (Kysor- and Four Star-related) shallow aquifer contamination that is less concentrated near the northeastern limit of the upper aquitard
- o a single-well system for the relatively concentrated VOC (Four Winns-related) shallow aquifer contamination
- o a three-well system for the areally extensive intermediate aquifer contamination
- o a single-well system for the small North Park Subdivision Plume
- o a single-well system to collect the thin plume of hexavalent chromium at the bottom of the intermediate aquifer

Table 7-4 summarizes the characteristics of each plume, the number of wells per plume, and the expected results of the extraction system, assuming a 90-percent removal efficiency.

7.3.1.1.1 Rexair Plume

The Rexair Plume is a concentrated, well-defined plume containing primarily TCE. By evaluating characteristics of the aquifer and the existing contaminants, it was determined that a single-well extraction system would be the most effective for cleanup. Figure 7-2 shows the Rexair Plume and the shape of the capture zone with the pumping well located at the leading edge of the plume between monitoring wells SLW-44 and SLW-51. Figure 7-3 is a cross sectional view of the contaminant plume based on screened auger field data used to determine extraction well screen and pump placement. The pumping well will be set at a depth of approximately 160 feet and screened the entire saturated thickness to this depth. A pumping rate of 135 gpm is necessary to create the capture zone

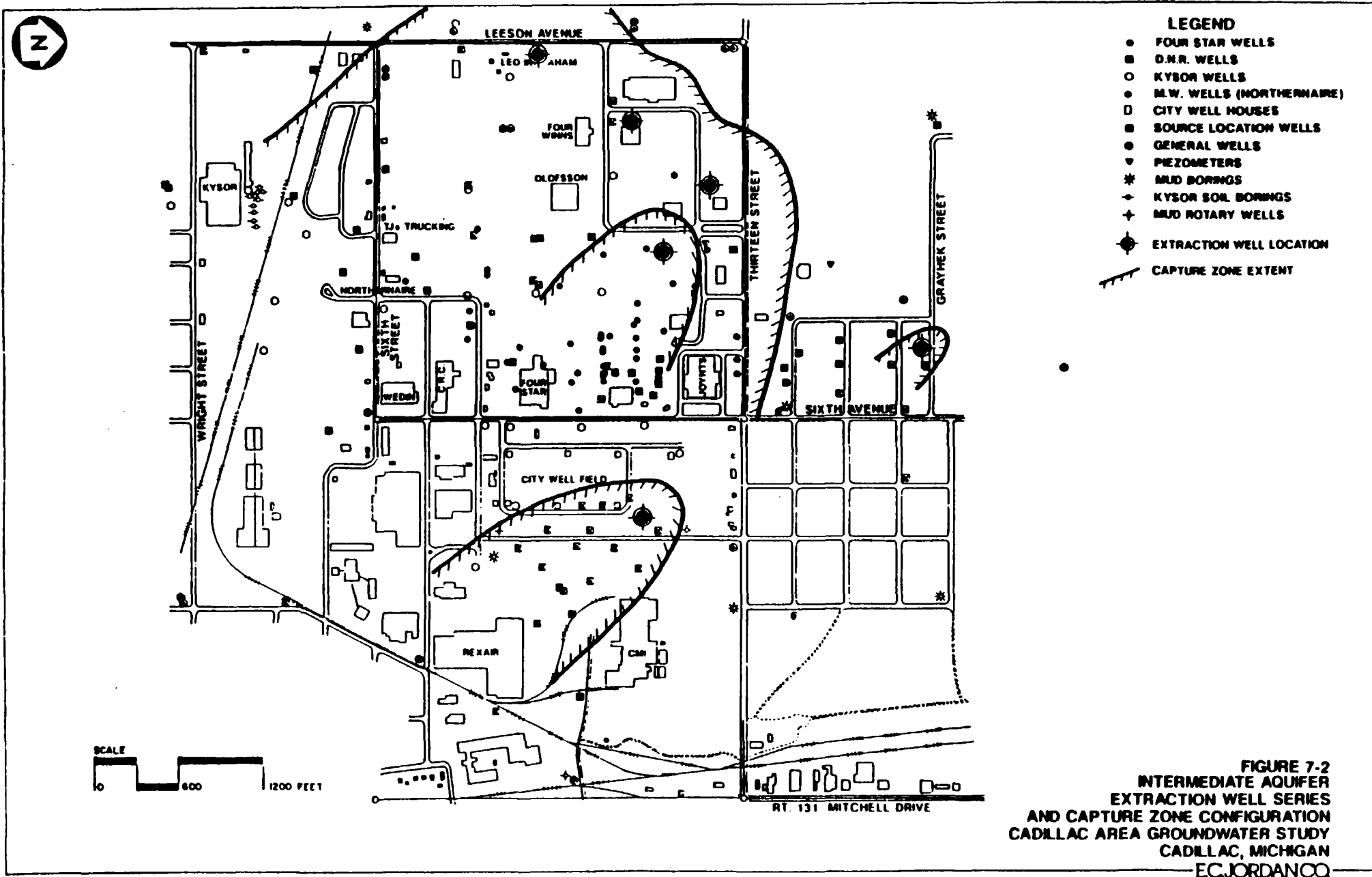
TABLE 7-4
EXTRACTION SYSTEM SUMMARY
CADILLAC AREA FEASIBILITY STUDY

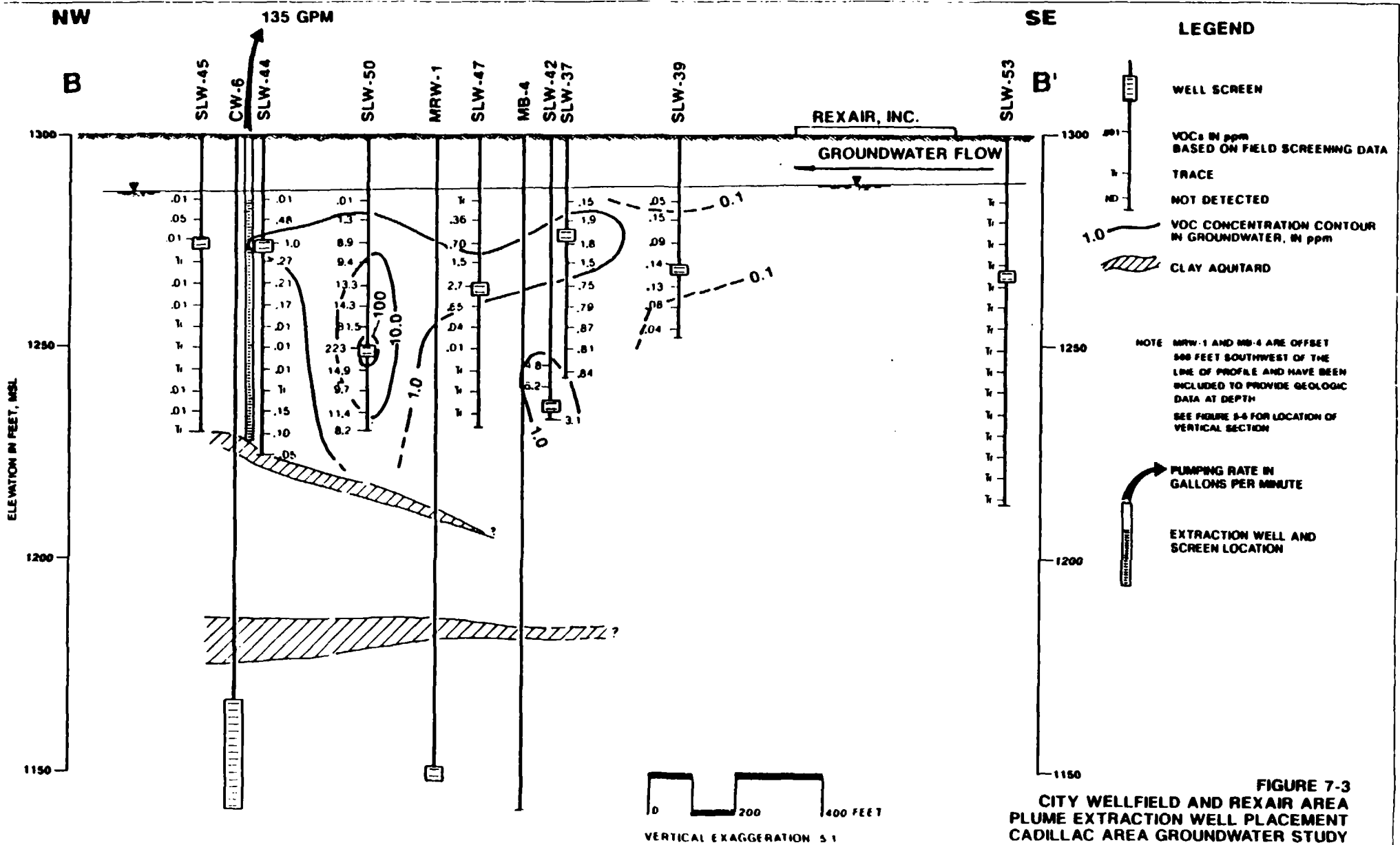
SERIES	SATURATED AQUIFER THICKNESS (ft.)	HYDRAULIC CONDUCTIVITY (cm/sec.)	HORIZONTAL HYDRAULIC GRADIENT	PLUME VELOCITY (ft./year) ¹	NO. OF WELLS IN SERIES	CAPTURE ZONE WIDTH (ft.)	PUMP RATE PER WELL (gpm)	TOTAL MILLION GALLONS PER YEAR	REQUIRED PUMPING TIME 1 PORE VOLUME (years)	REMOVAL TIME AT 90% EFFICIENCY (years)*	MILLION GALLONS TO TREAT AT 90% REMOVAL EFFICIENCY
Rexair	160	4.00E-02	1.10E-03	147	1	1,312	135	71	8.4	38	2,698
Kysor Shallow (A)	65	4.00E-02	1.50E-03	217	2	2,624	79	83	6.5	29	2,407
Kysor Shallow (B)	30	3.40E-02	3.30E-03	387	2	2,624	65	68	1.9	3.5	238
Four Winns	36	3.20E-02	3.80E-03	420	1	650	42	22	2.9	11	242
Kysor Intermediate	122	4.00E-02	9.10E-04	125	3	5,248	129	203	19.4	64	12,992
Hexavalent Chromium	30 ²	5.90E-02	9.30E-04	189	1	1,312	32	17	3.4	4	68
North Park	172	5.00E-02	6.80E-04	117	1	656	56	30	2.7	6.5	195

¹ Velocity calculated based on an assumed porosity of 0.3.

² The estimated thickness of the thin chromium plume is used instead of the total thickness of the aquifer.

* The extraction system has been evaluated for an assumed 90% efficiency of contaminant removal per volume of plume. The exact efficiency cannot be determined except under actual system operation. Efficiencies may be less than 100% due to intake of clean water, physiochemical responses, hydraulic behavior of the aquifer, and flushing by natural mechanisms. Other efficiencies were considered in the analyses, but the assumption of 90% seemed both reasonable and conservative.





(see Figure 7-2). The pump will be placed at an approximate depth of 100 feet below ground at the most concentrated zone of contamination.

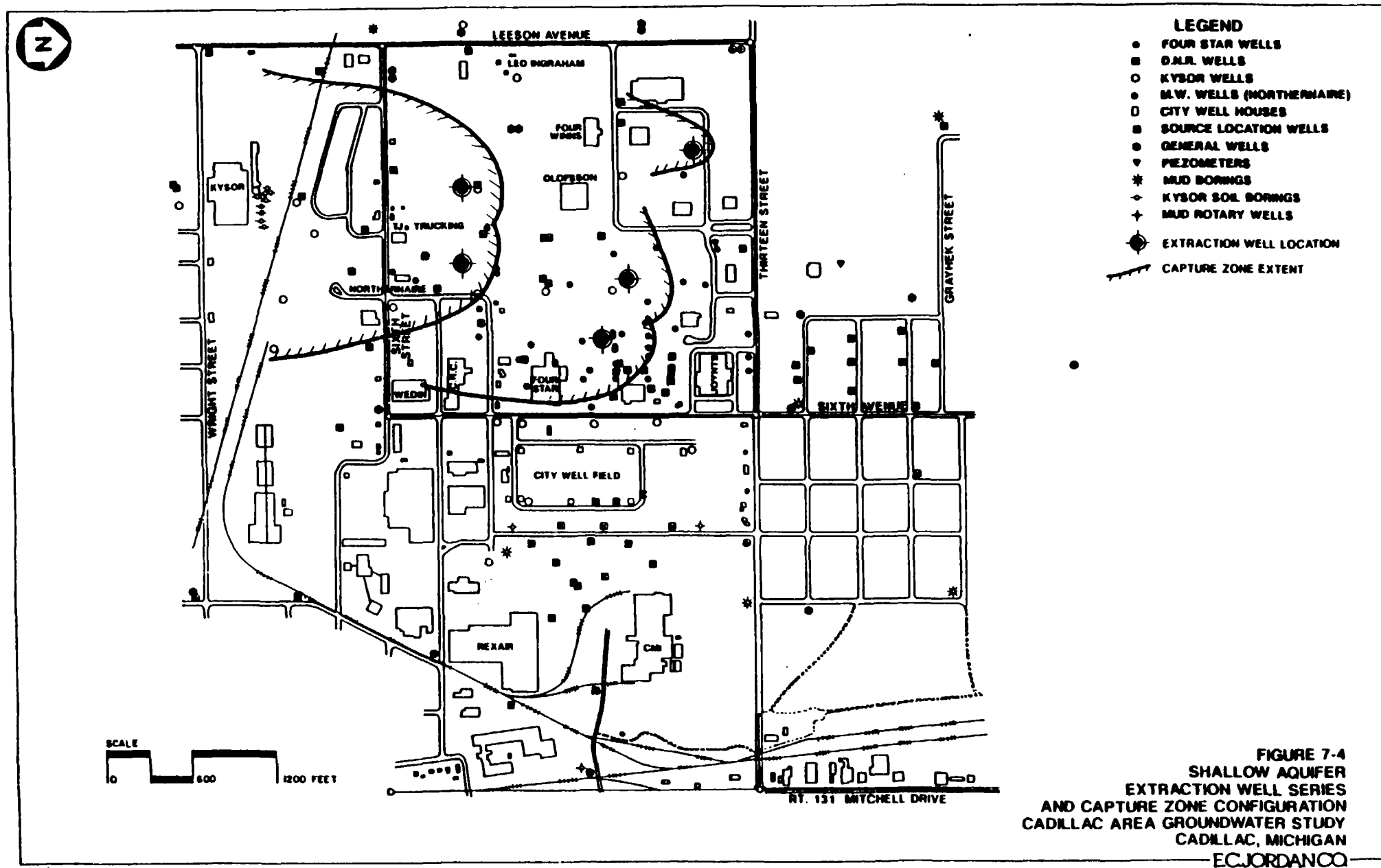
A range of contaminant recovery efficiencies is used to determine the number of pore volumes that will require pumping to reach TCLs. If it is assumed that after pumping each pore volume, 90 percent of the contaminant will be removed from the plume, then approximately 4.5 pore volumes will need to be pumped to decrease the contamination from 153 ppm to below detection levels of 0.005 ppm. Because it will take 8.4 years to extract one pore volume, it will require 38 years to reach the clean-up objectives for the Rexair Plume, assuming 90-percent contaminant removal efficiency. The percent of contaminant extraction for each well volume can only be estimated with currently available data. Better-defined estimates of time required for cleanup can only be determined after extraction and monitoring has started.

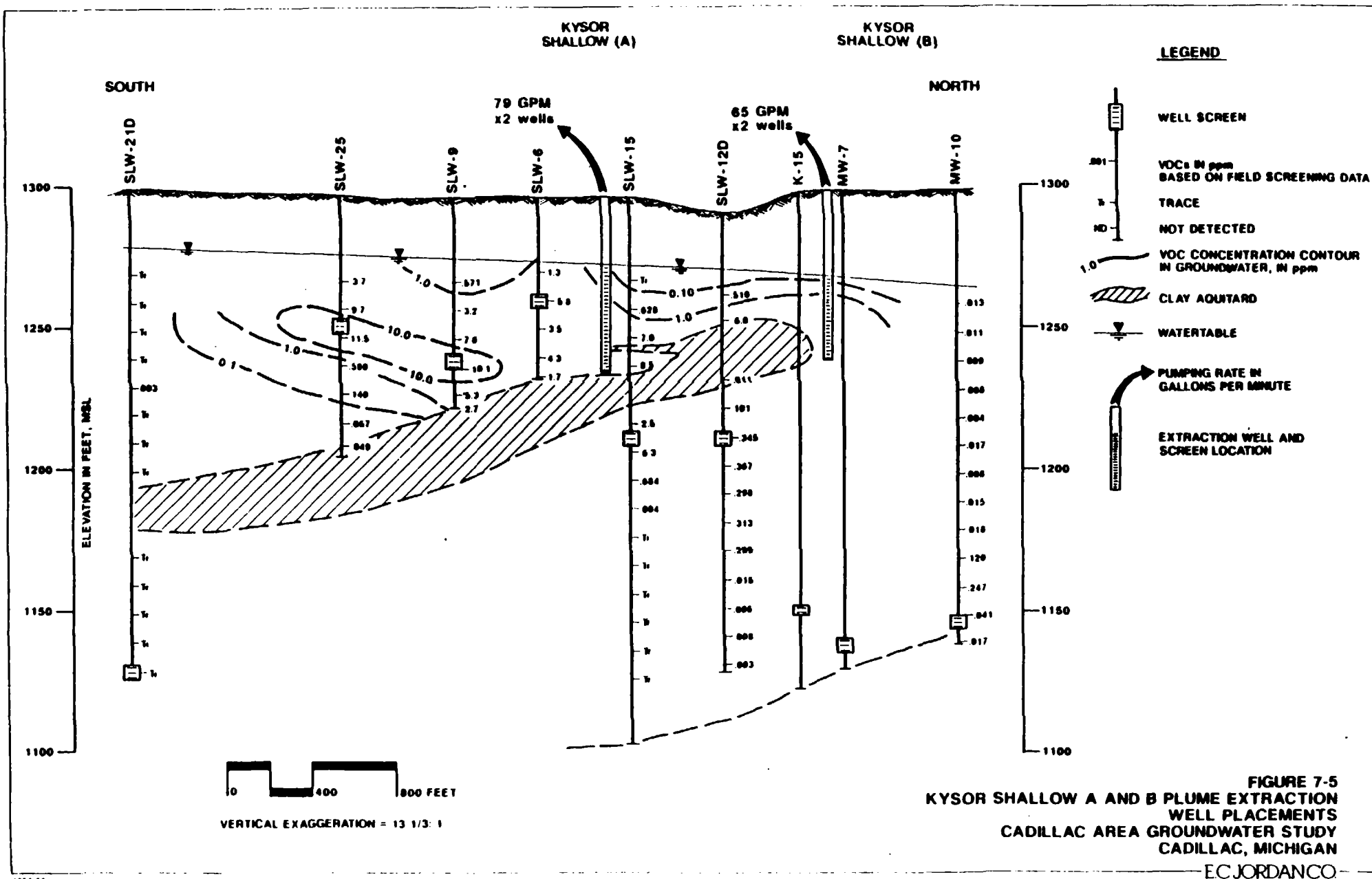
Approximately 71 million gallons per year of groundwater will be pumped; at 90-percent contaminant recovery efficiency, a total of 2.7 billion gallons will be pumped. This groundwater will be directed to the treatment facility located in the industrial park and ultimately discharged to the Clam River.

7.3.1.1.2 Concentrated VOC Shallow Plume (A)

Significant VOC contamination is found in the shallow aquifer where concentrations from 1987 sampling reached 115 ppm downgradient of known solvent dumping at Kysor. This subsection describes the pumping extraction system for the concentrated portion of the Kysor-related shallow VOC contamination. Separate pumping systems were designed for the shallow contamination where it merges with plumes from sources other than Kysor, and the intermediate aquifer contamination. The primary contaminants found in the concentrated VOC shallow plume include TCE, 1,1-DCE, 1,1,1-TCA, PCE, toluene, and xylene.

Due to the width of this shallow plume, a two-well pumping system is believed to be the most effective in capturing the contaminated groundwater. The two extraction wells will be placed at the leading edge of the contaminant plume, approximately 430 feet apart, and will be located just north of Sixth Street (Figure 7-4). Figure 7-5 is a cross sectional view of the contaminant plume based on screened auger field data used to determine extraction well screen and pump placement. Each well will be screened the entire saturated thickness to the clay aquitard at a depth of 50 to 75 feet. The pump will be placed at an approximate depth of 50 feet below ground at the most concentrated zone of contamination. Figure 7-4 shows in planar view the defined extent of the plume, the location of the extraction wells, and the associated capture zone. To achieve this capture zone configuration, each pumping well will need to pump at a rate of 79 gpm. The plume width is estimated at approximately 1,800 feet, with a length of 1,900 feet. The estimated time to extract one pore volume using the two pumping wells is 6.5 years. At a 90-percent recovery efficiency, it will require approximately 4.5 pore volumes and a total of 29 years to reach the TCLs. At a pumping rate of 83 million gallons per





year, a total of 2.4 billion gallons of groundwater will be pumped for treatment.

7.3.1.1.3 VOC Shallow Plume (B)

Less concentrated but more areally extensive shallow contamination by VOCs is present downgradient of the series of wells developed for the concentrated VOC shallow plume and extends northeastward to the limit of the upper clay aquitard. Much of this contamination is Kysor-related; however, at the eastern and western edges of the plume, VOC contributions by Four Star and Four Winns, respectively, are evident. A second series of shallow aquifer extraction wells was designed to pump this contamination situated near the northeastern edge of the aquifer.

Two pumping wells spaced about 400 feet apart are located along a northwest-southeast trending line in the central portion of the industrial park. These wells will extend to the bottom of the shallow, unconfined aquifer (approximately 55 feet deep) and will be screened the entire depth of the saturated zone. The pumps will be placed at a depth of 50 feet in the zone of highest contamination (Figure 7-5). The capture zone width (see Figure 7-4) required to collect the contaminated groundwater will be achieved by pumping each well at a rate of about 65 gpm. Estimating the plume length at approximately 1,100 feet, it will require one year to extract one pore volume from the shallow aquifer. At a 90-percent recovery efficiency, 3.5 years will be required to reach TCLs, and 238 million gallons of groundwater will be extracted.

7.3.1.1.4 Four Winns Shallow Plume

A separate extraction well for shallow aquifer contamination at the Four Winns property was designed because it is a localized area where contaminant concentrations are relatively high. Total VOC concentrations approach 20 ppm with TCA, TCE, acetone, and PCE as primary contaminants. Pumping at a rate of 42 gpm will create a capture zone of sufficient size to collect the contaminated groundwater. The well will be placed near Saunders Street (see Figure 7-4) at the edge of the upper clay aquitard to a depth of approximately 60 feet. The pump will be placed at an approximate depth of 45 feet below ground in the zone of highest contamination. Figure 7-6 is a cross sectional illustration of the contamination based on screened auger field data and extraction well pump and screen placement. At this pumping rate, it will require three years to collect one pore volume, assuming a plume length of 1,500 feet. At a 90-percent contaminant recovery efficiency, it will take 12 years to reach TCLs, and a total of 265 million gallons of groundwater will be extracted.

7.3.1.1.5 VOC Intermediate Plume

Due to migration through the upper clay aquitard, significant VOC contamination has reached the confined intermediate aquifer downgradient of the Kysor facility. Concentration levels were measured at 12 ppm during the 1987 sampling of groundwater at SLW-10D; the primary contaminants were TCE and t-1,2-DCE.

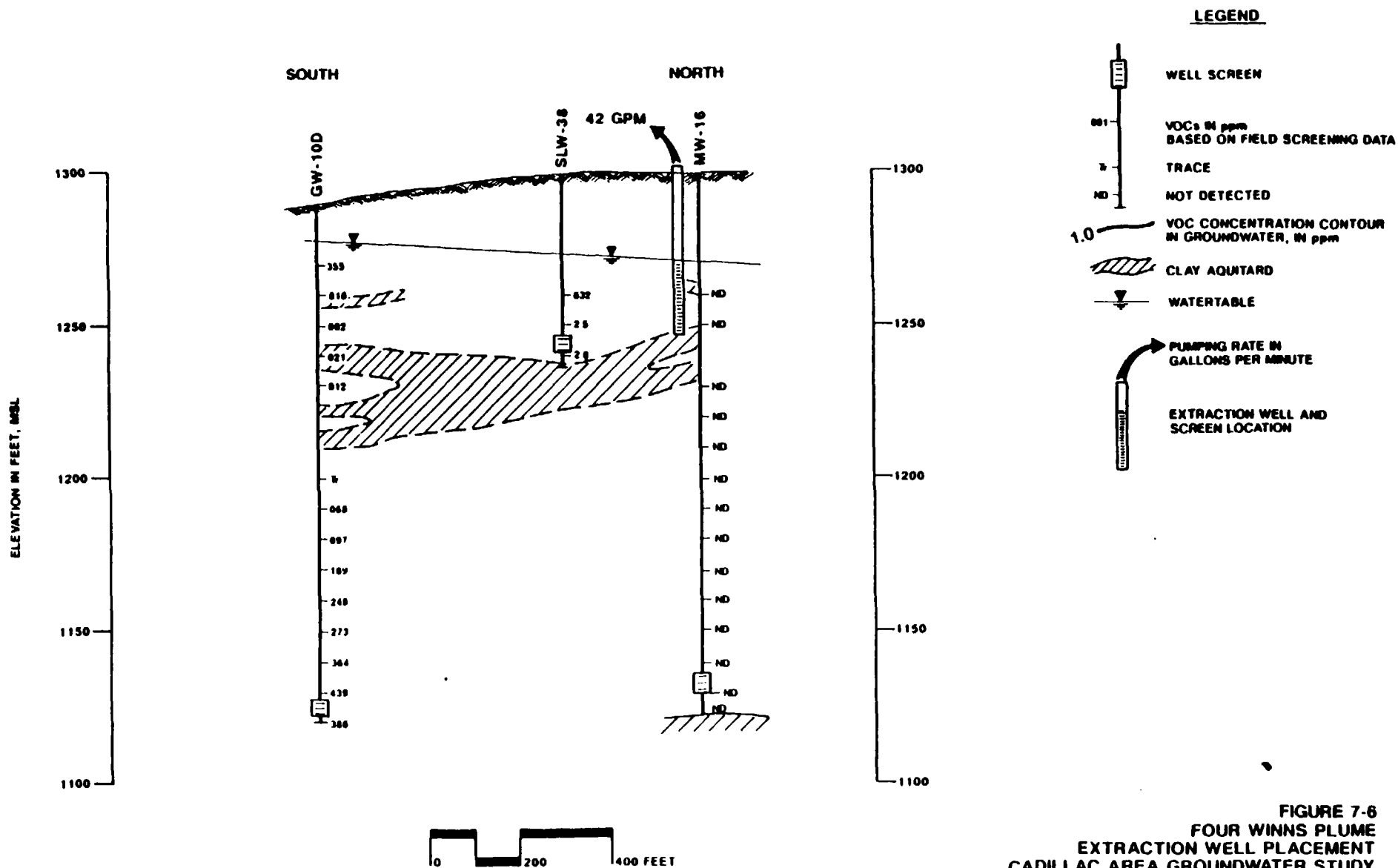


FIGURE 7-6
FOUR WINNS PLUME
EXTRACTION WELL PLACEMENT
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN
EC JORDAN CO

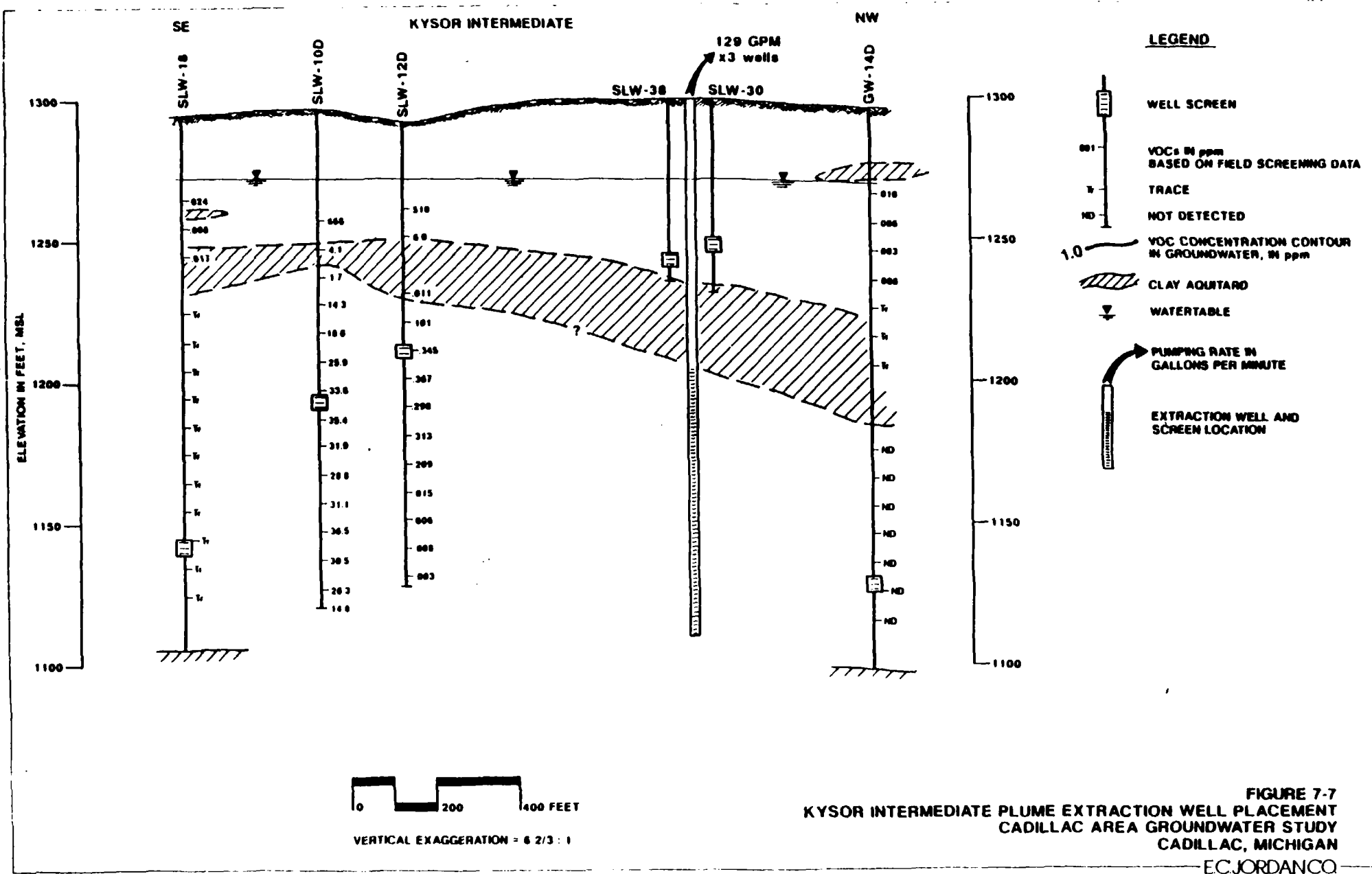
The contamination in this intermediate aquifer, although less concentrated, is areally extensive, requiring the development of a large capture zone. A pumping extraction system using three wells spaced approximately 795 feet apart was designed to capture the contaminated groundwater. The proposed locations of the wells are shown in Figure 7-2. To create the required capture zone width of approximately 5,200 feet, a pumping rate of 129 gpm from each well is required. Figure 7-7 is a cross sectional view of the contaminant plume used to determine extraction well screen and pump placement. The two southernmost pumping wells will be screened in the confined portion of the intermediate aquifer, with screens extending from approximately 60 to almost 200 feet below ground. The pumps will be placed at an approximate depth of 110 feet below ground at the zone of highest contamination. The northernmost pumping well will be located where the intermediate aquifer is unconfined, and will have a screened interval from 30 to 175 feet below ground. The pump will be placed at a depth of approximately 110 feet below ground at the zone of highest contamination. The plume length is estimated at 3,400 feet. At the 129-gpm pumping rate, it will require 19.4 years to extract one pore volume from the aquifer. Assuming a 90-percent contaminant recovery efficiency with each pore volume pumped, it will require 64 years to reach TCLs, and 13 billion gallons of groundwater will be pumped for treatment.

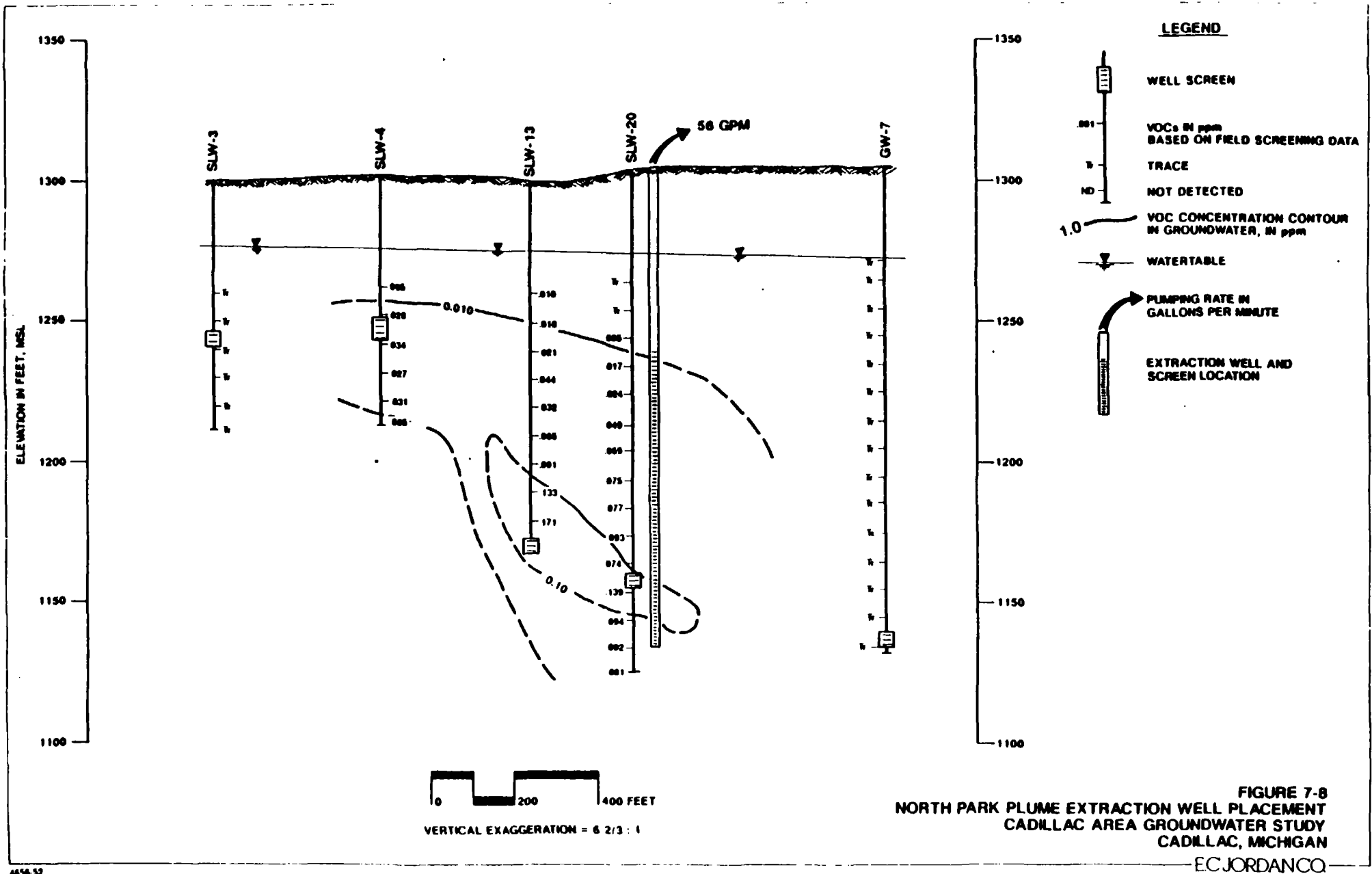
7.3.1.1.6 North Park Subdivision Plumes

The two groundwater plumes in the North Park Subdivision are relatively small with weak concentrations. Based on hydrogeologic data, these plumes may be extensions of Four Star- and Joynts-related contamination.

The two plumes are at discrete depths, with the more downgradient plume being the deepest and more concentrated. Figure 7-2 shows the plumes, as defined by the 1987 groundwater sampling. Figure 7-8 is a cross sectional view based on screened auger field data. The deepest plume has a concentration in the range of 0.600 ppm in SLW-20. Screened auger sampling of the groundwater was conducted to a depth of 180 feet below ground, at which point the augers could not be penetrated farther. These data indicate that the maximum contaminant concentration occurs at approximately 150 feet, but levels are still significant at the 180-foot depth. The shallower plume, defined by SLW-4 and SLW-2, is approximately 55 feet deep, with approximate concentrations of 0.090 ppm.

A one-well pumping system was designed to collect groundwater from the deeper plume. The well will be placed to a depth of 200 feet and screened the entire saturated thickness, beginning at approximately 30 feet below ground. The pump will be placed at a depth of approximately 150 feet below ground at the zone of highest contamination. The pumping rate will be 56 gpm to create a capture zone width of approximately 656 feet. One pore volume will be collected in approximately three years and, assuming a 90-percent contaminant recovery efficiency, 6.5 years will be required for cleanup. A total of 68 million gallons of groundwater will be extracted for treatment.





The concentration of the shallower plume is approximately 0.090 ppm. With time and mixing with clean groundwater, these contaminant levels should approach nondetectable levels. Future groundwater monitoring will determine the effectiveness of in-situ dilution and the need for installing an extraction system for this plume.

7.3.1.1.7 Hexavalent-Chromium Plumes

Two hexavalent-chromium plumes were identified in the Cadillac Industrial Park: one in the shallow aquifer and one in the intermediate aquifer. The shallow aquifer plume was detected during just one time period in MW-5, located 500 feet downgradient of the Northernnaire facility. The concentration of hexavalent chromium in MW-5 was 1.6 ppm in February 1986, and had decreased to 0.5 ppm during the May/June 1987 sampling episode. Hexavalent chromium was not detected in subsequent sampling episodes, nor was it detected in prior episodes. This intermittent detection suggests a small, rapidly moving plume that should eventually reach the northeastern limit of the shallow aquifer while becoming diluted by groundwater mixing. Because the exact position of this plume is unknown, an extraction system was not developed for the shallow aquifer.

The intermediate aquifer hexavalent-chromium plume has been studied and sampled since the Northernnaire groundwater investigation in 1985. The plume is only 10 to 20 feet thick and, at a depth of approximately 160 feet, is situated on top of the lower clay aquitard. Figure 7-9 is a cross sectional view of this plume. In 1986, chromium concentrations were in the 2.2-ppm range, but had decreased to 0.8 ppm during the 1987 sampling. The extent of the plume is well-defined, located in the central portion of the industrial park. The plume length is approximately 1,000 feet, elongated northwest-southeast in the direction of groundwater flow.

A one-well extraction system was designed to capture this deep hexavalent-chromium plume. The pumping well, located as shown in Figure 7-2, and in vertical profile in Figure 7-9, will be installed to a depth of 175 feet and screened from 145 to 175 feet. The short well screen will be used because of the plume thinness. The pump will be placed at a depth of approximately 165 feet below ground in the zone of highest contamination.

To create a capture zone width of approximately 1,312 feet, a pumping rate of 32 gpm will be required. It is estimated that 67 million gallons of groundwater and four years will be needed to extract the hexavalent chromium at a contaminant recovery efficiency rate of 90 percent.

7.3.1.2 Collection Piping. Figure 7-10 presents the plan view of the interconnection piping for the extraction wells. Approximately 9,150 feet of pipe would be required to interconnect the system. Force mains and gravity sewers would be used, depending on topography of the area where piping is to be laid. Submersible pumps would pump groundwater from each well into a manhole (when discharge from well flows into a gravity sewer), where it would then enter the collection piping. The system is designed to transport all flow to an enclosed wet well inside the treatment facility, which would be situated

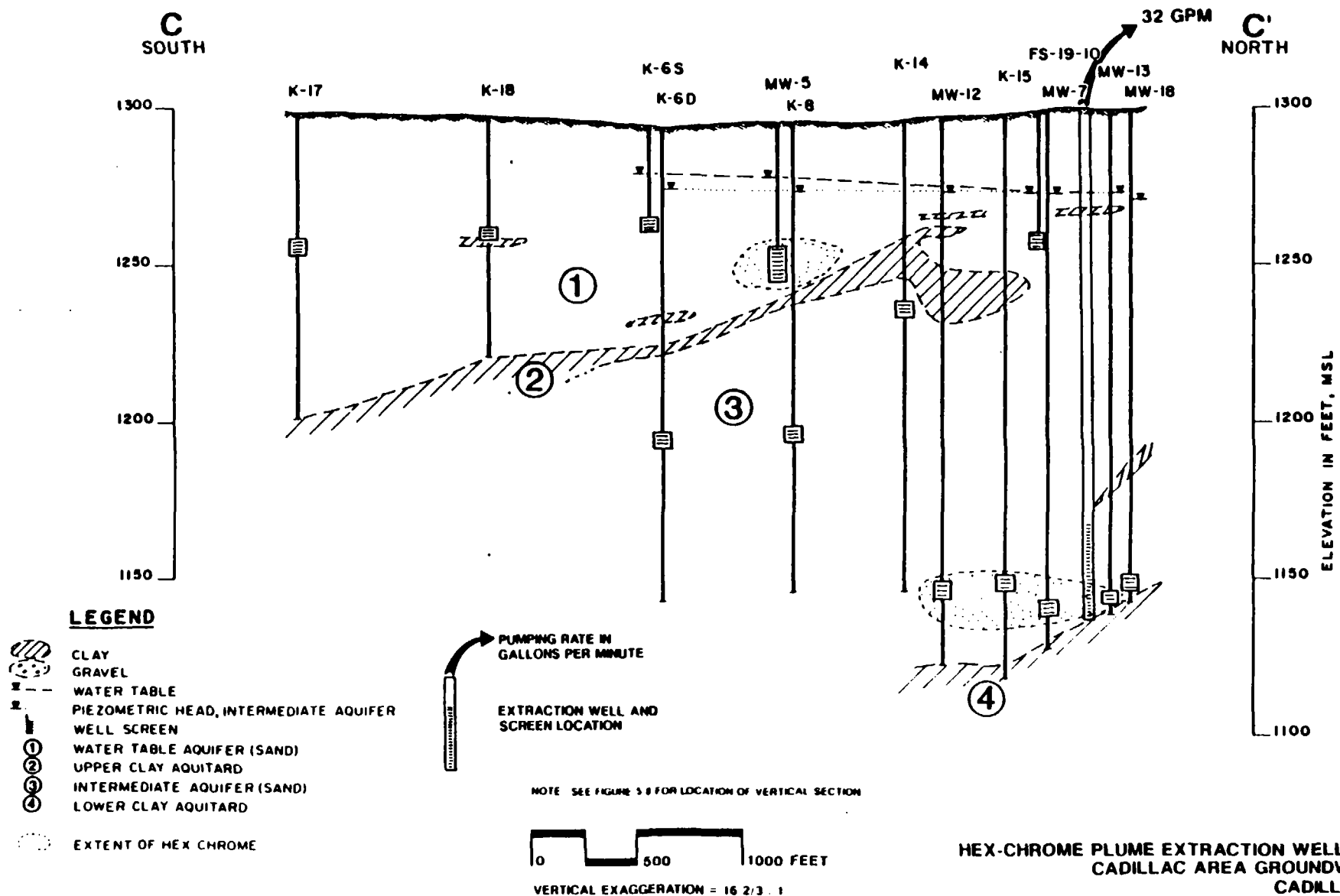
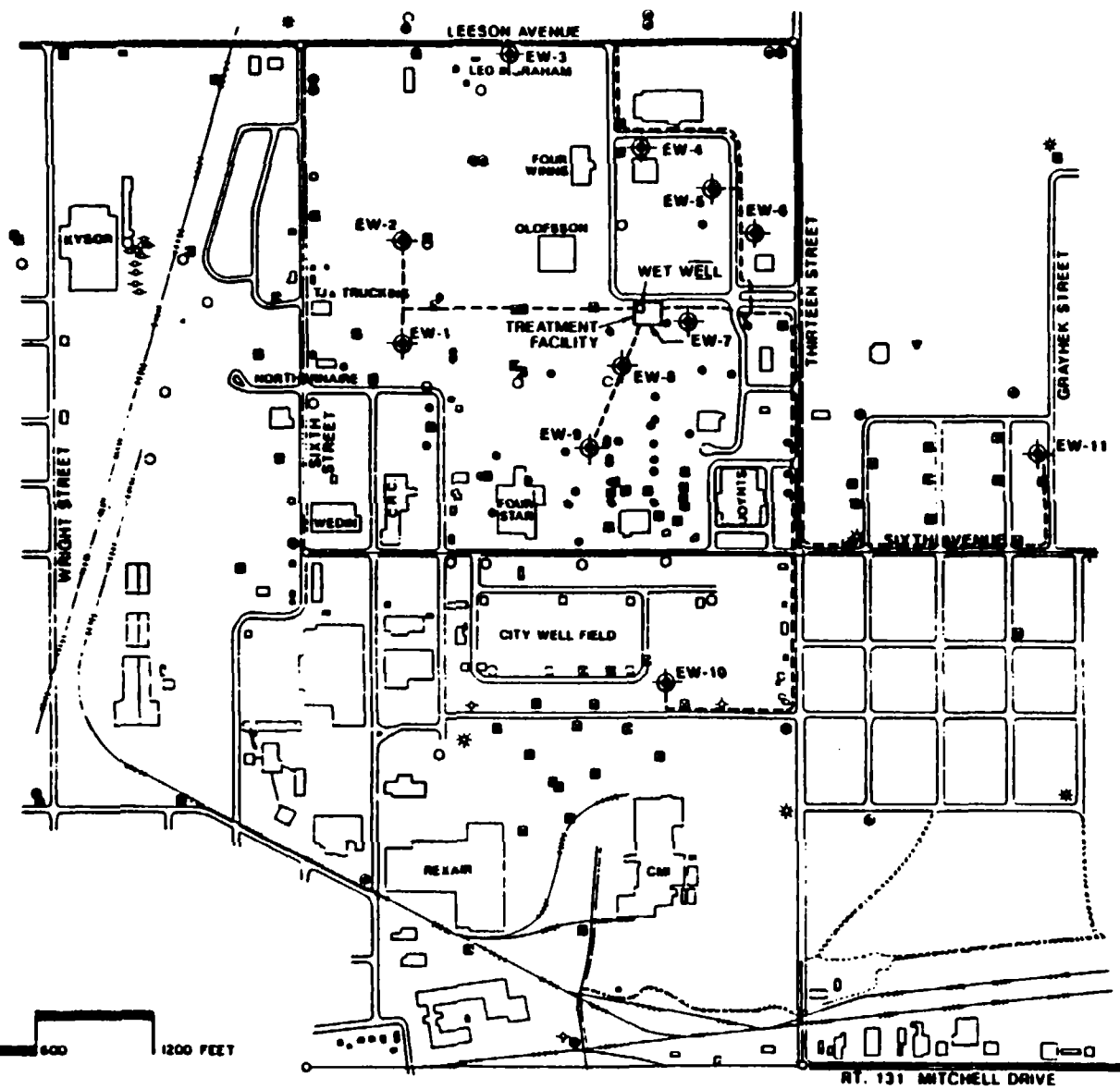


FIGURE 7-9
HEX-CHROME PLUME EXTRACTION WELL PLACEMENT
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN
EC JORDAN CO



- LEGEND**
- FOUR STAR WELLS
 - D.M.R. WELLS
 - KYSON WELLS
 - M.W. WELLS (NORTHERNAIRE)
 - CITY WELL HOUSES
 - SOURCE LOCATION WELLS
 - GENERAL WELLS
 - PEZOMETERS
 - * MUD BORINGS
 - ◆ KYSON SOIL BORINGS
 - ◆ MUD ROTARY WELLS
 - EW-1 ◆ EXTRACTION WELL LOCATION
 - COLLECTION PIPING

FIGURE 7-10
EXTRACTION SYSTEM COLLECTION PIPING
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN
ECJORDANCO

east of Holman Street, north of its intersection with Frisbee Street. The facility would house equipment for both VOC and hexavalent-chromium treatment systems (see Figure 7-10). If North Park Subdivision flow was added to the system, approximately 1,724 feet of additional piping would be required.

7.3.1.3 Monitoring of Groundwater Extraction Effectiveness. A groundwater sampling and analysis program was developed to evaluate the effectiveness of the remediation by groundwater extraction. The purposes of the monitoring program are to (1) assess the amount of contaminant reduction near the center and at the edges of each identified plume by sampling groundwater from monitoring wells at these locations; and (2) ensure that contamination is not migrating in the direction of regional groundwater flow. This would be accomplished by obtaining water level measurements to determine groundwater flow directions in the plume areas toward the extraction wells for each of the six proposed groundwater extraction systems. The North Park Subdivision Plume, for which the No-action Alternative is recommended, also will be monitored to assess dilution of the contamination by natural groundwater mixing.

Table 7-5 outlines the sampling program for each plume area. Wells installed for the remedial investigation program were identified for periodic sampling in this proposed monitoring program. Groundwater sampling will occur quarterly for the first year, after which the sampling will be reduced to an annual frequency. Based on expected clean-up rates, this sampling schedule should be adequate; however, it may be further refined, as needed, upon review of the actual performance of the installed systems. Groundwater elevation measurements of all monitoring wells, taken annually, will define the extent of the extraction well capture zones and determine if any contaminated groundwater is flowing past the wells. Flow rates of the extraction wells may then be adjusted to maintain the desired control without excessive pumping and collection of clean water.

The chemical analysis program will involve laboratory testing of the parameters specific to the contaminant plumes. The same parameters and test methods described for no-action monitoring will be used to monitor the effectiveness of the extraction system (see Section 7.2.1). Appendix G illustrates the monitoring well locations for monitoring the effectiveness of the extraction system.

7.3.2 Compliance with ARARs

There are no specific ARARs associated solely with groundwater extraction. The federal and Michigan Safe Drinking Water acts and rules govern the construction and use of wells for the supply of potable water. Part 8 of the Michigan act should be considered and MDPH should be consulted before final selection of all extraction, monitoring, and treatment systems. Depending on the location of the proposed extraction systems (e.g., on private property), prior approval of selected locations may be required by the state and the Town of Cadillac. Because the Kysor site is proposed for inclusion on the NPL, location of extraction and treatment systems within the Kysor site boundaries will not be subject to administrative requirements such as a formal permit and approval process.

TABLE 7-5
GROUNDWATER SAMPLING AND ANALYSIS PROGRAM
FOR EXTRACTION SYSTEM EFFECTIVENESS ASSESSMENT

CADILLAC AREA FEASIBILITY STUDY

EXTRACTION WELL SYSTEM	MONITORING WELL IDENTIFICATION		ANALYSIS
	CENTRAL PLUME LOCATION	PERIPHERAL PLUME LOCATION	
Rexair	SLW-41, 50	SLW-39, 45, 52, 46	VOA By #8010
Kysor Shallow (A)	SLW-6, 9, 25	SLW-11 MW-6	VOA By #8010 Acetone By #524.2
Kysor Shallow (B)	SLW-10S, 12S FS-1-10	FS-19, 20	VOA By #8010 Acetone By #524.2
Four Winns	SLW-38 FWMW-16	FWMW-7, 8 K-16D	VOA By #8010 Acetone By #524.2
Kysor Intermediate	SLW-10D, 12D GW-10D	SLW-7, 49 GW-9D, 14D	VOA By #8010
Hexavalent Chromium	MW-7, 9 K-15	MW-18	Hexavalent Chromium By #7197
North Park	No-action Alternative*		

*NOTE: The groundwater sampling and analysis program for the North Park
Subdivision is provided on Table 7-2 for the No-action Alternative.

When installing the groundwater extraction system, care should be taken that fugitive dust emissions do not exceed the PM10 standard (see Table 4-1 CAA NAAQS for particulate matter). For estimating fugitive dust, the guidelines in the EPA Superfund Exposure Evaluation Manual should be followed. The state should be consulted as to the requirements for compliance with any fugitive dust emission regulation.

Extraction of contaminated groundwater is consistent with the intent of the RCRA Corrective Action regulations (40 CFR 264.100) and the USEPA GWPS because contaminated groundwater will be removed from the ground. Appendix E identifies action-specific ARARs associated with this alternative. These ARARs are further detailed in Section 4-1 and summarized in Table 4-1.

7.3.3 Protection of Public Health and the Environment

This analysis of protection of public health and the environment assumes that the extraction system will capture the contaminated groundwater so that TCLs are achieved. It is also assumed that the sources of contamination to the aquifer plumes are no longer present. The extraction monitoring proposed in this FS is therefore crucial to ensure that the pumping system can achieve the remedial objectives for which it was designed.

The extraction system is expected to remove the contaminants in the groundwater so that, by dilution, groundwater contaminant concentrations meet the TCLs. These TCLs were set to meet the chemical-specific ARARs or, where not available, the appropriate risk levels required by the state. Because the health-based ARARs for PCE and methylene chloride are AWQC that are below analytical detection limits, the TCL was set at a 10^{-6} risk level. (USEPA will propose an MCL of 0.005 mg/l for PCE in January 1989. However, this MCL will not be enforceable until after public comment.)

As discussed previously, the possibility exists that concentrated pools of free product exist in both the unsaturated and saturated zones, which will greatly affect clean-up efforts. If concentrated pools exist, contaminant concentrations in the groundwater will increase shortly after the pumping system has shut down, and obtaining TCLs may not be possible. Through groundwater monitoring, the presence of these pools will be determined by the observed effectiveness of the extraction system after several years of pumping.

7.3.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

Groundwater extraction would reduce the mobility of groundwater contaminants by depressing the groundwater table and drawing water toward the extraction wells. The extraction systems were designed so that the cones of influence of the wells will capture enough contaminated groundwater to meet the TCLs.

7.3.5 Short-term Effectiveness

The operation of the extraction system is not expected to impact workers, although physical hazards during construction of the pumping wells would require the necessary precautions to ensure worker safety. Any wastewater generated during the drilling of the deep recovery wells considered hazardous

should be disposed of properly. No short-term impacts on public health are anticipated as a result of construction of the extraction wells.

Each extraction system for the six plumes would be designed to operate according to the number of years required to attain the groundwater TCLs in that plume. Operational times range from four years (i.e., the small hexavalent-chromium plume) to 64 years (i.e., the Kysor intermediate aquifer plume). Table 7-4 presents operational times required to achieve full protection for each plume. These estimates assume that the extraction systems would be 90-percent efficient in flushing contaminants from the plume.

If the North Park Subdivision Plume was included for extraction, a single well would pump for approximately 6.5 years to achieve groundwater TCLs.

7.3.6 Long-term Effectiveness

To evaluate the long-term effectiveness of extraction, the length of time to achieve TCLs is predicted to range from four years for the small hexavalent-chromium plume to 64 years for the Kysor intermediate aquifer plume. Therefore, although the extraction system designed for the Cadillac site will remediate the groundwater to the TCLs, potential risks to public health will exist during the remediation effort. However, no individuals are currently known to be drinking from contaminated wells. Institutional controls might be appropriate during the remediation period, banning any homeowners from utilizing domestic wells in areas of potential risk.

The total residual cancer risk associated with the final TCLs for the shallow plume equals 9.89×10^{-5} . For the Intermediate Plume, the total residual risk equals 8.59×10^{-5} .

The total residual risk associated with the Rexair Plume at TCLs is 2.64×10^{-6} . The total carcinogenic residual risks associated with the TCLs for these are within USEPA's target risk range of 10^{-7} to 10^{-4} . The Northernair Plume contains only one chemical of concern considered a risk to public health: hexavalent-chromium, which is noncarcinogenic and has an MCL of 0.05 mg/l. The TCL is set at this ARAR.

If the North Park Subdivision was added to the extraction system, groundwater would be extracted and collected for approximately 6.5 years to attain groundwater TCLs. There are two plumes at discrete depths in the North Park Subdivision. The deeper plume has the higher measured VOC concentration level. The one-well pumping system will only collect groundwater from the deeper plume. The remaining shallower plume has total VOC concentrations of approximately 0.090 ppm. Assuming that all the VOCs are TCE (based on the finding reported in the RI that the North Park Subdivision Plume contamination was mostly TCE), the incremental cancer risk in the shallower plume would be 2.83×10^{-5} , and the incremental risk in the deeper plume would be 1.88×10^{-4} . The incremental cancer risk in the shallower plume falls within the USEPA target risk range. Therefore, the decision to not extract the shallow plume in the North Park Subdivision is not expected to pose any significant public health risks.

Because of the presence of VOCs in the groundwater plumes, the extraction piping would be constructed of a resistant material (e.g., polyethylene) to maintain long-term effectiveness. The polyethylene pipe is expected to have a design life of 50 years. To protect against frost conditions, the piping would require a minimum cover depth of 5 feet. For areas where site topography would dictate cover depth to be less than 5 feet, an insulating material should be placed over the piping before backfilling. Manholes would be installed along the gravity sewer sections of the collection system at 300-foot intervals and at locations along the system where the piping route bends. A total of seven manholes would be utilized to inspect the integrity of the gravity system. Manholes would be lined with a resistant material (e.g., polyethylene), and locks and covers would be placed over the manholes to protect the public and the environment from the potential for direct contact with contaminated groundwater.

7.3.7 Implementability

Installation of extraction wells and collection piping is technically feasible at the Cadillac site. Many wells, including municipal production wells and monitoring wells, were installed successfully at the site. Drilling contractors and equipment are available for on-site use. Extraction well locations were selected in areas where installation would not interfere with the integrity of existing buildings, utilities, and other structures.

To install the collection piping, existing on-site utilities would have to be considered. Water, sewer, storm, natural gas, electrical, and telephone lines run underground throughout the site. Exact locations of these utilities would have to be identified before construction activities. The collection piping would have to be located at least 10 feet from existing water supply lines and on the opposite side of the street, where possible. The existing roadways generally have a 66-foot right-of-way, and several roads within the industrial park have a 10-foot utility right-of-way on either side of the road. These areas should be utilized to lay the piping. The appropriate rights-of-way would have to be attained prior to construction activities. Installation of collection piping would take approximately three months. Installation of the extraction wells would take approximately two weeks per well, or 22 weeks for 11 wells.

7.3.8 Costs

In this section, capital and annual costs are presented for the extraction system. In addition, the net present worth of extraction is presented. The costs listed herein were incorporated in the overall costs of the treatment alternatives that use extraction.

Capital Costs. The capital cost estimate for extraction includes all direct costs (i.e., construction and equipment costs). Indirect costs, such as engineering and contingency costs, are not presented herein, but are calculated as a percentage of the total capital costs for treatment alternatives that include extraction costs. The capital cost of the extraction system for the industrial park would be \$762,000. If remediation of the North Park Subdivision Plume was included in the remedial alternative, additional capital costs

for extraction would be \$122,000, for a total of \$884,000. A detailed cost estimate of extraction is presented in Appendix D.

Annual Costs. As discussed in Section 7.3, the estimated time for each of the seven extraction well series to pump the contaminated groundwater and reach TCLs is different. Therefore, annual costs will be assumed for different time frames, depending on the well series. Annual costs for the well series for the individual contaminant plumes are outlined in Appendix D. The duration of annual costs ranges from four to 64 years.

Present Worth Analysis. Based on the various performance periods of the individual well series and a 5-percent interest rate, the present worth of the annual costs for extraction is \$1,936,000. The net present worth, which includes capital costs, is \$2,820,000.

Sensitivity Analysis. A sensitivity analysis, assessing the effect that variations in the extraction system's efficiency will have on the present worth, is presented in Section 9.1.

7.3.9 Summary

The summary of the detailed analysis of extraction is presented within the summaries of the overall treatment alternatives, which include extraction (see Sections 7.6.9, 7.7.9, and 7.8.9).

7.4 DETAILED ANALYSIS OF DISCHARGE TECHNOLOGY

For the three on-site treatment alternatives (i.e., Alternatives GW-2, GW-3, and GW-4), effluent may be discharged to surface water. A system designed to discharge treated water to the Clam River is discussed in this section.

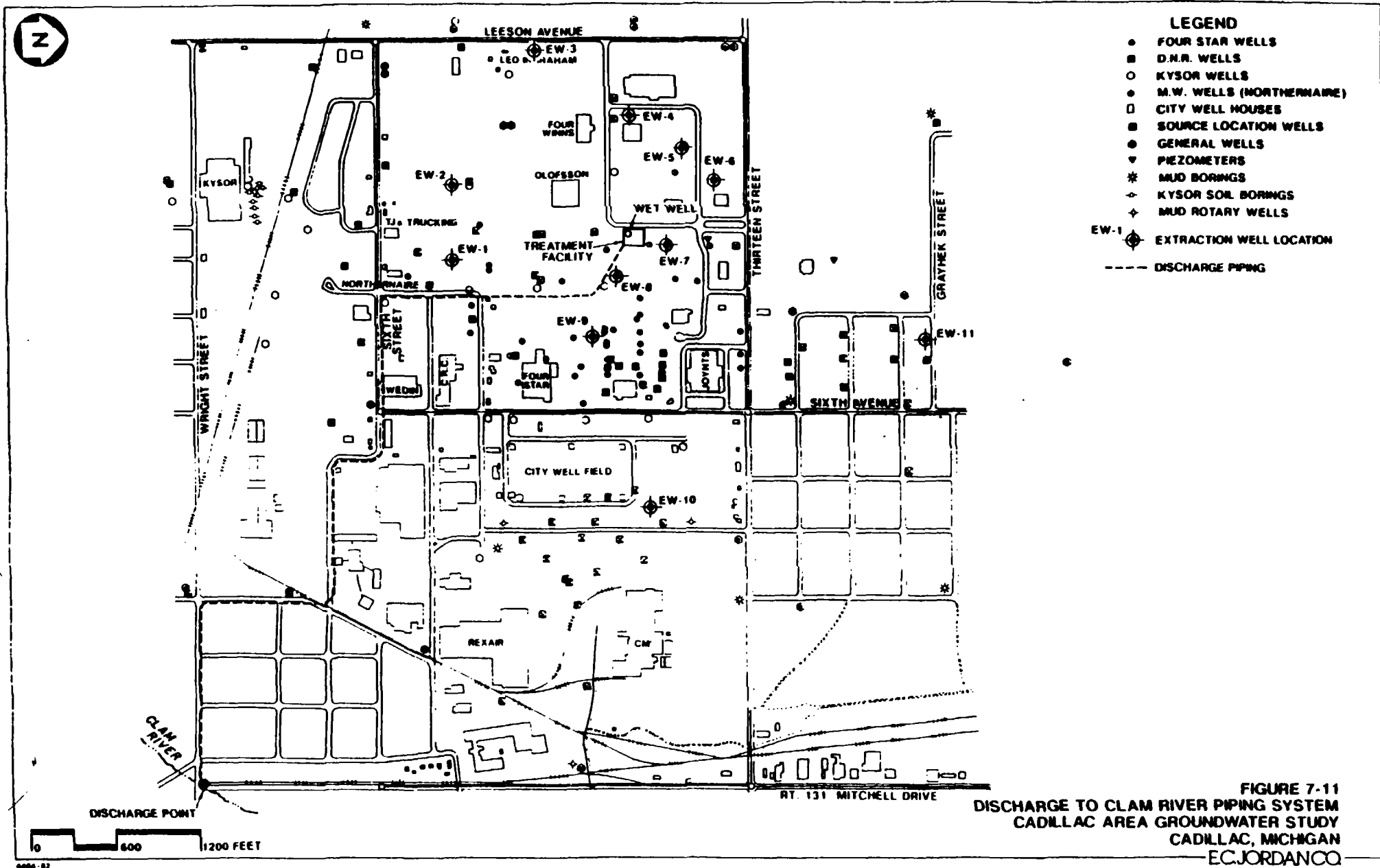
7.4.1 Description

As a component of the groundwater treatment alternatives, treated water can be discharged to the Clam River. Discharge to the river would require construction of a force main constructed of polyvinyl chloride (PVC) piping. Approximately 6,000 feet of piping would be required to discharge the treated water. The discharge point (Figure 7-11) would be located on the Clam River at its junction with River Street, on the northern side of River Street and the western side of Mitchell Street (U.S. Route 131).

If treatment of the North Park Subdivision was included in a treatment alternative, the discharge scenario would remain the same. The increase in treatment flow rate would be approximately 10 percent and would not require any significant changes in the preliminary design for the discharge system to the Clam River.

7.4.2 Compliance with ARARs

Discharges to surface water must comply with several federal and state requirements. Michigan is authorized to administer the NPDES permit program which



governs discharges to surface waters. Under the Michigan Wastewater Discharge Permit Rules, MDNR has established technology-based discharge levels for total VOCs and hexavalent chromium to the Clam River. All proposed groundwater treatment technologies at the Cadillac site are expected to reduce concentrations of VOCs and hexavalent chromium to the discharge levels established by MDNR. Because discharge will occur off-site, a Michigan Wastewater Discharge Permit must be obtained. In addition to obtaining a discharge permit and meeting the discharge limitations, routine completion of monitoring records must be performed in accordance to R 323, Part 21.

Prior to placement of piping connecting the groundwater extraction and treatment systems to the point of discharge at the Clam River, location approval and any rights-of-way must be obtained from the state and local governments wherever public and private properties may be impacted.

ARARs associated with the North Park Subdivision are the same as ARARs associated with the industrial park, except that due to the proximity of residential homes in the North Park Subdivision, approval of the location of extraction and treatment systems in this area must be obtained prior to construction of these facilities.

All on-site remedial activities must be conducted in compliance with OSHA Standards for the Hazardous Waste Operations (29 CFR 1910). Appendix E identifies action-specific ARARs associated with discharge. These ARARs are further detailed in Section 4.1 and summarized in Table 4-1.

7.4.3 Protection of Public Health and the Environment

The discharge system is designed to carry the treated groundwater to the discharge point on the Clam River. The effluent is expected to attain the appropriate standards for surface water discharges. The final discharge limitations have been set by MDNR.

The discharge limitations proposed for the Cadillac site by MDNR for TCE (i.e., 20 µg/l), one of the chemicals MDNR suggests for monitoring, is over 1,000 times less than the USEPA's AWQC for aquatic organisms (acute and chronic; 45,000 and 21,900 µg/l, respectively). The discharge limitations suggested by MDNR for total chromium and hexavalent chromium (52 and 6 µg/l) are also less than the USEPA AWQC for aquatic organisms (980 and 120 µg/l). MDNR's discharge limitation for 1,1,1-TCA (i.e., 10 µg/l), the second purgeable halocarbon recommended for monitoring, is over 1,000 times less than the AWQC for public health, assuming fish and water consumption (i.e., 18,400 µg/l).

7.4.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

Discharge of treated groundwater would not reduce the toxicity, mobility, or volume of groundwater contaminants. The discharge would occur following treatment of the groundwater, and serves only to transport flow to the discharge point on the Clam River.

7.4.5 Short-term Effectiveness

During construction of the force main system for the discharge technology, the appropriate worker safety precautions need to be followed. No community health impact would result directly from the construction or implementation of this remedial action in the short-term. During pipeline construction, there may be temporary effects on indigenous wildlife.

The discharge system for the treated groundwater would be designed to operate for 64 years, based on the time required to treat the most extensive plume; that is, the Kysor Intermediate Plume.

7.4.6 Long-term Effectiveness

By discharging the treated groundwater to the Clam River, potential impacts to public health due to ingestion of contaminated groundwater are eliminated. Discharge levels to the Clam River will be at or below MDNR discharge limitations. Because the Michigan standards prohibit toxic substances in surface water at levels that may be harmful to the public health, safety, or welfare, or plant and animal life, the public health and ecosystem should not be threatened by discharges to the Clam River.

Because the groundwater to be discharged would be a treated flow, the piping would not have to be a resistant material (e.g., polyethylene). The discharge piping would be constructed of PVC. To protect against frost conditions, the piping would require a minimum cover depth of 5 feet. For areas where site topography would dictate cover depth to be less than 5 feet, an insulating material should be placed over the piping prior to backfilling.

7.4.7 Implementability

To install the discharge piping, existing on-site utilities would have to be considered. Water, sewer, storm, natural gas, electric, and telephone lines run underground throughout the site. Exact locations of these utilities would have to be identified before construction activities. The discharge piping would have to be located at least 10 feet from existing water supply lines and on the opposite side of the street, when possible. The existing roadways generally have a 66-foot right-of-way, which should be utilized to lay the piping. A right-of-way would have to be obtained from the state where piping may cross existing railroad tracks (see Figure 7-11). Additionally, easements would have to be obtained where discharge piping would cross private property. Installation of the discharge piping would take approximately one to one and a half months.

7.4.8 Costs

In this section, capital and annual costs are presented for discharge. The costs listed herein were incorporated in the overall costs of the treatment alternatives, which include discharge as a component. The net present worth of the discharge is not used herein; however, it is calculated in conjunction with other general components of the treatment alternatives and is presented in the cost analyses section for those individual alternatives. If remediation of the

North Park Subdivision plumes was included in the remedial alternative, the costs would not increase. A detailed cost estimate of discharge is presented in Appendix D.

Capital Costs. The capital cost estimate for discharge includes all direct costs (i.e., construction and equipment costs). Indirect costs, such as engineering and contingency costs, are not presented herein; however, they are calculated as a percentage of total capital costs for the treatment alternatives, which include discharge costs. The capital cost of discharge is estimated at \$381,000.

Annual Costs. Annual O&M costs for discharge would be \$68,000.

7.4.9 Summary

The summary of the detailed analysis of discharge is presented within the summaries of the overall treatment alternatives, which include discharge (see Sections 7.6.9, 7.7.9, and 7.8.9).

7.5 DETAILED ANALYSIS OF HEXAVALENT-CHROMIUM TREATMENT OPTIONS

Two hexavalent-chromium treatment options were developed to address the Northernnaire groundwater contamination. The contaminants of concern include hexavalent chromium, total chromium (including hexavalent and trivalent chromium), and two VOC compounds (TCE and 1,1,1-TCA). These compounds must be treated to the surface water requirements prior to discharge.

7.5.1 Option A - Ion Exchange

This subsection evaluates ion exchange as a hexavalent-chromium treatment option.

7.5.1.1 Description. In the ion exchange process, toxic metal salts and ions are electrostatically bound to a solid resin material and removed from solution. The resulting residuals include spent resins that can be regenerated on-site by washing with caustic and acid solutions, or can be returned to the manufacturer for regeneration. On-site regeneration requires the disposal or treatment of spent regenerating solutions contaminated with metals.

The ion exchange system for the Cadillac area hexavalent-chromium plume (i.e., Northernnaire) would be designed to treat approximately 32 gpm extracted from the Northernnaire plume. The system components are presented in Figure 7-12. This groundwater treatment application would require four 21-by-70-inch tanks (i.e., two cation tanks and two anion tanks). One set of cation and anion tanks would be in service, and the other set in regeneration mode or on standby for continuous 32-gpm flow. Regeneration would consist of an acidic solution (e.g., hydrochloric acid) for the cationic tank, and a caustic solution (e.g., sodium hydroxide) for the anionic tank. Approximately 500 gallons of regeneration waste would be produced every day and would require off-site disposal. The exhaustion time is relatively short primarily because of other cations and anions detected in the groundwater, which would be exchanged onto the resins,

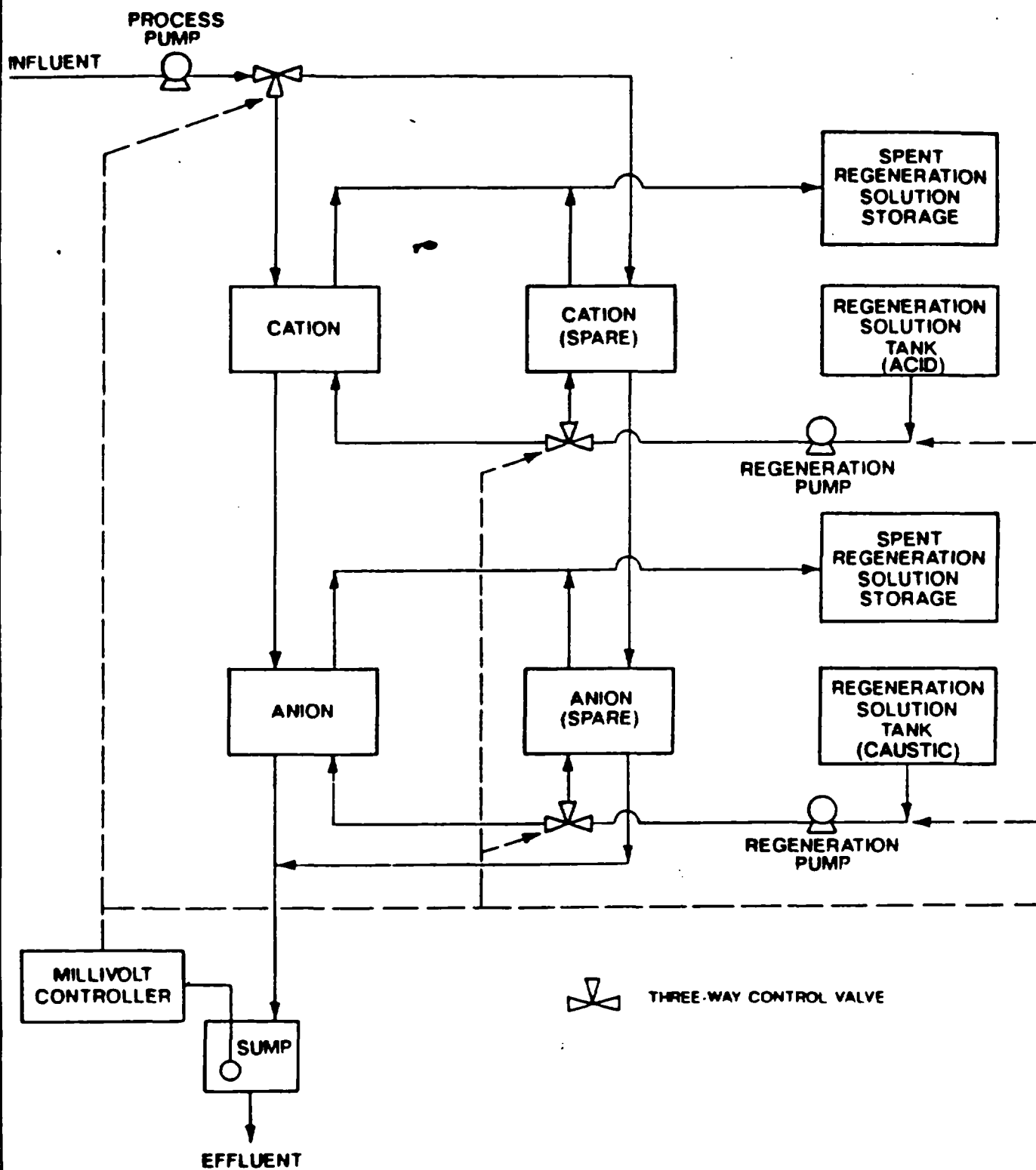


FIGURE 7-12
AC-1 : HEXAVALENT CHROMIUM - ION EXCHANGE
CADILLAC GROUNDWATER STUDY
CADILLAC, MICHIGAN

in addition to chromium ions. Because it is not possible to selectively exchange or precipitate chromium ions, the system previously described is analyzed in the following subsection. Effluent from the ion exchange treatment system would then pass to the VOC treatment system located within the treatment facility.

7.5.1.2 Compliance with ARARs. A liquid waste would be generated following application of this treatment technology. The liquid waste is considered hazardous due to the presence of hexavalent chromium and must be handled, transported, treated, and/or disposed of as a RCRA hazardous waste pursuant to 40 CFR (262 through 264) and Michigan Hazardous Waste Management Rules (R 299). Disposal must occur in a fully permitted RCRA facility operating in compliance with 40 CFR 264. Off-site transportation must be conducted by a licensed hazardous waste hauler in a permitted vehicle. The Michigan Air Pollution Control Regulations would not be applicable to this alternative because Ion Exchange is not considered a source of air pollutant emissions.

Construction and operation of the proposed treatment system is also subject to RCRA regulations for Environmental Performance of Miscellaneous Treatment Units (40 CFR 264, Subpart X). In general, these requirements state that the proposed treatment system should be protective of public health and the environment, and prevent releases and migration of contaminants to environmental media; waste analyses and trial tests should be performed; and all equipment and materials must be decontaminated prior to closure.

Groundwater monitoring under this alternative will be consistent with RCRA Corrective Action Monitoring (40 CFR 264.100). All on-site remedial activities must be conducted in compliance with OSHA Standards for Hazardous Waste Operations (29 CFR 1910).

Appendix E identifies the action-specific ARARs associated with this alternative. These ARARs are further detailed in Section 4.1 and summarized in Table 4-1.

7.5.1.3 Protection of Public Health and the Environment. Ion exchange would reduce the chromium ions to the levels required by MDNR limitations for discharge to the Clam River. Because state levels are below the MCLs and AWQC, discharge concentrations would be considered protective of public health and the environment.

7.5.1.4 Reduction in Toxicity, Mobility, or Volume of Contaminants. The ion exchange process would reduce toxicity of the groundwater by the removal of hexavalent chromium to meet the discharge requirements. However, the ion exchange process would produce 500 gallons of regeneration waste every day, which must be disposed of off-site. Approximately 16 cubic feet of spent resin would require replacement after the first and fourth years.

7.5.1.5 Short-term Effectiveness. Workers would be exposed to minimal construction hazards over an extremely short period during installation. The installation time for the equipment is only two to three days. During implementation, the acidic and caustic solutions used for regeneration of the resin

material on-site would pose potential risks to on-site workers and require enforcement of protective standards to avoid injury.

Community impacts are not expected from the ion exchange process. The only potential threat is from the on-site regeneration process where tanks of acids or caustics could be mishandled. This would present minimal risks, if any, to the community.

Because the hexavalent-chromium plume is relatively small, the time required to achieve full protection is estimated at four years, based on operational time of the extraction system. Therefore, the ion exchange process would be in operation on the Cadillac site for approximately four years, assuming 90-percent efficiency of the extraction system.

7.5.1.6 Long-term Effectiveness. The ion exchange process would effectively remove hexavalent chromium in the groundwater to reduce concentration levels below the MCL or AWQC before discharge. Residual risk would be associated with regeneration waste produced during the treatment process; however, these risks would be reduced by disposing of the waste off-site, in compliance with RCRA regulations.

The ion exchange process equipment would be constructed of a corrosion-resistant material such as fiberglass-reinforced plastic (FRP). Because the system would only need to run for four years, equipment is not expected to require replacement. To optimize the effectiveness of the system, a resin that would preferentially adsorb metal ions (specifically hexavalent chromium ions) should be selected. The resin in the exchange beds may have to be replaced after the first and fourth years of operation. To protect against adverse weather conditions, the system would be installed indoors. An operator would be on-site for 1 hour a day to ensure that the acid and caustic solution supplies are maintained and the process is functioning appropriately.

7.5.1.7 Implementability. Ion exchange is technically feasible for the Cadillac site. The process is a well-proven, conventional treatment for contaminated groundwater and wastewater. Skid-mounted equipment is readily available for on-site installation. Set-up time for a full-scale system is estimated at two to three days, assuming the rest of the system is in place. Prior to full-scale setup, a water analysis should be performed to determine cations and anions that would be exchanged on the resin to more accurately identify the rate of bed exhaustion.

7.5.1.8 Costs. In this section, capital and annual costs are presented for ion exchange. In addition, the net present worth of ion exchange is presented. The costs listed herein were incorporated in the overall costs of the treatment alternatives, which include a hexavalent-chromium treatment option as a component. A detailed cost estimate of ion exchange is presented in Appendix D.

Capital Costs. The capital cost estimate for ion exchange includes all direct costs (i.e., construction and equipment costs). Indirect costs, such as engineering and contingency costs, are not presented herein; however, they are calculated as a percentage of total capital costs for the treatment alterna-

tives, which include costs for ion exchange. The capital cost of ion exchange is estimated at \$271,000.

Annual Costs. Annual O&M costs for ion exchange would be \$352,000. Because the resin used in the ion exchange process will need to be changed after the first year of operation, an added O&M cost of \$2,735, which would need to be paid only once, is included.

Present Worth Analysis. Because the hexavalent-chromium in the Northernnaire Plume is expected to be extracted and remediated within four years, the present worth analysis is based on a performance period of four years. Using a 5-percent interest rate, the net present worth of ion exchange would be \$1,248,000. The present worth of the replacement resin, at a 5-percent interest rate, is \$3,000.

Sensitivity Analysis. A sensitivity analysis, assessing the effect that variations in the extraction system's efficiency and pumping duration will have on ion exchanges present worth, is presented in Section 9.1.

7.5.2 Option B - Carbon Adsorption

7.5.2.1 Description. The activated carbon matrix adsorbs organics and certain metals such as chromium. Activated carbon selectively adsorbs hazardous organic constituents by a surface attraction phenomenon that binds organic molecules to available carbon sites (i.e., internal pores). Carbon adsorption with granular activated carbon is usually accomplished in down-flow contactors, arranged in series to obtain high levels of removal and to increase operating times.

For the Cadillac site, this carbon adsorption system would be designed for an approximate flow of 32 gpm extracted from the Northernnaire plume. The system would consist of two carbon adsorption beds operating in series (Figure 7-13). Each bed would have a 4-foot diameter, an 11-foot height, and hold 2,000 pounds of carbon. The carbon adsorption system would produce 15,000 pounds of spent carbon per year for disposal.

7.5.2.2 Compliance with ARARs. When the hexavalent-chromium plume is extracted and treated with carbon adsorption, spent carbon will be generated. This spent carbon cannot be regenerated thermally, but must be handled for disposal or regeneration (using an acid solution) by a facility licensed to do so. The spent carbon must be transported, treated, and/or disposed of properly according to RCRA regulations (40 CFR, 262 through 264) and Michigan Hazardous Waste Management Regulations (R 299).

Transportation must occur via a licensed and permitted hazardous waste hauler and vehicle. Final disposal must occur in a fully permitted RCRA facility operating in compliance with 40 CFR 264.

Construction, operation, and groundwater monitoring activities would be subject to the same regulations identified for the hexavalent chromium ion exchange process (refer to Section 7.5.1.2).

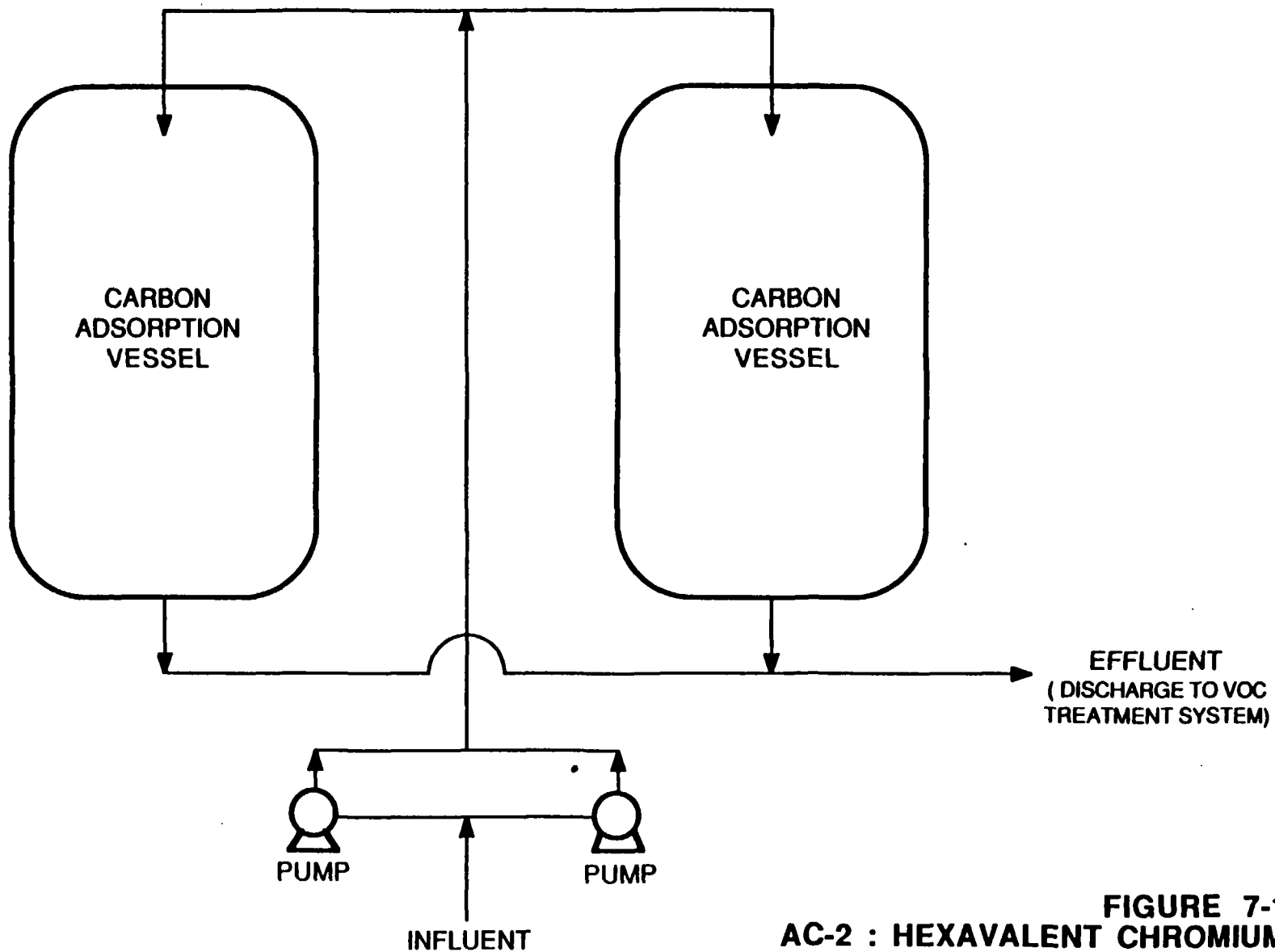


FIGURE 7-13
AC-2 : HEXAVALENT CHROMIUM -
CARBON ADSORPTION
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN
— E.C. JORDAN —

Appendix E identifies the action-specific ARARs associated with this alternative. These ARARs are further detailed in Section 4.1 and summarized in Table 4-1.

7.5.2.3 Protection of Public Health and the Environment. Protection of public health and the environment would be the same for the carbon adsorption treatment process as the ion exchange system (refer to Section 7.5.1.3).

7.5.2.4 Reduction in Toxicity, Mobility, or Volume of Contaminants. The carbon adsorption process would reduce the toxicity of groundwater by treating chromium ions to the surface water discharge requirements. Carbon adsorption would also reduce concentrations of the two VOC compounds detected in the Northern Plume. However, the effluent from the system would have to go through the VOC treatment system to achieve surface water discharge requirements. This process would produce approximately 15,000 pounds of spent carbon annually for disposal. Spent carbon from this process would not be thermally regenerable because hexavalent chromium binds tightly to carbon. Therefore, virgin carbon would be used for this process and spent carbon would be disposed of at a permitted off-site facility.

7.5.2.5 Short-term Effectiveness. The primary potential impacts to workers from carbon adsorption are from handling the spent carbon, which would be contaminated with metals and organic compounds. The spent carbon would be handled as a hazardous waste and transported off-site for disposal. Appropriate safety requirements must be met during handling of the wastes to ensure no adverse impacts to workers.

The community risks due to transport and handling of hazardous materials would be minimal, assuming appropriate handling of materials. In addition, hazardous wastes would not remain on-site.

Workers would be exposed to minimal construction hazards over an extremely short period during installation. The installation time for the equipment is two days.

The time required to achieve full protection (to meet TCLs for hexavalent chromium) is estimated to be the same as the ion exchange process (four years) assuming 90-percent efficiency of groundwater extraction.

7.5.2.6 Long-term Effectiveness. The carbon adsorption process would effectively reduce the chromium ion levels to appropriate levels below the MCL or AWQC before discharge to the Clam River. Risks associated with residuals would be present due to the handling of spent carbon produced during the treatment process; however, these risks would be reduced by disposing of the waste off-site in compliance with RCRA regulations.

The carbon adsorption process equipment would be constructed of a corrosion-resistant material (e.g., carbon steel). Because this system would only operate for four years, the process equipment is not expected to require replacement. To protect against adverse weather conditions, the system would be installed indoors. An operator would be on-site for 1 hour a day to ensure that the system is functioning properly.

7.5.2.7 Implementability. Like the ion exchange system, carbon adsorption is technically feasible for the Cadillac site. The process is well-proven for the treatment of contaminated groundwater. Skid-mounted equipment is readily available for on-site installation. Set-up time for the system is estimated at two days. Prior to full-scale implementation of this additional component, a bench-scale test should be performed to determine actual carbon usage rates for the system. Vendors and testing equipment are available to perform treatability tests.

7.5.2.8 Costs. In this subsection, capital, annual, and net present worth costs are presented for carbon adsorption in the same manner as the ion exchange costs (refer to Section 7.5.1.8). A detailed cost estimate of carbon adsorption is presented in Appendix D.

Capital Costs. The capital cost of carbon adsorption is estimated at \$140,000.

Annual Costs. Annual O&M costs for carbon adsorption are \$142,000.

Present Worth Analysis. Using a 5-percent interest rate, the net present worth of carbon adsorption is \$504,000.

Sensitivity Analysis. A sensitivity analysis, assessing the effect that variations in the extraction system's efficiency and pumping duration will have on the estimated present worth of carbon adsorption, is presented in Section 9.1.

7.5.3 Summary of Hexavalent-Chromium Treatments

Summaries of the detailed analyses of Option A (Ion Exchange) and Option B (Carbon Adsorption), using the evaluation criteria, are presented in Tables 7-6 and 7-7, respectively. Both technologies would treat the hexavalent chromium to meet discharge requirements and would provide protection of public health and the environment. The major difference is that ion exchange would produce approximately 200,750 gallons of hazardous regeneration waste every year, and carbon adsorption would produce approximately 15,000 pounds of spent carbon per year. These wastes would pose potential risks to on-site workers, requiring enforcement of protective standards.

From a practical standpoint, the carbon treatment is far less dangerous than the chemical-intensive ion exchange. In addition, both types of secondary waste would require off-site disposal. Due to the expense of off-site disposal costs and other general expenses, the annual costs and present worth of ion exchange are substantially higher than those for carbon adsorption. The annual costs of ion exchange are \$352,000, with a total present worth of \$1,248,000. Expected annual costs for the carbon adsorption system are \$142,000, and the present worth is \$504,000. Capital costs of ion exchange are also more expensive than for carbon adsorption (i.e., \$271,000 compared to \$140,000).

TABLE 7-6
SUMMARY OF DETAILED ANALYSIS
HEXAVALENT-CHROMIUM TREATMENT
OPTION A - ION EXCHANGE

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o Groundwater contaminated with hexavalent chromium would be treated to meet discharge requirements. - o The ion exchange treatment system will be designed and operated in compliance with all action-specific ARARs.
2. Protection of Public Health and the Environment	o Ion exchange would reduce chromium ions to meet discharge levels which would provide protection of public health and the environment.
3. Reduction of Toxicity, Mobility, or Volume	- o Ion exchange would reduce the volume of hexavalent chromium in the extracted groundwater. Toxicity of groundwater would be reduced due to chromium removal. - o The ion exchange process would produce 460 gallons of hazardous regeneration waste every 20 hours, requiring off-site disposal.
4. Short-term Effectiveness	- o No short-term community or worker health impacts would occur during installation of the treatment system, except for minor construction hazards. - o During implementation, the acid and caustic solutions used for regeneration would pose potential risks to on-site workers and require enforcement of protective standards. - o The length of time required to achieve full protection is estimated to be 4 years, assuming 90% efficiency of the extraction system.
5. Long-term Effectiveness	o Residual risk would be associated with the regeneration waste produced by the treatment. These risks would be reduced by disposal of waste off-site.

TABLE 7-6
(continued)
SUMMARY OF DETAILED ANALYSIS
HEXAVALENT-CHROMIUM TREATMENT
OPTION A - ION EXCHANGE

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
	- o Ion exchange would effectively treat and reduce the hexavalent chromium concentration levels in the water to surface water discharge levels. - o The resin in the exchange beds may have to be replaced after the first and fourth years of operation. However, the treatment process will most likely need to operate for only 4 years.
6. Implementation	- o Ion exchange is technically feasible for the Cadillac site. - o The technology has been demonstrated to treat hexavalent chromium. - o Equipment is readily available. - o Installation of ion exchange would take approximately 2 to 3 days.
7. Costs	- o Capital Costs = \$271,000 - o Annual Costs = \$352,000 - Cost of replacement resin after first year = \$2,735 - o Present Worth = \$1,248,000 - Present Worth of replacement resin = \$3,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

TABLE 7-6
(continued)
SUMMARY OF DETAILED ANALYSIS
HEXAVALENT-CHROMIUM TREATMENT
OPTION A - ION EXCHANGE

CADILLAC AREA FEASIBILITY STUDY

NOTES (continued):

A = The long-term uncertainties associated with land disposal.

B = The goals, objectives, and requirements of the SWDA.

C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.

D = Short- and long-term potential for adverse health effects from human exposure.

E = Long-term monitoring costs.

F = The potential for future remedial action costs if the alternative remedial action in question were to fail.

G = The potential threat to public health and the environment associated with excavation, transportation, and redisposal, or containment.

TABLE 7-7
SUMMARY OF DETAILED ANALYSIS
HEXAVALENT-CHROMIUM TREATMENT
OPTION B - CARBON ADSORPTION

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o Groundwater contaminated with hexavalent chromium would be treated to meet discharge requirements. - o The carbon adsorption treatment system will be designed and operated in compliance with all action-specific ARARs.
2. Protection of Public Health and the Environment	o Carbon adsorption would reduce chromium ions to meet discharge levels which would provide protection of public health and the environment.
3. Reduction of Toxicity, Mobility, or Volume	- o Carbon adsorption would reduce the volume of hexavalent chromium in the extracted groundwater. Toxicity of groundwater would be reduced due to chromium removal. - o Carbon adsorption would produce approximately 15,000 lbs/year of spent carbon requiring off-site disposal.
4. Short-term Effectiveness	- o No short-term community or worker health impacts would occur during installation of the treatment system, except for minor construction hazards. - o Potential impacts to workers exist from handling the spent carbon. Appropriate safety requirements must be met. - o The length of time required to achieve full protection is estimated to be 4 years, assuming 90% efficiency of the extraction system.
5. Long-term Effectiveness	- o Residual risk would be associated with the spent carbon produced by the carbon adsorption process. These risks would be reduced by disposal of waste off-site. - o Carbon adsorption would effectively treat and reduce the hexavalent chromium concentration levels in the water to surface water discharge levels.

TABLE 7-7
(continued)
SUMMARY OF DETAILED ANALYSIS
HEXAVALENT-CHROMIUM TREATMENT
OPTION B - CARBON ADSORPTION

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
	o Process equipment is not expected to require replacement.
6. Implementation	- o Carbon adsorption is technically feasible for the Cadillac site. - o The technology has been demonstrated to treat hexavalent chromium. - o Equipment is readily available. - o Installation of carbon adsorption would take approximately 2 days.
7. Costs	- o Capital Costs = \$140,000 - o Annual Costs = \$142,000 - o Present Worth = \$504,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

A = The long-term uncertainties associated with land disposal.

B = The goals, objectives, and requirements of the SWDA.

C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.

D = Short- and long-term potential for adverse health effects from human exposure.

E = Long-term monitoring costs.

F = The potential for future remedial action costs if the alternative remedial action in question were to fail.

G = The potential threat to public health and the environment associated with excavation, transportation, and redispisal, or containment.

7.6 DETAILED ANALYSIS OF ALTERNATIVE GW-2: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT, UV/OXIDATION, AND DISCHARGE

7.6.1 Description

Extraction. Extraction would be performed as described in Section 7.2.

Hexavalent-Chromium Treatment. Hexavalent-chromium treatment would be performed on groundwater extracted from the Northernnaire Plume by implementing either ion exchange or carbon adsorption (see Sections 7.5.1.1 and 7.5.2.1). Treated groundwater from the Northernnaire Plume would then be combined with all other extracted groundwater for further treatment, described in the following paragraphs.

UV/Oxidation. UV/Oxidation treatment chemically oxidizes organic compounds in water with a combination of UV light, ozone, and/or hydrogen peroxide. Ozone and hydrogen peroxide are powerful oxidants of organic compounds in water. Recent applications established that the use of UV light, in combination with ozone and/or hydrogen peroxide, can significantly enhance chemical degradation rates for certain organic compounds.

There are four main components of a UV/Oxidation system: an air compressor, an ozone generator, a reactor tank, and an electrical source for the UV lamp bulbs. The system utilizes a baffled UV/ozone contactor, within which the UV bulbs are mounted inside quartz sheets. Ozone is diffused into the bottom of the contactor, and allowed to make contact with the contaminated water in the presence of UV light. A manifold attached to the contactor collects any residual ozone from the process and catalytically converts it back to molecular oxygen. Hydrogen peroxide may be added to the influent stream prior to entering the contactor. Hydrogen peroxide and ozone are oxidants that can be utilized separately or in combination to achieve maximum oxidation conditions. Application of these oxidants depends on the contaminants to be treated and their ability to be oxidized.

The expenses associated with the system include electrical power (usually three-phase), supplies of liquid oxygen and hydrogen peroxide, daily system monitoring, and periodic maintenance. In addition, the system can be fully automatic, has no filtering or adsorption medium to dispose of or regenerate, and treats groundwater contaminated with VOCs without air emissions or generation of hazardous wastes.

The UV/Oxidation system for the Cadillac area groundwater would be designed to treat a total of approximately 940 gpm. The system components are presented in Figures 7-14 and 7-15. Extracted groundwater would be pumped to three 6,000-gallon baffled reactor vessels. Liquid oxygen would be supplied to two 400-pound ozone generators for the daily production of approximately 1,000 pounds of ozone. The generator would use 15,000 kWh/day of electrical power. Approximately 5 to 20 gpm of cooling water would be required by the generator. This cooling water could be supplied from the treated effluent stream. The retention time needed to treat the water to discharge requirements is expected to be 20 minutes. To reduce the amount of ozone required by the system and to maximize the oxidation potential of the system, 2,000 pounds a day of hydrogen

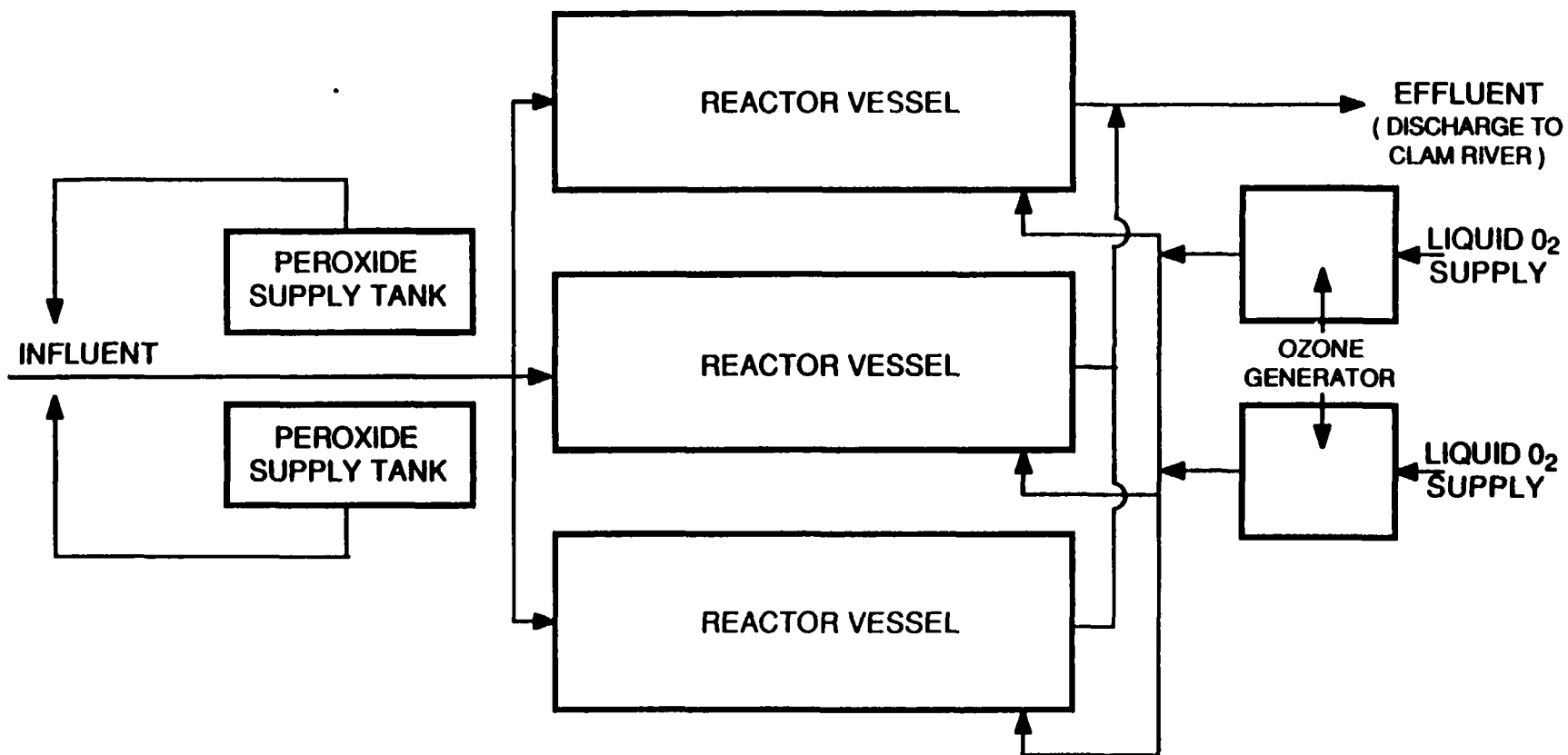


FIGURE 7-14
GW-2 : UV/OXIDATION, PLAN VIEW
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN
E.C. JORDAN

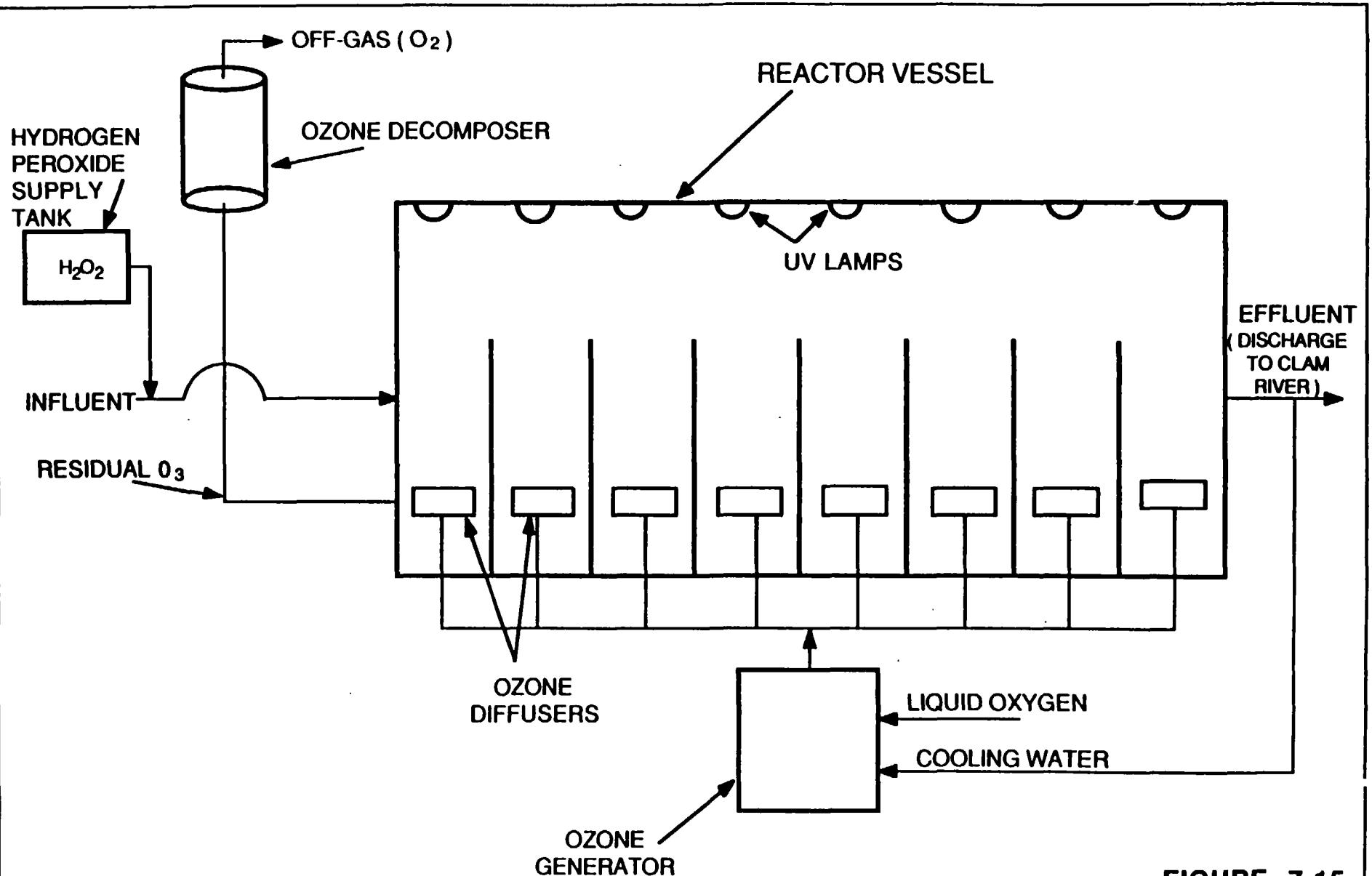


FIGURE 7-15
GW-2 : UV/OXIDATION, CROSS SECTION OF
SINGLE REACTOR
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

E.C. JORDAN

peroxide would be added to the influent stream before it enters the reactor vessel.

Discharge. Discharge of treated groundwater would be performed as described in Section 7.4.

7.6.2 Compliance with ARARs

Analysis of the ARARs associated with the extraction and discharge systems is discussed in Sections 7.3.2 and 7.4.2, respectively. Analysis of ARARs associated with the hexavalent-chromium treatment systems is discussed in Sections 7.5.1.2 and 7.5.2.2.

ARARs associated with the UV/Oxidation system may include the federal Clean Air Act (CAA) regulations (40 CFR 129) and the Michigan Air Pollution Control regulations (R 336, Part 201), and potential approvals required before locating the system on state, city, or private property. Relevant CAA regulations include the particulate matter standards. Under the Michigan Air Pollution Control regulations, the UV/Oxidation treatment system may be considered a source of air contamination and may require a permit for installation and operation (R 336, Rule 201). Michigan regulations limit fugitive dust (usually a problem during construction or excavation phases of remediation) and establish the maximum allowable emission rate from new sources of VOCs based on BACT (R 336, Rules 701 and 702). BACT is determined on a case-by-case basis through information submitted in the permit application. Furthermore, Michigan regulations prohibit the emission of air contaminants in quantities that will cause "injurious effects to human health or safety, animal life, plant life, of significant economic value, or property," or "unreasonable interference with the comfortable enjoyment of life and property" (R 336, Rule 901). Determination of what is considered "injurious" is outlined in the regulations.

The UV/Oxidation system is considered an extremely effective treatment and has no filtering or adsorption medium to dispose of or regenerate and treats groundwater contaminated with VOCs without air emissions or the generation of hazardous waste, and therefore appears to comply with all associated ARARs. MDNR may require proof, under Rule 901, that the proposed system will not be injurious to public health or welfare, or the environment prior to determining if the system will require a permit under R 336, Rule 201.

Construction and operation of the proposed treatment system is also subject to RCRA regulations for Environmental Performance of Miscellaneous Treatment Units (40 CFR 264, Subpart X). In general, these requirements state that the proposed treatment system should be protective of public health and the environment, and prevent releases and migration of contaminants to environmental media; waste analyses and trial tests should be performed; and all equipment and materials must be decontaminated prior to closure.

ARARs associated with the North Park Subdivision are the same as ARARs associated with the industrial park, except that due to the proximity of residential homes in the North Park Subdivision, approval of the location of extraction and treatment systems in this area must be obtained prior to construction of these facilities.

Groundwater monitoring under this alternative will be consistent with RCRA Corrective Action Monitoring (40 CFR 264.100). All on-site remedial activities must be conducted in compliance with OSHA Standards for Hazardous Waste Operations (29 CFR 1910).

Appendix E identifies the action-specific ARARs associated with this alternative. These ARARs are further detailed in Section 4.1 and summarized in Table 4-1.

7.6.3 Protection of Public Health and the Environment

UV/Oxidation, combined with the extraction, hexavalent-chromium treatment, and discharge technologies, would reduce existing risks to public health associated with the domestic use and drinking of contaminated groundwater. The evaluation of the extraction and discharge technologies' ability to provide protection of public health and the environment is described in Sections 7.3.3 and 7.4.3, respectively. The evaluations for the hexavalent-chromium treatment options are presented in Sections 7.5.1.3 and 7.5.2.3.

Through oxidation of organic compounds, the UV/Oxidation treatment alternative should effectively detoxify the VOCs in the extracted groundwater. The surface water discharge requirements would be met through this treatment system before effluent was released to the Clam River.

In summary, the overall alternative would meet ARARs applicable to remediating the groundwater and discharging to the Clam River.

7.6.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

UV/Oxidation, combined with the extraction, hexavalent-chromium treatment, and discharge technologies, would reduce the toxicity, mobility, and volume of groundwater contaminants. The UV/oxidation process would reduce the toxicity of groundwater contaminants by destructing contaminants to reduce VOC concentrations to the surface water discharge requirements. This process would not produce a secondary wastestream or air emissions requiring treatment. The off-gases emitted from this system would only include oxygen and carbon dioxide.

7.6.5 Short-term Effectiveness

No short-term community or worker health impacts should result from installation of the hexavalent-chromium treatment or the UV/Oxidation treatment component. During installation of the treatment systems, appropriate worker safety standards would be met. Potential risks to on-site workers during implementation are associated with the handling of hazardous substances and wastes generated by the hexavalent-chromium treatment. Appropriate safety requirements must be met.

The time required to achieve full protection is estimated to be 64 years. This estimate is based on the operation time of the extraction system developed for the Kysor Intermediate Plume, which is the most extensive plume. The estimate

also assumes that the extraction system would be 90-percent efficient in flushing contaminants from the plume.

If the North Park Subdivision were included in the groundwater treatment scheme, the overall treatment time would not have to be extended. This relatively small flow would be added to the main treatment system, and the extraction system would pump groundwater to the treatment facility for 6.5 years to achieve the groundwater TCLs.

7.6.6 Long-term Effectiveness

This alternative would effectively extract contaminated groundwater from the aquifer and reduce the contaminant concentration levels in the water to surface water discharge levels. Discussions of the long-term effectiveness of the extraction and discharge components of this groundwater alternative are presented in Sections 7.3.6 and 7.4.6. For the hexavalent-chromium treatments, the evaluation of long-term effectiveness is presented in Sections 7.5.1.6 and 7.5.2.6.

In summary, the extraction system is expected to capture the contaminated groundwater so that TCLs are achieved, resulting in attainment of ARARs and a total carcinogenic residual risk within USEPA's target range of 10^{-7} to 10^{-4} . In addition, the discharge levels will be at or below the Michigan discharge limitations. Because the UV/Oxidation groundwater treatment alternative does not produce a secondary wastestream to dispose of and there are no air emissions, this alternative would not cause any individuals to be exposed to hazardous materials. Furthermore, the alternative would effectively reduce the VOC concentration levels in the effluent to meet MDNR discharge limitations for surface water discharge; therefore, it is adequately protective of public health and environmental health. Residual risk would be associated with the secondary wastestreams produced by the hexavalent-chromium treatments; however, these risks would be reduced by disposal off-site in compliance with RCRA regulations.

To minimize corrosion during the operational life of the system, the UV/Oxidation process equipment would be constructed of a resistant material (e.g., stainless steel). The design life of the reactor vessels is expected to be 25 years; the ozone generator should be operational for at least 15 to 20 years; UV lamps would require annual replacement; and pumps may have to be replaced every five years. To protect against adverse weather conditions, this system would be installed indoors. The system would have to be monitored three times daily to maintain hydrogen peroxide supplies and take water samples, if necessary.

7.6.7 Implementability

The overall Alternative GW-2 is technically feasible for the Cadillac site. UV/Oxidation has been used to treat contaminated groundwater and equipment is available for on-site installation. A full-scale system could be installed in approximately two weeks, provided the electrical hookup and treatment building are available for use. Prior to full-scale implementation, a pilot-scale study should be conducted to obtain performance data for system requirements.

Vendors and equipment are available to perform UV/Oxidation treatability studies. However, installation of the overall alternative would take approximately three to six months, due to installation of the extraction system.

7.6.8 Costs

In this section, capital and annual costs are presented for Alternative GW-2, as well as the net present worth. The costs for Alternative GW-2 are listed in Tables 7-8 and 7-9 using ion exchange and carbon adsorption for hexavalent-chromium treatment, respectively.

Capital Costs. As discussed in Section 6.2, direct and indirect costs were estimated for this alternative. Engineering costs are assumed to be 10 percent, and contingency costs are assumed to be 25 percent. The capital cost of Alternative GW-2 using ion exchange for the hexavalent-chromium treatment is \$6,115,000. If carbon adsorption were used for the hexavalent chromium treatment, the total capital cost of Alternative GW-2 would be \$5,935,000.

Annual Costs. Annual O&M costs for the various components of Alternative GW-2 are presented in Tables 7-8 and 7-9. Annual costs for the UV/Oxidation treatment system are \$1,382,000 for the first 38 years, and \$914,000 for the remaining 26 years. Annual costs for the hexavalent-chromium treatment using ion exchange are \$352,000 or, if using carbon adsorption, \$142,000, for a duration of four years. Additional annual costs for general items are listed in the tables. Annual costs associated with extraction are not listed herein, but are accounted for in the total present worth.

Present Worth Analysis. Based on a 5-percent interest rate, the net present worth of Alternative GW-2 using ion exchange is \$38,473,000; using carbon adsorption, it is \$37,546,000.

Sensitivity Analysis. A sensitivity analysis, assessing the effect that variations in the extraction system's efficiency and pumping duration will have on the estimated present worth of Alternative GW-2, is presented in Section 9.1.

7.6.9 Summary

The detailed analysis of Alternative GW-2 is summarized in Table 7-10.

7.7 DETAILED ANALYSIS FOR ALTERNATIVE GW-3: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT, AIR STRIPPING, AND DISCHARGE

7.7.1 Description

For the treatment of the Cadillac area groundwater, the following two air-stripping systems could be implemented:

- o air stripping with vapor-phase carbon adsorption

TABLE 7-8
 GW-2 UV/OZONE/H2O2
 WITH HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

	DESCRIPTION	TOTAL
-----		-----
	CAPITAL COSTS	
	FIVE YEAR SITE REVIEW	\$
	MONITORING	
	BUILDING	646,00
	EXTRACTION	884,00
	UV/OZONE/H2O2 VOC TREATMENT SYSTEM	2,265,00
	HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE	271,00
	DISCHARGE	381,00
	ENGINEERING @ 10%	445,00
	CONTINGENCY @ 25%	1,223,00

1.	TOTAL CAPITAL COSTS	\$6,115,00
	PRESENT WORTH OF UV/OZONE/H2O2 VOC TREATMENT SYSTEM REPLACEMENT	
	AFTER 25 YEARS @ 5%	\$591,0
	AFTER 50 YEARS @ 5%	174,C

2.	TOTAL PRESENT WORTH OF VOC TREATMENT SYSTEM REPLACEMENT	\$765,C.
.	PRESENT WORTH OF REPLACEMENT RESIN FOR 4 YEARS @ 5%	\$3,00
	(FOR HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE)	
4.	PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION	\$1,936,00
	(TOTAL FOR ALL PLUMES)	

CONTINUED ON NEXT PAGE

TABLE 7-8 (CON'T)
 GW-2 UV/OZONE/H2O2
 WITH HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL

ANNUAL COSTS	
GENERAL ITEMS (DURATION OF 64 YEARS)	
FIVE YEAR SITE REVIEW	\$3,00
BUILDING	51,00
DISCHARGE	68,00

TOTAL OF ABOVE	\$122,00
HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE (DURATION OF 4 YEARS)	\$352,00
UV/OZONE/H2O2 TREATMENT SYSTEM	
FOR YEARS 1 - 38	\$1,382,00
FOR YEARS 39 - 64	914,00
MONITORING	
FOR YEAR 1	\$128,00
FOR YEARS 2 - 64	32,00
5. PRESENT WORTH OF GENERAL ITEMS FOR 64 YEARS @ 5%	\$2,333,00
6. PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%	\$1,248,00
PRESENT WORTH OF UV/OZONE/H2O2 TREATMENT SYSTEM	
7. FOR YEARS 1 - 38 @ 5%	\$23,311,00
8. FOR YEARS 39 - 64 @ 5%	2,058,00
PRESENT WORTH OF MONITORING	
9. FOR YEAR 1	\$122,00
10. FOR YEARS 2 - 64	582,00
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 10)	\$38,473,00

TABLE 7-9
 GW-2 UV/OZONE/H2O2
 WITH HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

	DESCRIPTION	TOTAL

	CAPITAL COSTS	
	FIVE YEAR SITE REVIEW	\$
	MONITORING	
	BUILDING	646,00
	EXTRACTION	884,00
	UV/OZONE/H2O2 VOC TREATMENT SYSTEM	2,265,00
	HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION	140,00
	DISCHARGE	381,00
	ENGINEERING @ 10%	432,00
	CONTINGENCY @ 25%	1,187,00

1.	TOTAL CAPITAL COSTS	\$5,935,00
	PRESENT WORTH OF UV/OZONE/H2O2 VOC TREATMENT SYSTEM REPLACEMENT	
	AFTER 25 YEARS @ 5%	\$591,00
	AFTER 50 YEARS @ 5%	174,00

2.	TOTAL PRESENT WORTH OF VOC TREATMENT SYSTEM REPLACEMENT	\$765,00
3.	PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION (TOTAL FOR ALL PLUMES)	\$1,936,00

CONTINUED ON NEXT PAGE

TABLE 7-9 (CON'T)
 GW-2 UV/OZONE/H2O2
 WITH HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION		TOTAL

ANNUAL COSTS		
GENERAL ITEMS (DURATION OF 64 YEARS)		
FIVE YEAR SITE REVIEW		\$3,00
BUILDING		51,00
DISCHARGE		68,00

TOTAL OF ABOVE		\$122,00
HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION (DURATION OF 4 YEARS)		\$142,00
UV/OZONE/H2O2 TREATMENT SYSTEM		
FOR YEARS 1 - 38		\$1,382,00
FOR YEARS 39 - 64		914,00
MONITORING		
FOR YEAR 1		\$128,00
FOR YEARS 2 - 64		32,00
4. PRESENT WORTH OF GENERAL ITEMS FOR 64 YEARS @ 5%		\$2,333,00
5. PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%		\$504,00
PRESENT WORTH OF UV/OZONE/H2O2 TREATMENT SYSTEM		
6. FOR YEARS 1 - 38 @ 5%		\$23,311,00
7. FOR YEARS 39 - 64 @ 5%		2,058,00
PRESENT WORTH OF MONITORING		
3. FOR YEAR 1		\$122,00
9. FOR YEARS 2 - 64		582,00
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 9)		\$37,546,00

TABLE 7-10
SUMMARY OF DETAILED ANALYSIS
GW-2: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
UV/OXIDATION, AND DISCHARGE

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o Drinking water standards would be met by the extraction of contaminated groundwater from the aquifer. - o Contaminated groundwater would be treated to meet discharge requirements. - o The hexavalent-chromium treatment and UV/oxidation systems would be designed and operated in compliance with all action-specific ARARs.
2. Protection of Public Health and the Environment	o GW-2 would reduce existing risks to public health associated with the domestic use and drinking of contaminated groundwater.
3. Reduction of Toxicity, Mobility, or Volume	- o GW-2 would reduce the toxicity and mobility of groundwater contaminants. - o Hexavalent-chromium treatment would produce a secondary wastestream requiring off-site disposal. - o UV/oxidation would not produce a secondary waste stream or air emissions.
4. Short-term Effectiveness	- o No short-term community or worker health impacts would occur during installation, except for potential minor construction hazards. - o During implementation of the hexavalent-chromium treatment, potential risks to on-site workers exist due to handling of hazardous substances and secondary wastes. Appropriate safety requirements must be met. - o The length of time required to achieve full protection is estimated at 64 years, assuming 90% efficiency of the extraction system.

TABLE 7-10
(continued)
SUMMARY OF DETAILED ANALYSIS
GW-2: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
UV/OXIDATION, AND DISCHARGE

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
5. Long-term Effectiveness	- o Residual risk would be associated with the secondary wastestreams produced by the hexavalent-chromium treatments. These risks would be reduced by disposal of waste off-site. - o No residual risk would be associated with UV/oxidation. - o Long-term groundwater monitoring would be conducted for 64 years and management of property is required. - o GW-2 would effectively extract contaminated groundwater from the aquifer and reduce the contaminant concentration levels in the water to surface water discharge levels. - o Potential for replacement of reactor vessels is expected at 25 years; ozone generator at 15 to 20 years; and, if ion exchange is utilized, the resin may have to be replaced after the first year of operation.
6. Implementation	- o GW-2 is technically feasible for the Cadillac site. - o Technologies associated with GW-2 have been demonstrated to treat the contaminants of concern. - o Equipment is readily available. - o Access to land for equipment installation must be obtained, including right-of-ways. - o Installation of UV/Oxidation system would take approximately 2 weeks. - o Installation of the overall alternative would take approximately 3 to 6 months. - o A pilot-scale study should be conducted.

TABLE 7-10
(continued)
SUMMARY OF DETAILED ANALYSIS
GW-2: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
UV/OXIDATION, AND DISCHARGE
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
7. Costs	- o Capital Costs - Using Ion Exchange = \$6,115,000 - Using Carbon Adsorption = \$5,935,000 - o Annual Costs (for UV/Oxidation treatment only) = \$1,382,000 (Years 1-38) \$914,000 (Years 39-64) - o Present Worth - Using Ion Exchange = \$38,473,000 - Using Carbon Adsorption = \$37,546,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

A = The long-term uncertainties associated with land disposal.

B = The goals, objectives, and requirements of the SWDA.

C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.

D = Short- and long-term potential for adverse health effects from human exposure.

E = Long-term monitoring costs.

F = The potential for future remedial action costs if the alternative remedial action in question were to fail.

G = The potential threat to public health and the environment associated with excavation, transportation, and redispisal, or containment.

- o air stripping with liquid-phase carbon adsorption and vapor-phase thermal treatment

The following paragraphs describe the components of the two systems.

Extraction. Extraction would be performed as described in Section 6.3.

Hexavalent-Chromium Treatment. Hexavalent-chromium treatment would be performed the same as for GW-2 (refer to Section 7.6.1).

Air Stripping with Vapor-phase Carbon Adsorption (Option A). Air-stripping is the mass transfer of VOCs from the liquid (water) phase to the gas (air) phase. This process is used to remove contaminants with relatively high volatility and low water solubility (e.g., chlorinated hydrocarbons and monoaromatics). Air stripping occurs in a packed tower, usually operated in a counter-current mode. By using the counter-current mode, the least-contaminated water contacts the cleanest air, thereby providing the most effective mass transfer. Air is supplied by a blower or compressor that is introduced to the bottom of the tower. The packing material functions to increase the area of contact between the air and the descending liquid waste. Treated effluent is collected in a sump at the bottom of the tower. Air-stripping processes requiring treatment of offgases generally use carbon adsorption to mitigate atmospheric discharge of contaminants. Air is supplied by a blower or compressor that is introduced to the bottom of the tower.

Carbon adsorption employs activated carbon to adsorb organics and certain metals. A surface attraction phenomenon binds organic molecules to available carbon sites (i.e., internal pores). Carbon adsorption with granular activated carbon is usually accomplished in down-flow contactors, arranged in series to obtain high levels of removal and to increase operating times. Carbon adsorption could be used as a polishing step in a treatment train, or for treatment of air-stripper off-gases.

The first air-stripping system for the Cadillac site would consist of two stripping columns operating in series. The columns would have a packing height of 30 feet and a diameter of 8 feet. Approximately 6,280 cubic feet per minute (cfm) of air providing a 50:1 air-to-water ratio would be required to strip VOCs from the groundwater. The first air stripper would be designed to remove 99 percent of the VOCs from the liquid phase. The second air stripper is needed to further reduce VOC concentrations to meet the liquid-phase effluent discharge requirements. The second stripper would also provide a 50:1 air-to-water ratio. Therefore, vapor-phase treatment is required for the first air stripper, which would be accomplished by a vapor-phase carbon adsorption unit. The vapor-phase treatment unit generally consists of two carbon vessels: one in operation and the other in standby mode. When the first vessel reaches exhaustion, the system automatically switches to the second vessel. The exhausted bed is regenerated in-situ by a thermal oxidizer, which obtains most of its heat from the desorption of VOC compounds from the carbon during the oxidation process. The system components are presented in Figure 7-16.

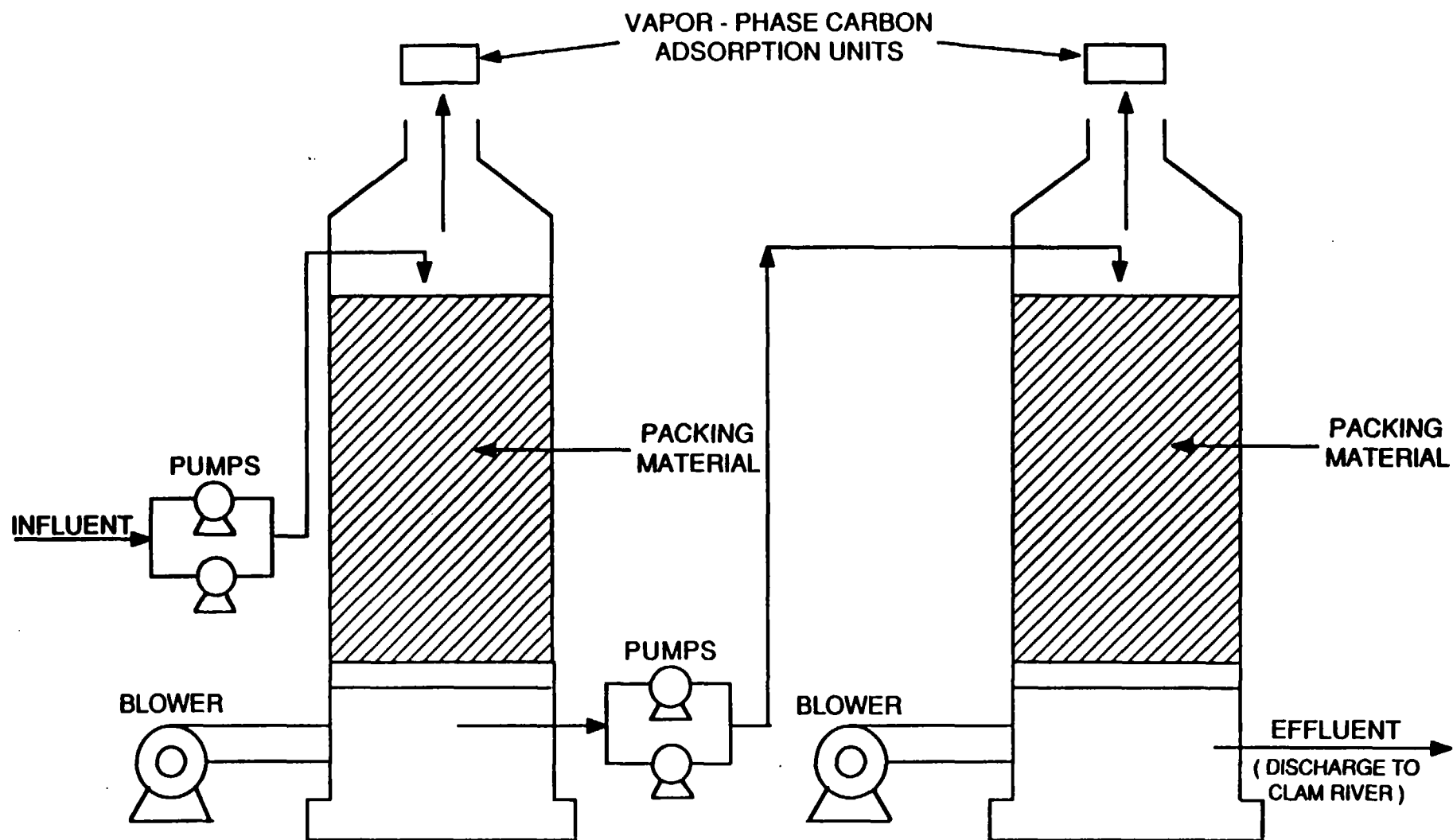


FIGURE 7-16
GW-3a : AIR STRIPPING WITH
VAPOR - PHASE CARBON ADSORPTION
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

E.C. JORDAN

Air Stripping with Liquid-Phase and Vapor-Phase Carbon Adsorption (Option B).

The second air-stripping system for the Cadillac site would consist of one air-stripping column followed by liquid-phase carbon adsorption treatment to meet the liquid-phase effluent discharge requirements. The system components are presented in Figure 7-17. The column would have a packing height of 30 feet and a diameter of 8 feet, and be constructed of a resistant material such as FRP; the packing material would be polypropylene rings. Approximately 6,280 cfm of air, providing a 50:1 air-to-water ratio, would be required to strip the VOCs from the groundwater. The contaminated vapor exiting the column would be treated by vapor-phase carbon adsorption, as described for the first air-stripping system. Water that discharges from the base of the stripper would be pumped to the carbon adsorption beds for treatment. Four carbon adsorption beds would be required for treatment. The beds would be arranged in pairs, each operating in series and treating half the flow from the air stripper. In addition, a spare transfer bed would be available when the carbon in a spent bed requires replacement. Each bed would have a 10-foot diameter and a 19-foot height, and would be constructed of a resistant material such as carbon-steel. Approximately 225,000 pounds of spent carbon would be produced per year for disposal.

Discharge. Discharge would be performed as described in Section 6.4.

7.7.2 Compliance with ARARs

Analysis of ARARs associated with extraction and discharge is discussed in Sections 7.3.2 and 7.4.2, respectively. Analysis of ARARs associated with the hexavalent-chromium treatment is discussed in Sections 7.5.1.2 and 7.5.2.2.

ARARs associated with air stripping and carbon adsorption are the same as those presented for GW-2 (refer to Section 7.6.2), except as noted below.

Because the air-stripping system produces off-gases that must be treated, a vapor-phase carbon adsorption unit is added to the system. The carbon adsorption unit would be designed to capture approximately 95 percent of the air-stripper off-gases prior to being emitted to the atmosphere. Subsequent emissions are not considered hazardous or injurious to public health or welfare, or to the environment and, therefore, meet the Michigan air regulations. However, spent carbon is considered hazardous, and must be handled as a hazardous waste, transported off-site in accordance to RCRA, DOT, and Michigan transportation regulations, and disposed of in an off-site, fully permitted, compliant RCRA facility (40 CFR 264). Suppliers of the carbon adsorption systems may also be contracted to handle regeneration of spent carbon beds and final disposal of spent carbon. In addition, other residuals or wastewater generated from the groundwater treatment operations is considered hazardous, and also must be transported according to RCRA, DOT, and Michigan transportation regulations (40 CFR 262, 49 CFR 173, and R 299 Part 3, respectively), and treated and disposed of in accordance with RCRA and Michigan regulation (40 CFR 264 and R 299).

ARARs identified for construction, operation, and groundwater monitoring activities and for the North Park Subdivision are the same as those identified for GW-2 (refer to Section 7.6.2).

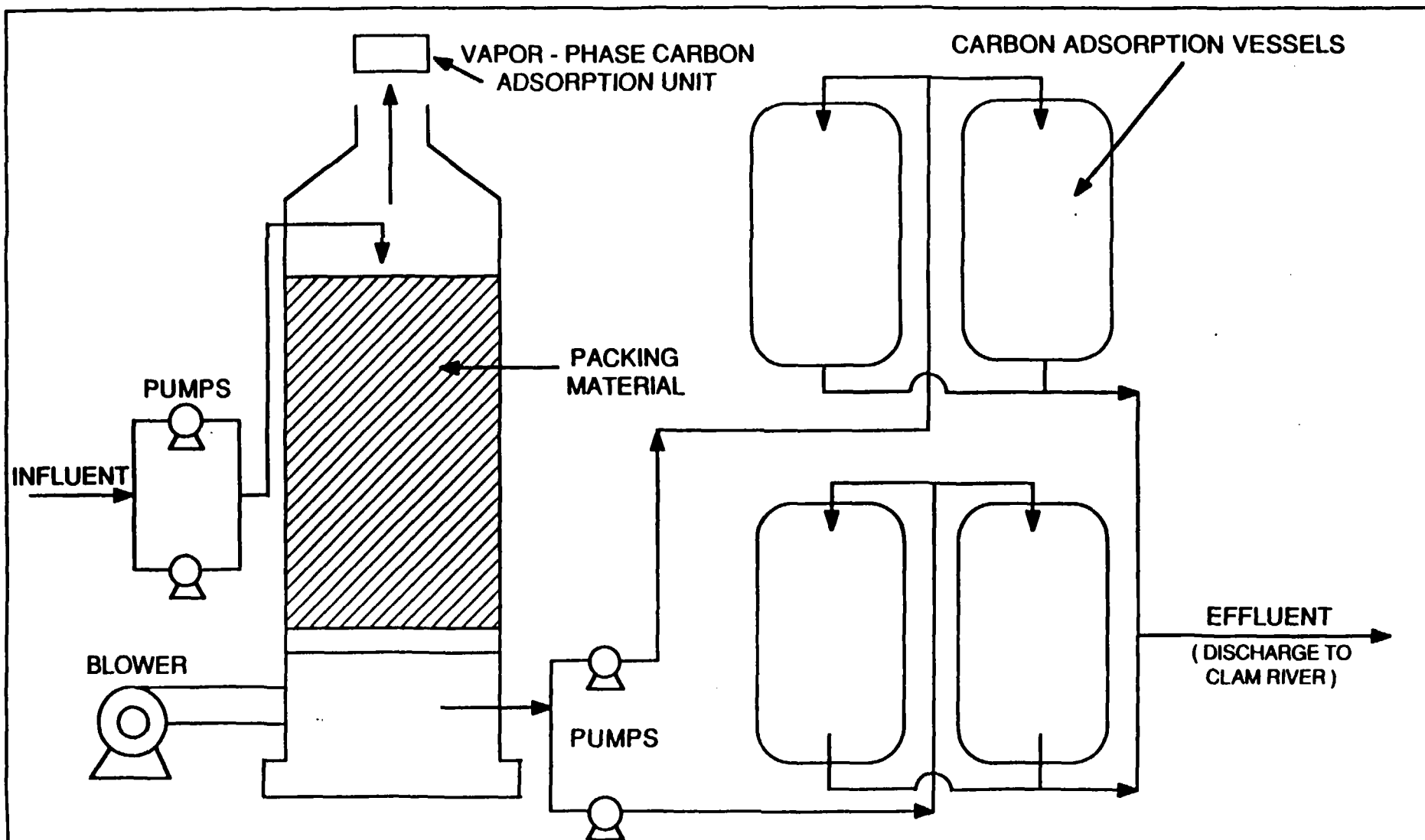


FIGURE 7-17
GW-3b : AIR STRIPPING WITH VAPOR - PHASE AND
LIQUID - PHASE CARBON ADSORPTION
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN
E.C. JORDAN

Appendix E identifies the action-specific ARARs associated with this alternative. These ARARs are further detailed in Section 4.1 and summarized in Table 4-1.

7.7.3 Protection of Public Health and the Environment

When combined with the extraction, hexavalent-chromium treatment, and discharge technologies, the air-stripping alternative would meet the applicable chemical-specific ARARs and reduce existing risks to public health associated with the domestic use and drinking of contaminated groundwater. The extraction system described in Section 7.3 will extract contaminated groundwater until the contaminants of concern in the remaining groundwater are at or below TCLs, which are set at ARARs and appropriate risk levels.

The air-stripping groundwater treatment alternative is designed to reduce VOC contaminant concentrations to MDNR surface water discharge requirements. Because the discharge limitations are set at levels protective of public health and the environment, no adverse impacts would occur due to exposure to the effluent.

The air emissions from the stripping process would be treated by carbon adsorption before release to the ambient environment. The vapor-phase carbon treatment would be designed to remove 95 percent of the air-stripper emissions. These emissions should meet the state requirement for treatment by the BACT and, therefore, are not expected to adversely impact public health or the environment.

7.7.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

Air stripping, combined with extraction, hexavalent-chromium treatment, and discharge technologies, would reduce the toxicity, mobility, and volume of groundwater contaminants. The air-stripping process would reduce the toxicity of groundwater by treating VOCs to surface water discharge requirements. The process would produce air emissions requiring treatment by carbon adsorption. A vapor-phase carbon adsorption unit would reduce the toxicity of VOCs in the off-gases by thermally destructing VOCs adsorbed to the carbon. The system would be designed to remove 95 percent of the VOCs in the off-gas. The vapor-phase unit would be regenerated in-situ by a thermal process and avoids the need for off-site disposal of spent vapor-phase carbon. The air-stripping system with liquid- and vapor-phase carbon adsorption would also produce spent carbon from the liquid-phase process. However, approximately 225,000 pounds per year of spent carbon would require off-site disposal from this process.

7.7.5 Short-term Effectiveness

The short-term effectiveness of the air stripping alternative is the same as GW-2, UV/oxidation (refer to Section 7.6.5). In addition, necessary worker safety measures should be taken in handling the contaminated spent carbon if the air stripping with liquid- and vapor-phase carbon adsorption is utilized. The spent carbon and packing materials should be appropriately handled as hazardous wastes to ensure that risks are not transferred from one site to another.

Low levels of VOCs not removed from the effluent airstream by vapor-phase carbon adsorption will be emitted by the air stripper. Monitoring of the VOC air concentration levels will be performed to ensure that no ARARs are exceeded and that measured levels would not pose risks to the community.

7.7.6 Long-term Effectiveness

In conjunction with the extraction, hexavalent-chromium treatments, and discharge components of this groundwater alternative, the air-stripping process would reduce VOC contaminant levels in the groundwater to meet MDNR discharge limitations before discharge to the Clam River; therefore, long-term protection of both public health and the environment would be provided. Discussions of the long-term effectiveness of the extraction and discharge components of this groundwater alternative are presented in Section 7.3.6 and 7.4.6. For the hexavalent-chromium treatments, evaluations of long-term effectiveness are presented in Sections 7.5.1.6 and 7.5.2.6.

Residual risk would be associated with the secondary wastestreams produced by the hexavalent-chromium treatments, and with the spent carbon if the liquid- and vapor-phase carbon adsorption was utilized. However, these risks would be reduced by disposal off-site in compliance with RCRA regulations.

To mitigate corrosion during the operational life of the system, the air-stripping equipment would be constructed of a resistant material (e.g., FRP). The stripping may require replacement every 25 to 30 years. The packing material would be polypropylene rings, which require replacement every 5 to 10 years. Carbon adsorption process equipment would be constructed of carbon steel to mitigate corrosion. The adsorption vessels would require replacement every 25 to 30 years and the resin lining of the adsorption vessels every 10 to 12 years. To protect against adverse weather conditions, the air-stripping system would be housed indoors. The first air-stripping system (i.e., with vapor-phase carbon adsorption) would be monitored approximately 2 hours daily, and the second air-stripping system (i.e., with vapor- and liquid-phase carbon adsorption) would be monitored approximately 3 hours daily to ensure the system is functioning properly. Prior to implementation of an air-stripping system, pilot-scale testing should be performed to provide performance data for system requirements.

7.7.7 Implementability

The air-stripping systems are technically feasible for the Cadillac site. Air stripping with vapor- and/or liquid-phase carbon adsorption has been used for groundwater remediation, and equipment is ready for on-site installation. The dual air-stripping system with vapor-phase carbon adsorption could be installed for full-scale use in approximately one and a half to two weeks. The air-stripping system with vapor- and liquid-phase carbon adsorption could be installed in approximately two weeks. As with GW-2, installation of the overall alternative would take approximately three to six months, due to installation of the extraction system.

7.7.8 Costs

7.7.8.1 Air Stripping with Vapor Phase Carbon Adsorption. In this section, the capital and annual costs are presented for Alternative GW-3 (Option A): extraction, hexavalent-chromium treatment, air stripping with vapor phase thermal treatment, and discharge. In addition, the net present worth of this alternative is presented. These costs are listed in Tables 7-11 and 7-12 for Alternative GW-3 (Option A) using ion exchange and carbon adsorption for hexavalent-chromium treatment, respectively. The capital and annual costs and the net present worths were derived using the same assumptions presented in Section 6.2.

Capital Costs. The capital cost of Alternative GW-3 (Option A) using ion exchange for hexavalent-chromium treatment is \$4,414,000. If carbon adsorption were used, the total capital cost is \$4,234,000.

Annual Costs. Annual O&M costs for the various components of Alternative GW-3 (Option A) are presented in Tables 7-11 and 7-12. The annual cost for air stripping with vapor-phase carbon adsorption is \$276,000 for the first 38 years, and \$188,000 for the remaining 26 years. Annual costs for hexavalent-chromium treatment using ion exchange are \$352,000, or \$142,000 if using carbon adsorption, for a duration of four years. Additional annual costs for general items are listed in Tables 7-11 and 7-12. Annual costs associated with extraction are not listed herein, but are accounted for in the total present worth.

Present Worth Analysis. The net present worth of Alternative GW-3 (Option A) using ion exchange is \$16,051,000, and \$15,124,000 using carbon adsorption.

Sensitivity Analysis. A sensitivity analysis, assessing the effect that variations in the extraction system's efficiency and pumping duration will have on the estimated present worth of Alternative GW-3, is presented in Section 9.1.

7.7.8.2 Air Stripping with Vapor-phase and Liquid-phase Carbon Adsorption. In this section, the capital and annual costs are presented for Alternative GW-3 (Option B): extraction, hexavalent-chromium treatment, air stripping with vapor-phase thermal treatment, liquid-phase carbon adsorption, and discharge. In addition, the net present worth of this alternative is presented. These costs are listed in Tables 7-13 and 7-14 for Alternative GW-3 (Option B) using ion exchange and carbon adsorption for hexavalent-chromium treatment, respectively. The capital and annual costs and the present worth of this alternative were derived using the assumptions presented in Section 6.2.

Capital Costs. The capital cost of Alternative GW-3 (Option B) using ion exchange for hexavalent-chromium treatment is \$4,959,000. If carbon adsorption is used for hexavalent-chromium treatment, the total capital cost is \$4,779,000.

Annual Costs. Annual O&M costs for the various components of Alternative GW-3 (Option B) are presented in Tables 7-13 and 7-14. The annual cost for air stripping with vapor- and liquid-phase carbon adsorption is \$524,000 for the first 38 years, and \$289,000 for the remaining 26 years. Annual costs for

TABLE 7-11
 GW-3(a) AIR STRIPPING
 WITH VAPOR-PHASE CARBON ADSORPTION AND
 HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

	DESCRIPTION	TOTAL
-----		-----
	CAPITAL COSTS	
	FIVE YEAR SITE REVIEW	\$0
	MONITORING	C
	BUILDING	646,000
	EXTRACTION	884,000
	AIR STRIPPING WITH VAPOR PHASE CARBON ADSORPTION	1,028,000
	HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE	271,000
	DISCHARGE	381,000
	ENGINEERING @ 10%	321,000
	CONTINGENCY @ 25%	883,000

1.	TOTAL CAPITAL COSTS	\$4,414,000
	PRESENT WORTH OF VOC AIR STRIPPING TREATMENT SYSTEM REPLACEMENT	
	AFTER 25 YEARS @ 5%	\$258,000
	AFTER 50 YEARS @ 5%	76,000

2.	TOTAL PRESENT WORTH OF VOC TREATMENT SYSTEM REPLACEMENT	\$334,000
3.	PRESENT WORTH OF REPLACEMENT RESIN FOR 4 YEARS @ 5% (FOR ION EXCHANGE ONLY)	\$3,000
4.	PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION (TOTAL FOR ALL PLUMES)	\$1,936,000

CONTINUED ON NEXT PAGE

TABLE 7-11 (CON'T)
 GW-3(a) AIR STRIPPING
 WITH VAPOR-PHASE CARBON ADSORPTION AND
 HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL

ANNUAL COSTS	
GENERAL ITEMS (DURATION OF 64 YEARS)	
FIVE YEAR SITE REVIEW	\$3,000
BUILDING	51,000
DISCHARGE	68,000

TOTAL OF ABOVE	\$122,000
HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE (DURATION OF 4 YEARS)	\$352,000
AIR STRIPPING TREATMENT SYSTEM	
FOR YEARS 1 - 38	\$276,000
FOR YEARS 39 - 64	188,000
MONITORING	
FOR YEAR 1	\$128,000
FOR YEARS 2 - 64	32,000
5. PRESENT WORTH OF GENERAL ITEMS 64 YEARS @ 5%	\$2,333,000
6. PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%	\$1,248,000
PRESENT WORTH OF AIR STRIPPING TREATMENT SYSTEM	
7. FOR YEARS 1 - 38 @ 5%	\$4,656,000
8. FOR YEARS 39 - 64 @ 5%	423,000
PRESENT WORTH OF MONITORING	
9. FOR YEAR 1	\$122,000
0. FOR YEARS 2 - 64	582,000
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 10)	\$16,051,000

TABLE 7-12
 GW-3(a) AIR STRIPPING
 WITH VAPOR-PHASE CARBON ADSORPTION AND
 HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

	DESCRIPTION	TOTAL
-----		-----
	CAPITAL COSTS	
	FIVE YEAR SITE REVIEW	\$0
	MONITORING	0
	BUILDING	646,000
	EXTRACTION	884,000
	AIR STRIPPING WITH VAPOR PHASE CARBON ADSORPTION	1,028,000
	HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION	140,000
	DISCHARGE	381,000
	ENGINEERING @ 10%	308,000
	CONTINGENCY @ 25%	847,000

1.	TOTAL CAPITAL COSTS	\$4,234,000
	PRESENT WORTH OF AIR STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	
	AFTER 25 YEARS @ 5%	\$258,000
	AFTER 50 YEARS @ 5%	76,000

2.	TOTAL PRESENT WORTH OF AIR STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	\$334,000
.	PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION (TOTAL FOR ALL PLUMES)	\$1,936,000

CONTINUED ON NEXT PAGE

TABLE 7-12 (CON'T)
 GW-3(a) AIR STRIPPING
 WITH VAPOR-PHASE CARBON ADSORPTION AND
 HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL

ANNUAL COSTS	
GENERAL ITEMS (DURATION OF 64 YEARS)	
FIVE YEAR SITE REVIEW	\$3,00
BUILDING	51,00
DISCHARGE	68,00

TOTAL OF ABOVE	\$122,00
HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION (DURATION OF 4 YEARS)	\$142,00
AIR STRIPPING TREATMENT SYSTEM	
FOR YEARS 1 - 38	\$276,00
FOR YEARS 39 - 64	188,00
MONITORING	
FOR YEAR 1	\$128,00
FOR YEARS 2 - 64	32,00
4. PRESENT WORTH OF GENERAL ITEMS FOR 64 YEARS @ 5%	\$2,333,00
5. PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%	\$504,00
PRESENT WORTH OF AIR STRIPPING TREATMENT SYSTEM	
6. FOR YEARS 1 - 38 @ 5%	\$4,656,00
7. FOR YEARS 39 - 64 @ 5%	423,00
PRESENT WORTH OF MONITORING	
8. FOR YEAR 1	\$122,00
9. FOR YEARS 2 - 64	582,00
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 9)	\$15,124,00

TABLE 7-13
 GW-3(b) AIR STRIPPING
 WITH VAPOR-PHASE AND
 LIQUID-PHASE CARBON ADSORPTION, AND
 HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

	DESCRIPTION	TOTAL
-----		-----
	CAPITAL COSTS	
	FIVE YEAR SITE REVIEW	\$
	MONITORING	
	BUILDING	646,00
	EXTRACTION	884,00
	AIR STRIPPING WITH VAP PH THERMAL TRT & LIQ PH CARBON ADSORP	1,424,00
	HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE	271,00
	DISCHARGE	381,00
	ENGINEERING @ 10%	361,00
	CONTINGENCY @ 25%	992,00

1.	TOTAL CAPITAL COSTS	\$4,959,00
	PRESENT WORTH OF AIR STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	
	AFTER 25 YEARS @ 5%	\$356,00
	AFTER 50 YEARS @ 5%	105,00

2.	TOTAL PRESENT WORTH OF AIR STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	\$461,00
3.	PRESENT WORTH OF REPLACEMENT RESIN FOR 4 YEARS @ 5% (FOR ION EXCHANGE)	\$3,000
4.	PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION (TOTAL FOR ALL PLUMES)	\$1,936,000

CONTINUED ON NEXT PAGE

TABLE 7-13 (CON'T)
 GW-3(b) AIR STRIPPING
 WITH VAPOR-PHASE AND
 LIQUID-PHASE CARBON ADSORPTION, AND
 HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL

ANNUAL COSTS	
GENERAL ITEMS (DURATION OF 64 YEARS)	
FIVE YEAR SITE REVIEW	\$3,00
BUILDING	51,00
DISCHARGE	68,00

TOTAL OF ABOVE	\$122,00
HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE (DURATION OF 4 YEARS)	\$352,00
AIR STRIPPING TREATMENT SYSTEM	
FOR YEARS 1 - 38	\$524,00
FOR YEARS 39 - 64	289,00
MONITORING	
FOR YEAR 1	\$128,00
FOR YEARS 2 - 64	32,00
5. PRESENT WORTH OF ABOVE FOR 64 YEARS @ 5%	\$2,333,00
6. PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%	\$1,248,00
PRESENT WORTH OF AIR STRIPPING TREATMENT SYSTEM	
7. FOR YEARS 1 - 38 @ 5%	\$8,839,00
8. FOR YEARS 39 - 64 @ 5%	650,00
PRESENT WORTH OF MONITORING	
9. FOR YEAR 1	\$122,00
10. FOR YEARS 2 - 64	582,00
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 10)	\$21,133,00

TABLE 7-14
 GW-3(b) AIR STRIPPING
 WITH VAPOR-PHASE AND
 LIQUID-PHASE CARBON ADSORPTION, AND
 HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

	DESCRIPTION	TOTAL
-----		-----
	CAPITAL COSTS	
	FIVE YEAR SITE REVIEW	\$
	MONITORING	
	BUILDING	646,000
	EXTRACTION	884,000
	AIR STRIPPING WITH VAP PH THERMAL TRT & LIQ PH CARBON ADSORP	1,424,000
	HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION	140,000
	DISCHARGE	381,000
	ENGINEERING @ 10%	348,000
	CONTINGENCY @ 25%	956,000

1.	TOTAL CAPITAL COSTS	\$4,779,000
	PRESENT WORTH OF AIR STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	
	AFTER 25 YEARS @ 5%	\$356,000
	AFTER 50 YEARS @ 5%	105,000

2.	TOTAL PRESENT WORTH OF AIR STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	\$461,000
3.	PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION (TOTAL FOR ALL PLUMES)	\$1,936,000

CONTINUED ON NEXT PAGE

TABLE 7-14 (CON'T)
 GW-3(b) AIR STRIPPING
 WITH VAPOR-PHASE AND
 LIQUID-PHASE CARBON ADSORPTION, AND
 HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION		TOTAL

ANNUAL COSTS		
GENERAL COSTS (DURATION OF 64 YEARS)		
FIVE YEAR SITE REVIEW		\$3,000
BUILDING		51,000
DISCHARGE		68,000

TOTAL OF ABOVE		\$122,000
HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION (DURATION OF 4 YEARS)		\$142,000
AIR STRIPPING TREATMENT SYSTEM		
FOR YEARS 1 - 38		\$524,000
FOR YEARS 39 - 64		289,000
MONITORING		
FOR YEAR 1		\$128,000
FOR YEARS 2 - 64		32,000
4. PRESENT WORTH OF GENERAL ITEMS FOR 64 YEARS @ 5%		\$2,333,000
5. PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%		\$504,000
PRESENT WORTH OF AIR STRIPPING TREATMENT SYSTEM		
5. FOR YEARS 1 - 38 @ 5%		\$8,839,000
7. FOR YEARS 39 - 64 @ 5%		650,000
PRESENT WORTH OF MONITORING		
3. FOR YEAR 1		\$122,000
3. FOR YEARS 2 - 64		582,000
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 9)		\$20,206,000

hexavalent-chromium treatment using ion exchange are \$352,000, or \$142,000 using carbon adsorption, for a duration of four years. Additional annual costs for general items are listed in Tables 7-13 and 7-14. Annual costs associated with extraction are not listed herein, but are accounted for in the total present worth.

Present Worth Analysis. The net present worth of Alternative GW-3 (Option B) using ion exchange is \$21,133,000, and \$20,206,000 using carbon adsorption.

Sensitivity Analysis. A sensitivity analysis, assessing the effect that variations in the extraction system's efficiency and pumping duration will have on the estimated present worth of Alternative GW-3 using vapor- and liquid-phase carbon adsorption, is presented in Section 9.1.

7.7.9 Summary

The detailed analysis of Alternative GW-3 is summarized in Table 7-15.

7.8 DETAILED ANALYSIS OF ALTERNATIVE GW-4: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT, STEAM STRIPPING, AND DISCHARGE

7.8.1 Description

Extraction. Extraction would be performed as described in Section 6.3.

Hexavalent-Chromium Treatment. Hexavalent-chromium treatment would be performed the same as for GW-2 (refer to Section 7.6.1).

Steam Stripping. Steam stripping is fundamentally comparable to air stripping because it also operates on the principle of mass transfer of organics from the liquid phase to the vapor phase. Steam is utilized when the volatility of the organics makes removal at ambient air temperatures difficult. As with air stripping, the process occurs in a packed tower, usually operated in a counter-current mode. By using the counter-current mode, the least-contaminated water contacts the cleanest steam, thereby providing the most effective mass transfer. The packing material in the column functions to increase the area of contact between the steam and the liquid waste. The contaminated steam, which passes out through the top of the column, may be condensed to a liquid and decanted to produce a concentrated condensed stream that must be treated and/or disposed at an off-site permitted facility.

The steam-stripping system for the Cadillac area groundwater would be designed to treat approximately 940 gpm. The system components are presented in Figure 7-18. Extracted groundwater would pass through a heat exchanger to increase the temperature and volatility of VOCs in the influent. Groundwater would be introduced to the top of the packed stripping column and steam would enter the bottom of the column. The column would have a packing height of 30 feet and a diameter of 5 feet, and would be constructed of a resistant material (e.g., stainless steel) to mitigate corrosion. The packing material would be polypropylene or stainless steel, depending on the corrosivity of the liquid stream. Approximately 5,900 pounds per hour of low-pressure steam would be

TABLE 7-15
SUMMARY OF DETAILED ANALYSIS
GW-3: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
AIR STRIPPING, AND DISCHARGE

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o Drinking water standards would be met by the extraction of contaminated groundwater from the aquifer. - o Contaminated groundwater would be treated to meet discharge requirements. - o The hexavalent-chromium treatment and air-stripping systems would be designed and operated in compliance with all action-specific ARARs.
2. Protection of Public Health and the Environment	o GW-3 would reduce existing risks to public health associated with the domestic use and drinking of contaminated groundwater.
3. Reduction of Toxicity, Mobility, or Volume	- o GW-3 would reduce the toxicity and mobility of groundwater contaminants. - o Hexavalent-chromium treatment would produce a secondary wastestream requiring off-site disposal. - o Air-stripping process would produce air emissions requiring additional treatment by carbon adsorption. - o Air stripping with liquid- and vapor-phase carbon adsorption would produce spent carbon requiring off-site disposal.
4. Short-term Effectiveness	- o No short-term community or worker health impacts would occur during installation, except for minor construction hazards. - o Potential risks to on-site workers exist due to handling of hazardous substances and secondary wastes generated from hexavalent-chromium treatment, and from spent carbon generated by the liquid- and vapor-phase carbon adsorption. Appropriate safety requirements must be met. - o The time required to achieve full protection is estimated at 64 years.

TABLE 7-15
(continued)
SUMMARY OF DETAILED ANALYSIS
GW-3: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
AIR STRIPPING, AND DISCHARGE

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
5. Long-term Effectiveness	- o Residual risk would be associated with the secondary wastestreams produced by the hexavalent-chromium treatments and the spent carbon from the liquid- and vapor-phase carbon adsorption. These risks would be reduced by treatment off-site. - o Residual risks posed by air emissions would be reduced by carbon adsorption. - o Long-term groundwater monitoring and management of property required. - o GW-3 would effectively extract contaminated groundwater from the aquifer and reduce the contaminant concentration levels to surface water discharge levels. - o Potential for replacement of polypropylene rings is expected at 5 to 10 years; lining of adsorption vessels at 10 to 12 years; and if ion exchange is utilized, the resin may have to be replaced after the first year of operation.
6. Implementation	- o GW-3 is technically feasible for the Cadillac site. - o Technologies associated with GW-3 have been demonstrated to treat the contaminants of concern. - o Equipment is readily available. - o Access to land for equipment installation must be obtained, including right-of-ways. - o Installation of air stripping with either carbon adsorption system would take approximately 1 to 2 weeks. - o Air emissions license would be required for air stripping. - o Pilot-scale testing should be performed. - o Installation of the overall alternative would take 3 to 6 months.

TABLE 7-15
(continued)
SUMMARY OF DETAILED ANALYSIS
GW-3: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
AIR STRIPPING, AND DISCHARGE
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
7. Costs - Air Stripping With Vapor Phase Carbon Adsorption(a)	- o Capital Costs - Using Ion Exchange = \$4,414,000 - Using Carbon Adsorption = \$4,234,000 - o Annual Costs (for air stripping only) = \$276,000 (yrs 1-38) \$188,000 (yrs 39-64) - o Present Worth - Using Ion Exchange = \$16,051,000 - Using Carbon Adsorption = \$15,124,000
Costs - Air Stripping With Vapor and Liquid Phase Carbon Adsorption(b)	- o Capital Costs - Using Ion Exchange = \$4,959,000 - Using Carbon Adsorption = \$4,779,000 - o Annual Costs (air stripping only) = \$524,000 (yrs 1-38) \$289,000 (yrs 39-64) - o Present Worth - Using Ion Exchange = \$21,133,000 - Using Carbon Adsorption = \$20,206,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

TABLE 7-15
(continued)
SUMMARY OF DETAILED ANALYSIS
GW-3: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
AIR STRIPPING, AND DISCHARGE
CADILLAC AREA FEASIBILITY STUDY

NOTES (continued):

- A = The long-term uncertainties associated with land disposal.
- B = The goals, objectives, and requirements of the SWDA.
- C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.
- D = Short- and long-term potential for adverse health effects from human exposure.
- E = Long-term monitoring costs.
- F = The potential for future remedial action costs if the alternative remedial action in question were to fail.
- G = The potential threat to public health and the environment associated with excavation, transportation, and redispisal, or containment.

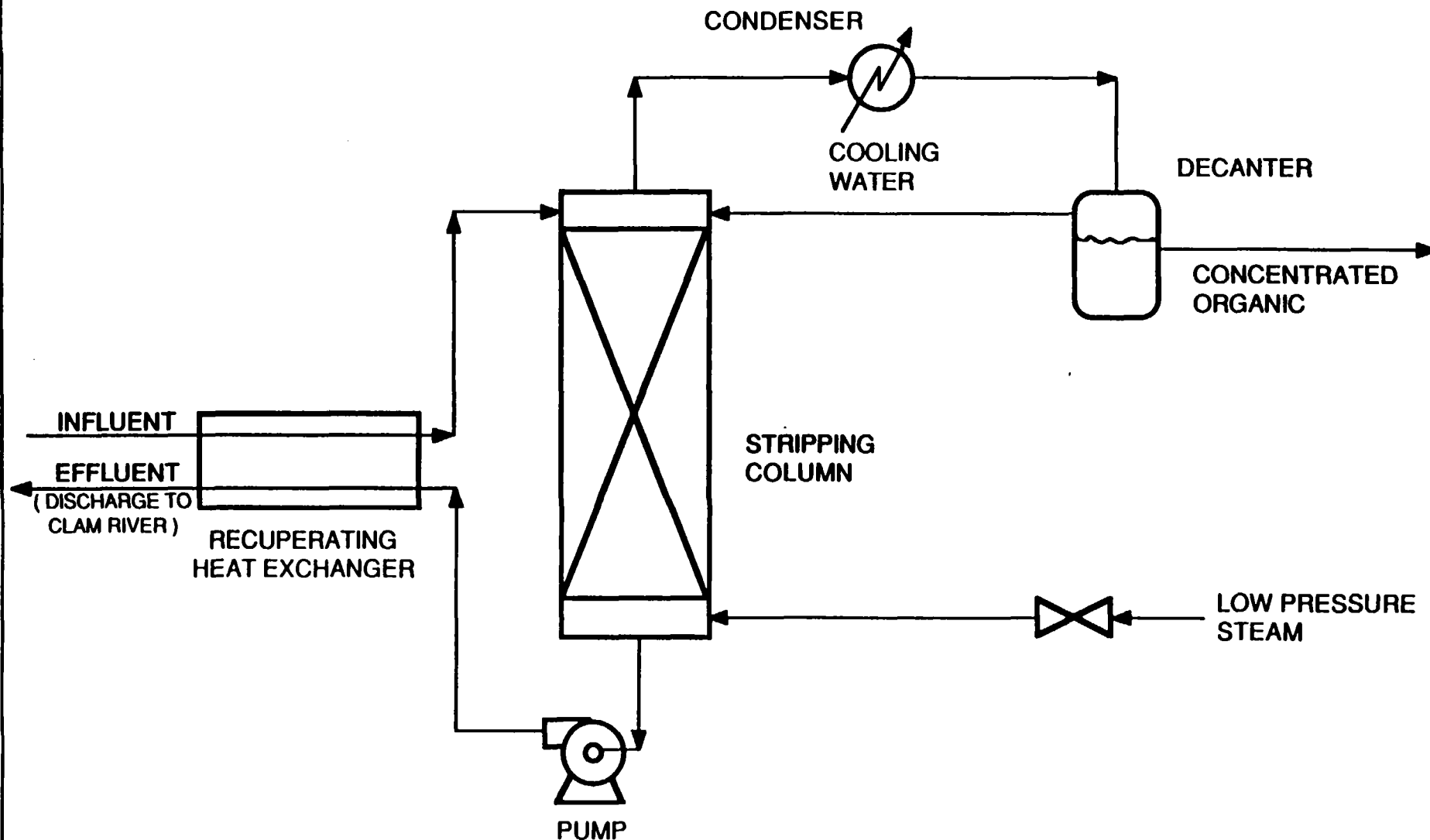


FIGURE 7-18
GW-4a : STEAM STRIPPING
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

E.C. JORDAN

required to strip VOCs from the groundwater. The contaminated steam passing out the top of the column would be condensed, and the concentrated organics (approximately 40 percent) would be separated in a decanter. Approximately 200 pounds per hour (i.e., 231 gallons/day) of concentrated organics would be produced for disposal. The dilute condensate from the decanter would be routed back to the stripping column for further treatment. Treated groundwater would be collected at the bottom of the column, pumped through the heat exchanger, and used to warm the influent flow.

To reduce the amount of electricity required to produce steam for the process, a vapor recompression system could be used (Figure 7-19). This process is similar to the straight steam-stripping process; however, a vapor compressor and reboiler are used in place of a condenser to produce steam, using treated groundwater from the base of the column and water vapor from the top.

Discharge. Discharge would be performed as described in Section 6.4.

7.8.2 Compliance with ARARs

Analysis of ARARs associated with extraction and discharge is discussed in Sections 7.3.2 and 7.4.2, respectively. Analysis of ARARs associated with the hexavalent-chromium treatment is discussed in Sections 7.5.1.2 and 7.5.2.2.

Thermal or steam stripping generates a condensed stream that is considered hazardous and that must be transported and disposed of pursuant to RCRA regulations (40 CFR 262 through 264) and Michigan Hazardous Waste Management Rules (R 299). Disposal must occur in a fully permitted RCRA facility operating in compliance with 40 CFR 264. Because the treatment system does not generate air pollutant emissions, federal and state air regulations are not considered ARARs.

ARARs associated with construction, operation, and groundwater monitoring activities and the North Park Subdivision are the same as those identified for GW-2 (refer to Section 7.6.2).

Appendix E identifies the action-specific ARARs associated with this alternative. These ARARs are further detailed in Section 4.1 and summarized in Table 4-1.

7.8.3 Protection of Public Health and the Environment

Steam stripping, combined with the extraction, hexavalent-chromium treatment, and discharge technologies, would achieve the same level of protection as GW-2 (refer to Section 7.6.3). Evaluations of the extraction and discharge technologies' ability to provide protection of public health and the environment are described in Sections 7.3.3 and 7.4.3, respectively. The evaluation for hexavalent-chromium treatment options is presented in Sections 7.5.1.1 and 7.5.2.1.

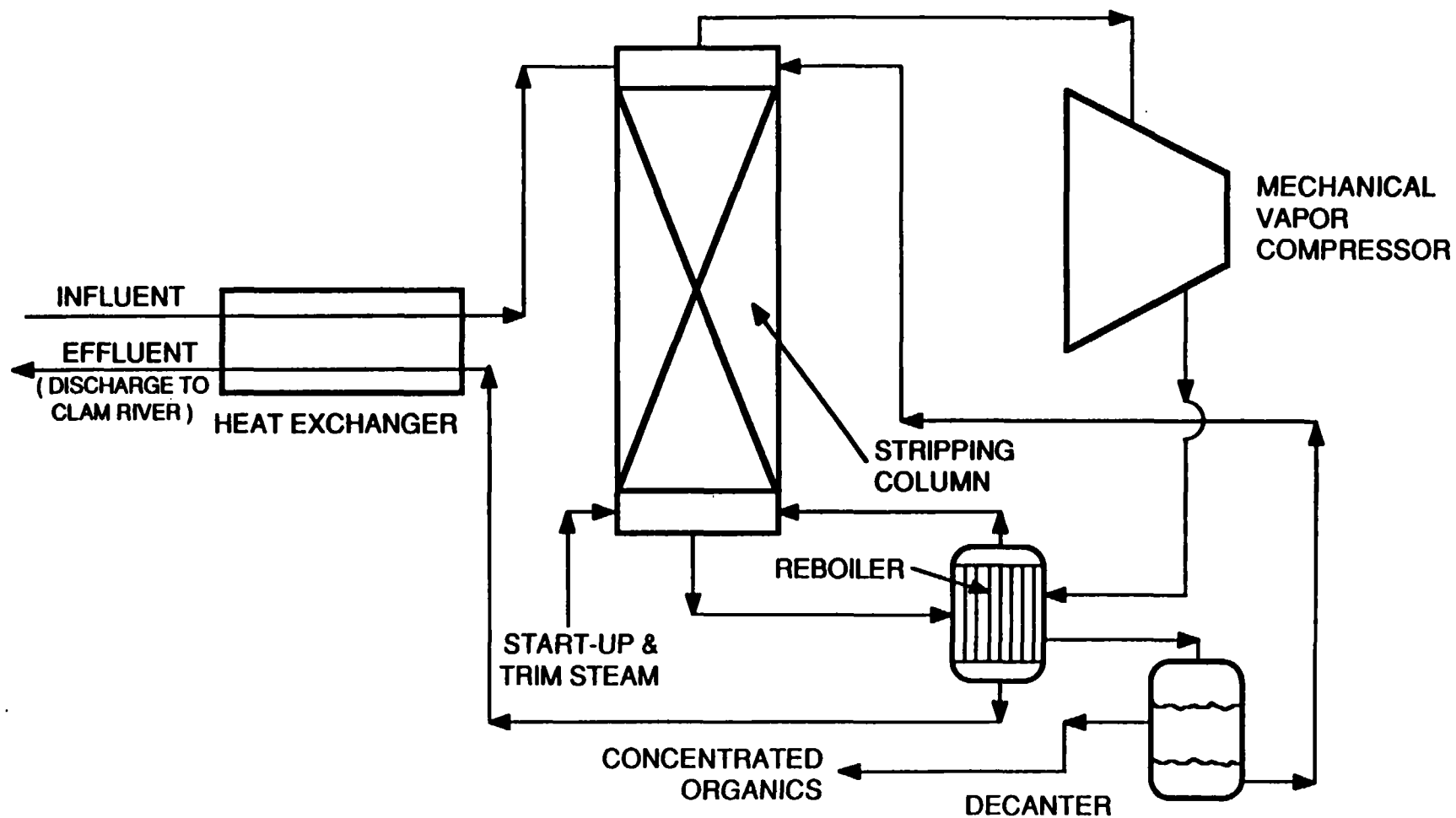


FIGURE 7-19
GW-4b : STEAM STRIPPING WITH
VAPOR RECOMPRESSION
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

7.8.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

Steam stripping, combined with extraction, hexavalent-chromium treatment, and discharge technologies, would reduce the toxicity, mobility, and volume of groundwater contaminants. Toxicity of the groundwater would be reduced by treating VOCs to the surface water discharge requirements. The process would produce approximately 231 gallons per day of a concentrated organic condensate requiring disposal. This is a high quantity estimate because it assumes that the highest organic concentrations detected in the plumes will be present continually in the influent flow.

7.8.5 Short-term Effectiveness

The short-term effectiveness of the steam stripping alternative is the same as GW-2 (refer to Section 7.6.5).

In addition, during implementation the necessary worker safety measures should be taken when handling the condensed steam from the packed column. This waste would contain concentrated levels of VOCs. Safety measures should also be taken when handling secondary wastes from the hexavalent-chromium treatment.

7.8.6 Long-term Effectiveness

Steam stripping, combined with extraction, hexavalent-chromium treatment, and discharge technologies, is capable of reducing VOCs in the groundwater pumped from the Cadillac plumes to nondetectable levels. When the effluent is discharged to the Clam River, it will not pose a potential threat to public health or the environment.

Long-term effectiveness of the extraction and discharge components of this groundwater alternative is discussed in Sections 7.3.6 and 7.4.6. For the hexavalent-chromium treatments, the long-term effectiveness is presented in Sections 7.5.1.6 and 7.5.2.6.

Residual risk would be associated with the secondary wastestreams produced by the hexavalent-chromium treatment and the steam-stripping process. However, these risks would be reduced by disposal off-site in compliance with RCRA regulations.

Steam-stripping process equipment would be constructed of a resistant material (e.g., stainless steel) to mitigate corrosion during operation life of the system. The design life of the major components (i.e., column, heat exchanger, and condenser) could be as low as 10 years. Smaller components (i.e., pumps, valves, and flow meters) may require replacement every five years. The main system could be insulated and set-up outdoors. It is recommended that the control valves (i.e., flow meters, bottoms pump level control, and decanter control) be housed in a small building to protect against adverse weather conditions.

7.8.7 Implementability

The steam-stripping process is technically feasible for the Cadillac site. Process equipment is available for on-site use, and the system is expected to run reliably once in operation. Ground level equipment (e.g., wiring, piping, controls, and smaller components) could be skid-mounted and transported for immediate setup at the site. The stripping column, support column (to contain components such as the condenser and decanter), and interconnection piping would require approximately three months for on-site installation. Prior to full-scale setup, treatability work should be performed to better define the system requirements and performance (i.e., steam flow rate, electrical consumption, compound removal efficiencies, and decanting potential). Installation of the overall alternative would take approximately three to six months, due to installation of the extraction system.

7.8.8 Costs

7.8.8.1 Steam Stripping. In this subsection, capital and annual costs are presented for Alternative GW-4 (Option A): extraction, hexavalent-chromium treatment, steam stripping, and discharge. In addition, the net present worth of this alternative is presented. These costs are listed in Tables 7-16 and 7-17 for Alternative GW-4 (Option A) using ion exchange and carbon adsorption for hexavalent-chromium treatment, respectively. The capital and annual costs and the net present worths were derived using the same assumptions used for GW-2 (refer to Section 7.6.8).

Capital Costs. The capital cost of Alternative GW-4 (Option A) using ion exchange for hexavalent-chromium treatment is \$4,238,000. If carbon adsorption were used, the total capital cost is \$4,058,000.

Annual Costs. Annual O&M costs for the various components of Alternative GW-4 (Option A) are presented in Tables 7-16 and 7-17. The annual cost for steam stripping is \$958,000 for the first 38 years, and \$448,000 for the remaining 26 years. Annual costs for hexavalent-chromium treatment using ion exchange are \$352,000, or \$142,000 using carbon adsorption, for a duration of four years. Additional annual costs for general items are listed in Tables 7-16 and 7-17. Annual costs associated with extraction are not listed herein, but are accounted for in the total present worth.

Present Worth Analysis. The net present worth of Alternative GW-4 (Option A) using ion exchange is \$28,571,000, or \$27,644,000 using carbon adsorption.

Sensitivity Analysis. A sensitivity analysis, assessing the effect that variations in the extraction system's efficiency and pumping duration will have on the estimated present worth of Alternative GW-4, is presented in Section 9.1.

7.8.8.2 Steam Stripping with Vapor Recompression. In this subsection, capital and annual costs are presented for Alternative GW-4 (Option B): extraction, hexavalent-chromium treatment, steam stripping, and discharge. In addition, the net present worth of this alternative is presented. These costs are listed in Tables 7-18 and 7-19 for Alternative GW-4 (Option B) using ion exchange and

TABLE 7-16
 GW-4(a) STEAM STRIPPING
 WITH HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL

CAPITAL COSTS	
FIVE YEAR SITE REVIEW	\$
MONITORING	
BUILDING	646,00
EXTRACTION	884,00
STEAM STRIPPING	900,00
HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE	271,00
DISCHARGE	381,00
ENGINEERING @ 10%	308,00
CONTINGENCY @ 25%	848,00

1. TOTAL CAPITAL COSTS	\$4,238,00
PRESENT WORTH OF VOC TREATMENT SYSTEM REPLACEMENT	
AFTER 10 YEARS @ 5%	\$384,00
AFTER 20 YEARS @ 5%	236,00
AFTER 30 YEARS @ 5%	145,00
AFTER 40 YEARS @ 5%	89,00
AFTER 50 YEARS @ 5%	55,00
AFTER 60 YEARS @ 5%	33,00

2. TOTAL PRESENT WORTH OF VOC TREATMENT SYSTEM REPLACEMENT	\$942,00
3. PRESENT WORTH OF REPLACEMENT RESIN FOR 4 YEARS @ 5%	\$3,00
(FOR ION EXCHANGE)	
4. PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION	\$1,936,00
(TOTAL FOR ALL PLUMES)	

CONTINUED ON NEXT PAGE

TABLE 7-16 (CON'T)
 GW-4(a) STEAM STRIPPING
 WITH HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION		TOTAL
ANNUAL COSTS		
GENERAL ITEMS (DURATION OF 64 YEARS)		
FIVE YEAR SITE REVIEW		\$3,00
BUILDING		51,00
DISCHARGE		68,00
TOTAL OF ABOVE		\$122,00
HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE (DURATION OF 4 YEARS)		\$352,00
STEAM STRIPPING TREATMENT SYSTEM		
FOR YEARS 1 - 38		\$958,00
FOR YEARS 39 - 64		448,00
MONITORING		
FOR YEAR 1		\$128,00
FOR YEARS 2 - 64		32,00
5. PRESENT WORTH OF GENERAL ITEMS FOR 64 YEARS @ 5%		\$2,333,00
6. PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%		\$1,248,00
PRESENT WORTH OF STEAM STRIPPING TREATMENT SYSTEM		
7. FOR YEARS 1 - 38 @ 5%		\$16,159,00
8. FOR YEARS 39 - 64 @ 5%		1,008,00
PRESENT WORTH OF MONITORING		
9. FOR YEAR 1		\$122,00
10. FOR YEARS 2 - 64		582,00
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 10)		\$28,571,00

TABLE 7-17
 GW-4(a) STEAM STRIPPING
 WITH HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION		TOTAL
-----		-----
CAPITAL COSTS		
	FIVE YEAR SITE REVIEW	\$
	MONITORING	
	BUILDING	646,00
	EXTRACTION	884,00
	STEAM STRIPPING	900,00
	HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION	140,00
	DISCHARGE	381,00
	ENGINEERING @ 10%	295,00
	CONTINGENCY @ 25%	812,00

1.	TOTAL CAPITAL COSTS	\$4,058,00
PRESENT WORTH OF STEAM STRIPPING VOC TREATMENT SYSTEM REPLACEMENT		
	AFTER 10 YEARS @ 5%	\$384,00
	AFTER 20 YEARS @ 5%	236,00
	AFTER 30 YEARS @ 5%	145,00
	AFTER 40 YEARS @ 5%	89,00
	AFTER 50 YEARS @ 5%	55,00
	AFTER 60 YEARS @ 5%	33,00

2.	TOTAL PRESENT WORTH OF STEAM STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	\$942,00
3.	PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION (TOTAL FOR ALL PLUMES)	\$1,936,00

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TABLE 7-17 (CON'T)
 GW-4(a) STEAM STRIPPING
 WITH HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION		TOTAL
ANNUAL COSTS		
GENERAL ITEMS (DURATION OF 64 YEARS)		
FIVE YEAR SITE REVIEW		\$3,00
BUILDING		51,00
DISCHARGE		68,00
TOTAL OF ABOVE		\$122,00
HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION (DURATION OF 4 YEARS)		\$142,00
STEAM STRIPPING TREATMENT SYSTEM		
FOR YEARS 1 - 38		\$958,00
FOR YEARS 39 - 64		448,00
MONITORING		
FOR YEAR 1		\$128,00
FOR YEARS 2 - 64		32,00
4. PRESENT WORTH OF ABOVE FOR 64 YEARS @ 5%		\$2,333,00
5. PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%		\$504,00
PRESENT WORTH OF STEAM STRIPPING TREATMENT SYSTEM		
6. FOR YEARS 1 - 38 @ 5%		\$16,159,00
7. FOR YEARS 39 - 64 @ 5%		1,008,00
PRESENT WORTH OF MONITORING		
8. FOR YEAR 1		\$122,00
9. FOR YEARS 2 - 64		582,00
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 9)		\$27,644,00

TABLE 7-18
 GW-4(b) STEAM STRIPPING
 WITH VAPOR RECOMPRESSION, AND
 HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL

CAPITAL COSTS	
FIVE YEAR SITE REVIEW	\$
MONITORING	
BUILDING	646,00
EXTRACTION	884,00
STEAM STRIPPING WITH VAPOR RECOMPRESSION	1,156,00
HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE	271,00
DISCHARGE	381,00
ENGINEERING @ 10%	334,00
CONTINGENCY @ 25%	918,00

1. TOTAL CAPITAL COSTS	\$4,590,00
PRESENT WORTH OF STEAM STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	
AFTER 10 YEARS @ 5%	\$537,00
AFTER 20 YEARS @ 5%	330,00
AFTER 30 YEARS @ 5%	202,00
AFTER 40 YEARS @ 5%	124,00
AFTER 50 YEARS @ 5%	76,00
AFTER 60 YEARS @ 5%	47,00

TOTAL PRESENT WORTH OF STEAM STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	\$1,316,00
3. PRESENT WORTH OF REPLACEMENT RESIN FOR 4 YEARS @ 5% (FOR ION EXCHANGE ONLY)	\$3,00
4. PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION (TOTAL FOR ALL PLUMES)	\$1,936,00

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TABLE 7-18 (CON'T)
 GW-4(b) STEAM STRIPPING
 WITH VAPOR RECOMPRESSION, AND
 HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL
ANNUAL COSTS	
GENERAL ITEMS (DURATION OF 64 YEARS)	
FIVE YEAR SITE REVIEW	\$3,00
BUILDING	51,00
DISCHARGE	68,00
TOTAL OF ABOVE	\$122,00
HEXAVALENT CHROMIUM TREATMENT BY ION EXCHANGE (DURATION OF 4 YEARS)	\$352,00
ANNUAL COSTS OF STEAM STRIPPING TREATMENT SYSTEM	
FOR YEARS 1 - 38	\$564,00
FOR YEARS 39 - 64	291,00
MONITORING	
FOR YEAR 1	\$128,00
FOR YEARS 2 - 64	32,00
5. PRESENT WORTH OF GENERAL ITEMS FOR 64 YEARS @ 5%	\$2,333,00
6. PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%	\$1,248,00
PRESENT WORTH OF STEAM STRIPPING TREATMENT SYSTEM	
7. FOR YEARS 1 - 38 @ 5%	\$9,513,00
9. FOR YEARS 39 - 64 @ 5%	655,00
PRESENT WORTH OF MONITORING	
9. FOR YEAR 1	\$122,00
3. FOR YEARS 2 - 64	582,00
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 10)	\$22,298,00

TABLE 7-19
 GW-4(b) STEAM STRIPPING
 WITH VAPOR RECOMPRESSION, AND
 HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

	DESCRIPTION	TOTAL
-----		-----
	CAPITAL COSTS	
	FIVE YEAR SITE REVIEW	\$
	MONITORING	
	BUILDING	646,00
	EXTRACTION	884,00
	STEAM STRIPPING WITH VAPOR RECOMPRESSION	1,156,00
	HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION	140,00
	DISCHARGE	381,00
	ENGINEERING @ 10%	321,00
	CONTINGENCY @ 25%	882,00

1.	TOTAL CAPITAL COSTS	\$4,410,00
	PRESENT WORTH OF STEAM STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	
	AFTER 10 YEARS @ 5%	\$537,00
	AFTER 20 YEARS @ 5%	330,00
	AFTER 30 YEARS @ 5%	202,00
	AFTER 40 YEARS @ 5%	124,00
	AFTER 50 YEARS @ 5%	76,00
	AFTER 60 YEARS @ 5%	47,00

2.	TOTAL PRESENT WORTH OF STEAM STRIPPING VOC TREATMENT SYSTEM REPLACEMENT	\$1,316,00
3.	PRESENT WORTH OF ANNUAL COSTS OF EXTRACTION (TOTAL FOR ALL PLUMES)	\$1,936,00

CONTINUED ON NEXT PAGE

TABLE 7-19 (CON'T)
 GW-4(b) STEAM STRIPPING
 WITH VAPOR RECOMPRESSION, AND
 HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL

ANNUAL COSTS	
GENERAL COSTS (DURATION OF 64 YEARS)	
FIVE YEAR SITE REVIEW	\$3,000
BUILDING	51,000
DISCHARGE	68,000

TOTAL OF ABOVE	\$122,000
HEXAVALENT CHROMIUM TREATMENT BY CARBON ADSORPTION (DURATION OF 4 YEARS)	\$142,000
STEAM STRIPPING TREATMENT SYSTEM	
FOR YEARS 1 - 38	\$564,000
FOR YEARS 39 - 64	291,000
MONITORING	
FOR YEAR 1	\$128,000
FOR YEARS 2 - 64	32,000
1. PRESENT WORTH OF GENERAL ITEMS FOR 64 YEARS @ 5%	\$2,333,000
PRESENT WORTH OF HEXAVALENT CHROMIUM TREATMENT FOR 4 YEARS @ 5%	\$504,000
PRESENT WORTH OF STEAM STRIPPING TREATMENT SYSTEM	
5. FOR YEARS 1 - 38 @ 5%	\$9,513,000
7. FOR YEARS 39 - 64 @ 5%	655,000
PRESENT WORTH OF MONITORING	
8. FOR YEAR 1	\$122,000
9. FOR YEARS 2 - 64	582,000
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 9)	\$21,371,000

carbon adsorption for hexavalent-chromium treatment, respectively. The capital and annual costs and net present worths were derived using the assumptions presented in Section 6.2.

Capital Costs. The capital cost of Alternative GW-4 (Option B) using ion exchange for hexavalent-chromium treatment is \$4,590,000. If carbon adsorption were used, the total capital cost is \$4,410,000.

Annual Costs. Annual O&M costs for the various components of Alternative GW-4 (Option A) are presented in Tables 7-18 and 7-19. The annual cost for steam stripping with vapor recompression is \$564,000 for the first 38 years, and \$291,000 for the remaining 26 years. Annual costs for hexavalent-chromium treatment using ion exchange are \$352,000, or \$142,000 using carbon adsorption, for a duration of four years. Additional annual costs for general items are listed in Tables 7-18 and 7-19. Annual costs associated with extraction are not listed herein, but are accounted for in the total present worth.

Present Worth Analysis. Based on a 5-percent interest rate, the net present worth of Alternative GW-4 (Option B) using ion exchange is \$22,298,000 and \$21,371,000 using carbon adsorption.

Sensitivity Analysis. A sensitivity analysis, assessing the effect that variations in the extraction system's efficiency and pumping duration will have on the estimated present worth of Alternative GW-4 with vapor recompression, is presented in Section 9.1.

7.8.9 Summary

The detailed analysis of Alternative GW-4 is summarized in Table 7-20.

7.9 DETAILED ANALYSIS OF AC-1: ALTERNATIVE WATER SUPPLY

7.9.1 Description

Additional Component AC-1 includes the search for and potential development of an alternate water supply for the City of Cadillac. Alternative AC-1 would be implemented if the removal of DNAPL compounds is not achieved by available remediation technologies. This alternative could be developed as a contingency option, whereby an alternative site is selected, but construction is not implemented unless the cleanup near the current wellfield is determined infeasible. The alternative could also be considered an interim measure, as the first phase of the remediation process. Alternative AC-1 would result in the development and use of a new wellfield, while contamination at the industrial park could be addressed, with the potential risk of human exposure to contaminated drinking groundwater being eliminated.

Efforts to Date to Locate Alternative Water Supplies. Contingency plans are required for all municipal water supplies. Contingency plans establish alternative water supplies and procedures for emergency conditions, water shortages, or contamination problems. The MDPH, Division of Water Supply (MDWS) requested an update to the existing contingency plan for the municipal water supply from

TABLE 7-20
SUMMARY OF DETAILED ANALYSIS
GW-4: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
STEAM STRIPPING, AND DISCHARGE
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o Drinking water standards would be met by the extraction of contaminated groundwater from the aquifer. - o Contaminated groundwater would be treated to meet discharge requirements. - o The hexavalent-chromium treatment and steam-stripping systems would be designed and operated in compliance with all action-specific ARARs.
2. Protection of Public Health and the Environment	o GW-4 would reduce existing risks to public health associated with the domestic use and drinking of contaminated groundwater.
3. Reduction of Toxicity, Mobility, or Volume	- o GW-4 would reduce the volume and mobility of groundwater contaminants. - o Hexavalent-chromium treatment would produce a secondary wastestream requiring off-site disposal. - o Steam-stripping would produce approximately 230 gallons/day of a concentrated organic condensate requiring off-site treatment.
4. Short-term Effectiveness	- o No short-term community or worker health impacts would occur during installation, except for minor construction hazards. - o Potential risks to on-site workers exist due to handling of hazardous substances and secondary wastes generated from hexavalent-chromium treatment, and from the condensed steam from the steam stripping. Appropriate safety requirements must be met.

TABLE 7-20
(continued)
SUMMARY OF DETAILED ANALYSIS
GW-4: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
STEAM STRIPPING, AND DISCHARGE
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
	o The length of time required to achieve full protection is estimated to be 64 years.
5. Long-term Effectiveness	- o Residual risk would be associated with the secondary wastestreams produced by the hexavalent-chromium treatments. These risks would be reduced by treatment off-site. - o Residual risk posed by organic condensate would be reduced by off-site treatment. - o Long-term groundwater monitoring and management of property required. - o GW-4 would effectively extract contaminated groundwater from the aquifer and reduce the contaminant concentration levels to surface water discharge levels. - o Potential for replacement of the major components of steam-stripping is expected after 10 years; and if ion exchange is utilized, the resin may have to be replaced after the first year of operation.
6. Implementation	- o GW-4 is technically feasible for the Cadillac site. - o Technologies associated with GW-4 have been demonstrated to treat the contaminants of concern. - o Equipment is readily available. - o Access to land for equipment installation must be obtained, including right-of-ways. - o Installation of steam stripping would take approximately 3 months. - o Treatability studies should be performed to better define system requirements and performance. - o Installation of the overall alternative would take 3 to 6 months.

TABLE 7-20
(continued)
SUMMARY OF DETAILED ANALYSIS
GW-4: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
STEAM STRIPPING, AND DISCHARGE

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
7. Costs - Steam Stripping(a)	- o Capital Costs - Using Ion Exchange = \$4,238,000 - Using Carbon Adsorption = \$4,058,000 - o Annual Costs (steam stripping only) = \$958,000 (Years 1-38) \$448,000 (Years 39-64) - o Present Worth - Using Ion Exchange = \$28,571,000 - Using Carbon Adsorption = \$27,644,000
Costs - Steam Stripping with Vapor Recompression(b)	- o Capital Costs - Using Ion Exchange = \$4,590,000 - Using Carbon Adsorption = \$4,410,000 - o Annual Costs (steam stripping w/vapor compression) = \$564,000 (Years 1-38) \$291,000 (Years 39-64) - o Present Worth - Using Ion Exchange = \$22,298,000 - Using Carbon Adsorption = \$21,371,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

TABLE 7-20
(continued)
SUMMARY OF DETAILED ANALYSIS
GW-4: EXTRACTION, HEXAVALENT-CHROMIUM TREATMENT,
STEAM STRIPPING, AND DISCHARGE

CADILLAC AREA FEASIBILITY STUDY

NOTES (continued):

A = The long-term uncertainties associated with land disposal.

B = The goals, objectives, and requirements of the SWDA.

C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.

D = Short- and long-term potential for adverse health effects from human exposure.

E = Long-term monitoring costs.

F = The potential for future remedial action costs if the alternative remedial action in question were to fail.

G = The potential threat to public health and the environment associated with excavation, transportation, and redisposal, or containment.

the City of Cadillac due to observed groundwater contamination within the industrial park. The City of Cadillac updated the 1983 contingency plan in the fall of 1987 and submitted it to MDWS.

The updated plan included proposals to decrease pumping at the existing wellfield and identify existing industrial water supply sources that could be temporarily upgraded for use as a potable water supply. The updated plan also identified a new water supply location approximately 1 mile west of the existing wellfield on city-owned property earmarked for development as an industrial park. No extensive geologic/hydrologic studies have been performed on the proposed alternative water supply location. The proposed location is close to existing piping and connections and is expected to yield an adequate water supply; however, there is concern over the location in an industrial park. The City of Cadillac may consider reserving land for no development to prevent the recurrence of contaminants in the water supply.

The search for and ultimate development of an alternate water supply could be accomplished through a phased investigative approach. The investigation would include the following elements:

- o Perform a thorough search of existing geologic data and review topographic maps and aerial photographs to define existing water sheds and geologic soils conducive to producing high quantities of groundwater.
- o Review past and current land use maps to locate areas appropriate for extracting high quality groundwater for human consumption.
- o Prioritize a list of potentially useable sites based on the accumulated data.
- o Begin a field investigation to collect geologic and hydrogeologic data on the potentially most-suitable site, including deep borehole drilling and monitoring well installation to collect soil and groundwater samples to characterize the aquifer.
- o Conduct a series of aquifer stress tests to determine aquifer permeability, transmissivity, storativity, specific yield and capacity.
- o Design and construct the wellfield to specifications capable of producing an adequate supply of water.
- o Develop a strict wellfield preservation program outlining and restricting future uses of the site and adjacent areas to ensure integrity of the groundwater supply water quality.
- o Sample upgradient groundwater sources periodically for quality analysis.

7.9.2 Compliance with ARARs

ARARs associated with location and development of an alternative water supply are contained in the Michigan Safe Drinking Water Act and Rules and the authority vested in the MDPH. The MDPH is authorized to protect and regulate current and future drinking water supplies. Any investigation for alternate water supplies should be performed in conjunction with the MDPH and the town in question.

Appendix E identifies the action-specific ARARs associated with the alternate water supply. These ARARs are further detailed in Section 4.1 and Table 4-1.

7.9.3 Protection of Public Health and the Environment

Development of an alternate water supply for the City of Cadillac would reduce existing risks to public health posed by the domestic use and drinking of contaminated groundwater under the Cadillac Industrial Park. Future risks would be prevented by an adequate wellfield protection program for the new site. Remediation of existing contamination would still be required to provide full protection of public health and the environment. Existing contamination would continue to pose a threat through use of private domestic wells, unless the alternate water supply was combined with a treatment alternative, and the extraction system effectively removed the contaminated groundwater.

7.9.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

The alternate water supply would not reduce the toxicity, mobility, or volume of groundwater contaminants, unless it was combined with a treatment alternative.

7.9.5 Short-term Effectiveness

Protection from drinking contaminated groundwater would be achieved as soon as the new wellfield was brought on-line. Installation of a new wellfield would take approximately 13 to 14 months. No short-term community or worker health impacts would occur during installation or implementation of the alternate water supply.

7.9.6 Long-term Effectiveness

To maintain long-term effectiveness, a program designed to maintain the groundwater quality of the new water-supply site is required. Strict regulations should be placed on land use in the vicinity of the wellfield, and monitoring of any potential generators or users of hazardous substances within the identifiable water shed of the aquifer should be imposed. Monitoring wells placed upgradient of the supply well field should be installed and tested at least biannually.

The high level of effectiveness that can be achieved by developing a new municipal wellfield can be maintained indefinitely, provided adequate land use regulations and enforcement procedures are established. Residual risk would be associated with the existing groundwater contamination and the level and

success of treatment combined with implementation of an alternate water supply. Institutional controls should be implemented to prevent exposure through private wells.

7.9.7 Implementability

The technical feasibility of searching for and developing an alternate wellfield is proven and practical. Operational reliability of technologies should be at least that of the existing wellfield, with possible improvements by using technologies advanced since development of the original wellfield.

Installation of pumping wells is a proven and practical technology, and design and construction should not be complicated for the development of a new wellfield in the Cadillac area. Installation of the wells would take approximately four weeks per well or 24 weeks for six wells.

The limiting factor for new wellfield development will be land availability and acquisition in the Cadillac area. Once a suitable site has been selected, the actual acreage required will not be excessive.

7.9.8 Costs

In this section, capital and annual costs, as well as the present worth analysis, are presented for the alternate water supply. These costs are presented in Table 7-21. The detailed cost estimate outlining the individual costs of the components is presented in Appendix D. Because the alternate water supply would be an additional component of either the No-action Alternative or a treatment alternative, the costs presented herein should be added to the cost of whichever alternative with which the alternate water supply would be combined. The alternate water supply would be considered an additional cost if it was determined that the groundwater could not be remediated by extraction due to the presence of DNAPLs, or if it was determined that an alternate water supply should be used during remediation.

This cost estimate assumes the alternate water supply would be constructed and operated similarly to the existing municipal wellfield. Because it is uncertain where the alternate water supply would be located, it was also assumed that water mains would be approximately 12,000 feet long.

Capital Costs. Direct and indirect costs were estimated for this additional component. Direct costs include hydrogeology, engineering, and equipment costs (i.e., locating new wellfield, new wells, power supply, well building, and undeveloped design details). Assuming engineering costs would be 10 percent and contingency costs would be 25 percent, the total capital costs would be \$5,019,000.

Annual Costs. Annual O&M costs were not calculated because it is assumed that these costs would not differ from current annual costs of the existing system, and therefore would not be an added cost if the alternate water supply was implemented.

TABLE 7-21
AC-1 ALTERNATE WATER SUPPLY
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	TOTAL
CAPITAL COSTS	
ALTERNATE WATER SUPPLY	\$3,650,000
ENGINEERING @ 10%	365,000
CONTINGENCY @ 25%	1,004,000
TOTAL CAPITAL COSTS	\$5,019,000
ANNUAL COSTS	
ALTERNATIVE WATER SUPPLY	\$0
TOTAL ANNUAL COSTS	\$0
 PRESENT WORTH OF TOTAL ANNUAL COSTS FOR 64 YEARS @ 5%	 \$0
 PRESENT WORTH FOR 64 YEARS @ 5%	 \$5,019,000

Present Worth Analysis. Because there would not be any annual costs other than those already assumed by the existing water supply, the net present worth of the alternate water supply is \$5,019,000.

7.9.9 Summary

The detailed analysis of Alternative AC-1 is summarized in Table 7-22.

TABLE 7-22
SUMMARY OF DETAILED ANALYSIS
AC-1 ALTERNATE WATER SUPPLY
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o An investigation for an alternate water supply would be performed in compliance with Michigan Department of Health regulations. - o The new source of drinking water would meet drinking water standards.
2. Protection of Public Health and the Environment	o AC-1 would reduce existing risks to public health associated with domestic use and drinking of contaminated groundwater. However, existing contamination could still pose a threat through the use of domestic wells.
3. Reduction of Toxicity, Mobility, or Volume	o The alternate water supply would not reduce the toxicity, mobility, or volume of groundwater contaminants, unless it was combined with a treatment alternative.
4. Short-term Effectiveness	- o No short-term health impacts would occur during installation or implementation. - o Protection would be achieved as soon as the new well was brought on-line.
5. Long-term Effectiveness	- o Residual risk would be associated with the existing groundwater contamination, and the level and success of treatment combined with AC-1. - o Long-term effectiveness would be maintained by land-use controls and an aquifer protection plan.
6. Implementation	- o AC-1 is technically feasible. - o Technologies associated with AC-1 are readily available and reliable. - o Execution of AC-1 would take approximately 4 weeks per well (i.e., 24 weeks for 6 wells). - o The limiting factor of AC-1 would be land availability.

TABLE 7-22
(continued)
SUMMARY OF DETAILED ANALYSIS
AC-1 ALTERNATE WATER SUPPLY
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
7. Costs	- o Capital Costs = \$5,019,000 - o Annual Costs = \$0 - o Present Worth = \$5,019,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

A = The long-term uncertainties associated with land disposal.

B = The goals, objectives, and requirements of the SWDA.

C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.

D = Short- and long-term potential for adverse health effects from human exposure.

E = Long-term monitoring costs.

F = The potential for future remedial action costs if the alternative remedial action in question were to fail.

G = The potential threat to public health and the environment associated with excavation, transportation, and redispposal, or containment.

8.0 DETAILED ANALYSIS OF SOIL ALTERNATIVES

8.1 PRESENTATION OF INFORMATION

The soil alternatives that passed the initial screening and that will undergo the detailed analysis include SC-1, No-Action Alternative; SC-2, Surficial Capping and Subsurface Containment Wall; SC-6, Low-temperature Thermal Stripping; SC-7, In-situ Soil Flushing; SC-9, In-situ Vacuum Extraction; and SC-11, On-site Incineration. Each detailed analysis section compares the alternative to the nine criteria discussed in Section 6.0, excluding state and community acceptance (see Section 6.2). Also addressed is the alternatives' applicability to both the Kysor and Four Winns sites in general terms, noting any site-specific differences or requirements for the individual areas.

The total cost of each alternative assumes that the alternative will be implemented at both sites, and considers the possibility of using the same piece of treatment equipment at both sites when applicable. Site-specific costs are itemized and presented in Appendix D. The total cost of each alternative is presented in the detailed analysis for that alternative. As discussed in Section 6.3, the following types of costs are presented: capital costs, annual costs, and present worth.

8.2 ALTERNATIVE SC-1: NO-ACTION

The following subsections describe the detailed analysis of the No-action Alternative for soil remediation of the Kysor and Four Winns sites.

8.2.1 Description

Under this alternative, no remedial response activity would be performed at the sites to remediate on-site soils. Direct contact hazards could potentially be prevented by deed restrictions and by limiting physical access with a fence or other barrier. The following actions are proposed for this alternative:

- o establishment of institutional controls limiting the future use of the site (e.g., deed and land restrictions)
- o development of a plan outlining periodic site monitoring, inspections, and maintenance of the site
- o installation of a perimeter surface barrier (e.g., a fence) in the immediate area of soil contamination to limit site access
- o five-year site review

Institutional controls would be necessary legal instruments in a No-action Alternative to restrict future site use for the protection of public health. These controls would restrict site activities of the current site owners/operators, as well as alert future owners to potential site-related risks.

Existing groundwater monitoring wells in the vicinity of each site would be periodically sampled (annually), and a laboratory analysis of the groundwater would be conducted on a long-term basis (30 years). Annual site inspections would be performed. The site inspections would consist of visual observations of existing site conditions, any operations conducted on-site, and a check for unexpected activities.

As part of the No-action Alternative, a contingency plan would be developed to outline specific actions to be taken as site conditions warrant. The contingency plan would describe specific maintenance procedures and/or remedial actions to be implemented that would be considered capable of remediating site conditions identified during periodic site inspections.

The installation of a fence or physical barrier would limit access by zoning off affected soils and limiting the potential for contact with surface and subsurface soils via excavation activities. It is assumed the institutional controls would be effective indefinitely.

Because the No-action Alternative results in hazardous contaminants remaining at the site, a five-year review of the site would be required, as stipulated in Section 121 of SARA.

The Four Winns facility is currently fenced to restrict access. Additional fencing at Kysor and in the immediate area of the affected soils at Four Winns would need to be installed. Modifications to the existing deed, noting the presence of hazardous materials on the two properties, should be made. A monitoring program, inspection/maintenance program, and contingency plan would be implemented.

8.2.2 Compliance with ARARs

The No-action Alternative at the Cadillac site would not be consistent with the goal and intent of CERCLA because contaminated soil would be left in place. However, the No-action Alternative would incorporate a USEPA/state-approved post-closure groundwater monitoring program at the site (40 CFR 264, Subpart F). According to RCRA closure regulations (40 CFR 264.119), deeds to all contaminated properties must include a notice stating the type and location of hazardous waste contained on the property and identify all land use restrictions. All remedial activities must be conducted in accordance with OSHA Standards for Hazardous Waste Operations.

Appendix E identifies action-specific ARARs associated with this remedial alternative. These ARARs are further detailed in Section 5.1 and summarized in Table 5-1.

8.2.3 Protection of Public Health and the Environment

Under the No-action Alternative for soils at the Cadillac site, the VOC concentrations in the soils at both Kysor and Four Winns would remain above TCLs. Because there are no health-based chemical-specific ARARs for soils, and because surficial soil contamination was determined to be essentially nonexistent at Kysor and minimal at Four Winns, the soil TCLs were based solely on the

potential for soil contamination to continue to leach into the shallow aquifer. TCLs for soils at Kysor and Four Winns were established by using the groundwater TCLs for the same chemicals found in the soils, and back-calculating to a "safe" concentration in the soil using the USEPA OLM.

Kysor Site. At the Kysor site, the baseline RA identified 1,1,1-TCA, TCE, and xylene as chemicals of concern in on-site soils, and widespread VOC contamination in the shallow and intermediate aquifers, which could pose a significant risk to human receptors and the environment. Under the No-action Alternative, the established TCLs for the on-site soils would not be met.

The No-action Alternative would also not result in an improvement to the natural environment over current site conditions at Kysor described in the baseline RA. Direct contact exposures by terrestrial organisms (i.e., birds, small mammals, reptiles, amphibians, and terrestrial invertebrates) are not expected because the soil contamination at Kysor is located at least 6 feet below grade.

In summary, the No-action Alternative does little to prevent present or future exposure to chemicals in soil or groundwater; does not result in attainment of soil TCLs; and does not result in any improvement to the natural environment over current conditions. Therefore, this alternative does not adequately protect public health or the environment.

Four Winns Site. Under the No-action Alternative for soils at the Four Winns site, the soil TCL will not be met. The baseline RA identified methylene chloride in Four Winns soils at depths from zero to 23 feet, which could pose a risk to human and environmental receptors through direct contact and indirectly through groundwater contamination. Direct contact by humans and large mammals would be prevented by restricting access with fencing and land restrictions. However, smaller terrestrial animals and organisms could gain access to, and have direct contact with, contaminated soils, which may cause indirect food chain exposures to larger mammals via ingestion. This alternative would not result in an improvement to the natural environment over current conditions described in the baseline RA, and would continue to be a source of groundwater contamination.

8.2.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

This alternative would not reduce the volume, toxicity, or mobility of soil contamination at either the Kysor or Four Winns site. The on-site soils would continue to be a source of groundwater contamination, through continual leaching. The sandy soils present at both sites will do little to control infiltration of precipitation or groundwater movement.

8.2.5 Short-term Effectiveness

Under this alternative, no remedial activities would be performed at either site, except for site inspections and monitoring activities. Implementation of the No-action Alternative would not pose any short-term impacts to the community. However, the contaminated soils would continue to pose a threat to the community by the leaching of contaminants to groundwater.

During the proposed site inspections and monitoring activities, the necessary worker health and safety measures must be followed to prevent exposures while on-site.

8.2.6 Long-term Effectiveness

The No-action Alternative for Cadillac soils poses potential significant risks to public health through continued leaching of VOCs to the groundwater. Groundwater monitoring will be required to assess the groundwater contaminant plume, but would not remediate the source of contamination.

The installation of site fencing, coupled with institutional land controls, would result in a low reliability for preventing exposure to the public and the environment to on-site chemicals over the long term. This alternative will require annual inspection and maintenance of the perimeter fencing because it is susceptible to vandalism and the effects of the natural environment.

Kysor Site. The specific chemicals of concern at the Kysor site are TCE, toluene, 1,1,1-TCA, and xylene. Kysor soils were identified in the RI as a source of VOC contamination to the groundwater downgradient of the Kysor site. Therefore, the No-action Alternative is likely to pose significant risks to public health through VOC contamination in the groundwater. The current impact to terrestrial flora and fauna is nonexistent because no surficial contamination of the soils exists.

Four Winns Site. Although the sampling data for Four Winns soils is not sufficient to assess fate and transport, it is assumed (based on observed concentrations in groundwater, and MDNR reports of spills and leaks) that the concentration of VOCs in the soils warrants remediation. The No-action Alternative is likely to pose significant risks to public health through continued leaching of VOC contamination to the groundwater. If the physical barriers to contaminated areas are destroyed, there would be potential for future human exposure to VOCs through dermal contact, which could pose risks to health. The Four Winns site, which has some surficial contamination, would be of more concern than the Kysor site where contamination is subsurface.

8.2.7 Implementability

There is little difficulty in implementing and monitoring this alternative. Wells to monitor migration of chemicals in groundwater have already been installed at the site. Well drilling contractors and supplies are available locally, if any wells require replacement in the future.

Implementation of this alternative would not change the physical or chemical nature of the contaminants, and therefore would not limit future response options. The regulatory requirements for deed restrictions could easily be addressed.

Selection of this alternative would require implementation of a long-term monitoring program, including the quality and movement of the groundwater. Site inspections should also be conducted to observe site changes and identify

problems. Degradation of the perimeter fencing and missing signs should be repaired and replaced as soon as possible.

8.2.8 Costs

In this subsection, capital and annual costs, as well as the present worth analysis, are presented for the No-action Alternative. These costs, presented in Table 8-1, are accurate to -30 to +50 percent. As mentioned previously, the costs assume that the alternative will be implemented at both sites. Site-specific costs are presented in Appendix D.

Capital Costs. Capital costs associated with the No-action Alternative would include costs for installing a chainlink fence and gate and for preparing a contingency plan. Total capital costs for implementing no-action at both the Kysor and Four Winns sites are \$49,000. Site-specific costs are presented in Appendix D.

Annual Costs. Annual costs are O&M costs associated with post-construction activities, necessary to provide continued effectiveness of the remedial action. For the No-action Alternative, annual O&M costs would include costs for conducting the five-year site review and annual inspections and for maintaining the fencing. Costs associated with groundwater monitoring are assumed to be included in the groundwater alternatives. These costs are detailed in Appendix D. Annual costs associated with the No-action Alternative are \$18,000 per year.

Present Worth Analysis. The present worth analysis evaluates costs that occur over different time periods by discounting future costs to the present. The net present worth (in 1988 dollars) of implementing the No-action Alternative at both sites is \$326,000. This value was calculated assuming a 5-percent interest rate and a 30-year performance period.

8.2.9 Summary

The detailed analysis of SC-1 is summarized in Table 8-2.

8.3 ALTERNATIVE SC-2: SURFICIAL CAPPING AND SUBSURFACE CONTAINMENT WALL

The following subsections describe the detailed analysis of the Surficial Capping and Subsurface Containment Wall Alternative for soil remediation of the Kysor and Four Winns sites.

8.3.1 Description

This alternative consists of encapsulating those portions of the site that contain soils exhibiting chemical concentrations exceeding the TCLs, and utilizing a combination impermeable cap and subsurface wall system. The reduction of infiltration and redirection of groundwater flow can be effective in reducing the vertical and horizontal migration of hazardous chemicals.

TABLE 8-1
SC-1 NO-ACTION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	TOTAL
CAPITAL COSTS	
FIVE YEAR SITE REVIEW	\$
INSPECTION	
FENCING	29,00
CONTINGENCY PLAN	20,00
TOTAL CAPITAL COSTS	----- \$49,00
ANNUAL COSTS	
FIVE YEAR SITE REVIEW	\$3,00
INSPECTION	13,00
FENCING	• 2,00
CONTINGENCY PLAN	
TOTAL ANNUAL COSTS	----- \$18,00
PRESENT WORTH FOR 30 YEARS @ 5%	\$326,00

TABLE 8-2
SUMMARY OF DETAILED ANALYSIS
SC-1: NO-ACTION

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o The No-action Alternative would not be consistent with the goal and intent of CERCLA. - o RCRA groundwater monitoring requirements would be met. - o RCRA closure regulations requiring deed notifications would be met.
2. Protection of Public Health and the Environment	- o The No-action Alternative would not result in an improvement to the natural environment over the current site conditions described in the baseline risk assessment. - o Concentrations of VOCs in the soils at Kysor and Four Winns would remain above the TCLs. - o The No-action Alternative would not prevent the VOCs in the soils at Kysor and Four Winns from continued leaching to groundwater. - o Risks to humans through direct contact with Four Winns soils would be reduced by restricting access via fencing and land restrictions. - o Smaller terrestrial animals and organisms could gain access to and come in direct contact with Four Winns' soils.
3. Reduction of Toxicity, Mobility, or Volume	o No reduction in toxicity, mobility, or volume.
4. Short-term Effectiveness	o There would be no significant risks to community, workers, or the environment associated with implementation.
5. Long-term Effectiveness	- o Long-term risks to public health through continued leaching of VOCs to the groundwater. - o Long-term groundwater monitoring and annual inspection and maintenance of site fencing would be required.

TABLE 8-2
(continued)
SUMMARY OF DETAILED ANALYSIS
SC-1: NO-ACTION

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
	o Site fencing and institutional land controls would not reliably prevent exposures to the public and environment over the long term.
6. Implementation	- o Groundwater monitoring equipment is readily available. - o No construction activities would be required for groundwater monitoring because the wells have been installed. - o Perimeter fencing can be readily installed. - o Site inspections can be readily performed.
7. Costs	- o Capital Costs = \$49,000 - o Annual Costs = \$18,000 - o Present Worth = \$326,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

A = The long-term uncertainties associated with land disposal.

B = The goals, objectives, and requirements of the SWDA.

C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.

D = Short- and long-term potential for adverse health effects from human exposure.

E = Long-term monitoring costs.

F = The potential for future remedial action costs if the alternative remedial action in question were to fail.

G = The potential threat to public health and the environment associated with excavation, transportation, and redispisal, or containment.

Capping is a process used to cover waste materials to prevent contact with the land surface and groundwater. An impermeable cover coupled with an efficient site drainage system can eliminate infiltration caused from precipitation and site run-on. The design of the cap and selection of capping materials (e.g., synthetic membranes and/or natural materials) are based on the wastes being covered, availability of materials, local climate and hydrogeology, and the intended future use of the site. This technology is simple in nature and widely accepted as an effective method to control infiltration, thereby reducing leaching potential.

Subsurface containment walls are low-permeability barriers used to contain, capture, or redirect groundwater flow. Types of containment walls vary from the common slurry wall to sheet piling; site characteristics dictate the specific containment wall. Slurry walls are constructed in an excavated trench, while sheet piling is driven into place using a pile driver. These walls must be tied into a low-permeable underlying layer (e.g., clay or bed-rock) to achieve containment. Proper placement and construction reduces the amount of uncontaminated groundwater entering the site from the upgradient direction, thereby reducing the amount of leachate generated and controlling off-site migration.

Surficial capping and subsurface containment walls are usually used together to prevent or significantly reduce plume development. Groundwater collection wells placed through the cap can be used to remove contaminated groundwater and relieve hydrostatic pressure. Monitoring wells placed on the exterior of the cell can be used to monitor the integrity of the cell. This alternative is depicted in Figure 8-1.

Kysor Site. A circumferential slurry wall with a surficial cap installation would isolate the contaminated soils. This would require 650 linear feet of slurry wall tied into the clay layer 90 feet below grade. Infiltration would be controlled by grading the site and by using approximately 20,000 square feet of a multilayer cap to cover the area encircled by the barrier wall. The cap system would consist of 24 inches of low-permeability (i.e., 1.0×10^{-7}) clay, followed by a synthetic liner. The liner would be covered with a minimum 12-inch permeable drainage layer and 24 inches of soil, which would support a topsoiled vegetative layer to reduce the potential for soil erosion.

Four Winns Site. A circumferential slurry wall with a surficial cap at the Four Winns site would require 680 linear feet of slurry wall tied into the clay layer 65 feet below grade. Infiltration would be controlled by grading the site and by using 20,000 square feet of a multilayer cap to cover the area encircled by the barrier wall.

8.3.2 Compliance with ARARs

ARARs associated with capping and a subsurface containment wall include RCRA closure and post-closure regulations (40 CFR 264, Subpart G), RCRA landfill closure requirements (40 CFR, Subpart N), and Michigan Hazardous Waste Management Rules (R 299, Rule 619). The cap design appears to conform to the regulations. Post-closure groundwater monitoring, deed notifications, and land use

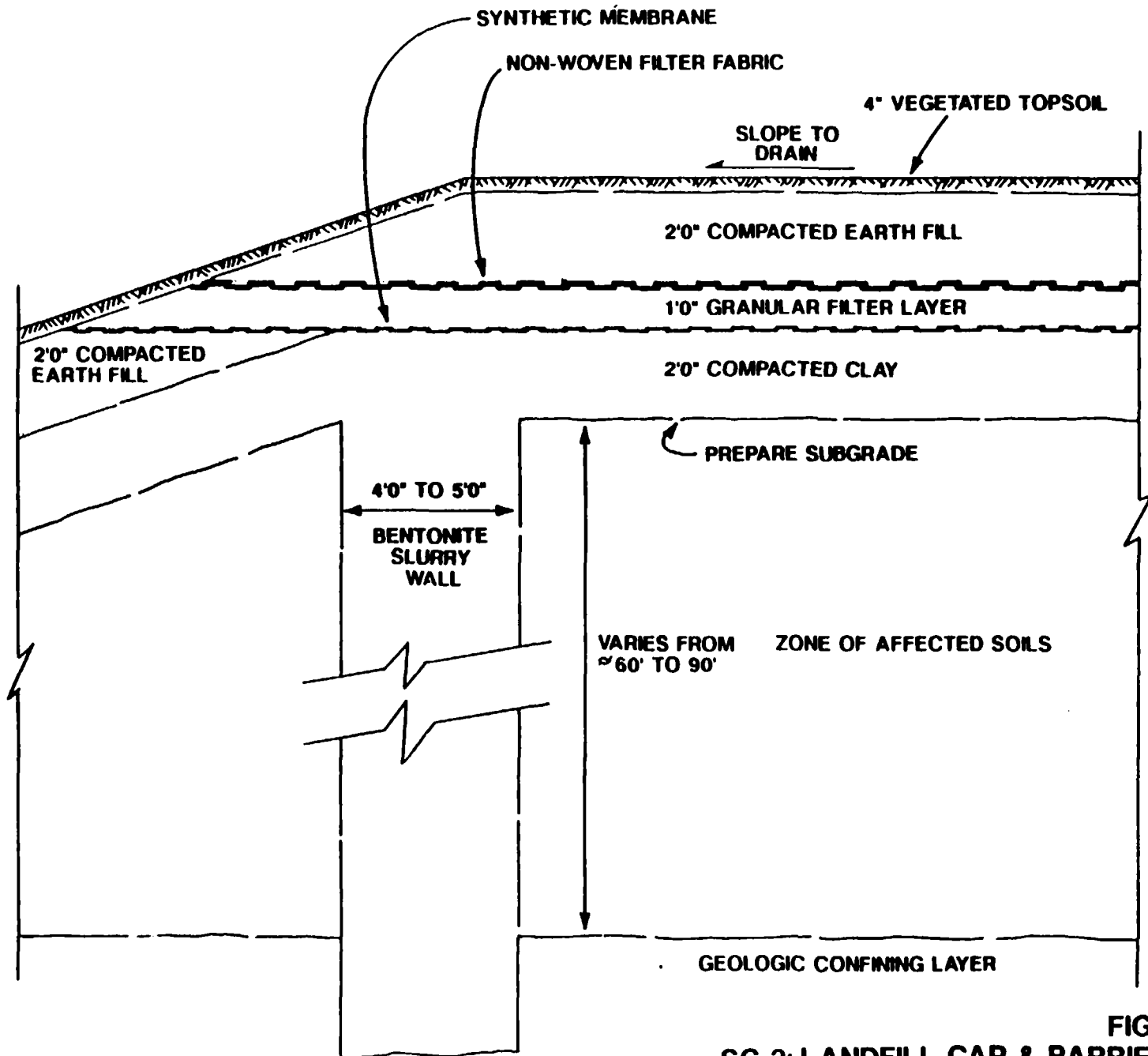


FIGURE 8-1
SC-2: LANDFILL CAP & BARRIER WALL
CADILLAC AREA FEASIBILITY STUDY
CADILLAC, MICHIGAN

restrictions will be implemented in compliance with RCRA and Michigan regulations (40 CFR 264.100 and 119, and R 299, Rule 619).

The RCRA and Michigan location requirements (40 CFR 264.18 and R 299, Rule 603) and the RCRA and Michigan facility design and operating standards (40 CFR 264 and R 299, Rule 604) apply to the temporary location of the treatment unit, as well as any final cover systems. In general, the unit may not be located within 200 feet of a fault that was displaced in Holocene time, and must be built and maintained to withstand discharge from a 24-hour, 25-year storm and control water volume from a 24-hour, 100-year storm.

There are no ARARs specifically governing the design or installation of the subsurface containment wall. Excavation of the trench will occur outside the zone of contamination; therefore, no contaminated soil will be generated (i.e., moved and brought to the surface). Any soil generated during excavation will be used on-site as clean fill and/or in the slurry mixture.

All on-site activities must be conducted in compliance with OSHA Standards for Hazardous Waste Operations. Groundwater monitoring conducted under this alternative will be consistent with RCRA Corrective Action Monitoring (40 CFR 264.100).

Appendix E identifies action-specific ARARs associated with this remedial alternative. These ARARs are further detailed in Section 5.1 and summarized in Table 5-1.

8.3.3 Protection of Public Health and the Environment

The installation of a low-permeability subsurface containment wall and surficial cap system is designed to prevent infiltration of surface water and the subsequent leaching of soil contaminants into the surrounding groundwater. Therefore, this alternative would reduce the risk to public health posed by the leaching of contaminants in the soils to groundwater. However, Alternative SC-2 would not result in the attainment of soil TCLs, and does not provide a permanent remedy or improve the natural environment.

8.3.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

A soil/bentonite slurry wall and multilayer cap system could effectively isolate and reduce the mobility of the contaminants at both sites. However, this technique does nothing to treat or destroy the contaminants, nor does it reduce the volume of waste materials. It is only useful to temporarily control the source of the contaminants and restrict their mobility.

Because this containment alternative is used to isolate the contaminants without altering the physical or chemical characteristics, any of the permanent treatment technologies under consideration could be implemented in the future. However, because site preparation for this alternative requires additional fill material added at both sites, an increase of the total material to be handled during a final remedy would be greater.

8.3.5 Short-term Effectiveness

During site preparation and construction of the subsurface containment wall and placement of the cap, on-site workers could be exposed to dermal contact with contaminated soil or inhalation of VOCs. Appropriate worker safety and health practices would need to be strictly enforced.

The implementation of Alternative SC-2 would also generate fugitive dust and VOC emissions from the on-site contamination. Because on-site workers could inhale air emissions, dust control measures should be employed and the appropriate level of personal protection utilized by workers during all construction activities. Likewise, particulate and VOC air monitoring should be conducted concurrently with construction activities to ensure that relevant threshold limit values (TLVs), NAAQS for particulates, or Michigan Air Pollution Control Commission (MAPCC) regulations are not exceeded. Because procurement and construction of this alternative is estimated at 12 months, protection provided by this alternative would be achieved in one year.

The implementation of this alternative at the Four Winns site would result in fugitive dust and methylene chloride air emissions. On-site workers may inhale air emissions and have dermal contact during containment wall installation and site surface preparation. Therefore, dust control measures should be used during construction activities. The appropriate level of personal protection for on-site workers should be predetermined and utilized during construction activities. Likewise, air monitoring for methylene chloride and particulates should also be conducted.

8.3.6 Long-term Effectiveness

Surficial capping and subsurface containment walls provide engineering controls that reduce the potential risk for direct contact with soil and fugitive dust. The containment system isolates the contaminated soil, and minimizes the amount of contaminants leaching to the groundwater. Over a number of years, additional degradation of groundwater is possible and periodic groundwater monitoring would be necessary to track movement of the contaminant plume.

Because the contamination is not actually removed from the site, there is potential for risks if the containment structures fail to operate as designed. The containment system alternative could fail if the cap or containment wall loses its integrity from weathering, breaching by vegetation or animal borrowing, or soil erosion. The degree of failure to which this alternative could affect public health and the environment depends on the extent of damage sustained by the cap and subsurface containment wall. The possibility for the failure of the containment system can be minimized through perpetual monitoring and periodic maintenance to ensure that failures will not go unnoticed and unreported. The surficial cap and subsurface containment wall would probably need replacement following the anticipated design life of 20 to 30 years.

If properly designed and installed, a surficial cap and containment wall system is recognized as a technically feasible method for isolating contaminated soils and redirecting groundwater. The materials used in construction of the barrier system must be compatible with the site contaminants and hydrogeology. Strong

acids or organics can reduce the life of a soil-bentonite slurry wall. Sheet piling has interlocking edges to facilitate installation; however, the edges allow the passage of water, making the wall ineffective below the water table. Groundwater quality should be monitored after installation to assess integrity of the barrier. Differential hydrostatic pressures should be periodically measured across the wall. A variation in the hydrostatic pressure on opposing sides of the containment wall could cause the wall to breach. Frequent inspections of the cap system are also necessary. A regular inspection program of the system should be initiated and maintained during the life of the cell, including maintenance of the cap (e.g., mowing and seeding), as well as inspecting for signs of erosion or subsidence failure. Every five years, an in-depth investigation is required to assure the integrity of the contained cell, as required by SARA Section 121.

8.3.7 Implementability

The surficial capping and subsurface containment wall alternative would be easily implementable at either the Kysor or Four Winns site. Both sites are relatively flat and accessible to construction equipment. Large open areas exist for mixing backfill materials and stockpiling solids removed in trenching operations. A soil/bentonite slurry wall is inexpensive to install, has a low permeability rate, and is considered compatible with the wastes at both sites.

To be effective as a barrier, the subsurface containment wall must extend into an impermeable underlying geologic layer. The deeper the impermeable geologic layer, the larger the equipment needed for excavation. A shallow wall is relatively inexpensive to construct. However, the confining layers below these sites are from 60 to 90 feet below grade, which would require considerably more expensive specialty equipment.

When excavating a deep trench, the slurry is introduced during excavation to keep the side walls from collapsing. This technique is difficult when trying to reliably excavate into an underlying layer 90 feet below grade. The saturated soils removed from the trench during installation of the wall are potentially contaminated. Depending on the nature of the contaminants, these soils could present a significant threat to site workers and adjacent receptors; special handling of the contaminated soils may be required. The Kysor and Four Winns sites are free of subsurface obstructions, structures, and other major obstacles, which will facilitate installation of the cap system and final grading of the site.

It would take approximately three months to select a contractor and mobilize, and six to 12 months to install a barrier wall and surficial cap at either site. Additional time may be required for wet or cold weather. This alternative is appealing from a construction standpoint as it contains the waste source while handling a minimum amount of contaminated material.

The materials needed for construction are readily available. Sources for clay and fill exist in the nearby community. Synthetic liners can be purchased and delivered promptly. Several contractors experienced in the installation of slurry walls and multilayer caps provide service in the area.

8.3.8 Costs

In this subsection, capital and annual costs are presented for the Surficial Capping and Subsurface Containment Wall Alternative. In addition, the net present worth of this alternative is presented, assuming it would be implemented at both sites. These costs are listed in Table 8-3; site-specific costs are presented in Appendix D. The capital and annual costs and the net present worth were derived using the assumptions presented in Section 6.3.

Capital Costs. 10 percent, and contingency costs were assumed to be 25 percent. The capital cost of the Surficial Cap and Subsurface Wall Alternative is \$2,024,000.

O&M Costs. Annual O&M costs for the Surficial Capping and Subsurface Containment Wall Alternative are \$21,000.

Present Worth Analysis. Based on a 30-year performance period and a 5-percent interest rate, the net present worth of the Surficial Capping and Subsurface Containment Wall Alternative is \$2,347,000.

8.3.9 Summary

The detailed analysis of Alternative SC-2 is summarized in Table 8-4.

8.4 DETAILED ANALYSIS OF ALTERNATIVE SC-6: EXCAVATION AND LOW-TEMPERATURE THERMAL STRIPPING

8.4.1 Description

This alternative consists of the following remedial technologies to address the identified response activities for soil remediation of the Kysor and Four Winns sites. This alternative is depicted in Figure 8-2.

Excavation of Contaminated Soils. All soils contaminated with VOCs above TCLs will be excavated and staged for on-site low-temperature thermal stripping. Both the Kysor and Four Winns sites have VOC-contaminated soil zones down to the water table.

Low-temperature Thermal Stripping. All excavated material will be treated on-site to remove VOC contamination by thermal stripping. A mobile low-temperature thermal stripping unit will be temporarily installed at each site to thermally treat the contaminated soils, and the treated soil will be back-filled and disposed of on-site. The desorbed gaseous contaminants will be treated either captured or thermally destroyed in an afterburner emission control unit.

The low-temperature thermal stripping alternative is considered a treatment alternative for either the Kysor site or the Four Winns Site, or both.

Low-temperature thermal stripping is a treatment method that uses temperatures typically 150 to 700°F to remove adsorbed contaminants from the soils. The

TABLE 8-3
SC-2 SURFICIAL CAPPING & SUBSURFACE WALL
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	TOTAL
CAPITAL COSTS	
FIVE YEAR SITE REVIEW	\$
INSPECTION	
SURFACE CAP & SUBSURFACE WALL	1,472,00
ENGINEERING @ 10%	147,00
CONTINGENCY @ 25%	405,00
TOTAL CAPITAL COSTS	<u>\$2,024,00</u>
ANNUAL COSTS	
FIVE YEAR SITE REVIEW	\$3,00
INSPECTION	13,00
SURFACE CAP & SUBSURFACE WALL	5,00
TOTAL ANNUAL COSTS	<u>\$21,00</u>
PRESENT WORTH FOR 30 YEARS @ 5%	\$2,347,00

TABLE 8-4
SUMMARY OF DETAILED ANALYSIS
SC-2: SURFICIAL CAPPING, SUBSURFACE CONTAINMENT WALL
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o RCRA closure and post-closure regulations and landfill closure requirements should be met. - o On-site activities would be conducted in compliance with OSHA Standards for Hazardous Waste Operations. - o Groundwater monitoring would be consistent with RCRA Corrective Action Monitoring.
2. Protection of Public Health and the Environment	- o Would not attain the target levels for soils at Kysor and Four Winns. - o Would not provide a permanent remedy. - o Would reduce risk to public health and environment due to leaching of VOCs in soil to the groundwater.
3. Reduction of Toxicity, Mobility, or Volume	o Would reduce mobility of VOCs in soils at Kysor and Four Winns.
4. Short-term Effectiveness	o On-site workers could be exposed to fugitive dust, air emissions, contaminated soils, and groundwater during site preparation and construction at Kysor and Four Winns.
5. Long-term Effectiveness	- o The containment system would minimize the amount of contaminants leaching to the groundwater; however, additional degradation of groundwater is possible. - o The surficial cap and subsurface containment wall may require replacement after 20 to 30 years.
6. Implementability	- o SC-2 is technically feasible for the Cadillac site. - o A surficial cap and subsurface containment wall would be readily implementable.

TABLE 8-4
(continued)
SUMMARY OF DETAILED ANALYSIS
SC-2: SURFICIAL CAPPING, SUBSURFACE CONTAINMENT WALL
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
	- o Equipment is readily available. - o Contractor selection, mobilization, and installation of SC-2 would take approximately 9 to 15 months.
7. Costs	- o Capital Costs = \$2,024,000 - o Annual Costs = \$21,000 - o Present Worth = \$2,347,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

A = The long-term uncertainties associated with land disposal.

B = The goals, objectives, and requirements of the SWDA.

C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.

D = Short- and long-term potential for adverse health effects from human exposure.

E = Long-term monitoring costs.

F = The potential for future remedial action costs if the alternative remedial action in question were to fail.

G = The potential threat to public health and the environment associated with excavation, transportation, and redispisal, or containment.

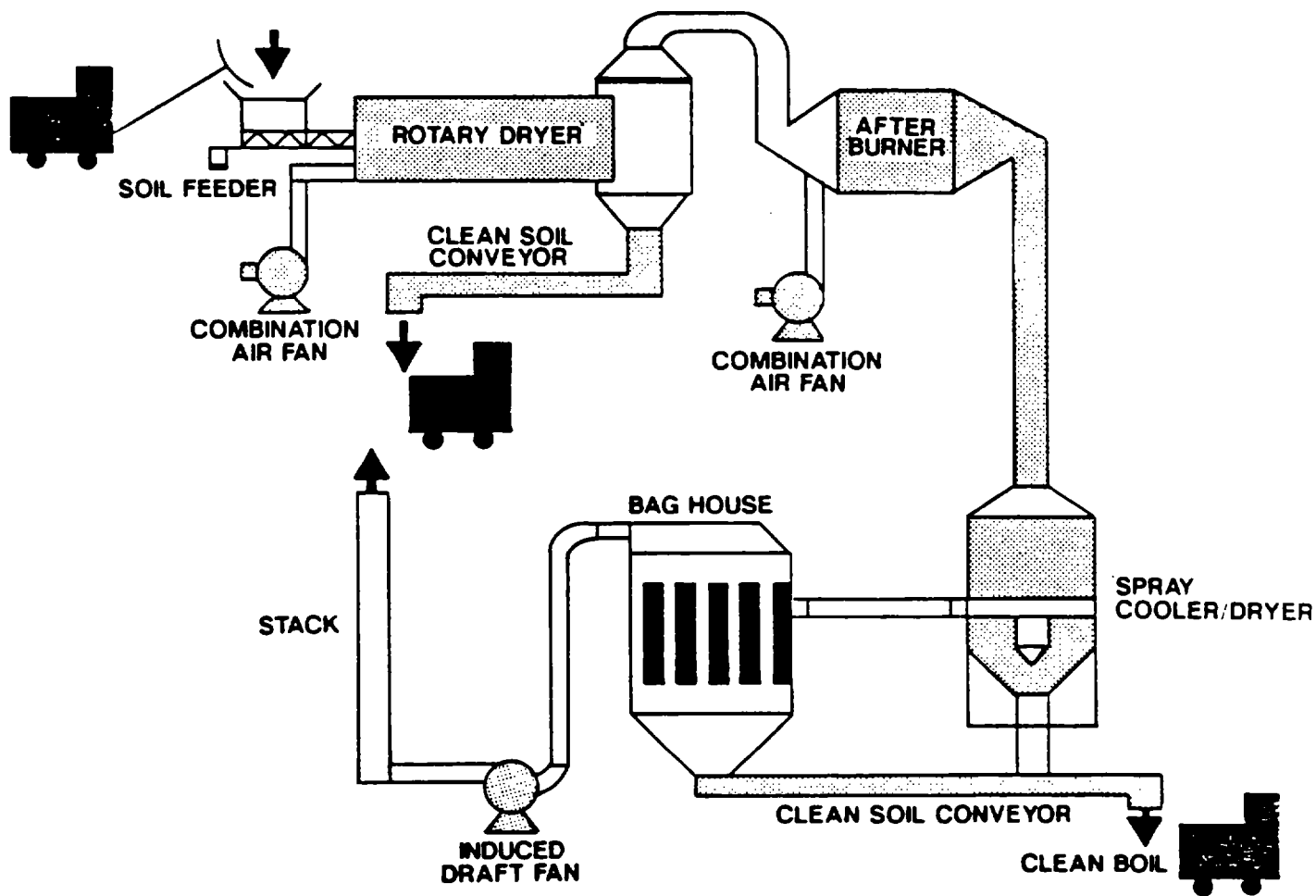


FIGURE 8-2
SC-6: LOW TEMPERATURE STRIPPING
CADILLAC AREA FEASIBILITY STUDY
CADILLAC, MICHIGAN
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thermal stripping unit consists of a process feed system, large rotating drum (kiln), and a temperature-controlled burner system. The thermal stripping unit is operated under controlled conditions, and will generate emissions such as particulates, water vapor, VOCs, and products of incomplete combustion. To prevent these emissions from being released into the environment, gaseous emission control equipment will be required. Process emissions are typically controlled using an afterburner to thermally destroy the contaminants, or vapor-phase carbon adsorption. If vapor-phase activated charcoal canisters are employed in the emission control, the canisters will be transported and treated off-site.

A typical on-site low-temperature thermal-stripping unit operates most efficiently on a 24-hour-per-day continuous schedule, and has feed rates ranging from 25 to 200 tons per day. Therefore, a limited supply of contaminated material may require stockpiling during low-temperature thermal-stripping operating hours. However, stockpiling of waste material would be kept to a minimum.

Kysor Site. The field implementation of the low-temperature thermal-stripping alternative at the Kysor site would include excavation of contaminated soils and stockpiling, as well as construction activities for siting and operating the on-site low-temperature thermal treatment unit.

The likely place to install the low-temperature thermal-stripping system would be northwest of the Kysor building and south of the railroad right-of-way, where a gravel access road from Leeson Avenue currently exists. Topography of this industrial area is relatively level and should provide easy installation of the concrete support pads for the thermal treatment facility. The site work areas would require improvements to support the thermal treatment facility and ancillary activities. Outdoor illumination for multiple-shift operation should also be considered.

Four Winns Site. The field implementation of Alternative SC-6 at the Four Winns site would include excavation of contaminated soils, stockpiling contaminated soils, and construction, siting, and operation of the on-site thermal treatment unit.

The southeastern corner of the Four Winns site is a likely location to install the low-temperature thermal-stripping system. Topography of this industrial area is relatively level and should provide easy installation of the concrete support pads for the thermal treatment facility. The site work areas would require improvements to support the thermal treatment facility and ancillary activities. Outdoor illumination for multiple-shift operation should also be considered.

8.4.2 Compliance with ARARs

Several federal and state ARARs are associated with thermal stripping. Under Environmental Performance Standards for Miscellaneous Units (40 CFR 264, Subpart X), the alternative must be protective of public health and the environment, and must prevent releases and migration of contaminants to environmental media.

The RCRA and Michigan location requirements are the same as those identified for SC-2 (refer to Section 8.3.2).

Relevant CAA regulations include a standard limiting the emission of particulate matter (40 CFR 129) (see Table 5-1, CAA NAAQS PM10 Standard). This standard must not be exceeded during excavation, construction, and operation of the treatment system. For estimating fugitive dust, the guidelines in the EPA Superfund Exposure Evaluation Manual should be followed. The state should be consulted as to the compliance with any fugitive dust emission regulation. Under MAPCC regulations, the low-temperature thermal-stripping system is considered a source of air contamination and, therefore, a permit will be required for both construction and operation of the system, according to R 336, Rule 201. The Michigan regulations also limit fugitive dust and establish maximum allowable emission rates from new VOC sources, based on BACT (R 336, Rules 701 and 702). BACT is determined on a case-by-case basis through information submitted in the permit application. Although remedial actions carried out entirely on-site (i.e., on a CERCLA site such as the Kysor site) are not required to obtain permits or comply with administrative requirements (e.g., obtaining permits), application for an air emissions permit in the case of the Cadillac site is considered necessary to comply with the substantive requirements of the Michigan Air Pollution Control Act Rules. In addition, Michigan regulations prohibit the emissions of air contaminants in quantities that will cause "injurious effects to human health and welfare, and the environment" (R 336, Rule 901). Information submitted during the permit application is reviewed to determine where injurious effects may occur.

If residuals and processed soil are considered hazardous, they must be handled, transported, and disposed of as RCRA hazardous wastes according to RCRA regulations (40 CFR 262 through 264) and Michigan Hazardous Waste Regulations (R 299, Rule 613), including transportation via a Michigan-licensed hauler and vehicle and disposal in a fully permitted RCRA-compliant facility.

If contaminated soil at the Four Winns and/or Kysor site is determined by USEPA and MDNR to be a RCRA-listed solvent hazardous waste, the RCRA Land Ban (40 CFR 268) becomes applicable. The Land Ban prohibits the land disposal of hazardous wastes and residuals not meeting specified treatment standards. Certain solvent-contaminated wastes (i.e., RCRA F001-F005 spent solvents) were banned from land disposal on November 8, 1986, if treatment standards (specified in 40 CFR 268 Subpart D, Table CCWE) were not achieved. Solvent-contaminated soil and debris resulting from a CERCLA response action was granted an extension until November 8, 1990. USEPA and MDNR will need to determine whether the contaminated soil at the Kysor and Four Winns sites is considered a RCRA-listed solvent waste (e.g., the result of a RCRA-listed solvent waste spill). If the soil is not considered a RCRA waste, then the Land Ban is neither applicable nor relevant and appropriate. The proposed treatment will most likely meet Land Ban treatment and disposal requirements.

Because contaminated soil would be excavated, the site must be closed in compliance with RCRA and Michigan closure regulations (40 CFR 264, Subpart G, and R 299, Rule 613). Deed notifications and land use restrictions must be implemented, as well as long-term post-closure care and maintenance of the property.

Groundwater monitoring conducted under this alternative will be consistent with RCRA Corrective Action Monitoring (40 CFR 264.100). All remedial activities must be conducted in accordance with OSHA Standards for Hazardous Waste Operations.

Appendix E identifies action-specific ARARs associated with this remedial alternative. These ARARs are further detailed in Section 5.1 and summarized in Table 5-1.

8.4.3 Protection of Public Health and the Environment

The excavation and treatment of on-site soils using low-temperature thermal stripping would effectively reduce public health and environmental risks associated with contaminated soils; remove the source of groundwater contamination; improve the natural environment; and provide a permanent remedy for the site. The implementation of this alternative would result in the attainment of soil TCLs.

The low-temperature thermal stripper produces air emissions that would present a potential route of human exposure (through inhalation by workers on-site and other downwind receptors). The gas-phase activated charcoal canisters or thermal afterburner used for emission control are expected to control VOC emissions so that the air quality standards for particulates, VOCs, nitrogen dioxide, and carbon monoxide would be met. A pilot test of the system would provide evidence of the system's efficiency. During pilot testing, there is potential for short-term releases that do not meet the required air standards; however, due to the small scale of a pilot test, this would not be expected to pose significant risks. In addition, periodic monitoring of air emissions during the treatment process would significantly reduce the probability of emission control equipment failure.

Due to more extensive and complex waste-handling activities, potential for adverse effects to on-site workers and the general public would be considered greater with this alternative than for Alternatives SC-1 and SC-2. Certain activities associated with this alternative (i.e., excavation and temporary stockpiling of contaminated soils; management of emission control streams; and stockpiling and disposal of treated soil) increase the likelihood of health risks to on-site workers via direct contact and inhalation of dusts, vapors, and gases. Therefore, workers should use the appropriate level of protection during construction, excavation, and treatment process activities. Risk to the environment is also potentially increased by erosion of contaminated soils and fugitive dust emissions generated during excavation and treatment activities. Proper implementation of erosion and sediment control measures, and staging the work effort to expose excavated soils and wastes to the ambient environment for the shortest time possible, will reduce the likelihood of adverse environmental effects. Dust control measures should be used during site remediation, and ambient air monitoring for particulates and organics should be conducted to evaluate compliance with NAAQS and MAPCC regulations, with appropriate corrective actions taken, as necessary.

In summary, Alternative SC-6 would reduce the baseline public health risks associated with the contaminated soils and potential for continued leaching of

contamination to the groundwater; improve the natural environment; and provide a permanent remedy for the site. However, because of the extent of waste-handling activities, the potential for adverse environmental effects is greater for this alternative than for Alternatives SC-1 and SC-2. Therefore, careful management of remedial activities and emission abatement measures will reduce the probability of adverse effects.

8.4.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

The toxicity, mobility, and volume of contaminated materials would be reduced by thermal stripping. Low-temperature thermal stripping is an effective technology for treating contaminated soils from the unsaturated zone. It is recognized as a reliable method for the removal of VOCs and semivolatile organic compounds (SVOCs). VOCs such as those at the Kysor and Four Winns sites can be effectively removed using this process. The sandy soils present are suitable for treatment using this technology and, following treatment, would be disposed of as a RCRA waste at an off-site RCRA facility, or delisted and backfilled on-site. The successful implementation of this alternative would be considered permanent at either site. The treated soils, if nonhazardous and delisted, would be available as fill material. Properly implemented, this alternative will eliminate the need for future remedial actions.

8.4.5 Short-term Effectiveness

The excavation of soils would pose potential risks to workers from exposure to VOCs by dermal contact or inhalation of dusts or vapors. Therefore, protective worker practices would have to be followed. Air monitoring would have to be performed to ensure appropriate respiratory protection and compliance with air standards. During the moving of soils, fugitive dust emissions containing VOC contamination are possible and control measures may be needed to reduce exposures.

During the thermal-stripping process itself, there would be greater potential for community exposures due to releases of the gaseous products of incomplete combustion and VOC emissions. The off-gas treatment equipment (e.g., activated charcoal canisters and thermal afterburners) would have to be sufficient to ensure that appropriate air standards for VOC air pollutants are not exceeded.

Stockpiling of contaminated soils is required to allow 24-hours-per-day continuous operation. This stockpiling poses some potential risks to workers, but the appropriate RCRA requirements for waste piles would have to be met. Furthermore, the residue streams would require hazardous waste management to prevent any unnecessary exposures to workers or the community.

The basic calculation for estimating the time required to achieve the on-site remedial response objectives associated with this alternative is governed by the treatment unit feed rate, hours of operation, and amount of unit downtime.

Kysor Site. The time anticipated to achieve the soil remedial response objectives at the Kysor site would be approximately 14 months. It is estimated that five months would be necessary to treat the 13,200 cy of excavated contaminated

soil, once the low-temperature thermal-stripping unit is placed in full operation. This estimate conservatively assumes a feed rate of 10 cy per hour and the treatment unit operating 24 hours per day, with three days of treatment per week. This estimate is intended for use in evaluating remedial alternatives, and will be further refined during the design phase of this project.

Four Winns Site. The time necessary for completing the preliminary activities prior to full-scale operation at the Four Winns site is estimated at eight to 10 months, the same for the Kysor site. The total time required to achieve the soil remedial response objectives is estimated at 15 months, after the treatment unit begins full operation. This estimate assumes 16,100 cy of contaminated soils, a feed rate of 10 cy per hour, and the low-temperature thermal stripper operating 24 hours per day over a three-day work week. Again, the time estimate is intended for use in evaluating remedial alternatives, and will be more accurately determined during the conceptual design phase of this project.

8.4.6 Long-term Effectiveness

This soil remediation alternative permanently reduces the VOC contamination of soils through excavation and thermal stripping of the contaminants. Therefore, the source of contamination from the soils at the Kysor and Four Winns sites through potential leaching would be eliminated.

Because the TCLs for soils would be met, residual risk to public health due to inadvertent consumption of drinking water contaminated by leachate from these soils would not be significant. Because the soil TCLs are derived from health- and risk-based chemical-specific ARARs for groundwater, the protection of public health would be attained, assuming groundwater remediation is completed.

The excavation and treatment of on-site soils using low-temperature thermal stripping would provide engineering controls that would remove VOC contamination from site soils and be recovered via carbon adsorption or destroyed via on-site afterburner. Alternative SC-6 would offer a permanent remedy that has been proven for soils contaminated with VOCs, and would eliminate future potential risks posed by the soil contaminants. This alternative would not require long-term management of the site, and only limited monitoring would be needed to verify achievement of the soil TCLs.

The process will generate residue streams (e.g., air, water, and solids) which will require hazardous waste management such as emissions control or off-site treatment. If the excavated soil at either site was determined to be a RCRA-listed waste or a substantially similar, treated soil will either have to be delisted prior to backfilling on-site or treated to comply with Land Ban treatment levels and disposed of in a RCRA landfill.

8.4.7 Implementability

Implementation of a low-temperature thermal-stripping system is a two-staged process: (1) mobilization and installation of the thermal-stripping unit; and (2) excavation of the contaminated soil and operation of the thermal processing unit.

Excavation of contaminated soils at either the Kysor or Four Winns site does not present any major implementation constraints. Both sites are accessible for use of heavy equipment; no surface clearing or movement of utilities is anticipated. The side walls of the excavation may require stabilization, such as sheet pile or shoring to avoid cave-in; therefore, increasing the volume of contaminated soil. Chemicals present in excavated soils will begin to volatilize when exposed to the atmosphere. This could pose a threat to workers and adjacent receptors.

The Kysor and Four Winns sites would require little site preparation for installation of an on-site thermal-stripping process unit. Both sites have large, relatively flat open areas for construction of process equipment. Construction requires a level well-compacted area; however, a concrete pad may be preferable. The treatment units, transported on semitrailers, are designed to be assembled quickly for on-site installation. The utilities required for the operation of the thermal-stripping system are available at both the Kysor and Four Winns sites.

If a thermal afterburner is used for off-gas control, the off-gas from the process system requires several treatment steps to achieve an acceptable emissions level. The exhaust gas from the afterburner is discharged at flow rates up to 40,000 cfm, through a multistep emission control system. The gas first enters a spray cooling tower where the temperature is reduced from 1,400°F to approximately 400°F. Effluent from the cooling tower is treated for reuse, or removed for treatment off-site. The gas then enters a baghouse, where the particulate level is reduced to less than 0.02 grains per dry cubic foot of gas. Solids removed in the baghouse are collected and analyzed for disposal. The gas then leaves the baghouse, is drawn through an induced draft fan, and is exhausted through the stack to the atmosphere. Carbon adsorption canisters can be used to collect any organics that may persist. Carbon adsorption canisters are not expected to be necessary for the Four Winns site; however, they may be required for the VOCs at the Kysor site.

The treated soil must also be monitored. Low-temperature thermal stripping will not treat inorganics (i.e., metals) and some heavier organics. The handling and disposition of the treated soils should conform with appropriate regulations. If the treated soil is found to be hazardous, additional treatment steps may be necessary. However, the levels of inorganics and organics present at these sites indicate the treated soils should be characteristically nonhazardous.

These units are designed to operate on a homogeneous wastestream. Variations in soil moisture or BTU value can have an impact on operating temperatures and fuel consumption. These fluctuations can also impact the emission control systems. Material handling for the thermal stripper can be a difficult, time-consuming operation.

There are several advantages of this alternative over Alternative SC-11, On-site Incineration (refer to Section 8.7). A thermal-stripping system operates at a much lower process temperature, which reduces many equipment problems, such as refractory life associated with incineration units. The low-temperature thermal-stripping unit can also reduce auxiliary fuel

consumption by allowing the temperature to be adjusted for the particular wastestream. The soils contaminated with methylene chloride at the Four Winns site would only require a low temperature for stripping and combustion. The less volatile contaminants at the Kysor site would likely require higher process temperatures. The most significant advantage of this alternative over on-site incineration is evidenced in the process feed rate. The thermal-stripping system can handle up to 700 tons per day, while even large incinerators are limited to approximately 100 tons per day. This difference in feed rates can greatly reduce the number of days on-site and the cost of remediation.

Several contractors offer on-site thermal-stripping services. The process is based on demonstrated technologies for processing and drying soils. Emission control and water treatment equipment is readily available. A treatability study should be conducted to determine the optimum configuration for system operation. Pilot testing would be necessary to evaluate the effectiveness and reliability of the thermal treatment unit before full operation can be implemented at either site. During the pilot-testing, actual on-site contaminated soils will be processed through the thermal treatment unit. Air pollution control tests will be conducted to determine compliance with air quality standards.

8.4.8 Costs

Capital and annual costs and the net present worth of the alternative were derived using the assumptions presented in Section 6.3. In this subsection, capital and annual costs are presented for the Low-temperature Thermal-stripping Alternative. In addition, the net present worth of this alternative is presented, assuming it would be implemented at both sites. These costs are listed in Table 8-5. Site-specific costs are presented in Appendix D.

Capital Costs. The capital cost of the Low-temperature Thermal-stripping Alternative is \$17,214,000.

Annual Costs. Because there would not be any hazardous substances, pollutants, or contaminants remaining on-site, a five-year site review would not be required under SARA Section 121. In addition, because this alternative would be implemented in approximately five to six months, there would be no associated annual costs.

Present Worth Analysis. The present worth of this alternative would equal the capital costs, because no annual costs are associated with low-temperature thermal stripping. Therefore, present worth would equal \$17,214,000.

8.4.9 Summary

The detailed analysis of Alternative SC-6 is summarized in Table 8-6.

TABLE 8-5
 SC-6 LOW TEMPERATURE THERMAL STRIPPING
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	TOTAL
CAPITAL COSTS	
LOW TEMPERATURE THERMAL STRIPPING	\$12,519,00
ENGINEERING @ 10%	1,252,00
CONTINGENCY @ 25%	3,443,00
TOTAL CAPITAL COSTS	\$17,214,00
ANNUAL COSTS	
LOW TEMPERATURE THERMAL STRIPPING	\$
TOTAL ANNUAL COSTS	\$
PRESENT WORTH FOR 30 YEARS @ 5%	\$17,214,00

TABLE 8-6
SUMMARY OF DETAILED ANALYSIS
SC-6: EXCAVATION, LOW-TEMPERATURE THERMAL STRIPPING
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o RCRA Interim Standards for chemical, physical, and biological treatment may be generally applied to the Cadillac site. - o RCRA Performance Standards for Miscellaneous Units would be met. RCRA and Michigan Location Standards and facility design and operating standards would be met. - o RCRA and Michigan regulations for handling, transporting, and disposing hazardous waste must be met. - o Applicable air regulations would be met.
2. Protection of Public Health and the Environment	- o SC-6 should achieve soil TCLs. - o SC-6 would reduce baseline public health risks associated with contaminated soil and the potential for continued leaching of VOCs from soil to groundwater. - o Because the alternative would require extensive waste handling activities, the potential for adverse environmental effects increases. - o SC-6 would result in an improvement to the natural environment over the current site conditions described in the baseline risk assessment. - o SC-6 would provide a permanent remedy.
3. Reduction of Toxicity, Mobility, or Volume	o The toxicity, mobility, and volume of VOCs in the soils would be permanently reduced.
4. Short-term Effectiveness	o Excavation would pose potential risks to workers through dermal contact or inhalation of dusts or vapors.

TABLE 8-6
(continued)
SUMMARY OF DETAILED ANALYSIS
SC-6: EXCAVATION, LOW-TEMPERATURE THERMAL STRIPPING
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
	- o Thermal-stripping process would require off-gas treatment to reduce potential risk to the community due to the release of products of incomplete combustion. - o Time to achieve full protection is expected to be 15 months for Kysor soils and 16 months for Four Winns soils.
5. Long-term Effectiveness	- o Long-term risk to public health due to exposure to groundwater contaminated by leaching from the soils would not be significant. - o SC-6 provides a permanent remedy for the soil contamination. - o Long-term management would not be required.
6. Implementability	- o SC-6 is readily implementable for the Cadillac site. - o Equipment is readily available. - o Treatability studies should be performed to determine optimum configuration for system operation.
7. Costs	- o Capital Costs = \$17,214,000 - o Annual Costs = \$0 - o Present Worth = \$17,214,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

A = The long-term uncertainties associated with land disposal.

TABLE 8-6
(continued)
SUMMARY OF DETAILED ANALYSIS
SC-6: EXCAVATION, LOW-TEMPERATURE THERMAL STRIPPING
CADILLAC AREA FEASIBILITY STUDY

NOTES (continued):

- B = The goals, objectives, and requirements of the SWDA.
- C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.
- D = Short- and long-term potential for adverse health effects from human exposure.
- E = Long-term monitoring costs.
- F = The potential for future remedial action costs if the alternative remedial action in question were to fail.
- G = The potential threat to public health and the environment associated with excavation, transportation, and redispisal, or containment.

8.5 DETAILED ANALYSIS OF ALTERNATIVE SC-7: IN-SITU SOIL FLUSHING

8.5.1 Description

This alternative consists of implementing the following remedial technology to address the identified response activities for soil remediation of the Kysor and Four Winns sites. This alternative is depicted in Figure 8-3.

In-situ Soil Flushing. Soils in the unsaturated zone are contaminated with VOCs. In-situ soil flushing would be conducted to remove the contaminants of concern from the soil. The soil-flushing system would consist of infiltration piping installed above the zone of contaminated soils. A source of flushing water (municipal water) will be provided at each site. The flushate would be infiltrated through the affected soils and subsequently discharge to groundwater. The flushate would be collected using groundwater recovery wells and treated at the on-site groundwater treatment facility proposed for groundwater remediation. During the treatment process, soil and groundwater sampling and analyses would be conducted periodically to monitor treatment effectiveness. Post-treatment soil sampling would be conducted to verify the removal effectiveness of the soil-flushing process.

Soil-flushing systems generally consist of a system of pipes to introduce the flushate into the zone of affected soils, one or more groundwater recovery wells, transport piping, a water treatment system, and provision for discharge of the treated water. Several water treatment technologies are available for removing VOC contamination from groundwater (see Section 4.0). The design and implementation of a soil-flushing system must be coordinated with the selected groundwater treatment alternative.

The achievable level of treatment varies and depends on the contact of the flushing solution with the contaminants, the appropriateness of the flushing solutions for the removal of contaminants, and the hydraulic conductivity of the soil. Soil-flushing systems can be enhanced by using surfactants or chemical additives; however, it requires careful management and knowledge of reactions that may adversely affect the treatment system.

Kysor Site. The implementation of the In-situ Soil-Flushing Alternative at the Kysor site would require provision of a source of flushate (i.e., water), construction of the infiltration system, an earthen cover, and implementation of a groundwater treatment remedy including groundwater recovery wells and VOC treatment.

The soil-flushing system would be installed directly over the zone of affected soils at the Kysor site. Topography of this area is relatively level, and would promote installation of the system. A water treatment system would be installed in an area dictated by the selected groundwater treatment alternative.

Four Winns Site. Field implementation of the soil-flushing alternative at the Four Winns site would include the same concerns associated with the Kysor site.

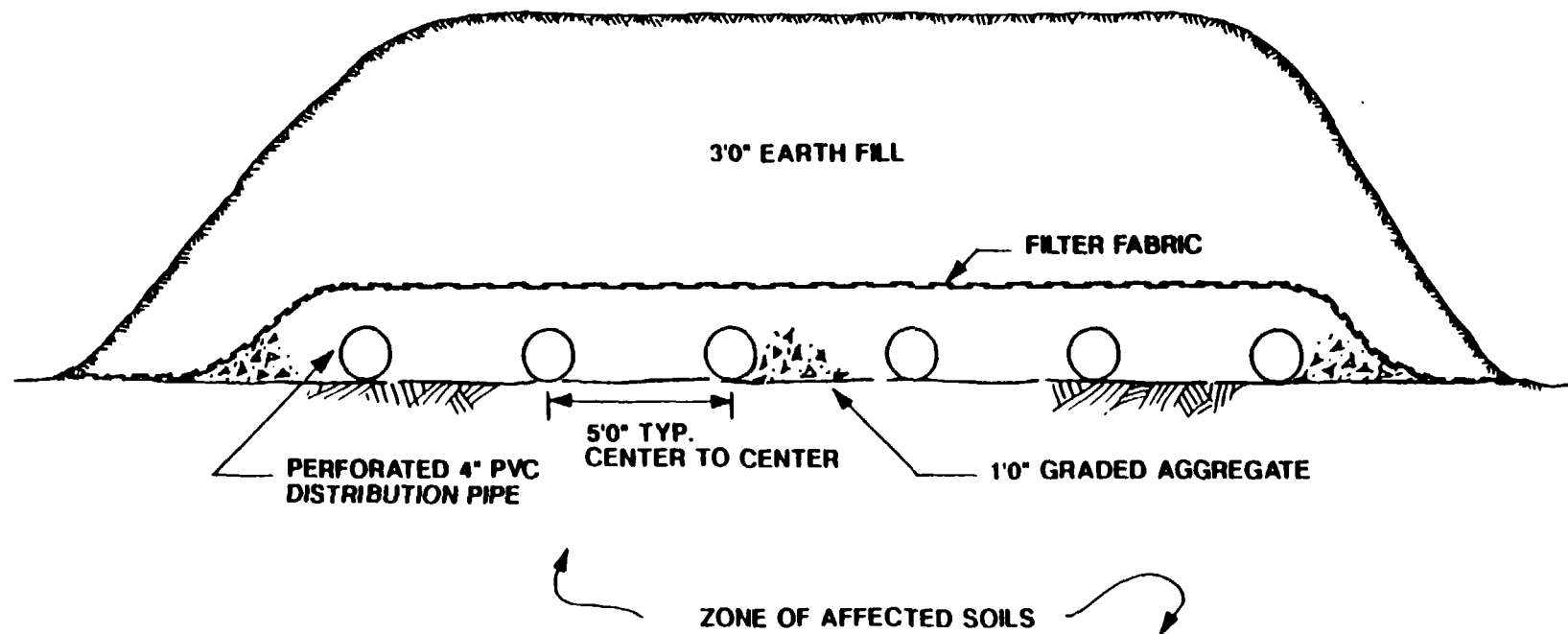


FIGURE 8-3
SC-7: IN SITU SOIL FLUSHING
CADILLAC AREA FEASIBILITY STUDY
CADILLAC, MICHIGAN
EC 1000-0000

8.5.2 Compliance with ARARs

In-situ soil flushing should be conducted in compliance with RCRA regulations for Environmental Performance (Miscellaneous Treatment Units, 40 CFR 264, Subpart X). Although these RCRA regulations are not unit- or process-specific and do not contain technical requirements, the regulations do require that the protection of public health and the environment be ensured; releases and migration of contaminants to the environmental media prohibited; waste analyses and trial tests performed; and all equipment decontaminated upon completion or closure.

No air emissions are created and no contaminated soils are removed; therefore, a Michigan air emission permit, and RCRA and Michigan Hazardous Waste generator, transport, disposal, and closure regulations are not considered ARARs. However, a Michigan groundwater discharge permit (R 323, Part 22) may be required because this action will generate contaminated leachate below ground that will enter the groundwater (and eventually be pumped and treated). Because this alternative utilizes clean water as a flushate to discharge to the ground, it is unclear whether this action will fall under MDNR's jurisdiction for groundwater discharges.

Groundwater monitoring conducted under this alternative will be consistent with RCRA Corrective Action Monitoring (40 CFR 264.100). All remedial activities must be conducted in accordance with OSHA Standards for Hazardous Waste Operations.

Appendix E identifies action-specific ARARs associated with this remedial alternative. These ARARs are further detailed in Section 5.1 and summarized in Table 5-1.

8.5.3 Protection of Public Health and the Environment

The in-situ soil-flushing technology has been proven successful in hydrogeologic conditions similar to those at the Kysor and Four Winns sites. Therefore, it can be assumed that VOC concentration levels in the soil would be reduced to the TCLs.

The In-situ Soil-Flushing Alternative would reduce the baseline public health risks associated with chemicals in soil potentially leaching into groundwater; would result in attainment of site TCLs; and would provide a permanent remedy for the site. The In-situ Soil-Flushing Alternative, in theory, would result in attainment of TCLs by dissolving and mobilizing organic chemicals into groundwater for subsequent collection and treatment. Failure of this alternative would occur if the groundwater recovery system does not adequately capture the flushed contaminants, or if the flushate does not uniformly infiltrate the affected area, and significant pockets of contaminants remain.

8.5.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

Implementation of this alternative would remove soil contamination found at both the Kysor and Four Winns sites. The extraction process would result in a substantial reduction of volume and mobility of contaminants in soil. It is

effective in removing water-soluble contaminants, which would then be collected and treated using one of the groundwater treatment alternatives. In-situ soil-flushing systems are conceptually simplistic in nature and have limitations due to the lack of process control in the subsurface environment. Design of the extraction/injection system is critical because of the potential for contaminants to remain in the subsurface soils and/or groundwater.

8.5.5 Short-term Effectiveness

There would be potential worker risks during the construction of groundwater extraction wells, pumps, and transport piping. Appropriate safety and health regulations would need to be followed to protect workers from adverse impacts. The water from the in-situ soil flushing would need to be treated for VOCs by the same treatment system selected for groundwater. Therefore, the short-term impacts for that alternative would also apply.

If the soil-flushing system was enhanced with chemical additives or surfactants, then precautions would be taken to avoid significant risks to workers handling the chemicals.

Construction of this alternative presents the least risk of any of the containment or treatment alternatives to on-site workers during implementation because it involves no soil excavation and has limited potential for contact with contaminated soils. It poses a high potential for risk to the environment and groundwater receptors because the migration of contaminants into groundwater is being promoted.

The Four Winns site has surface soil contamination; therefore, during construction and implementation of Alternative SC-7, additional health and environmental precautions should be instituted to prevent on-site workers and terrestrial wildlife from direct contact with contaminated soils.

An accurate time required to achieve the soil remedial response objectives cannot be estimated without site-specific operational data. This is due to the unknowns such as desorption/removal efficiency of chemicals from the soil to the flushate; possibility of the channelling of flushate away from the zone of contaminated soils; uncertain stability of the flushate in the soils; and efficiency of the groundwater extraction system. However, a preliminary estimate was developed based on anticipated removal efficiency, resulting in an implementation period of two years for the Kysor site and one year for the Four Winns site.

8.5.6 Long-term Effectiveness

No long-term maintenance or monitoring of a system is necessary. The soils would be permanently remediated to a level that would no longer pose significant potential risks due to human ingestion of groundwater contaminated through leaching. Protection of public health assumes that groundwater contamination is also remediated.

The In-situ Soil-Flushing Alternative will provide engineering controls that should remove and reduce VOC contamination from the site soils in the

unsaturated soil zone, and allow the contaminants to be collected by groundwater recovery for subsequent treatment via one of the groundwater treatment alternatives. Alternative SC-7 has been effectively demonstrated for the remediation of gasoline spills in sandy or granular soils, similar to soils at the Kysor site. The effectiveness of this technology at more complex waste sites is not as proven. This alternative should reduce the future potential risks posed by the VOC contamination, and not require long-term management of the site. Only limited monitoring would be needed to verify achievement of the soil treatment TCLs.

Typically, in-situ soil-flushing systems are sensitive to freezing and operate inefficiently during periods of cold weather. Provisions can be made for all-weather operation, which would include installation of an earthen surficial cap system above the infiltration piping to prevent freezing.

8.5.7 Implementability

Implementation of the In-situ Soil-Flushing Alternative consists of installation and operation of a soil-flushing system. The equipment required for a soil-flushing system is readily available from vendors and equipment manufacturers, and can be easily installed. It is estimated that six to 12 months would be necessary to select a vendor, construct the in-situ soil-flushing treatment system, and conduct pilot-scale field testing. A contractor would install a series of infiltration pipes overlain by a coarse aggregate drainage layer, filter fabric, and vegetated soil cover. This cover system would allow for operation of the soil-flushing system throughout periods of inclement weather.

Following installation, system operation would commence by introducing flushate into the soil zone, collecting the flushate using recovery wells, and treating it to remove the contaminants. The flushate would consist of water provided by the municipal water supply. Detergents or additives could be introduced if higher removal efficiencies are desired. Water can be used as the flushate to remove water-soluble organics. Medium-solubility organics, such as those at the Kysor and Four Winns sites, can be effectively removed from soils by water-flushing.

The infiltration system would be monitored to determine the optimal flow rate and proper distribution of the flushate. The geometry of the infiltration system will be considered during system design, and is dependent on several variables, including type of soils, depth to groundwater, anticipated movement of leachate, surface structures, and type of contaminants. The hydrogeology of both the Kysor and Four Winns site is well-suited to in-situ soil flushing. The sandy soils are highly permeable, and site access would not restrict installation of the infiltration system.

The flushate collected from the soil-contacting process can be treated and recycled by operating an on-site treatment process. Removal of the extracted contaminants from recovered groundwater would be accomplished by using conventional treatment systems suited for the contaminants present. The organics in the soils at the Kysor and Four Winns sites could be effectively removed using conventional groundwater technologies.

A major advantage of this alternative, compared to other treatment options, is that the contaminated soils are treated in situ. The contaminated material would not have to be excavated prior to treatment, which significantly reduces material handling costs, as well as adverse health and safety risks to workers.

Increased contamination of groundwater is a major concern. Nonhomogeneities or uncertainties in the hydrogeologic system can allow flushate to pass uncollected, further contaminating groundwater. Therefore, this alternative should be implemented in conjunction with a groundwater extraction and treatment alternative that is closely monitored to prevent expanding groundwater degradation.

Site-specific pilot testing will be necessary to evaluate the effectiveness and reliability of the soil treatment unit before full operation can be implemented at either site. During pilot testing, measurements will be taken to determine groundwater and flushate recovery rates, treatment requirements, and the expected duration of the soil treatment effort.

8.5.8 Costs

In this subsection, capital and annual costs are presented for the In-situ Soil-Flushing Alternative. In addition, the net present worth of this alternative is presented, assuming it would be implemented at both sites. These costs are listed in Table 8-7. Site-specific costs are presented in Appendix D.

Because this alternative would be combined with one of the groundwater treatment alternatives, the costs for extraction and treatment are not presented herein. However, the costs would be equal to those presented for the groundwater treatment alternative with which in-situ soil flushing would be combined. Capital and annual costs and the present worth of the alternative were derived using the assumptions presented in Section 6.3.

Capital Costs. The capital cost of the In-situ Soil-Flushing Alternative is \$308,000.

Annual Costs. Annual O&M costs for the In-situ Soil-Flushing Alternative are \$52,000.

Present Worth Analysis. Because there would not be any hazardous substances, pollutants, or contaminants remaining on-site, a five-year site review would not be required under SARA Section 121. Because this alternative would be implemented in approximately two years at the Kysor site and one year at the Four Winns site, the present worth was based on a two-year and a one-year performance period, respectively, and a 5-percent interest rate. The total present worth of this alternative is \$455,000.

8.5.9 Summary

The detailed analysis of Alternative SC-7 is summarized in Table 8-8.

TABLE 8-7
SC-7 IN-SITU SOIL FLUSHING
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION		TOTAL
-----		-----
CAPITAL COSTS		
	IN-SITU SOIL FLUSHING	\$224,00
	ENGINEERING @ 10%	22,00
	CONTINGENCY @ 25%	62,00
1.	TOTAL CAPITAL COSTS	----- \$308,00
ANNUAL COSTS		
	IN-SITU SOIL FLUSHING	\$52,00
	TOTAL ANNUAL COSTS	----- \$52,00
PRESENT WORTH OF ANNUAL COSTS AT		
2.	KYSOR SITE FOR 2 YEARS @ 5%	\$97,00
3.	FOUR WINNS SITE FOR 1 YEARS @ 5%	\$50,00
TOTAL PRESENT WORTH OF THIS ALTERNATIVE (ITEMS 1 - 3)		\$455,00
(THIS DOES NOT INCLUDE EXTRACTION & TREATMENT OF FLUSHING WATER)		

TABLE 8-8
SUMMARY OF DETAILED ANALYSIS
SC-7: IN-SITU SOIL FLUSHING

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o Appropriate RCRA requirements for treatment units and groundwater monitoring would be met. - o Michigan groundwater discharge permit requirements would be met, if applicable.
2. Protection of Public Health and the Environment	- o Soil TCLs would be achieved. - o Baseline public health risks would be reduced by minimizing leaching of VOCs to groundwater. - o SC-7 would present lowest risk to on-site workers during installation and implementation.
3. Reduction of Toxicity, Mobility, or Volume	- o SC-7 would reduce the volume and mobility of soil contaminants. Implementation would increase the toxicity and mobility of groundwater contaminants. - o The toxicity of the contaminated groundwater would be reduced by implementing one of the groundwater treatments.
4. Short-term Effectiveness	- o Potential risks to workers during construction of groundwater extraction wells, pumps, and transport piping. - o Time to achieve full protection cannot be estimated without site-specific operational data.
5. Long-term Effectiveness	- o Long-term maintenance would not be required. - o Limited monitoring would be required to verify the effectiveness of the treatment. - o Potential of further groundwater contamination if all flushate cannot be collected.
6. Implementability	o Approximately 6 to 12 months would be necessary to select a vendor, construct the treatment system, and perform a pilot-scale study.

TABLE 8-8
(continued)
SUMMARY OF DETAILED ANALYSIS
SC-7: IN-SITU SOIL FLUSHING

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
7. Costs	- o Capital Costs = \$308,000 - o Annual Costs = \$52,000 - o Present Worth = \$455,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

A = The long-term uncertainties associated with land disposal.

B = The goals, objectives, and requirements of the SWDA.

C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.

D = Short- and long-term potential for adverse health effects from human exposure.

E = Long-term monitoring costs.

F = The potential for future remedial action costs if the alternative remedial action in question were to fail.

G = The potential threat to public health and the environment associated with excavation, transportation, and redispisal, or containment.

8.6 DETAILED ANALYSIS OF ALTERNATIVE SC-9: IN-SITU VACUUM EXTRACTION

8.6.1 Description

This alternative consists of the following remedial technology to address the identified response activities for soil remediation at the Kysor and Four Winns sites. This alternative is depicted in Figure 8-4.

In-situ Vacuum Extraction. The vacuum extraction system is an in-situ soil treatment method that uses a series of wells to circulate soil gas within the unsaturated zone of contaminated soil. Wells are installed to the water table and screened. A gas induction blower system forces clean gases into the unsaturated zone and soil pore space. At the same time, the vacuum extraction system is withdrawing contaminated gases from the unsaturated zone. The purging gases volatilize the adsorbed VOCs from the soil, where they are treated by emission control equipment before being exhausted to the atmosphere. Generally, vacuum extraction systems can be implemented with relatively conventional equipment, labor, and materials.

Kysor Site. Field implementation of the In-situ Vacuum Extraction Alternative at the Kysor site would include installing a series of induction and extraction wells, control trenches, header piping, air-circulation equipment, and air pollution control equipment.

The likely place to install the vacuum extraction and emission control system would be out from the northwestern corner of the Kysor building, and south of the gravel access road leading from Leeson Avenue. Topography of this area is relatively flat and open for the installation of concrete pads to support the vacuum extraction and emission control equipment. The site areas would require improvements to support the vacuum extraction system, pumping, air pollution control equipment, and ancillary activities.

The vacuum extraction well system would consist of a series of induction and extraction wells designed to extract VOC contamination from the soils. The proper spacing of these wells will be determined from pilot testing at the Kysor site; however, a 50-by-50-foot horizontal grid is typical, considering the extent and depth of contamination.

Four Winns Site. Field implementation of the In-situ Vacuum Extraction Alternative at the Four Winns site would be similar to the Kysor site. The southeastern corner of the Four Winns site is a likely location to install the in-situ vacuum extraction and emission control system. Topography of this area is relatively flat and open for installation of concrete pads to support the vacuum extraction and emission control equipment. The site work areas would require improvements to support the vacuum extraction system, air pollution control equipment, and ancillary activities.

8.6.2 Compliance with ARARs

CAA ARARs include particulate matter standards (40 CFR 50), which must not be exceeded during construction and operation of the treatment system. Under the Michigan Air Pollution Control Regulations, the in-situ vacuum extraction

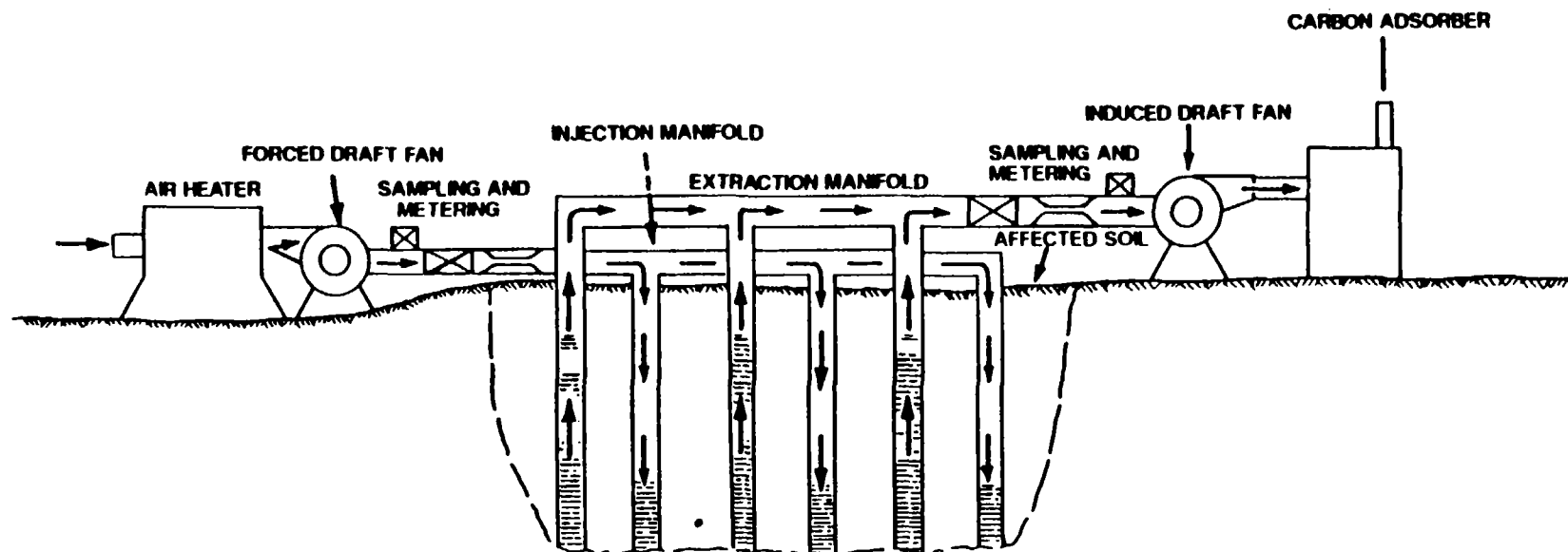


FIGURE 8-4
SC-9: IN SITU VACUUM EXTRACTION
CADILLAC AREA FEASIBILITY STUDY
CADILLAC, MICHIGAN

SC 100000000

system is considered a source of air contamination; therefore, a permit will be required for both construction and operation of the system, according to R 336, Rule 201. The Michigan regulations also limit fugitive dust and establish maximum allowable emission rates from new VOC sources, based on BACT (R 336, Rules 701 and 702). BACT is determined on a case-by-case basis through information submitted in the permit application. Remedial actions carried out entirely on-site (i.e., on a CERCLA site such as the Kysor site) are not required to obtain permits or comply with administrative requirements, such as obtaining permits. However, in the case of the Cadillac site, application for an air emissions permit is considered necessary to comply with the substantive requirements of the Michigan Air Pollution Control Act Rules. In addition, the Michigan regulations prohibit the emissions of air contaminants in quantities that would cause "injurious effects to human health and welfare, and the environment" (R 336, Rule 901). Information submitted during the permit application is reviewed to determine where injurious effects may occur.

RCRA ARARs for miscellaneous units are the same as for SC-7 (refer to Section 8.5.2).

The RCRA and Michigan location requirements (40 CFR 264.18 and R 299, Rule 603) and the RCRA and Michigan facility design and operating standards (40 CFR 264 and R 299, Rule 604) apply to the temporary location of the treatment unit, as well as any final cover systems. In general, the unit may not be located within 200 feet of a fault that was displaced in Holocene time, and must be located outside the 100-year floodplain.

Spent carbon from the treatment will be considered hazardous waste. The carbon may either be regenerated on-site, or collected and regenerated off-site by a licensed facility. If carbon is not generated, it must be handled as a hazardous waste and transported via a Michigan-licensed hazardous waste hauler and licensed vehicle to a permitted RCRA disposal facility, in compliance with 40 CFR 264.

Groundwater monitoring conducted under this alternative will be consistent with RCRA Corrective Action Monitoring (40 CFR 264.100). All remedial activities must be conducted in accordance with OSHA Standards for Hazardous Waste Operations.

Appendix E identifies action-specific ARARs associated with this remedial alternative. These ARARs are further detailed in Section 5.1 and summarized in Table 5-1.

8.6.3 Protection of Public Health and the Environment

The in-situ vacuum extraction system would be pilot-tested to ensure that it could meet the TCLs for soils. Implementation of the In-situ Vacuum Extraction Alternative would reduce the baseline public health risk associated with organic chemicals present in soils; improve the natural environment of the site; and result in a permanent remedy. It is anticipated the soil TCLs would be attained because the site possesses the ideal soil type (i.e., sandy and porous) to facilitate the in-situ vacuum extraction treatment process.

Failure of this alternative could occur if the soil porosity is not homogeneous, which would cause the vacuum to channel away from the soil-sorbed contamination, or if the off-gas treatment process for extracted soil gas fails to reduce air emissions to acceptable levels to meet potential ARARs. Pilot testing and careful air monitoring during implementation should reduce the possibility of failure.

In summary, this alternative should adequately reduce public health risks, attain TCLs, provide a permanent remedy, and improve the natural environment of the site.

8.6.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

The treatment process would result in a substantial reduction of toxicity, mobility, and volume of contaminants in the soils at both sites. The VOCs contaminating the soil would be removed and, through off-gas control, either contained for disposal or thermally destroyed.

8.6.5 Short-term Effectiveness

During the construction and installation phase of the system, worker risks would be associated with the drilling of extraction wells. On-site workers could also be exposed to chemicals via dermal contact and inhalation of volatilized chemicals. Therefore, proper worker safety measures would be enforced to avoid health risks. Air emissions generated during implementation of the vacuum extraction system would be controlled.

Kysor Site. The time to achieve the soil remedial response objectives is not easily estimated; however, according to vendors, it is anticipated to take two years. The extent of project duration depends on the extraction/removal efficiency of the chemicals in soil; pattern of soil gas movement within the soil matrix; and capacity and limitations of the final treatment system configuration.

Four Winns Site. The site-specific short-term effectiveness description for Alternative SC-9 is the same as the Kysor site, with two exceptions. The Four Winns site has surface soil contamination; therefore, during construction and implementation of a vacuum extraction alternative, additional health and environmental controls should be instituted to prevent on-site workers and terrestrial wildlife from direct contact with contaminated soils. The estimated duration of treatment at the Four Winns site is estimated at less than one year.

8.6.6 Long-term Effectiveness

At both the Kysor and Four Winns sites, the In-situ Vacuum Extraction Alternative is considered an effective approach for reducing VOC concentrations in the soils to TCLs. The technology would provide a permanent reduction in the VOC levels and, subsequently, a long-term reduction in risks to public health.

As long as any hazardous materials generated on-site are decontaminated or transported appropriately to an off-site facility, there appear to be no persistent adverse effects to public health or the environment.

The In-situ Vacuum Extraction Alternative would provide engineering controls that would remove and reduce VOC contamination from on-site subsurface soils and allow them to be recovered using off-gas treatment via carbon adsorption, or destroyed via an on-site afterburner. Alternative SC-9 is considered a permanent remedy, and is most effective in sandy or granular soils similar to those at the Kysor and Four Winns sites. This alternative would not require long-term management of the site, and only limited monitoring would be needed to verify compliance with the soil TCLs.

8.6.7 Implementability

Soil gas collection systems are sophisticated well systems connected by a series of manifold pipes. The geometry of the system depends on several variables, including depth to groundwater, type of soils, surface structures, and type of contamination present. The hydrogeology of both the Kysor and Four Winns sites is well-suited to the vacuum extraction of soil gas. The sandy soils present are highly permeable. The relatively shallow water table limits the cost of well installation. Site access would not restrict placement of wells.

The contaminants present at these sites are relatively well-suited for vacuum extraction. Methylene chloride at the Four Winns site would be an excellent candidate for removal via vacuum extraction because of its relatively high vapor pressure. Compounds with a lower vapor pressure, such as the xylene at the Kysor site, can also be extracted, but would likely require a longer treatment period.

The extracted soil gases are collected through the manifold for treatment. The gaseous wastes can be treated thermally to render contaminants less hazardous, or treated chemically or physically to remove and concentrate the contaminants. Treatment options available include adsorption, flaring, and incineration. All three of these treatment methods are considered available, conventional, and proven technologies for off-gas treatment. Adsorption is a process of concentrating contaminants from one medium onto another. It has been successfully used for years by municipal water treatment facilities for the removal of organics using granulated carbon. Flaring is a process that thermally oxidizes substances by exposing them to an open flame. The oil and gas industry commonly uses this flaring process for the destruction of refinery off-gas. Incineration is a treatment method that thermally oxidizes substances using high temperatures under controlled conditions. Incineration has been used by waste disposal firms for years in the destruction of organic materials.

The major advantage of this alternative, compared to other treatment alternatives, is that the waste is treated in situ. Contaminated material would not have to be excavated prior to treatment. However, the surface area of each of the Kysor and Four Winns sites may require capping to effectively remove the near-surface contaminants. This alternative would represent a permanent

remedy. It would eliminate any need for a future remedial action to deal with contaminated soils.

Site-specific pilot testing would be necessary to evaluate the effectiveness and reliability of the soil treatment unit before full operation can be implemented at either site. During pilot testing, measurements will be taken to determine soil gas recovery rates, actual system design, and expected duration of the soil treatment effort. In addition, it is likely that field testing of the vapor-phase carbon canister, or afterburner emission control equipment, will be required. The pilot-testing results will be employed to more accurately predict the performance of the selected control equipment, and its ability to sufficiently reduce emissions to comply with air-quality standards.

If the pilot testing indicates marginal results, the vacuum extraction system can be enhanced by the following techniques: heating the air being blown into the soil; supplementing the induced gases with steam; and/or installing a temporary clay cap to seal the treatment zone to prevent the intrusion of surface water and air into the extraction wells. Post-treatment soil sampling will be conducted to verify effectiveness of the vacuum extraction process.

8.6.8 Costs

In this subsection, capital and annual costs are presented for the In-situ Vacuum Extraction Alternative. In addition, the net present worth of this alternative is presented, assuming it would be implemented at both sites. These costs are listed in Table 8-9. Site-specific costs are presented in Appendix D. The capital and annual costs and the net present worth of this alternative were derived using the assumptions presented in Section 6.3.

Capital Costs. The capital cost of the In-situ Vacuum Extraction Alternative is \$1,854,000.

Annual Costs. Because there would not be any hazardous substances, pollutants, or contaminants remaining on-site, a five-year site review would not be required under SARA Section 121. In addition, because actual operation of this alternative would take approximately a year and a half at the Kysor site and nine months at the Four Winns site, there would be no associated annual costs. Any operational costs were considered a capital cost.

Present Worth Analysis. The present worth of this alternative would equal the capital costs, because no annual costs are associated with in-situ vacuum extraction. Therefore, present worth is \$1,854,000.

8.6.9 Summary

The detailed analysis of Alternative SC-9 is summarized in Table 8-10.

TABLE 8-9
SC-9 IN-SITU VACUUM EXTRACTION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

		DESCRIPTION	TOTAL
CAPITAL COSTS			
	IN-SITU VACUUM EXTRACTION		\$1,348,00
	ENGINEERING @ 10%		135,00
	CONTINGENCY @ 25%		371,00
1.	TOTAL CAPITAL COSTS		\$1,854,00
ANNUAL COSTS			
	IN-SITU VACUUM EXTRACTION		\$
	TOTAL ANNUAL COSTS		\$
TOTAL PRESENT WORTH OF THIS ALTERNATIVE			\$1,854,00

TABLE 8-10
SUMMARY OF DETAILED ANALYSIS
SC-9: IN-SITU VACUUM EXTRACTION

CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o RCRA Interim Standards for chemical, physical, and biological treatment may be generally applied to the Cadillac site. - o RCRA Performance Standards for Miscellaneous Units would be met. RCRA and Michigan Location Standards and facility design and operating standards would be met. - o RCRA and Michigan regulations for handling, transporting, and disposing hazardous waste must be met. - o Applicable air regulations would be met.
. Protection of Public Health and the Environment	- o SC-9 should achieve soil TCLs. - o SC-9 would reduce baseline public health risks associated with contaminated soil and the potential for continued leaching of VOCs from soil to groundwater. - o Because waste handling activities would be extensive, the potential for adverse environmental effects is increased. - o SC-9 would result in an improvement to the natural environment over the current site conditions described in the baseline risk assessment. - o SC-9 would provide a permanent remedy.
. Reduction of Toxicity, Mobility, or Volume	o The mobility and volume of VOCs in the soils would be permanently reduced.
. Short-term Effectiveness	o Drilling of extraction wells would pose potential risks to workers through dermal contact or inhalation of dusts or vapors.

TABLE 8-10
(continued)
SUMMARY OF DETAILED ANALYSIS
SC-9: IN-SITU VACUUM EXTRACTION
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
	- o In-situ vacuum extraction process would control air emissions to reduce potential risk to the community. - o Time to achieve full protection is expected to be 2 years for Kysor soils and 1 year for Four Winns soils.
5. Long-term Effectiveness	- o SC-9 would provide a permanent remedy for the soil contamination. - o Long-term management would not be required.
6. Implementability	- o SC-9 is readily implementable for the Cadillac site. - o Equipment is readily available. - o Pilot-scale testing would have to be performed to determine optimum configuration for system operation.
7. Costs	- o Capital Costs = \$1,854,000 - o Annual Costs = \$0 - o Present Worth = \$1,854,000

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

TABLE 8-10
(continued)
SUMMARY OF DETAILED ANALYSIS
SC-9: IN-SITU VACUUM EXTRACTION
CADILLAC AREA FEASIBILITY STUDY

NOTES (continued):

- A = The long-term uncertainties associated with land disposal.
- B = The goals, objectives, and requirements of the SWDA.
- C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.
- D = Short- and long-term potential for adverse health effects from human exposure.
- E = Long-term monitoring costs.
- F = The potential for future remedial action costs if the alternative remedial action in question were to fail.
- G = The potential threat to public health and the environment associated with excavation, transportation, and redisposal, or containment.

8.7 ALTERNATIVE SC-11: ON-SITE INCINERATION

The following subsections describe the detailed analysis of the On-site Incineration Alternative for soil remediation of the Kysor and Four Winns sites. This alternative is depicted in Figure 8-5.

8.7.1 Description

This alternative consists of the following remedial technologies to address the identified response activities for the Kysor and Four Winns sites:

Excavation of Contaminated Soils. Soils contaminated with VOCs above the established TCLs would be excavated and staged for on-site incineration. Both the Kysor and Four Winns sites have VOC-contaminated soil zones extending to the water table.

On-site Incineration. Excavated material would be incinerated on-site for the treatment of VOCs by thermal destruction. A mobile incinerator would be installed at each site to thermally treat the contaminated soils. If delisted and considered nonhazardous, the residual process ash will be disposed of at each site, using backfilling methods. The on-site incineration alternative is considered a treatment alternative for either the Kysor or Four Winns site, or both.

Thermal destruction is a treatment method that uses high temperature to oxidize contaminants under controlled conditions, thereby degrading a substance into products that generally include carbon dioxide, water vapor, sulfur dioxide, hydrochloric acid gases, and process ash. The hazardous products of the thermal destruction/incineration unit (e.g., particulates, sulfur dioxide, nitrogen oxide, hydrochloric acid, and products of incomplete combustion) require air pollution control equipment to prevent release into the environment. Thermal destruction methods can be used to destroy organic contaminants in liquid, gaseous, and solid wastestreams.

A typical on-site incinerator unit operates most efficiently on a 24-hour-per-day continuous schedule, and has feed rates ranging from 25 to 200 tons per day. Therefore, stockpiling of a limited supply of contaminated material may be required during incinerator operating hours. However, stockpiling of waste material would be kept to a minimum to minimize uncontrolled losses of VOCs from the excavated soils. In addition to stockpile management, the process would generate residue streams (e.g., air, water, and solids) that would require additional management. On-site incinerators have been permitted and successfully demonstrated in compliance with performance standards set forth in 40 CFR 264, Subpart O.

Site-specific trial burns would be necessary to evaluate the effectiveness and reliability of the thermal treatment unit before full operation can be implemented at either site. During the trial burns, actual on-site contaminated soils would be processed through the thermal treatment unit. Air pollution control tests would be conducted to determine compliance with the 40 CFR 264 standards. At the same time, the process ash would be tested for EP Toxicity to determine if metal concentrations are characteristically hazardous.

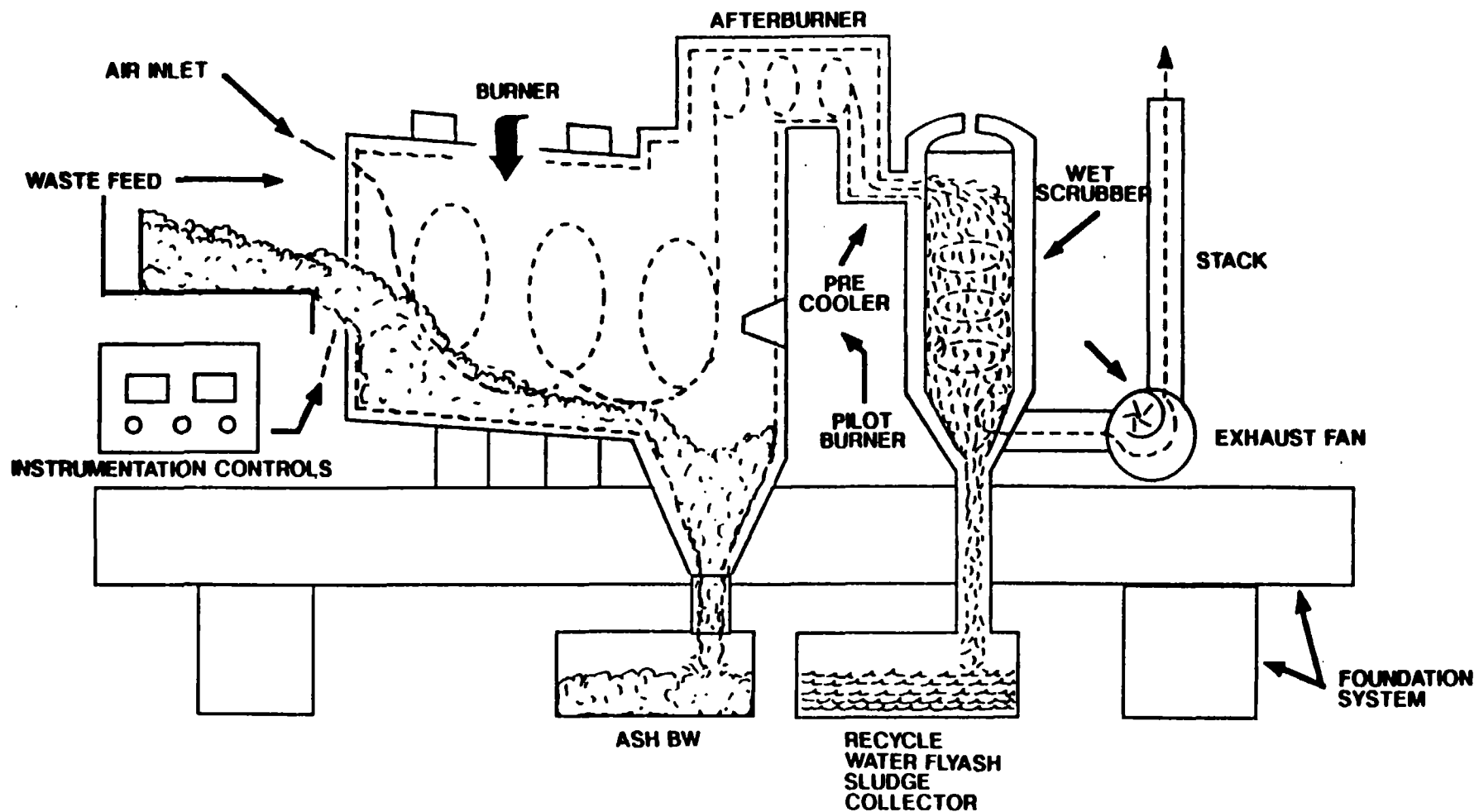


FIGURE 8-5
SC-11: ON-SITE INCINERATION
CADILLAC AREA FEASIBILITY STUDY
CADILLAC, MICHIGAN

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The process residue wastestreams (e.g., air, water and, solids) would require additional management. If excavated soil or the process ash is considered hazardous waste, the proper disposal method would be required either off-site, or constructed on-site. If the excavated soil at either site is determined to be a RCRA-listed waste or substantially similar, treated soil would either have to be delisted prior to backfilling on-site, or treated to comply with Land Ban treatment levels and disposed of in a RCRA landfill.

Kysor Site. The likely place to install the incinerator system would be northwest of the Kysor building and south of the railroad right-of-way, where a gravel access road from Leeson Avenue currently exists. Topography of this area is relatively flat and open for installation of an incineration facility. The site work areas would require improvements to support the thermal treatment facility and ancillary activities. Outdoor illumination for multiple-shift operation should also be considered.

Four Winns Site. The southeastern corner of the Four Winns site is a likely location to install the incinerator system. Topography of this area is relatively flat and open for installation of an incineration facility. The site work areas would require improvements to support the thermal treatment facility and ancillary activities. Outdoor illumination for multiple-shift operation should also be considered.

8.7.2 Compliance with ARARs

This alternative would require the excavation and incineration of contaminated soil, and the transport of incineration residues and treated soil and ash to an off-site RCRA landfill. Because this alternative proposes to excavate and treat contaminated soils, RCRA and Michigan closure and post-closure regulations (40 CFR 264, Subpart G, and R 299, Rule 613) are considered ARARs. These regulations include post-closure care and monitoring, deed notifications, and land-use restrictions (if deemed necessary).

CAA ARARs identified for SC-9 are the same for this alternative (refer to Section 8.6.2). In addition, the CAA NAAQS for particulate matter (i.e., the PM10 standard) is applicable to excavation and incineration. These standards are summarized in Table 5-1.

In addition, the Michigan regulations prohibit the emissions of air contaminants in quantities that will cause "injurious effects to human health and welfare, and the environment" (R 336, Rule 901). Information submitted during the permit application is reviewed to determine if injurious effects may occur. The Michigan air rules also contain specific requirements limiting particulate matter emissions from incinerators. Table 31 in Part 3 of R 336 summarizes the prohibitions. Incinerators with a capacity of 100 pounds or less per hour have a maximum allowable emission of 0.65 pounds of particulates per 1,000 pounds of gas; incinerators with a capacity over 100 pounds per hour have a maximum allowable emission limit of 0.30.

The mobile incinerator must be a RCRA/Michigan-permitted incinerator and must comply with RCRA and Michigan requirements for incinerators (40 CFR 264, Subpart O, and R 299, Rule 623 through R 299, Rule 626), as follows:

- o demonstrated destruction and removal efficiencies (DREs) of 99.9999 percent, relative to the hazardous organic constituents in the waste feed;
- o hydrochloric acid emissions not to exceed 1.8 kilograms per hour (kg/hr), or 1 percent of the chlorine content in the waste feed, whichever is greater;
- o particulate emissions not to exceed 180 milligrams per cubic meter dry air (mg/m3), corrected to 7-percent oxygen;
- o required parameters, including combustion temperature, waste feed rate, an indicator of combustion gas velocity, and carbon monoxide;
- o controlled fugitive emissions;
- o automatic cutoff system to stop waste feed when operating conditions deviate; and
- o the following Michigan minimum design criteria:

WASTE TYPE	COMBUSTION ZONE TEMPERATURE (°C)	RETENTION TIME (SECONDS)	OXYGEN IN EXITING COMBUSTION GASES (%)
Halogenated aromatic hydrocarbons	1,200	2	3
All other combustible hazardous constituents	1,000	2	2

The RCRA and Michigan ARARs for location, facility design, and operation identified for SC-9 are the same for this alternative.

If excavated soil or processed ash is temporarily stored on-site, storage must be secure and the facility should be constructed to withstand a 100-year flood. If storage exceeds 90 days, RCRA and Michigan storage facility requirements may become applicable (40 CFR 264 and R 299). Waste-handling procedures must follow RCRA and Michigan standards for generators and transporters of hazardous wastes. Specifically, the soil must be stored in RCRA-conforming containers or in a RCRA storage facility. Off-site transport of treated soils must be via Michigan-approved and licensed haulers and vehicles.

If the contaminated soil at the Four Winns and/or Kysor sites is determined by USEPA and MDNR to be a RCRA-listed solvent hazardous waste, the RCRA Land Ban becomes applicable (40 CFR 268). The Land Ban prohibits land disposal of hazardous wastes and residuals not meeting specified treatment standards. Certain solvent-contaminated wastes (i.e., RCRA F001-F005 spent solvents) were

banned from land disposal on November 8, 1986, if treatment standards (specified in 40 CFR 268, Subpart D, Table CCWE) were not achieved. Solvent-contaminated soil and debris resulting from a CERCLA response action was granted an extension until November 8, 1990. USEPA and MDNR will need to determine whether the contaminated soil at the Kysor and Four Winns sites is considered a RCRA-listed solvent waste (e.g., the result of a RCRA-listed solvent waste spill). If the soil is not considered a RCRA waste, then the Land Ban is neither applicable nor relevant and appropriate. The proposed treatment will most likely meet Land Ban treatment and disposal requirements.

Groundwater monitoring conducted under this alternative will be consistent with RCRA Corrective Action Monitoring (40 CFR 264.100). All remedial activities must be conducted in accordance with OSHA Standards for Hazardous Waste Operations.

Appendix E identifies action-specific ARARs associated with this remedial alternative. These ARARs are further detailed in Section 5.1 and summarized in Table 5-1.

8.7.3 Protection of Public Health and the Environment

On-site incineration has been shown to be effective in reducing the toxicity of VOCs through oxidation at high temperatures. If appropriate pilot tests confirm this effectiveness, VOC concentrations in the soil at the Kysor and Four Winns sites would likely meet the TCLs.

The excavation and on-site incineration of contaminated soils above TCLs would effectively eliminate the baseline public health and environmental risks associated with the soil contamination and the potential for continued leaching into the groundwater; result in attainment of the soil TCLs; improve the natural environment; and provide a permanent remedy for site contamination.

If properly implemented, the on-site incineration alternative would result in environmental benefits superior to the no-action, containment, soil-flushing, and vacuum extraction alternatives, because it completely destroys all soil contaminants. However, it could potentially have a greater adverse health and environmental effect due to the more extensive and complex operational, excavation and soil-handling, and air quality requirements.

The Four Winns site has surface soil contamination; therefore, during construction and implementation of Alternative SC-11, additional health and environmental controls should be instituted to prevent on-site workers and terrestrial wildlife from direct contact with contaminated surface soils.

8.7.4 Reduction in Toxicity, Mobility, or Volume of Contaminants

Incineration is an available conventional technology that has been well-demonstrated as an option for the destruction of solvents, oils, refinery wastes, and polychlorinated biphenyls (PCBs). On-site applications have been found to be 99.9999-percent efficient in the removal of organic compounds. Incineration of the soils at the Kysor or Four Winns site, which are contami-

nated with VOCs, would eliminate the soil contamination and a potential continuing source of groundwater contamination.

This alternative would be considered a permanent remedy at either site. The toxicity, mobility, and volume of contaminated material would be irreversibly reduced. However, concentrations of heavy metals in the treated soil may be significantly higher than untreated soil. With proper handling of the incinerator ash and scrubber water, incineration would eliminate the need for additional remedial actions.

8.7.5 Short-term Effectiveness

During excavation of the soil at the Kysor and Four Winns sites, workers could potentially be exposed to contaminants by dermal contact or inhalation of dusts or vapors. Personal worker protection requirements need to be enforced and air monitoring of VOCs and particulates should ensure that no air emission concentrations exceed appropriate standards.

The excavation and incineration of contaminated soils would generate air emissions that may present a potential route of human exposure (via inhalation by on-site workers and downwind receptors) and a potential adverse environmental risk. Air pollutants would include products of thermal destruction, such as particulates, sulfur dioxide, nitrogen oxide, hydrochloric acid, carbon monoxide, and VOC emissions. Air pollution control equipment must be monitored to ensure there are no releases to the environment that exceed relevant standards. The possibility of failure of the excavation and incinerator emissions exceeding potential ARARs would be significantly reduced during implementation of this alternative if (1) dust, erosion, and sediment control measures are instituted during excavation and stockpiling activities; (2) on-site workers use appropriate levels of personal protection during construction, excavation, stockpiling, and incineration operations; and (3) ambient air and incinerator emission control equipment is carefully monitored to evaluate compliance with NAAQS and MAPCC regulations.

Following treatment, inorganics in the soil would be more concentrated in the ash and may require additional treatment. Air treatment media would need to be properly handled and potentially disposed of as hazardous waste.

The estimated time required to complete the on-site treatment activities, associated with Alternative SC-11, is generally determined and controlled by the specific incineration unit feed rate, hours of operation, and amount of unit downtime. However, it is anticipated that one year would be necessary to treat the 13,200 cy (18,000 tons) of contaminated soil once the incinerator is fully operational. This scenario conservatively assumes the incineration unit to possess a feed rate of 5 tons per hour, and to be operated 24 hours per day, three days per week. This estimate is intended for use in evaluating remedial alternatives, and will be further refined during the design phase. Therefore, the time anticipated to achieve remedial response objectives for the Kysor site is approximately 22 months.

Four Winns Site. The anticipated time necessary to complete the initial on-site activities prior to full incinerator operation is the same as the Kysor

site: approximately 10 to 12 months. The estimated time required to complete the on-site treatment activities associated with Alternative SC-11 is governed by the same variables discussed for the Kysor site. However, it has been calculated that 61 weeks would be required to incinerate the 16,100 cy (22,000 tons) of excavated soil. This calculation assumes the mobile incinerator has a 5-ton-per-hour feed rate, and is operated 24 hours per day during each three days of operation per work week. As at the Kysor site, this estimate is intended for use in evaluating the remedial alternatives, and will be more accurately determined during the design phase of the project. Therefore, the estimated time calculated to achieve the remedial response objectives for the Four Winns site is 24 months.

8.7.6 Long-term Effectiveness

The incineration of soils at the Kysor and Four Winns sites would be effective in destroying VOC contaminants to eliminate any possible source of pollution to the groundwater from these site soils. The organics are permanently destroyed during this process, thereby reducing any residual risks due to soil contamination.

Assuming that all residues from incineration and the ash were properly disposed of, no adverse public health or environmental impacts are predicted to occur due to this alternative.

The excavation and on-site incineration of contaminated soils would provide engineering controls that would destroy VOC contamination in soils and eliminate future risk to public health and the environment. Alternative SC-11 would offer a remedy that is a highly reliable, permanent remediation of VOC contamination of the soil, and would require only limited monitoring to verify attainment of the soil TCLs. Alternative SC-11 would not require long-term management of the site because the contaminants would be destroyed.

8.7.7 Implementability

Implementation of on-site incineration involves both complete excavation of the contaminated soils and mobilization, construction, and operation of the mobile on-site incinerator. It is estimated that 10 to 12 months would be required to select a mobile incinerator contractor; make necessary site preparations and install the on-site mobile incinerator; conduct start-up testing to evaluate the performance and demonstrate the effectiveness of the incinerator; and obtain air discharge permits from the MAPCC.

Excavation activities do not appear to present any extraordinary problems at either the Kysor or Four Winns site. Both sites are accessible for heavy excavating equipment. No surface clearing or movement of utilities is necessary. The side walls may require some type of stabilization, such as sheet pile, during excavation to avoid cave-in, thereby avoiding the increase of the volume of contaminated soils.

Soils removed during excavation are considered contaminated. The chemicals present will begin to volatilize when exposed to the surface, which could pose a significant threat to workers and adjacent receptors; therefore, proper

safety measures should be taken. The Kysor and Four Winns sites would require little preparation for on-site incineration. Both have large flat open areas available for construction purposes.

Trailer-mounted mobile rotary kiln incinerators can be fully operational within 24 hours of arrival on-site. However, the low feed rates limit these units to small low-volume incineration projects. Wastestreams the size of those at the Kysor or Four Winns site could be handled more efficiently with a larger transportable unit. These units are transported on several semi-trailers and are erected on-site. Feed rates of the larger units approach 100 tons per day (4 to 5 tons per hour), while smaller mobile units are limited to 20 tons per day. A major limitation of the large transportable incinerators is the high cost of mobilization, installation, and start up. Refractory insulation is often damaged during transport, requiring relining. Foundations must be poured and the unit erected upon arrival. Alternative fuel sources must also be available for wastes having low BTU values and emergency operations.

Test burns are necessary during the start-up phase to determine operational parameters. The off-gas from an incinerator must be discharged to an emissions control system, typically a wet gas scrubber, to prevent the emissions of inorganic acids. The effluent from the gas scrubber consists primarily of water and dissolved hydrochloric acid. This wastewater must either be treated on-site for reuse or transported off-site for treatment. Wastewater flow rates can be as high as 10 gpm, depending on the type of incinerator.

The residue from the combustion chamber must also be monitored. Because incinerators do not destruct inorganics (metals), high levels of inorganics in the wastestream may result in a residue that is a hazardous waste. The residue or ash must be collected and analyzed. Nonhazardous ash may be stockpiled, delisted, and used as fill material. Hazardous ash must be transported off-site for treatment or disposal, unless an on-site alternative is available.

During implementation of this alternative, several variables that must be controlled: (1) proper temperatures must be maintained in the combustion chambers; (2) auxiliary fuels and combustion gases must be regulated to assure complete combustion; (3) emission control systems must maintain an acceptable level of particulates in the stack gas; and (4) wastewater must be treated to neutralize inorganic acids.

On-site incineration often presents a material handling challenge. Most incinerators will accept a limited variation in waste feed. The waste may need to be introduced to the combustion chamber in drums, as a high-solids sludge or a low-solids liquid. There may be a minimum or maximum BTU value. Any variations in chemical composition may affect operation of the emissions control system. Preparing the waste for incineration can be a time-consuming operation.

Several contractors offer on-site incineration services that would be appropriate for either the Kysor or Four Winns site. Various scrubber systems are available; on-site water treatment capabilities are also available. A treatability study should be conducted to determine the optimum system configuration.

8.7.8 Costs

In this subsection, capital and annual costs are presented for the On-site Incineration Alternative. In addition, the net present worth of this alternative is presented, assuming it would be implemented at both sites. These costs are listed in Table 8-11. Site-specific costs are presented in Appendix D. The capital and annual costs and the net present worth of the alternative were derived using the assumption presented in Section 6.3.

Capital Costs. The capital cost of the On-site Incineration Alternative is \$28,755,000.

Annual Costs. Because there would not be any hazardous substances, pollutants, or contaminants remaining on-site, a five-year site review would not be required under SARA Section 121. In addition, because this alternative would be implemented in approximately eight to 10 months, there would be no associated annual costs.

Present Worth Analysis. The present worth of this alternative would equal the capital costs, because no annual costs are associated with On-site Incineration. Therefore, present worth is \$28,755,000.

8.7.9 Summary

The detailed analysis of Alternative SC-11 is summarized in Table 8-12.

TABLE 8-11
SC-11 ON-SITE INCINERATION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION		TOTAL
-----		-----
CAPITAL COSTS		
	ON-SITE INCINERATION	\$20,913,00
	ENGINEERING @ 10%	2,091,00
	CONTINGENCY @ 25%	5,751,00
1.	TOTAL CAPITAL COSTS	----- \$28,755,00
ANNUAL COSTS		
	ON-SITE INCINERATION	\$
	TOTAL ANNUAL COSTS	----- \$
TOTAL PRESENT WORTH OF THIS ALTERNATIVE		\$28,755,00

TABLE 8-12
SUMMARY OF DETAILED ANALYSIS
SC-11: EXCAVATION, ON-SITE INCINERATION
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
1. Compliance with ARARs	- o RCRA and Michigan Standards for Incinerators would be met. - o RCRA and Michigan closure and post-closure requirements would be met. - o RCRA and Michigan regulations for storing, handling, transporting, and disposing of hazardous waste must be met. - o Applicable Michigan air regulations would be met. - o RCRA Land Ban requirements would be met.
2. Protection of Public Health and the Environment	- o SC-11 would achieve soil TCLs. - o SC-11 would reduce baseline public health risks associated with contaminated soil and the potential for continued leaching of VOCs from soil to groundwater. - o Because waste handling activities would be extensive, the potential for adverse environmental effects is increased. - o SC-11 would result in an improvement to the natural environment over the current site conditions described in the baseline risk assessment. - o SC-11 would provide a permanent remedy.
3. Reduction of Toxicity, Mobility, or Volume	o The toxicity, mobility, and volume of VOCs in the soils would be permanently reduced.
4. Short-term Effectiveness	o Excavation would pose potential risks to workers through dermal contact or inhalation of dusts or vapors.

TABLE 8-12
(continued)
SUMMARY OF DETAILED ANALYSIS
SC-11: EXCAVATION, ON-SITE INCINERATION
CADILLAC AREA FEASIBILITY STUDY

CRITERIA	ASSESSMENT
	- o Incineration process would require off-gas treatment to reduce potential risk to the community due to the release of products of incomplete combustion. - o Time to achieve full protection is expected to be 17 months for Kysor soils and 19 months for Four Winns soils.
5. Long-term Effectiveness	- o Long-term risk to public health due to exposure to groundwater contaminated by leaching from the remaining soils would not be significant. - o SC-11 provides a permanent remedy for the soil contamination. - o Long-term management would not be required. - o Limited monitoring would be required to verify achievement of TCLs.
6. Implementability	- o SC-11 is readily implementable for the Cadillac site. - o Equipment is readily available. - o SC-11 would have the greatest equipment problems of all the alternatives. - o A test burn and treatability studies would be performed to determine optimum configuration for system operation.
7. Costs	- o Capital Costs = \$28,755,000 - o Annual Costs = \$0 - o Present Worth = \$28,755,000

TABLE 8-12
(continued)
SUMMARY OF DETAILED ANALYSIS
SC-11: EXCAVATION, ON-SITE INCINERATION

CADILLAC AREA FEASIBILITY STUDY

NOTES:

These evaluation criteria to be used in the remedy selection process were adapted from USEPA OSWER Directive No. 9355.0-21, "Additional Interim Guidance for FY'87 Records of Decision"; July 24, 1987. The criteria generally correspond to the following statutory factors in CERCLA, as amended, Section 121(b)(1)(A through G):

- A = The long-term uncertainties associated with land disposal.
- B = The goals, objectives, and requirements of the SWDA.
- C = The persistence, toxicity, mobility, and propensity to bioaccumulate of the hazardous substances and their constituents.
- D = Short- and long-term potential for adverse health effects from human exposure.
- E = Long-term monitoring costs.
- F = The potential for future remedial action costs if the alternative remedial action in question were to fail.
- G = The potential threat to public health and the environment associated with excavation, transportation, and redispisal, or containment.

9.0 COMPARISON OF REMEDIAL ALTERNATIVES FOR THE CADILLAC SITE

In this section, the four groundwater alternatives and the alternate water supply are compared on the basis of public health risk reduction, attainment of ARARs, costs, and permanence of the remedy. The six soil alternatives are also compared on these factors.

9.1 COMPARISON OF GROUNDWATER REMEDIAL ALTERNATIVES

Table 9-1 presents the comparison of the groundwater remedial alternatives. As shown, the No-action Alternative has the lowest net worth, but does not provide a permanent remedy, nor does it reduce public health risks posed by the domestic use and drinking of contaminated groundwater. Drinking water standards would not be met. The three treatment alternatives would all result in a permanent remedy and would reduce public health risks posed by domestic use and drinking of contaminated groundwater. Drinking water standards and discharge requirements would be met. The major differences between these alternatives are the present worths and the quantity and type of secondary wastestreams generated from the different processes. The alternate water supply would reduce the public health risks, because the source of drinking water would change. However, complete protection from risk would depend on whether the alternate water supply is combined with a treatment alternative, and whether the extraction system can effectively remove the contaminated groundwater. Table 9-2 is the sensitivity analysis of the effect of variations in the extraction system's efficiency and treatment duration of the present worth of each groundwater treatment remedial alternative.

9.2 SUMMARY OF SOIL REMEDIAL ALTERNATIVES

Table 9-3 presents the comparison of the soil remedial alternatives. The No-action Alternative has the lowest present worth, but would not prevent VOCs in the soil from continual leaching to the groundwater, nor would TCLs be met. The one containment alternative would prevent continued leaching of VOCs but would not remediate soils to TCLs, nor would it provide a permanent remedy. Under the four treatment alternatives, the toxicity, mobility, and volume of the contaminants in the soil would be permanently reduced. Differences in present worth, quantity, and type of secondary wastestreams; potential risk to site workers; and ARARs are listed in Table 9-3.

TABLE 9-1
COMPARISON OF GROUNDWATER ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	PUBLIC HEALTH AND THE ENVIRONMENT	ARARS	COST	PERMANENT REMEDIES
GW-1: No-Action	Public health risks due to the domestic use and drinking of contaminated groundwater would not be reduced.	Drinking Water Standards would not be met.	Total PW = \$727,000	No; remedial activities would not be implemented.
GW-2: Extraction, Hex-chrome Treatment, UV/oxidation, Discharge	Public health risks due to the domestic use and drinking of contaminated groundwater would be reduced.	Drinking water standards would be met.	Total PW = \$38,380,000 (using Ion Exchange)	Yes; volume of contaminants in groundwater would be reduced by extraction. Toxicity of contaminated groundwater would be reduced through treatment.
	Secondary wastestream would be produced by the hexavalent chromium treatment.	Discharge requirements would be met.	Total PW = \$37,453,000 (using Carbon Adsorption)	
	No secondary wastestream would be generated by the UV/Oxidation process.	The treatment systems would be designed and operated in compliance with ARARs.		
GW-3(a): Extraction, Hex-chrome Treatment, Air Stripping with Vapor-Phase Carbon Adsorption, and Discharge	Public health risks due to the domestic use and drinking of contaminated groundwater would be reduced.	Drinking water standards would be met.	Total PW = \$15,970,000 (using Ion Exchange)	Yes; volume of contaminants in groundwater would be reduced by extraction. Toxicity of contaminated groundwater would be reduced through treatment.
	Secondary wastestreams would be produced by the hexavalent chromium treatment.	Discharge requirements would be met.	Total PW = \$15,043,000 (using Carbon Adsorption)	
	95 percent of the VOC emissions from the air stripper would be removed by the vapor phase carbon adsorption treatment.	The treatment systems would be designed and operated in compliance with ARARs.		
	In-situ process avoids need for off-site disposal of spent carbon.			
GW-3(b): Extraction, Hex-chrome Treatment, Air Stripping with Liquid- and Vapor-Phase Carbon Adsorption	Public health risks due to the domestic use and drinking of contaminated groundwater would be reduced.	Drinking water standards would be met.	Total PW = \$21,040,000 (using Ion Exchange)	Yes; volume of contaminants in groundwater would be reduced by extraction. Toxicity of contaminated groundwater would be reduced through treatment.
	Secondary wastestreams would be produced by the hexavalent chromium treatment.	Discharge requirements would be met.	Total PW = \$20,113,000 (using Carbon Adsorption)	
	95 percent of the VOC emissions from the air stripper would be removed by the vapor phase carbon adsorption treatment.	The treatment systems would be designed and operated in compliance with ARARs.		
	Approximately 225,000 pounds per year of spent carbon would be produced for off-site disposal.			

TABLE 9-1
(continued)
COMPARISON OF GROUNDWATER ALTERNATIVES (

CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	PUBLIC HEALTH AND THE ENVIRONMENT	ARARS	COST	PERMANENT REMEDIES
GW-4(a): Extraction, Hex-chrome Treatment, Steam Stripping, Discharge	Public health risks due to the domestic use and drinking of contaminated groundwater would be reduced. Secondary wastestream would be produced by the hexavalent chromium treatment. Steam stripping would produce approximately 231 gallons per day of a concentrated organic condensate requiring disposal.	Drinking water standards would be met. Discharge requirements would be met. The treatment systems would be designed and operated in compliance with ARARs.	Total PW = \$28,478,000 (using Ion Exchange) Total PW = \$27,551,000 (using Carbon Adsorption)	Yes; volume of contaminants in groundwater would be reduced by extraction. Toxicity of contaminated groundwater would be reduced through treatment.
GW-4(b): Extraction, Hex-chrome Treatment, Steam Stripping with Vapor Recompression, and Discharge	Public health risks due to the domestic use and drinking of contaminated groundwater would be reduced. Secondary wastestreams would be produced by the hexavalent chromium treatment. Steam stripping would produce approximately 231 gallons per day of a concentrated organic condensate requiring off-site disposal.	Drinking water standards would be met. Discharge requirements would be met. The treatment systems would be designed and operated in compliance with ARARs.	Total PW = \$22,205,000 (using Ion Exchange) Total PW = \$21,278,000 (using Carbon Adsorption)	Yes; volume of contaminants in groundwater would be reduced by extraction. Toxicity of contaminated groundwater would be reduced through treatment.
AC-1: Alternate Water Supply	Public health risks due to the domestic use and drinking of contaminated groundwater would be reduced, because the source has changed. Risk would continue to exist from drinking contaminated groundwater through private wells, unless AC-1 is considered with a treatment alternative, and the extraction system can effectively remove the contaminated groundwater.	Drinking water standards would be met unless conditioned with a treatment alternative.	Total PW = \$5,019,000	No; groundwater would remain contaminated unless remedial activities are implemented. Alternate water supply would need to be protected from contamination.

TABLE 9-2
EFFECT OF VARIATIONS IN THE EXTRACTION
SYSTEM'S EFFICIENCY OF THE NET PRESENT
WORTH OF THE GROUNDWATER TREATMENT ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	NET PRESENT WORTH AT 90% EFFICIENCY (millions)	NET PRESENT WORTH AT 70% EFFICIENCY (millions)	COST DIFFERENCE ¹ (percent)
GW-2: UV/Oxidation			
with Ion Exchange	\$38.473	\$41.970	+9.1
with Carbon Adsorption	37.546	40.427	+7.7
GW-3(a): Air Stripping with Vapor Phase Carbon Adsorption			
with Ion Exchange	16.051	17.906	+11.6
with Carbon Adsorption	15.124	16.393	+8.4
GW-3(b): Air Stripping with Vapor and Liquid Phase Carbon Adsorption			
with Ion Exchange	21.133	23.488	+11.1
with Carbon Adsorption	20.206	21.965	+8.7
GW-4(a): Steam Stripping			
with Ion Exchange	28.571	31.766	+11.2
with Carbon Adsorption	27.644	30.223	+9.3
GW-4(b): Steam Stripping with Vapor Recompression			
with Ion Exchange	22.298	24.780	+11.1
with Carbon Adsorption	21.371	23.237	+8.7

NOTE:

¹ Cost difference is based on additional annual O&M costs, equipment replacement costs, and interest rates accrued over a longer time period due to variations in the extraction system's efficiency.

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TABLE 9-3
SUMMARY OF ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	PUBLIC HEALTH AND THE ENVIRONMENT	ARARS	COST	PERMANENT REMEDIES
SC-1: No-Action	Concentrations of VOCs in the soils would remain above TCLs. Would not prevent the VOCs in the soils from continual leaching to the groundwater. Risks to humans through direct contact with Four Winn's soils would be reduced via fencing and land restrictions.	The no-action alternative would not be consistent with the goal and intent of CERCLA. RCRA groundwater monitoring requirements would be met.	Total PW = \$326,000	No; remedial activities would not be implemented.
SC-2: Surficial Capping Subsurface Containment Wall	Risks to public health due to the leaching of VOCs in soil to groundwater would be reduced. Concentrations of VOCs in the soils would remain above TCLs. On-site workers could be exposed to fugitive dust, air emissions, and contaminated soils and groundwater.	RCRA closure and post-closure regulations and landfill closure requirements would be met. Groundwater monitoring would be consistent with RCRA Corrective Action monitoring.	Total PW = \$2,347,000	No; volume of contaminants would not be reduced. Toxicity of contaminated soils would not be reduced. The surficial cap and subsurface containment walls would require replacement after 20 to 30 years.
SC-6: Extraction, Low-temperature Thermal Stripping	Risks to public health due to the leaching of VOCs in soil to groundwater would be reduced. Soil TCLs would be met. This alternative would result in an improvement to the natural environment over current site conditions. Excavation would pose potential risks to on-site workers through dermal contact or inhalation of dust and vapors. Thermal stripping process would release products of incomplete combustion requiring off gas treatment. Air emissions control will be required.	The management of hazardous waste, and the design and operation of treatment process would be in compliance with ARARs. Applicable air regulations would be met. RCRA land ban requirements would be met.	Total PW = \$17,214,000	Yes; the mobility and volume of VOCs in the soils would be permanently reduced. The toxicity of the contaminated soils would be permanently reduced.

TABLE 9-3
(continued)
SUMMARY OF SOIL ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	PUBLIC HEALTH AND THE ENVIRONMENT	ARARS	COST	PERMANENT REMEDIES
SC-7: In-Situ Soil Flushing	Risks to public health due to the leachability of VOCs in soil to groundwater would be reduced. Soil TCLs would be met. Potential of further groundwater contamination if all flushate cannot be collected. SC-7 would present lowest risk to on-site workers during installation and implementation. This alternative would result in an improvement to the natural environment over current site conditions.	Appropriate RCRA requirements for treatment and groundwater monitoring would be met. Michigan groundwater discharge permit requirements would be met, if applicable.	Total PW = \$455,000	Yes; the volume and mobility of VOCs in the soils would be permanently reduced. The toxicity of the contaminated groundwater would be reduced by implementing one of the groundwater treatments.
SC-9: In-situ Vacuum Extraction	Risks to public health due to the leaching of VOCs in the soil to groundwater would be reduced. Soil TCLs would be met. This alternative would result in an improvement to the natural environment over current site conditions. Drilling of extraction wells would pose potential risks to workers through dermal contact or inhalation of dust or vapors. Air emissions control would be required.	The management of hazardous waste and the operation of the treatment process would be in compliance with ARARs. Applicable air regulations would be met.	Total PW = \$1,854,000	Yes; the mobility and volume of VOCs in the soil would be permanently reduced. The toxicity of the contaminated soils would be permanently reduced.

TABLE 9-3
(continued)
SUMMARY OF SOIL ALTERNATIVES
CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	PUBLIC HEALTH AND THE ENVIRONMENT	ARARS	COST	PERMANENT REMEDIES
SC-11: Excavation, On-site Incineration	Risks to public health due to the leaching of VOCs in the soil to groundwater would be reduced. Soil TCLs would be met. This alternative would result in an improvement to the natural environment over the current conditions. Incineration process would require off-gas treatment to reduce release of product of incomplete combustion. Excavation would pose potential risks to workers through dermal contact or inhalation of dust or vapors. Incineration would provide the most effective treatment because it has a demonstrated destruction efficiency of 99.9999%.	RCRA and Michigan standards for incinerators would be met. The management of hazardous waste and the design and operation of treatment process would be in compliance with ARARs. Applicable air regulations would be met. RCRA Land Ban requirements would be met.	Total PW = \$28,755,000	Yes; the mobility and volume of VOCs in the soils would be permanently reduced. The toxicity of the contaminated soils would be permanently reduced.

GLOSSARY OF ACRONYMS

ARARs	Applicable or Relevant and Appropriate Requirements
AWQC	Ambient Water Quality Criteria
BACT	best available control technology
BAT	best available technology
BDATs	best demonstrated available treatment technologies
BPJ	best professional judgment
CAA	Clean Air Act
CAG	Carcinogen Assessment Group
CDI	Chronic Daily Intake
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (the Superfund statute)
cfm	cubic feet per minute
CFR	Code of Federal Regulations
CLP	Contract Laboratory Program
CRDL	Contract Required Detection Level
CWA	Clean Water Act
cy	cubic yards
DCA	dichloroethane
DCE	dichloroethylene
DNAPL	dense non-aqueous phase liquid
DOT	Department of Transportation
DRE	destruction and removal efficiency
FFS	Focused Feasibility Study
FRP	fiberglass-reinforced plastic
FS	Feasibility Study
GC	gas chromatograph
gpm	gallons per minute
GWPS	Groundwater Protection Strategy (USEPA)
HEA	Health Effects Assessment
HSWA	Hazardous and Solid Waste Amendments
kg/hr	kilograms per hour
MAPCC	Michigan Air Pollution Control Commission
MC	methylene chloride
MCL	Maximum Contaminant Level
MCLG	Maximum Contaminant Level Goal
MDNR	Michigan Department of Natural Resources
MDPH	Michigan Department of Public Health
MDWS	Michigan Division of Water Supply
mg/l	milligrams per liter
MGWQ	Michigan Groundwater Quality Rules

NAAQS	National Ambient Air Quality Standards
NCP	National Contingency Plan
NESHAPs	National Emission Standards for Hazardous Air Pollutants
NPDES	National Pollutant Discharge Elimination System
NPL	National Priority List
OLM	Organic Leaching Model
O&M	operation and maintenance
OSHA	Occupational Safety and Health Administration
OSWER	Office of Solid Waste and Emergency Response
PCBs	polychlorinated biphenyls
PCE	tetrachloroethylene
POHC	Principal Organic Hazardous Constituent
POTWs	Publicly Owned Treatment Works
ppb	parts per billion
ppm	parts per million
PVC	polyvinyl chloride
RA	Risk Assessment
RCRA	Resource Conservation and Recovery Act
RI	Remedial Investigation
ROD	Record of Decision
SARA	Superfund Amendments and Reauthorization Act of 1986
SDWA	Safe Drinking Water Act
SVOC	semivolatile organic compound
t-1,2-DCE	trans-1,2-dichloroethylene
TBC	To Be Considered
TCA	trichloroethane
TCE	trichloroethylene
TCL	target clean-up level
TCLP	Toxicity Characteristic Leaching Procedure
TLV	threshold limit value
USEPA	U.S. Environmental Protection Agency
UV	ultraviolet
VOA	volatile organic analyte
VOC	volatile organic compound

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APPENDIX A
CHEMICAL-SPECIFIC ARARS

APPENDIX A
CHEMICAL-SPECIFIC ARARS
CADILLAC AREA FEASIBILITY STUDY

SITE FEATURE	REQUIREMENT	REQUIREMENT SYNOPSIS	CONSIDERATION IN THE FS
GROUNDWATER			
Federal Applicable Requirements	Safe Drinking Water Act (SDWA) - Maximum Contaminant Levels (MCLs) (40 CFR 141.11 - 141.16)	MCLs have been promulgated for several common organic and inorganic contaminants. These levels regulate the concentration of contaminants in public drinking water supplies, and are considered applicable for groundwater aquifers used for drinking water.	When the risks to public health due to consumption of groundwater are assessed, concentrations of contaminants should be compared to their MCLs.
Federal Relevant and Appropriate Requirements	SDWA - Maximum Contaminant Level Goals (MCLGs)	MCLGs are health-based criteria that are relevant and appropriate for drinking water sources, as a result of SARA, when extraordinary risks exist. These goals are available for several organic and inorganic contaminants.	Groundwater concentrations of noncarcinogenic contaminants should be compared to their MCLGs when extraordinary risks exist.
	Federal Ambient Water Quality Criteria (AWQC)	Federal AWQC are health-based criteria that have been developed for 95 carcinogenic and noncarcinogenic compounds.	AWQC should be used in characterizing public health risks due to contaminant concentrations in groundwater.
State Applicable Requirements	Michigan (MWRL), Act 245, R 323 Part 22, Groundwater Quality Rules	These rules establish allowable discharge levels to protect Michigan groundwater quality in all usable aquifers as potential water supplies.	Contaminated groundwater must be treated to nondetectable levels before being recharged to groundwater.
Federal Advisories and Guidance to be Considered	Health Advisories (USEPA Office of Drinking Water)	Health advisories are estimates of risks due to consumption of contaminated drinking water; they consider noncarcinogenic effects only.	Health advisories should be considered for contaminants in groundwater that is used for drinking water.
	USEPA Risk Reference Doses (RfDs)	RfDs are dose levels developed by USEPA for noncarcinogenic effects.	USEPA RfDs are used to characterize risks due to exposure to contaminants in groundwater, as well as other media.
	USEPA Carcinogen Assessment Group Potency Factors	Potency Factors are developed by USEPA from Health Effects Assessments (HEA) or evaluations by the Carcinogen Assessment Group (CAG).	USEPA Carcinogenic Potency Factors are used to complete the individual incremental cancer risk resulting from exposure to contaminants.

APPENDIX B

JUSTIFICATION FOR ELIMINATING RECHARGE TO GROUNDWATER IN THE SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

Several technologies for disposing of the treated groundwater have been considered, including discharging to the Clam River, discharging to a POTW, or recharging the groundwater back to the groundwater systems. The two methods of recharge considered for the Cadillac site involve using deep injection wells or discharging the treated water to a surface water retention pond and allowing the water to infiltrate to the groundwater aquifer. As discussed in Section 3.0, the screening consists of evaluating technologies to ensure that they are suitable for the location and conditions at the Cadillac site, and applicable to the contaminants and environmental media. Therefore, the impact of these two methods of recharge on the existing groundwater aquifer system has been evaluated through computer modeling, and considers site geology and hydrogeology. The level of treatment required for the contaminants discharged to the particular environmental media is also considered. Based on results of the computer modeling and the stringency of the effluent standards for discharging to groundwater, recharge was eliminated as a remedial technology. This appendix summarizes results of the computer modeling.

INJECTION WELLS

A potential advantage of injecting the treated groundwater into the aquifer is that the injected water could be used to increase the hydraulic gradient, and therefore the groundwater velocity, by placing an injection well upgradient of the extraction well and near the trailing edge of the plume. The injected water could potentially create a groundwater mound, increasing the horizontal hydraulic gradient and allowing the extraction well to pump at a faster rate without increasing the capture zone width of this well. The effects of injection at the Cadillac site were evaluated through the use of computer groundwater flow models.

Several scenarios for injection of water were investigated for extraction of the Rexair plume and the Kysor shallow aquifer plume. These plumes are relatively isolated from the other extraction system and may be modeled independently.

In both instances, an injection well was placed directly upgradient of the extraction well(s) and near the trailing edge of the plume. The intent was to place the injection well within the capture zone of the extraction well so that the water would be "recycled" and the extraction rate could be increased as a consequence of the greater hydraulic gradient resulting from the groundwater mounding at the injection well. Pumping scenarios included (1) injection at the same rate as extraction at each plume, and (2) doubling the extraction rate over the injection rate.

The primary purpose of the modeling was to determine if the extraction rate could be increased in these specific plumes by injecting treated groundwater back into the aquifer, thereby decreasing the cleanup time. However, the computer modeling revealed that the relatively high hydraulic conductivity and transmissivity of the Cadillac Industrial Park aquifer allows the injected groundwater to disperse rapidly so that a significant groundwater mound is not created and the hydraulic gradients are not substantially increased. Therefore, the injection well will have little effect on groundwater flow in the

vicinity of the extraction well. As a result, increasing the extraction rate increases the capture zone width so that not only contaminated groundwater will be pumped, but clean groundwater also will be pumped and unnecessarily treated.

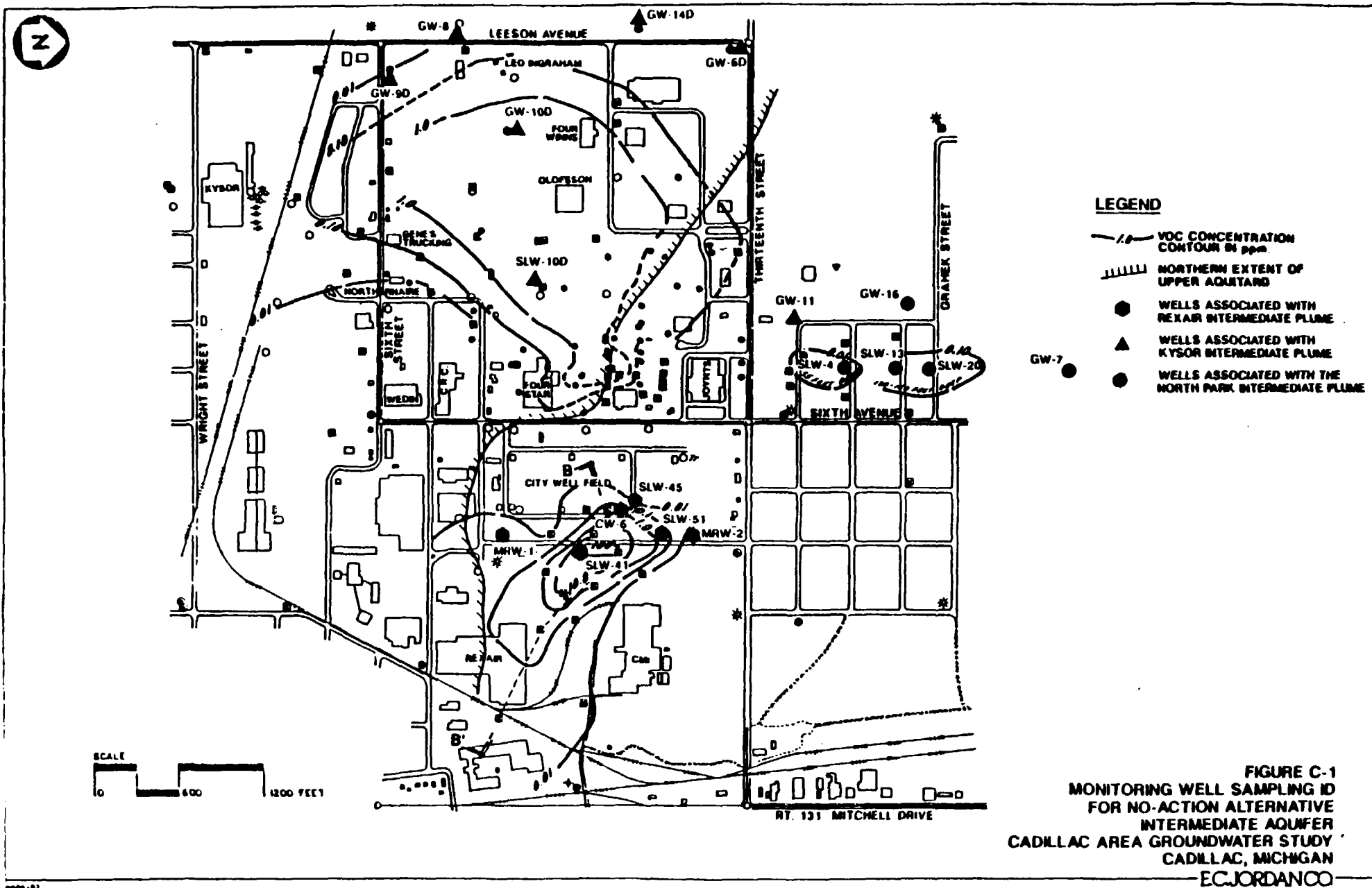
In summary, due to the Cadillac site's hydrogeologic conditions, recharging the groundwater by injection wells would not achieve the benefit of increasing extraction rate. In fact, it could result in the pumping and treatment of uncontaminated groundwater. In addition, recharging the groundwater will require treatment of the contaminants in the groundwater to nondetect levels, while discharging to surface water requires protective but less stringent discharge levels.

INFILTRATION

MDNR investigated by computer modeling the feasibility of discharging the treated water to a surface water retention pond located at the southeastern corner of Thirteenth Street and Leeson Avenue. Infiltration rates of the surface water to the soils was estimated to be on the order of 300 gpm. Modeling suggested that the infiltration may cause existing groundwater flow directions and plume positions to shift and result in adverse effects on designed extraction systems (K. Kincara, personal communication, June 1988).

APPENDIX C

MONITORING WELL LOCATIONS
FOR THE NO-ACTION ALTERNATIVE



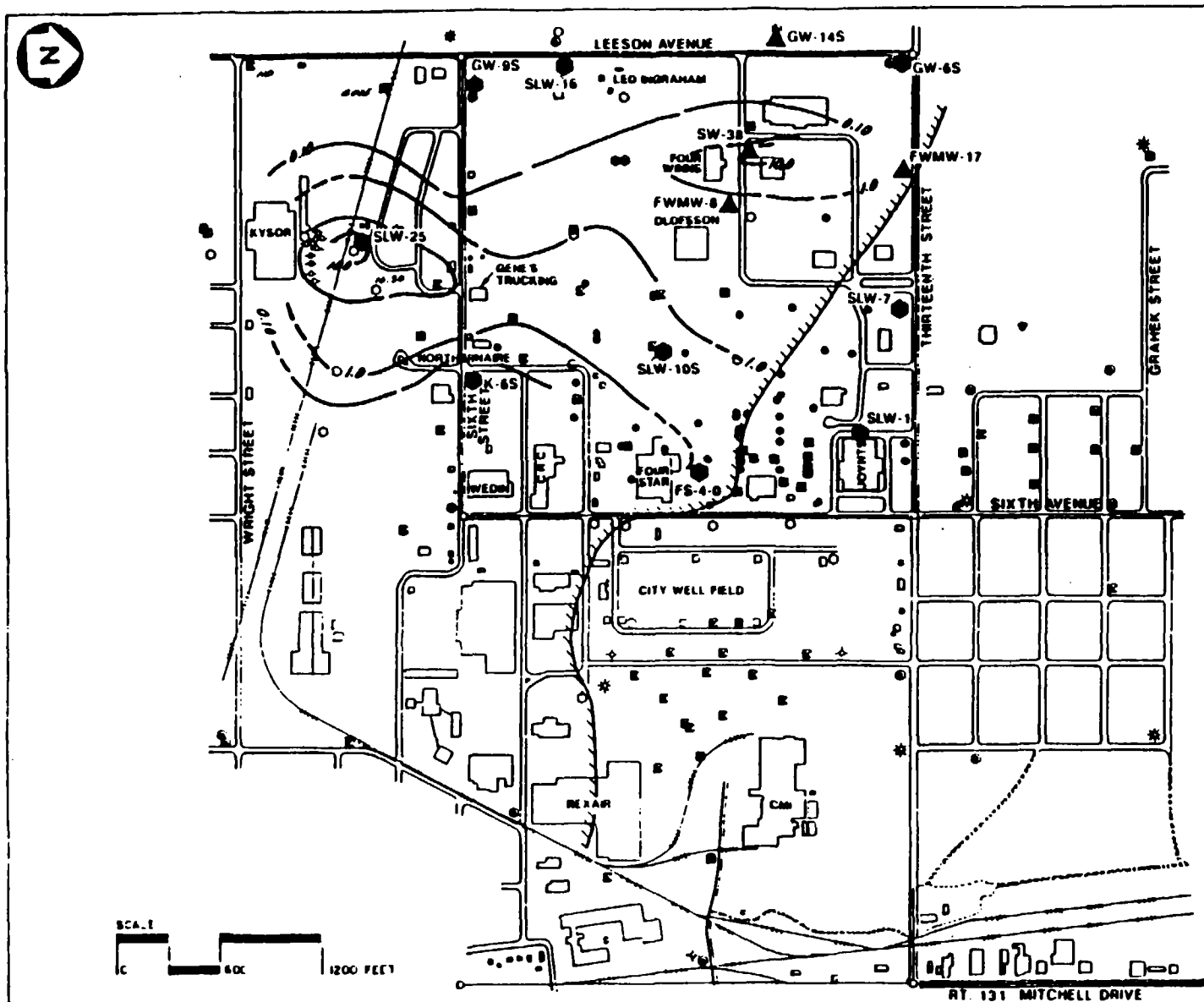


FIGURE C-2
MONITORING WELL SAMPLING ID
FOR NO-ACTION ALTERNATIVE
SHALLOW AQUIFER
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN
EC JORDAN CO

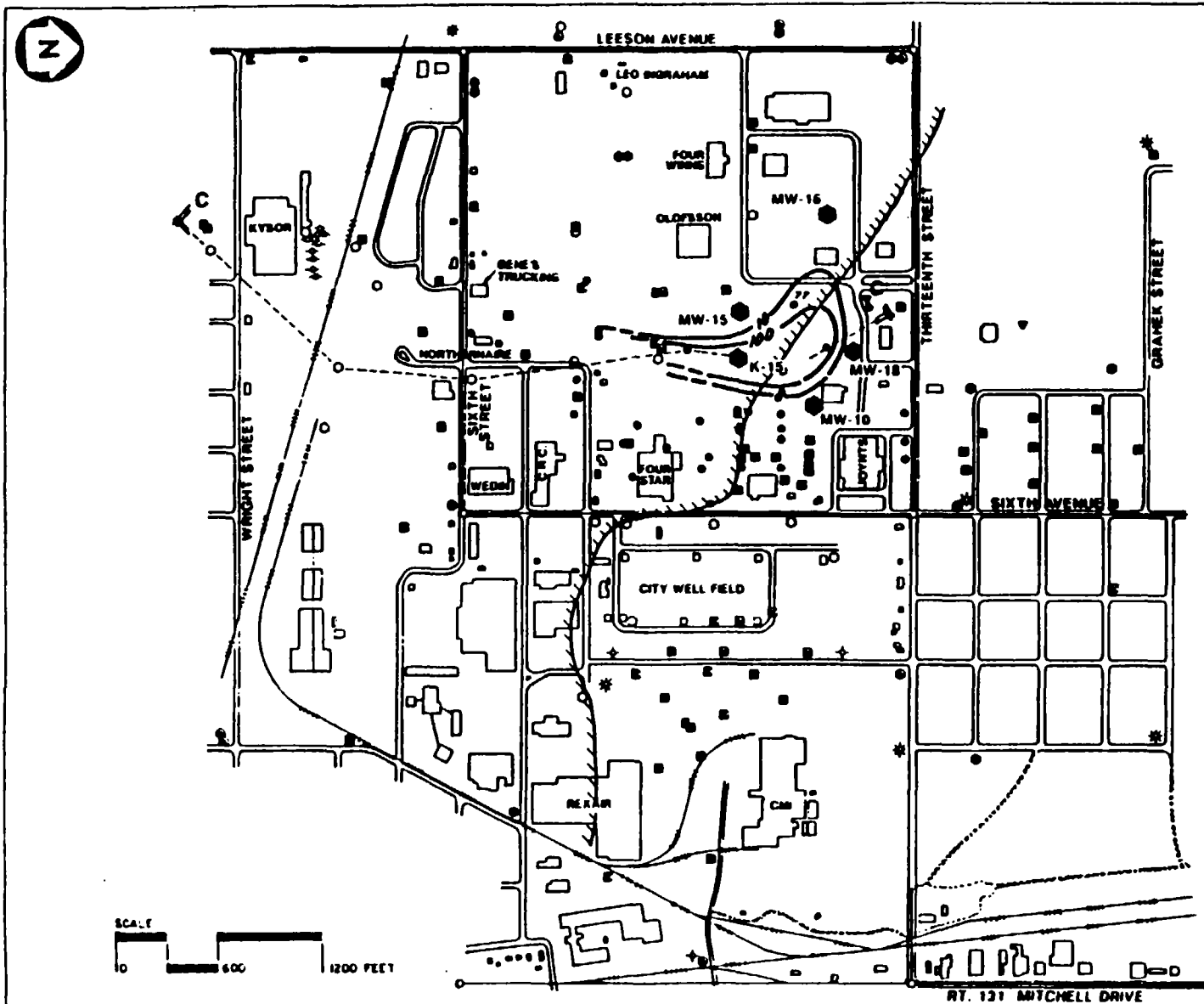


FIGURE C-3
 MONITORING WELL SAMPLING ID
 FOR NO-ACTION ALTERNATIVE
 HEX-CHROME
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN
 EC.JORDANCO

APPENDIX D
DETAILED COST ESTIMATES

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APPENDIX D
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

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TABLE D-1
 FIVE YEAR SITE REVIEW
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				\$
ANNUAL COSTS				
REVIEW OF EXISTING DATA	80	MHR	60.00	\$4,80
REVIEW OF AVAILABLE TREATMENT TECHNOLOGIES	40	MHR	60.00	2,40
REPORT PREPARATION AND REVIEW	80	MHR	60.00	4,80
UNDEVELOPED DESIGN DETAILS				3,00
TOTAL COST EVERY FIVE YEARS				\$15,00
ANNUALIZED COST OF FIVE YEAR SITE REVIEW				\$3,00

TABLE D-2
MONITORING
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				\$
ANNUAL COSTS				
MONITORING FOR GW-1				
LABOR	1	LS	10,167.00	\$10,167.00
VEHICLES	1	LS	779.00	779.00
AIR FARE	4	RT	700.00	2,800.00
MEALS & LODGING	1	LS	1,516.00	1,516.00
OTHER DIRECT COSTS	1	LS	975.00	975.00
SAMPLE COLLECTION & ANALYSIS EQUIP	1	LS	1,127.00	1,127.00
HEALTH & SAFETY EQUIPMENT	1	LS	540.00	540.00
EXPENDABLES, MISC, CONTAINERS	1	LS	1,488.00	1,488.00
ANALYSIS	1	LS	8,393.00	8,393.00
UNDEVELOPED DESIGN DETAILS				7,210.00
TOTAL PER MONITORING EVENT				\$35,000.00
TOTAL ANNUAL MONITORING - YEAR 1 (4 EVENTS)				\$140,000.00
TOTAL ANNUAL MONITORING - YEAR 2 - 64 (1 EVENT)				\$35,000.00
MONITORING FOR GW-2, GW-3, GW-4				
LABOR	1	LS	9,099.00	\$9,099.00
VEHICLES	1	LS	689.00	689.00
AIR FARE	4	RT	700.00	2,800.00
MEALS & LODGING	1	LS	1,328.00	1,328.00
OTHER DIRECT COSTS	1	LS	975.00	975.00
SAMPLE COLLECTION & ANALYSIS EQUIP	1	LS	988.00	988.00
HEALTH & SAFETY EQUIPMENT	1	LS	473.00	473.00
EXPENDABLES, MISC, CONTAINERS	1	LS	1,488.00	1,488.00
ANALYSIS	1	LS	8,154.00	8,154.00
UNDEVELOPED DESIGN DETAILS				6,000.00
TOTAL PER MONITORING EVENT				\$32,000.00
TOTAL ANNUAL MONITORING - YEAR 1 (4 EVENTS)				\$128,000.00
TOTAL ANNUAL MONITORING - YEAR 2 - 64 (1 EVENT)				\$32,000.00

TABLE D-3
EXTRACTION SYSTEM
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
EW-1 TO WET WELL				
WELL & PUMP	1	LS	10,000.00	\$10,000.00
4" PE PIPE, 10' DEEP	260	LF	25.00	6,500.00
6" PE PIPE, 10' DEEP	1250	LF	26.00	32,500.00
4" 45 DEG EL, PE	2	EA	100.00	200.00
4"x4"x6" WYE, PE	1	EA	200.00	200.00
EW-2 TO WET WELL				
WELL & PUMP	1	LS	15,000.00	15,000.00
4" PE PIPE, 10' DEEP	200	LF	25.00	5,000.00
4" 45 DEG EL, PE	2	EA	100.00	200.00
EW-3 TO EW-4				
WELL & PUMP	1	LS	35,000.00	35,000.00
6" PE PIPE, 10' DEEP	840	LF	26.00	21,840.00
6" 45 DEG EL, PE	4	EA	100.00	400.00
EW-4 TO GRAVITY SEWER				
WELL & PUMP	1	LS	12,000.00	12,000.00
8" PE PIPE, 10' DEEP	160	LF	30.00	4,800.00
8" 45 DEG EL, PE	2	EA	150.00	300.00
GRAVITY SEWER TO EW-5				
6" PE PIPE, 10' DEEP	635	LF	26.00	16,510.00
6" 45 DEG EL, PE	2	EA	100.00	200.00
EW-5 & EW-6 TO GRAVITY SEWER				
WELL & PUMP - EW-5	1	LS	40,000.00	40,000.00
WELL & PUMP - EW-6	1	LS	40,000.00	40,000.00
3" PE PIPE, 10' DEEP	118	LF	22.00	2,596.00
6" PE PIPE, 10' DEEP	18	LF	26.00	468.00
10" PE PIPE, 10' DEEP	340	LF	34.00	11,560.00
EW-6 TO "T" SECTION				
10" PE PIPE, 10' DEEP	325	LF	34.00	11,050.00
10"x10"x12" TEE, PE	1	EA	400.00	400.00
10" 90 DEG EL, PE	2	EA	200.00	400.00
EW-7 TO WET WELL				
WELL & PUMP	1	LS	35,000.00	35,000.00
4" PE PIPE, 10' DEEP	300	LF	25.00	7,500.00
4" 45 DEG EL, PE	2	EA	100.00	200.00
TOTAL THIS SHEET				\$309,820.00

TABLE D-3 (CON'T)
EXTRACTION SYSTEM
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
TOTAL PREVIOUS SHEET				\$309,82
"T" SECTION TO WET WELL				
12" PE PIPE, 10' DEEP	510	LF	38.00	19,38
WET WELL, 10' DIA x 20' DP, PE LINED, DUPLEX SUBMERSIBLE PUMPS	1	LS	40,000.00	40,00
EW-9 TO EW-8				
WELL & PUMP	1	LS	9,000.00	9,00
4" PE PIPE, 10' DEEP	442	LF	25.00	11,05
EW-8 TO WET WELL				
WELL & PUMP	1	LS	9,000.00	9,00
4" PE PIPE, 10' DEEP	20	LF	25.00	50
6" PE PIPE, 10' DEEP	340	LF	26.00	8,84
EW-10 TO "T" SECTION				
WELL & PUMP	1	LS	32,000.00	32,00
6" PE PIPE, 10' DEEP	2963	LF	26.00	77,03
6" 45 DEG EL, PE	6	EA	100.00	60
MANHOLES, 4' DIA x 12' DEEP, PE LINED	13	EA	4,000.00	52,00
WELL UTILITIES (PER WELL)				
ACCESS ROAD & PARKING AREA	800	SF	0.50	40
GRAVEL PAVING - 12" DEEP				
FENCING - CHAIN LINK	80	LF	15.00	1,20
GATE, 10' WIDE	1	EA	800.00	80
POWER - CONDUIT ON POLE	25	LF	10.00	25
POWER CABLE, UNDERGROUND	200	LF	5.00	1,00
MOUNTING POST FOR CONTROLS	1	LS	200.00	20
METER SOCKET	1	LS	150.00	15
TOTAL FOR ONE WELL				4,00
TOTAL FOR 10 WELLS				40,00
UNDEVELOPED DESIGN DETAILS				152,76
TOTAL FOR INDUSTRIAL PARK				\$762,00

TABLE D-3 (CON'T)
EXTRACTION SYSTEM
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
FOR NORTH PARK				
DELETE THE FOLLOWING				
EW-10 TO "T" SECTION				
WELL & PUMP	1	LS	32,000.00	(\$32,00
6" PE PIPE, 10' DEEP	2963	LF	26.00	(77,03
6" 45 DEG EL, PE	6	EA	100.00	(60
ADD THE FOLLOWING				
EW-10 TO "Y" SECTION				
WELL & PUMP	1	LS	32,000.00	32,00
6" PE PIPE, 10' DEEP	1627	LF	26.00	42,30
6" 45 DEG EL, PE	4	EA	100.00	40
EW-11 TO "Y" SECTION				
WELL & PUMP	1	LS	20,000.00	20,00
3" PE PIPE, 10' DEEP	1720	LF	22.00	37,84
3" 45 DEG EL, PE	2	EA	100.00	20
"Y" SECTION TO "T" SECTION				
10" PE PIPE, 10' DEEP	1340	LF	34.00	45,56
10"x3"x6" WYE, PE	1	EA	350.00	35
10" 45 DEG EL, PE	2	EA	200.00	40
MANHOLES, 4' DIA x 12' DEEP, PE LINED	6	EA	4,000.00	24,00
UTILITIES FOR 1 WELL				4,00
UNDEVELOPED DESIGN DETAILS				24,58
TOTAL FOR NORTH PARK				\$122,00
TOTAL EXTRACTION SYSTEM - NORTH PARK PLUS INDUSTRIAL PARK				\$884,00

TABLE D-3 (CON'T)
EXTRACTION SYSTEM
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
O&M COSTS FOR INDIVIDUAL CONTAMINANT PLUMES				
KYSOR/FOUR STAR, EW-1, EW-2				
ASSUME EACH WELL REQUIRES 5 Kw FOR A TOTAL OF 10 Kw				
10Kw x 24Hr/Day x 365 Day/Yr =	87600	KwHr	0.08	\$7,00
MAINTENANCE @ 5% OF CAPITAL COSTS				
EW-1 TO WET WELL	49400			
EW-2 TO WET WELL	20200			
PUMP UTILITIES 2 EA @ 4000	8000			
	77600			
	x1.25			
	97000	x.05		4,85
UNDEVELOPED DESIGN DETAILS				2,96
TOTAL ANNUAL O&M FOR KYSOR/FOUR STAR PLUME				\$14,82
PRESENT WORTH OF ANNUAL O&M FOR KYSOR/FOUR STAR PLUME FOR 29 YEARS @ 5%				\$224,00
KYSOR INTERMEDIATE, EW-3, EW-4, EW-6				
ASSUME EACH WELL REQUIRES 15 Kw FOR A TOTAL OF 45 Kw				
45Kw x 24Hr/Day x 365 Day/Yr =	394200	KwHr	0.08	\$31,53
MAINTENANCE @ 5% OF CAPITAL COSTS				
EW-3 TO EW-4	57240			
EW-4 TO GRAVITY SEWER	17100			
EW-6 TO GRAVITY SEWER	47312			
EW-6 TO "T"	11850			
"T" TO WET WELL	59380			
MANHOLE, 8 EA @ 4000	32000			
PUMP UTILITIES 3 EA @ 4000	12000			
	236882			
	x1.25			
	296000	x.05		14,80
UNDEVELOPED DESIGN DETAILS				11,58
TOTAL ANNUAL O&M FOR KYSOR INTERMEDIATE PLUME				\$57,92
PRESENT WORTH OF ANNUAL O&M FOR KYSOR INTERMEDIATE PLUME FOR 64 YEARS @ 5%				\$1,107,00

TABLE D-3 (CON'T)
EXTRACTION SYSTEM
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
O&M COSTS FOR INDIVIDUAL CONTAMINANT PLUMES				
FOUR WINNS, EW-5				
ASSUME WELL REQUIRES 5 Kw				
5Kw x 24Hr/Day x 365 Day/Yr =	43800	KwHr	0.08	\$3,504
MAINTENANCE @ 5% OF CAPITAL COSTS				
EW-5 TO GRAVITY SEWER	47312			
GRAVITY SEWER TO EW-5	16710			
MANHOLE 3 EA @ 4000	12000			
PUMP UTILITIES	4000			
	80022			
	x1.25			
	100000	x.05		5,000
UNDEVELOPED DESIGN DETAILS				2,124
TOTAL ANNUAL O&M FOR FOUR WINNS PLUME				\$10,634
PRESENT WORTH OF ANNUAL O&M FOR FOUR WINNS PLUME				\$88,000
FOR 11 YEARS @ 5%				
HEXCHROME, EW-7				
ASSUME WELL REQUIRES 5 Kw				
5Kw x 24Hr/Day x 365 Day/Yr =	43800	KwHr	0.08	\$3,504
MAINTENANCE @ 5% OF CAPITAL COSTS				
EW-7 TO WET WELL	42700			
PUMP UTILITIES	4000			
	46700			
	x1.25			
	58000	x.05		2,900
UNDEVELOPED DESIGN DETAILS				1,600
TOTAL ANNUAL O&M FOR HEXCHROME PLUME				\$8,000
PRESENT WORTH OF ANNUAL O&M FOR HEXCHROME PLUME				\$28,000
FOR 4 YEARS @ 5%				

TABLE D-3 (CON'T)
EXTRACTION SYSTEM
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
O&M COSTS FOR INDIVIDUAL CONTAMINANT PLUMES				
KYSOR SHALLOW, EW-8 & EW-9				
ASSUME EACH WELL REQUIRES 5 Kw FOR A TOTAL OF 10 Kw				
10Kw x 24Hr/Day x 365 Day/Yr =	87600	KwHr	0.08	\$7,008
MAINTENANCE @ 5% OF CAPITAL COSTS				
EW-9 TO EW-8	20050			
EW-8 TO WET WELL	18340			
PUMP UTILITIES 2 EA @ 4000	8000			
MANHOLES 2 EA @ 4000	8000			
	54390			
	x1.25			
	68000	x.05		3,400
UNDEVELOPED DESIGN DETAILS				2,600
TOTAL ANNUAL O&M FOR KYSOR SHALLOW PLUME				\$13,010
PRESENT WORTH OF ANNUAL O&M FOR KYSOR SHALLOW PLUME FOR 3.5 YEARS @ 5%				\$35,000
REXAIR, EW-10				
ASSUME WELL REQUIRES 15 Kw				
15Kw x 24Hr/Day x 365 Day/Yr =	131400	KwHr	0.08	\$10,512
MAINTENANCE @ 5% OF CAPITAL COSTS				
EW-10 TO "T"	109638			
PUMP UTILITIES	4000			
	113638			
	x1.25			
	142000	x.05		7,100
UNDEVELOPED DESIGN DETAILS				4,400
TOTAL ANNUAL O&M FOR REXAIR PLUME				\$22,015
PRESENT WORTH OF ANNUAL O&M FOR REXAIR PLUME FOR 38 YEARS @ 5%				\$371,000

TABLE D-3 (CON'T)
EXTRACTION SYSTEM
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
O&M COSTS FOR INDIVIDUAL CONTAMINANT PLUMES				
NORTH PARK, EW-11				
ASSUME WELL REQUIRES 10 Kw				
10Kw x 24Hr/Day x 365 Day/Yr =	87600	KwHr	0.08	\$7,00
MAINTENANCE @ 5% OF CAPITAL COSTS				
TOTAL CAPITAL COSTS	97414			

	97414			
	x1.25			

	122000	x.05		6,10
UNDEVELOPED DESIGN DETAILS				3,27

TOTAL ANNUAL O&M FOR NORTH PARK PLUME				\$16,38
PRESENT WORTH OF ANNUAL O&M FOR NORTH PARK PLUME				\$83,00
FOR 6.5 YEARS @ 5%				
SUMMARY OF PRESENT WORTHS OF VARIOUS PLUMES ANNUAL O&M COSTS				
KYSOR/FOUR STAR				\$224,00
KYSOR INTERMEDIATE				1,107,00
FOUR WINNS				88,00
HEXCHROME				28,00
KYSOR SHALLOW				35,00
REXAIR				371,00
NORTH PARK				83,00

TOTAL OF PRESENT WORTHS OF VARIOUS PLUMES ANNUAL O&M COSTS				\$1,936,00

TABLE D-4
TREATMENT BUILDING
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
GRADING	833	CY	1.50	\$1,25
FENCING - CHAIN LINK	600	LF	15.00	9,00
GATE, 20' WIDE	2	EA	1,500.00	3,00
PAVING - ASPHALT	5,200	SF	2.50	13,00
POWER - POLE	2	EA	1,000.00	2,00
3 PHASE CABLE	100	LF	5.00	50
TRANSFORMER		BY UTILITY		
WATER SUPPLY - 2" PVC	100	LF	20.00	2,00
SANITARY SEWER - MANHOLE	2	EA	1,250.00	2,50
12" DIA RCP PIPE	100	LF	25.00	2,50
CONCRETE PAD	400	SF	3.00	1,20
BUILDING	8,000	SF	60.00	480,00
UNDEVELOPED DESIGN DETAILS				129,05
TOTAL CAPITAL COSTS				\$646,00
O&M COSTS				
MISCELLANEOUS POWER & HEATING	109,500	KwHr	0.08	\$8,76
MAINTENANCE @ 5% OF CAPITAL COSTS	1	LS	32,300.00	32,30
UNDEVELOPED DESIGN DETAILS				9,94
TOTAL O&M COSTS				\$51,00

TABLE D-5
DISCHARGE SYSTEM
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				
14" DIA PVC PIPE, 8' DEEP	6,000	LF	45.00	\$270,00
14" DIA PVC ELBOW	13	EA	300.00	3,90
DUPLEX PUMP STATION, 1000GPM	1	EA	30,000.00	30,00
HEAD WALL AT CLAM RIVER	1	EA	500.00	50
UNDEVELOPED DESIGN DETAILS				76,60
TOTAL CAPITAL COSTS				<u>\$381,00</u>
ANNUAL O&M COSTS				
POWER (ASSUME 50 HP)	438,000	KwHr	0.08	\$35,04
MAINTENANCE @ 5% OF CAPITAL COSTS	1	LS	19,050.00	19,05
UNDEVELOPED DESIGN DETAILS				13,91
TOTAL O&M COSTS				<u>\$68,00</u>

TABLE D-6
 UV/OZONE/H2O2 TREATMENT
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
UV/OZONE/H2O2 TREATMENT SYSTEM	1	LS	1,600,000	\$1,600,00
INCLUDES THE FOLLOWING:				
REACTOR VESSELS - 6000 GAL, 3 EA.				
LIQUID OXYGEN TANK				
OZONE GENERATORS - 400 LB, 2 EA.				
OZONE DIFFUSERS				
ULTRAVIOLET LAMPS				
HYDROGEN PEROXIDE (H2O2) CONTACTORS - 2 EA.				
PUMPS				
INSTALLATION	480	HR	25.00	12,00
PROCESS MECHANICAL WORK	1	LS	100,000.00	100,00
PROCESS ELECTRICAL WORK	1	LS	100,000.00	100,00
UNDEVELOPED DESIGN DETAILS				453,00
TOTAL CAPITAL COST				\$2,265,00
PRESENT WORTH OF TREATMENT SYSTEM REPLACEMENT AFTER 25 YEARS @ 5%				\$591,00
PRESENT WORTH OF TREATMENT SYSTEM REPLACEMENT AFTER 50 YEARS @ 5%				\$174,00
ANNUAL O&M COSTS FOR YEARS 1 - 38				
UV/OZONE/H2O2 TREATMENT SYSTEM	494,000	1000 GAL	2.00	988,000
INCLUDES THE FOLLOWING:				
ELECTRICITY - 15000 KWHr/DAY				
ULTRAVIOLET LAMP REPLACEMENT				
HYDROGEN PEROXIDE - 2000 LB/DAY				
LIQUID OXYGEN - 150 LB/DAY				
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	13,250.00	13,250
OPERATOR @ 40 HR/WK	2,080	HR	25.00	52,000
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,000
UNDEVELOPED DESIGN DETAILS				276,750
TOTAL ANNUAL O&M COSTS FOR YEARS 1 - 38				\$1,382,000

TABLE D-6 (CON'T)
 UV/OZONE/H2O2 TREATMENT
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
ANNUAL O&M COSTS FOR YEARS 39 - 64				
UV/OZONE/H2O2 TREATMENT SYSTEM	198,000	1000 GAL	3.10	613,80
INCLUDES THE FOLLOWING:				
ELECTRICITY - 15000 KWHr/DAY				
ULTRAVIOLET LAMP REPLACEMENT				
HYDROGEN PEROXIDE - 800 LB/DAY				
LIQUID OXYGEN - 60 LB/DAY				
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	13,250.00	13,25
OPERATOR @ 40 HR/WK	2,080	HR	25.00	52,00
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,00
UNDEVELOPED DESIGN DETAILS				182,95
TOTAL ANNUAL O&M COSTS FOR YEARS 39 - 64				\$914,00

TABLE D-7
AIR STRIPPING WITH VAPOR-PHASE CARBON ADSORPTION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
AIR STRIPPING WITH VAPOR-PHASE CARBON ADSORPTION SYSTEM INCLUDING: STRIPPING COLUMNS - 8' DIA, 40' HIGH, 2 EA. CARBON ADSORPTION VESSELS	1	LS	700,000.00	\$700,00
SYSTEM INSTALLATION	360	MHR	25.00	9,00
PUMP 1000 GPM, 100 TDH	2	EA	6,700.00	13,40
PROCESS MECHANICAL WORK	1	LS	50,000.00	50,00
PROCESS ELECTRICAL WORK	1	LS	50,000.00	50,00
UNDEVELOPED DESIGN DETAILS				205,60
TOTAL CAPITAL COST				<u>\$1,028,00</u>
PRESENT WORTH OF TREATMENT SYSTEM REPLACEMENT AFTER 25 YEARS @ 5%				\$258,00
PRESENT WORTH OF TREATMENT SYSTEM REPLACEMENT AFTER 50 YEARS @ 5%				\$76,00
ANNUAL O&M COSTS FOR YEARS 1 - 38				
TREATMENT SYSTEM ELECTRICAL	1,750,000	KwHr	0.08	\$140,00
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	7,650.00	7,65
OPERATOR @ 16 HR/WK	832	HR	25.00	20,80
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,00
UNDEVELOPED DESIGN DETAILS				55,55
TOTAL ANNUAL O&M COSTS FOR YEARS 1 - 38				<u>\$276,00</u>

TABLE D-7 (CON'T)
 AIR STRIPPING WITH VAPOR-PHASE CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
ANNUAL O&M COSTS FOR YEARS 39 - 64				
TREATMENT SYSTEM ELECTRICAL	875,000	KwHr	0.08	\$70.00
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	7,650.00	7.65
OPERATOR @ 16 HR/WK	832	HR	25.00	20.80
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52.00
UNDEVELOPED DESIGN DETAILS				37.55
TOTAL ANNUAL O&M COSTS FOR YEARS 39 - 64				\$188.00

TABLE D-8
AIR STRIPPING WITH LIQUID-PHASE
AND VAPOR-PHASE CARBON ADSORPTION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
AIR STRIPPING WITH LIQUID-PHASE AND VAPOR-PHASE CARBON ADSORPTION STRIPPING COLUMN - 8' DIA, 40' HIGH, 1 EA. CARBON ADSORPTION VESSELS - 10' DIA, 19' HIGH, 4 EA. CARBON TRANSFER VESSEL - 10' DIA, 19' HIGH, 1 EA.	1	LS	965,000.00	\$965,000
SYSTEM INSTALLATION	480	MHR	25.00	12,000
PUMP 500 GPM, 100 TDH	2	EA	5,900.00	11,800
PROCESS MECHANICAL WORK	1	LS	75,000.00	75,000
PROCESS ELECTRICAL WORK	1	LS	75,000.00	75,000
UNDEVELOPED DESIGN DETAILS				285,200
TOTAL CAPITAL COST				\$1,424,000
PRESENT WORTH OF TREATMENT SYSTEM REPLACEMENT AFTER 25 YEARS @ 5%				\$356,000
PRESENT WORTH OF TREATMENT SYSTEM REPLACEMENT AFTER 50 YEARS @ 5%				\$105,000
ANNUAL O&M COSTS FOR YEARS 1 - 38				
TREATMENT SYSTEM ELECTRICAL	1,312,500	KwHr	0.08	\$105,000
REPLACEMENT CARBON	225,000	LS	1.00	225,000
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	10,887.50	10,888
OPERATOR @ 20 HR/WK	1,040	HR	25.00	26,000
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,000
UNDEVELOPED DESIGN DETAILS				105,112
TOTAL ANNUAL O&M COSTS FOR YEARS 1 - 38				\$524,000

TABLE D-8 (CON'T)
 AIR STRIPPING WITH LIQUID-PHASE
 AND VAPOR-PHASE CARBON ADSORPTION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
ANNUAL O&M COSTS FOR YEARS 39 - 64				
TREATMENT SYSTEM ELECTRICAL	656,250	KwHr	0.08	\$52,50
REPLACEMENT CARBON	90,000	LS	1.00	90,00
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	10,887.50	10,88
OPERATOR @ 20 HR/WK	1,040	HR	25.00	26,00
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,00
UNDEVELOPED DESIGN DETAILS				57,61
TOTAL ANNUAL O&M COSTS FOR YEARS 39 - 64				\$289,00

**TABLE D-9
STEAM STRIPPING
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN**

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
STEAM STRIPPING INCLUDING:	1	LS	500,000.00	\$500,000
HEAT EXCHANGER				
STRIPPING COLUMN - 5' DIA, 30' HIGH, 1 EA.				
CONDENSER				
DECANTER				
PUMPS				
SYSTEM INSTALLATION	800	MHR	25.00	20,000
PROCESS MECHANICAL WORK	1	LS	100,000.00	100,000
PROCESS ELECTRICAL WORK	1	LS	100,000.00	100,000
UNDEVELOPED DESIGN DETAILS				180,000
TOTAL CAPITAL COST				\$900,000
PRESENT WORTH OF TREATMENT SYSTEM REPLACEMENT				
AFTER 10 YEARS @ 5%				\$384,000
AFTER 20 YEARS @ 5%				236,000
AFTER 30 YEARS @ 5%				145,000
AFTER 40 YEARS @ 5%				89,000
AFTER 50 YEARS @ 5%				55,000
AFTER 60 YEARS @ 5%				33,000
TOTAL COST OF TREATMENT SYSTEM REPLACEMENT				\$942,000
ANNUAL O&M COSTS FOR YEARS 1 - 38				
STEAM GENERATION	50,000	1000 GAL	10.00	\$500,000
CONDENSATE DISPOSAL				
TRANSPORTATION	1	LS	11,025.00	11,025
DISPOSAL	84,315	GAL	2.00	168,630
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	13,750.00	13,750
OPERATOR @ 16 HR/WK	832	HR	25.00	20,800
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,000
UNDEVELOPED DESIGN DETAILS				191,795
TOTAL ANNUAL O&M COSTS FOR YEARS 1 - 38				\$958,000

TABLE D-9 (CON'T)
 STEAM STRIPPING
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
ANNUAL O&M COSTS FOR YEARS 39 - 64				
STEAM GENERATION	20,000	1000 GAL	10.00	\$200,00
CONDENSATE DISPOSAL				
TRANSPORTATION	1	LS	4,720.00	4,72
DISPOSAL	33,726	GAL	2.00	67,45
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	13,750.00	13,75
OPERATOR @ 16 HR/WK	832	HR	25.00	20,80
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,00
UNDEVELOPED DESIGN DETAILS				89,27
TOTAL ANNUAL O&M COSTS FOR YEARS 39 - 64				\$448,00

TABLE D-10
STEAM STRIPPING WITH VAPOR RECOMPRESSION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
STEAM STRIPPING INCLUDING: HEAT EXCHANGER STRIPPING COLUMN - 5' DIA, 30' HIGH, 1 EA. MECHANICAL VAPOR COMPRESSOR REBOILER DECANTER PUMPS	1	LS	700,000.00	\$700,000.00
SYSTEM INSTALLATION	1000	MHR	25.00	25,000.00
PROCESS MECHANICAL WORK	1	LS	100,000.00	100,000.00
PROCESS ELECTRICAL WORK	1	LS	100,000.00	100,000.00
UNDEVELOPED DESIGN DETAILS				231,000.00
TOTAL CAPITAL COST				\$1,156,000.00
PRESSENT WORTH OF TREATMENT SYSTEM REPLACEMENT				
AFTER 10 YEARS @ 5%				\$537,000.00
AFTER 20 YEARS @ 5%				330,000.00
AFTER 30 YEARS @ 5%				202,000.00
AFTER 40 YEARS @ 5%				124,000.00
AFTER 50 YEARS @ 5%				76,000.00
AFTER 60 YEARS @ 5%				47,000.00
TOTAL COST OF TREATMENT SYSTEM REPLACEMENT				\$1,316,000.00
ANNUAL O&M COSTS FOR YEARS 1 - 38				
STEAM GENERATION	18,500	1000 GAL	10.00	\$185,000.00
CONDENSATE DISPOSAL				
TRANSPORTATION	1	LS	11,025.00	11,025.00
DISPOSAL	84,315	GAL	2.00	168,630.00
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	14,050.00	14,050.00
OPERATOR @ 16 HR/WK	832	HR	25.00	20,800.00
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,000.00
UNDEVELOPED DESIGN DETAILS				112,490.00
TOTAL ANNUAL O&M COSTS FOR YEARS 1 - 38				\$564,000.00

TABLE D-10 (CON'T)
 STEAM STRIPPING WITH VAPOR RECOMPRESSION
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
ANNUAL O&M COSTS FOR YEARS 39 - 64				
STEAM GENERATION	7,400	1000 GAL	10.00	\$74,00
CONDENSATE DISPOSAL				
TRANSPORTATION	1	LS	4,720.00	4,72
DISPOSAL	33,726	GAL	2.00	67,45
MAINTENANCE @ 5% OF CAPITAL COSTS (LESS TREATMENT SYSTEM)	1	LS	14,050.00	14,05
OPERATOR @ 16 HR/WK	832	HR	25.00	20,80
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,00
UNDEVELOPED DESIGN DETAILS				57,97
TOTAL ANNUAL O&M COSTS FOR YEARS 39 - 64				\$291,00

TABLE D-11
 HEXAVALENT CHROMIUM - ION EXCHANGE
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
ION EXCHANGE SYSTEM INCLUDING:	1	LS	33,600.00	\$33,600
ANION TANKS - 2 EA.				
CATION TANKS - 2 EA.				
REGENERATION SOLUTION TANKS - 2 EA.				
PROCESS PUMPS				
REGENERATION PUMPS				
CONTROL EQUIPMENT				
SYSTEM INSTALLATION	1	LS	25,000.00	25,000
PUMPS 50 GPM, 50 TDH	3	EA	1,900.00	5,700
TANKS - 150 GAL REGEN WASTE SOLUTION	1	EA	300.00	300
300 GAL REGEN WASTE SOLUTION	1	EA	500.00	500
1000 GAL DEIONIZED WATER	1	EA	1,500.00	1,500
PROCESS MECHANICAL WORK	1	LS	100,000.00	100,000
PROCESS ELECTRICAL WORK	1	LS	50,000.00	50,000
UNDEVELOPED DESIGN DETAILS				54,400
TOTAL CAPITAL COST				\$271,000
O&M ANNUAL COSTS				
HYDROCHLORIC ACID (CATION REGEN)	7,008	GAL	0.75	\$5,256
SODIUM HYDROXIDE (ANION REGEN)	1,752	GAL	1.09	1,910
OFF-SITE DISPOSAL - REGENERATION WASTE	1	LS	188,100.00	188,100
MAINTENANCE @ 5% OF CAPITAL COSTS	1	LS	13,550.00	13,550
OPERATOR @ 16 HR/WK	832	HR	25.00	20,800
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,000
UNDEVELOPED DESIGN DETAILS				70,380
TOTAL ANNUAL O&M COSTS				\$352,000
REPLACEMENT RESIN COSTS	1	CHG	2,735.00	2,735
(CHANGE AFTER YEAR 1 & EVERY 4 YEARS AFTER)				
PRESENT WORTH OF REPLACEMENT RESIN FOR 4 YEARS @ 5% INCLUDING UNDEVELOPED DESIGN DETAILS				3,000

TABLE D-12
HEXAVALENT CHROMIUM - CARBON ADSORPTION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CARBON ADSORPTION SYSTEM INCLUDING: CARBON ADSORPTION VESSELS - 4' DIA, 11' HIGH, 2 EA. ASSOCIATED PIPING	1	LS	45,000.00	\$45,000
PUMP 50 GPM, 50 TDH	1	EA	1,900.00	1,900
TANK 500 GAL	1	EA	750.00	750
pH PROBE & CONTROLLER	1	LS	2,500.00	2,500
SYSTEM INSTALLATION	480	MHR	25.00	12,000
PROCESS MECHANICAL WORK	1	LS	25,000.00	25,000
PROCESS ELECTRICAL WORK	1	LS	25,000.00	25,000
UNDEVELOPED DESIGN DETAILS				27,850
TOTAL CAPITAL COST				\$140,000
O&M COSTS				
CARBON	15,000	LB	1.00	\$15,000
SPENT CARBON DISPOSAL	1	LS	19,000.00	19,000
MAINTENANCE @ 5% OF CAPITAL COSTS	1	LS	7,000.00	7,000
OPERATOR @ 16 HR/WK	832	HR	25.00	20,800
MONITORING - WEEKLY SAMPLING	52	WK	1,000.00	52,000
UNDEVELOPED DESIGN DETAILS				28,200
TOTAL O&M COSTS				\$142,000

TABLE D-13
 ALTERNATE WATER SUPPLY
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				
LOCATE NEW WELL FIELD	1	LS	235,000.00	\$235,000
NEW WELLS	7	EA	140,000.00	980,000
POWER SUPPLY (700 HP TO WELLS)	12,000	LF	20.00	240,000
WATER MAINS - 12" DIA	2,000	LF	45.00	90,000
30" DIA	10,000	LF	120.00	1,200,000
WELL BUILDING	7	EA	25,000.00	175,000
UNDEVELOPED DESIGN DETAILS				730,000
TOTAL CAPITAL COSTS				\$3,650,000
ANNUAL O&M COSTS - BREAK EVEN WITH RESPECT TO OLD SYSTEM				\$

TABLE D-14
CONTINGENCY PLAN
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				
PREPARE CONTINGENCY PLAN	250	HR	65.00	\$16,25
UNDEVELOPED DESIGN DETAILS				3,75
TOTAL ANNUAL COSTS				----- \$20,00

ANNUAL O&M COSTS

\$

TABLE D-15
INSPECTION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS			\$
ANNUAL O&M COSTS INSPECTOR	4 DAY	400.00	\$1,60
GROUNDWATER SAMPLING	6 EA	1,500.00	9,00
UNDEVELOPED DESIGN DETAILS			2,40
TOTAL ANNUAL COSTS			\$13,00

TABLE D-16
FENCING
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				
KYSOR SITE				
CHAIN LINK FENCE	700	LF	15.00	\$10,50
GATE	1	EA	600.00	60
FOUR WINNS SITE				
CHAIN LINK FENCE	750	LF	15.00	11,25
GATE	1	EA	600.00	60
UNDEVELOPED DESIGN DETAILS				6,05
TOTAL CAPITAL COSTS				<u>\$29,00</u>
ANNUAL O&M COSTS				
MAINTENANCE @ 5% OF CAPITAL COSTS	1	LS	1,450.00	\$1,45
UNDEVELOPED DESIGN DETAILS				55
TOTAL ANNUAL COSTS				<u>\$2,00</u>

TABLE D-17
 SURFACE CAP & SUBSURFACE WALL
 PRELIMINARY COST ESTIMATE
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				
SURFACE CAP				
KYSOR SITE	18,750	SF	4.00	\$75,00
FOUR WINNS SITS	18,900	SF	4.00	75,60
SUBSURFACE WALL				
KYSOR SITE	58,500	SF	10.00	585,00
FOUR WINNS SITS	44,200	SF	10.00	442,00
UNDEVELOPED DESIGN DETAILS				294,40
TOTAL CAPITAL COSTS				\$1,472,00
ANNUAL O&M COSTS				
MOWING, SEEDING, FERTILIZING, ETC.	1	LS	5,000.00	\$5,00

TABLE D-18
LOW TEMPERATURE THERMAL STRIPPING
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				
EXCAVATE SOIL (LEVEL C)				
KYSOR SITE	13,200	CY	7.50	\$99,00
FOUR WINNS SITE	16,100	CY	7.50	120,75
THERMAL STRIPPING INCLUDING FEED, PROCESSING, SAMPLING	29,300	CY	150.00	4,395,00
MOVE STRIPPING EQUIPMENT TO FOUR WINNS SITE	1	LS	150,000.00	150,00
DISPOSAL OF TREATED SOIL & BAGHOUSE RESIDUE				
TRANSPORTATION				
TREATED SOIL	29,300	CY	40.00	1,172,00
BAGHOUSE RESIDUE	1,500	CY	40.00	60,00
DISPOSAL COSTS	30,800	CY	120.00	3,696,00
IMPORTED BACKFILL	32,230	CY	10.00	322,30
UNDEVELOPED DESIGN DETAILS				2,503,95
TOTAL CAPITAL COSTS				\$12,519,00

ANNUAL O&M COSTS

\$

TABLE D-19
IN-SITU SOIL FLUSHING
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				
4" DIA PERFORATED PVC PIPE ON GROUND	3,445	LF	3.00	\$10,33
4" DIA SOLID PVC PIPE ON GROUND	520	LF	2.50	1,30
4" DIA PVC TEE	94	EA	80.00	7,52
4" DIA PVC EL	16	EA	50.00	80
2" DIA PVC PRESSURE PIPE IN TRENCH	300	LF	18.00	5,40
4" DIA PVC PRESSURE PIPE IN TRENCH	500	LF	20.00	10,00
VALVES, FLOW METER, MANHOLE, ETC	4	EA	2,500.00	10,00
AGGREGATE FILL	770	CY	20.00	15,40
FILTER FABRIC	18,900	SF	0.08	1,51
EARTH FILL	2,310	CY	10.00	23,10
4" TOPSOIL	18,900	SF	0.20	3,78
SEED, MULCH, FERTILIZER	18,900	SF	0.04	75
UNDEVELOPED DESIGN DETAILS				22,09
TOTAL KYSOR SITE				\$112,00
FOUR WINNS SITE SAME AS KYSOR SITE				112,00
TOTAL CAPITAL COSTS				\$224,00
ANNUAL O&M COSTS				
MAINTENANCE @ 5% KYSOR SITE TOTAL	1	LS	5,600.00	\$5,600
WATER	12	MON	3,000.00	36,000
UNDEVELOPED DESIGN DETAILS				10,400
TOTAL ANNUAL COSTS				\$52,000

TABLE D-20
IN-SITU VACUUM EXTRACTION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				
VACUUM EXTRACTION INCLUDING				
DESIGN, OPERATION, OFF-GAS TREATMENT				
KYSOR SITE	13,200	CY	50.00	\$660,00
FOUR WINNS SITE	16,100	CY	25.00	402,50
DISPOSAL OF CARBON				
TRANSPORTATION	100	CY	40.00	4,00
DISPOSAL COSTS	100	CY	120.00	12,00
UNDEVELOPED DESIGN DETAILS				269,50
TOTAL CAPITAL COSTS				\$1,348,00

ANNUAL O&M COSTS

\$

TABLE D-21
ON-SITE INCINERATION
PRELIMINARY COST ESTIMATE
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	TOTAL
CAPITAL COSTS				
ON-SITE INCINERATION INCLUDING				
EXCAVATE SOILS, FEED,				
PROCESSING, SAMPLING, RESIDUE DISPOSAL				
KYSOR SITE	13,200	CY	400.00	\$5,280,00
FOUR WINNS SITE	16,100	CY	400.00	6,440,00
DISPOSAL OF TREATED SOIL				
TRANSPORTATION				
KYSOR SITE	13,200	CY	40.00	528,00
FOUR WINNS SITE	16,100	CY	40.00	644,00
DISPOSAL COSTS				
KYSOR SITE	13,200	CY	120.00	1,584,00
FOUR WINNS SITE	16,100	CY	120.00	1,932,00
IMPORTED BACKFILL				
KYSOR SITE	14,520	CY	10.00	145,20
FOUR WINNS SITE	17,710	CY	10.00	177,10
UNDEVELOPED DESIGN DETAILS				4,182,70
TOTAL CAPITAL COSTS				<u>\$20,913,00</u>

ANNUAL O&M COSTS

\$

APPENDIX E

SUMMARY OF ACTION-SPECIFIC ARARS
FOR GROUNDWATER AND SOIL
REMEDIAL ALTERNATIVES

TABLE E-1
ACTION-SPECIFIC ARARS FOR GROUNDWATER REMEDIAL ALTERNATIVES

ALTERNATIVES	No Action	Hex. Chrom. Trmt.- Ion Exchange	Hex. Chrom Trmt.- Carbon Adsorption	UV/Oxidation (GW-2)	Air Stripping, Carbon Adsorption (GW-3)	Steam Stripping (GW-4)	Alternate Water Supply (AC-1)
ARARS							
1. CWA - NPDES Regs. (40 CFR 122)		X	X	X	X	X	
2. RCRA Facility and Location Stds. (40 CFR 264 A-E)		X	X	X	X	X	
3. RCRA Groundwater Monitoring (40 CFR 264 F)	X	X	X	X	X	X	X
4. RCRA Closure and Post- Closure (40 CFR 264 G)							
5. RCRA Generator Regs. (40 CFR 262)		X	X		X	X	
6. RCRA Transport. Regs. (40 CFR 263)		X	X		X	X	
7. RCRA Subtitle D Solid Waste Regs. (40 CFR 241-257)							
8. CAA-NAAQS (40 CFR 50)		X	X	X	X	X	
9. RCRA Regs Envir. Perf. and Chem/Physical/Bio Trmt. (40 CFR 264 X, 40 CFR 265 Q)		X	X	X	X	X	
10. Mich. Wastewater Disch. Permits (R 323, Part 21)		X	X	X	X	X	

TABLE E-1 (continued)
ACTION-SPECIFIC ARARS FOR GROUNDWATER REMEDIAL ALTERNATIVES

ALTERNATIVES	No Action	Hex. Chrom. Trmt.- Ion Exchange	Hex. Chrom Trmt.- Carbon Adsorption	UV/Oxidation (GW-2)	Air Stripping, Carbon Adsorption (GW-3)	Steam Stripping (GW-4)	Alternate Water Supply (AC-1)
ARARS							
11. Mich. Groundwater Quality Regs. (R 232, Part 22)							
12. Mich. Liquid Industrial Waste Disposal (MCL c.323)							
13. MHWAR Monitoring Regs. (R 299, Rules 611-612)	X	X	X	X	X	X	X
14. MHWAR Closure/Post- Closure Regs. (R 299, Rule 613)							
15. MHWAR Generator Regs. (R 299, Part 3)		X	X		X	X	
16. MHWAR Facility Design, Oper. & Location Regs. (R 299, Rules 603-604)		X	X	X	X	X	
17. MHWAR Const. & Oper. Permits (R 299, Part 5)							
18. MHWAR Transportation Regs. (R 299, Part 4)		X	X		X	X	
19. Mich. Air Poll. Control (R 336)				X	X		
20. OSHA Stds. (29 CFR 1910)	X	X	X	X	X	X	X

TABLE E-2
ACTION-SPECIFIC ARARS FOR SOIL REMEDIAL ALTERNATIVES

Alternatives	No Action	Surf. Capping, Subsurface Containment Wall (SC-2)	Low Temp Thermal Stripping (SC-6)	In-situ Soil Flushing (SC-7)	In-situ Vacuum Extraction (SC-8)	On-Site Incineration (SC-11)
ARARS						
1. RCRA Landfill Regs. (40 CFR 264 N)		X				
2. RCRA Facility Site Location Monitoring Regs. (40 CFR 264 A-F)		X	X	X	X	X
3. RCRA Incinerator Regs. (40 CFR 264 O)						X
4. RCRA Storage Regs. (40 CFR 264 I)						X
5. RCRA/DOT Transportation Regs. (40 CFR 263) (49 CFR 171-179)			X		X	X
6. RCRA Land Disposal Regs. (40 CFR 268)			X			X
7. RCRA Subtitle D Solid Waste Regs. (40 CFR 241-257)						
8. RCRA Stds. for Envir. Perf. and Phy/Chem/Bio Trmt. (40 CFR 264 X, 265 O)			X	X	X	
9. CAA-NAQS (40 CFR 50)		X	X		X	X
10. MMWR Facility Design, Oper. & Location Stds. (R 299, Rules 603-604)		X	X	X	X	X

TABLE E-2 (continued)
ACTION-SPECIFIC ARARS FOR SOIL REMEDIAL ALTERNATIVES

Alternatives	No Action	Surf. Capping, Subsurface Containment Wall (SC-2)	Low Temp Thermal Stripping (SC-6)	In-situ Soil Flushing (SC-7)	In-situ Vacuum Extraction (SC-8)	On-Site Incineration (SC-11)
ARARS						
11. MHWAR Monitoring Regs. (R 299, Rules 611-612)	X	X	X	X	X	X
12. MHWAR Closure/Post Closure (R 299, Rule 613)			X			X
13. MHWAR Container Regs. (R 299, Rule 614)			X			X
14. MHWAR Landfill Regs. (R 299, Rule 619)		X				
15. MHWAR Incinerator Regs. (R 299, Rule 623-626)						X
16. MHWAR Construction & Oper. Permits (R 299, Part 5)						X
17. MHWAR Generator Regs. (R 299, Part 3)			X		X	X
18. MHWAR Transporter Regs. (R 299, Part 4)			X		X	X
19. Mich Air Poll, Control Regs. (R 336)			X		X	X
20. Mich. Groundwater Quality Regs. (R 232, Part 22)				X		
21. OSHA Stds. (29 CFR 1910)	X	X	X	X	X	X

APPENDIX F

DEVELOPMENT OF THE GROUNDWATER EXTRACTION SYSTEM

Development of a pumping system to extract contaminated groundwater requires integrating several hydrogeologic factors of the aquifer (e.g., porosity, hydraulic conductivity, vertical and horizontal gradients, contaminant thickness and extent, and location of confining layers) into a series of analytical expressions. These equations were then applied to determine the radius of influence a pumping well will have when placed in a selected area of the contaminated aquifer. These equations, combined with developed type curves for one-, two-, and three-well pumping systems were used at the Cadillac site for contaminant plume interception and collection.

The formulas and type curves used for the pumping system development at the Cadillac site appear in a 1986 article entitled, "Capture-zone Type Curves: A Tool for Aquifer Cleanup" (Javandel and Tsang, 1986). The type curves were used to determine the optimal number and placement of the pumping well(s) for contaminated groundwater extraction. With a location selected, a pumping rate in gpm could be estimated that would create a capture zone of appropriate size for the associated plume. Once a pumping rate is determined, a time estimate for removing one pore volume of water from the aquifer can be estimated based on plume length. Then, using an expected efficiency of contaminant removal with each pore volume, the number of pore volumes required to flush the aquifer and reach specified contaminant TCLs can be estimated. Finally, a time estimate for cleanup can be determined based on pumping rate and the total number of pore volumes that require pumping. This appendix describes the application of the formulas and type curves to each identified contaminant plume in the Cadillac Industrial Park.

For a single-well pumping system, the equation defining the type curve is as follows:

$$y = Q/2BU - Q/2(\pi)BU * \text{TAN}^{-1} (Y/X)$$

WHERE:

Q = pumping well discharge rate
 B = aquifer thickness
 U = regional groundwater flow velocity (Darcian =
 hydraulic conductivity (K) * horizontal
 hydraulic gradient (i))
 PI = 3.14159

The appropriate-sized capture zone for each plume is selected based on the parameter Q/BU, from which the well discharge rate Q (the only remaining unknown variable) can be estimated.

Pumping systems for each contaminant plume in the industrial park were considered for groundwater extraction. Type curves were selected based on plume width, as defined by 1987 groundwater sampling of the monitoring wells and contaminant TCLs established in the RA. Because TCE is the most common compound found particularly in the monitoring wells defining the outer reaches of the plumes, the TCL of .005 ppm for TCE, which is also its minimum detection level, was used to determine capture zone width. Selecting capture zone type curves that were too small would result in some contaminated groundwater flowing past the extraction well, whereas capture zones that were too large

would result in the pumping and treating of clean groundwater beyond the defined extent of the plume. Once the appropriate width of the capture zone was determined, pumping rates were estimated.

In all, the following seven extraction well series were designed to pump the contaminated groundwater at the Cadillac Industrial Park:

- o a single-well system for the Rexair Plume
- o a two-well system for the concentrated portion of the VOC (Kysor-related) shallow aquifer plume
- o a two-well system for the VOC (Kysor- and Four Star-related) shallow aquifer contamination that is less concentrated near the northeastern limit of the upper aquitard
- o a single-well system for the relatively concentrated VOC (Four Winns-related) shallow aquifer contamination
- o a three-well system for the areally extensive intermediate aquifer contamination
- o a single-well system for the small North Park Subdivision Plume
- o a single-well system to collect the thin plume of hexavalent chromium at the bottom of the intermediate aquifer

REXAIR PLUME

The Rexair Plume, located near the east-central edge of the industrial park, is a concentrated, well-defined plume containing primarily TCE. TCLs for this compound were set at the compound's CLP detection level of 0.005 ppm. Plume concentrations were measured as high as 153 ppm at its center at a depth of 100 feet in monitoring well SLW-50, and 100 ppm at a depth of 140 feet in SLW-41. Contamination may exist at greater depths, but as described in the RI, auger drilling could not be advanced through a dense gravel and silt layer beginning at a depth of approximately 130 feet. Based on screened auger sampling data, the saturated contaminated thickness is estimated to be 160 feet. The hydraulic conductivity of the aquifer in this area of the industrial park is 4.0×10^{-2} cm/sec and the horizontal gradient is 1.1×10^{-3} cm/sec. Groundwater flow direction is to the northwest and the aquifer is unconfined.

Using these aquifer and contaminant characteristics, it was determined that a one-well extraction system would be the most effective for cleanup. Figure 7-2 shows the Rexair Plume and the shape of the capture zone, with the pumping well located at the leading edge of the plume between SLW-44 and SLW-51. The pumping well will be set at a depth of approximately 160 feet and screened the entire saturated thickness to this depth.

Using these parameters and the formula given in Section 2.1, a pumping rate of 135 gpm is necessary to create the capture zone as shown in Figure 7-2. The

appropriately sized type curve is selected and the parameter Q/BU is determined by placing the single-well type curve (to the same scale as the Cadillac Industrial Park base map) underneath the base map where the Rexair Plume has been contoured and defined. For the Rexair Plume:

$$Q/BU = 1,312 \text{ feet (capture zone width)}$$

WHERE:

$$B = 160 \text{ feet (saturated aquifer thickness)}$$

$$U = K_i = (1.3 \times 10^{-3} \text{ ft/sec}) * (1.1 \times 10^{-3} \text{ ft/ft}) \\ = 1.4 \times 10^{-6} \text{ ft/sec}$$

THEREFORE:

$$Q = (1,312 \text{ ft}) * (160 \text{ ft}) * (1.4 \times 10^{-6} \text{ ft/sec}) \\ = 0.30 \text{ cf/sec, or 135 gpm}$$

The plume, elongated in a northwestern direction, has an estimated length of 1700 feet. Using this and an integrated equation of groundwater horizontal flow rate under existing conditions, and subtracting the increase in flow rate as groundwater approaches the pumping well, an approximate time for collecting one pore volume of groundwater from the contaminated aquifer can be determined. The equation used for this estimation is as follows:

$$T = L/V - d/V * \ln(L/d + 1)$$

WHERE:

$$T = \text{time in years}$$

$$L = \text{plume length}$$

$$V = \text{plume velocity} = K_i/n \text{ where } n = \text{soil porosity, and}$$

$$d = Q/(2(\pi)BK_i)$$

For the Rexair Plume:

$$L = 1,700 \text{ ft}$$

$$V = (1.4 \times 10^{-6} \text{ ft/sec}) / 0.3 = 147 \text{ ft/year}$$

$$d = (0.3 \text{ cf/sec}) / 2(3.14)(160)(1.4 \times 10^{-6} \text{ ft/sec}) \\ = 213 \text{ feet}$$

THEREFORE:

$$T = 11.6 \text{ years} - 3.2 \text{ years} = 8.4 \text{ years}$$

Once the pumping system is designed (using the formulas described in Appendix B), a range of contaminant recovery efficiencies is used to determine the number of pore volumes that will require pumping to reach TCLs, including recovery efficiencies of 70, 80, and 90 percent. If it is assumed that after pumping each pore volume, 70 percent of the contaminant will be removed from the plume volume, then approximately 8.5 pore volumes will need to be pumped to decrease the contamination from 153 ppm to below detection levels of 0.005 ppm. Because it will take 8.4 years to extract one pore volume, it will require 72 years to reach the TCLs for the Rexair Plume. If 90 percent of the contamination is extracted with each pore volume, a total time of 40 years will be needed to reach the contaminant TCLs. The percent of contaminant extraction for each well volume can only be estimated with currently available data. Better-defined estimates of time required for cleanup can only be determined after extraction and monitoring has started. It is believed that efficiencies of removal will be relatively high (90 percent or possibly better, depending on aquifer response), because the aquifers are fairly homogeneous with low potentials for retardation. Calculations for the lower efficiencies have also been included to indicate the sensitivity of the analysis. Final comparisons used in the FS were based on the 90-percent value as a reasonable, but probably still conservative estimate of anticipated removal efficiencies.

Using these estimates for the one-well pumping system for extraction of the Rexair Plume, approximately 71 million gallons per year of groundwater will be pumped; at 70-percent contaminant recovery efficiency, a total of 5.1 billion gallons will be pumped. This groundwater will be directed to the treatment facility located in the industrial park and ultimately discharged to the Clam River.

CONCENTRATED VOC SHALLOW PLUME (A)

VOC contamination was detected in both the shallow and intermediate aquifers below the Cadillac Industrial Park. The most significant contamination is found in the shallow aquifer where concentrations from 1987 sampling reached 115 ppm downgradient of known solvent dumping at the Kysor site. Lower solvent concentrations were detected in the intermediate aquifer, which reflect the ability of the solvents to pass through the clay aquitard as a result of downward vertical gradients between aquifers, and the denser-than-water nature of the solvents. This subsection describes the pumping extraction system for only the concentrated portion of the shallow VOC contamination. Separate pumping systems were designed for the shallow contamination where it merges with plumes from sources other than the Kysor site, and the intermediate aquifer contamination.

The primary contaminants found in the concentrated VOC shallow plume include TCE, 1,1-DCE, 1,1,1-TCA, PCE, toluene, and xylene. The bottom of the shallow aquifer is defined by a southwestward dipping clay aquitard varying in thickness from about 2 to 22 feet and occurring at a depth of 35 to 105 feet. The water table is approximately 20 to 30 feet below ground and the saturated thickness of the aquifer ranges from about 46 to 86 feet. The hydraulic conductivity in this area of the shallow aquifer is estimated to be 4.2×10^{-2}

cm/sec and the horizontal gradient is 1.5×10^{-3} cm/cm. Groundwater flow is northeasterly in the direction of decreasing saturated thickness.

Due to the width of this shallow plume, a two-well pumping system is believed to be more effective in capturing the contaminated groundwater. The two extraction wells will be placed at the leading edge of the contaminant plume and approximately 426 feet apart, as shown in Figure 7-3. Each well will be screened the entire saturated thickness to the clay aquitard at a depth of 50 to 75 feet. Figure 7-3 shows the defined extent of the plume, the location of the extraction wells, and the associated capture zone. To achieve this capture zone configuration, each pumping well will need to pump at a rate of 79 gpm. The plume width is estimated at approximately 1,800 feet and has a length of 1,900 feet. The estimated time to extract one pore volume using the two pumping wells is 6.5 years.

To estimate time to achieve TCLs, contaminant recovery efficiencies of 70, 80, and 90 percent per pore volume were used. At a 70-percent recovery, it will require just over eight pore volumes and a total of 54 years to reach the TCLs. At a pumping rate of 83 million gallons per year, a total of 4.5 billion gallons of groundwater will be pumped for treatment.

VOC SHALLOW PLUME (B)

(Section 2.2.2 discusses the concentrated portion of the Kysor-related shallow plume and the extraction wells located just north of Sixth Street to capture this contamination.) Less concentrated but more areally extensive shallow contamination by VOCs is present downgradient of the series of wells developed for the concentrated VOC shallow plume and extends northeastward to the limit of the upper clay aquitard. Much of this contamination is Kysor-related, but at the eastern and western edges of the plume, VOC contributions by Four Star and Four Winns (respectively) are evident. A second series of shallow aquifer extraction wells was designed to pump this contamination situated near the northeastward edge of the aquifer.

Two pumping wells spaced 404 feet apart were located along a northwest-southeast trending line in the central portion of the industrial park. These wells will extend to the bottom of the shallow, unconfined aquifer (approximately 45 feet deep) and will be screened the entire distance. The capture zone width required to collect the contaminated groundwater will be achieved by pumping each well at a rate of 65 gpm (see Figure 7-3). Estimating the plume length to be approximately 1,100 feet, one year will be required to extract one pore volume from the shallow aquifer. At a 90-percent recovery efficiency, 3.5 years will be required to reach TCLs, and 238 million gallons of groundwater will be extracted.

FOUR WINNS SHALLOW PLUME

A separate extraction well for shallow aquifer contamination at the Four Winns property was designed because it is a localized area where contaminant concentrations are relatively high. Total VOC concentrations approach 20 ppm, with

TCA, TCE, acetone, and PCE being the primary contaminants. Pumping at a rate of 42 gpm will create a capture zone of sufficient size to collect the contaminated groundwater. The well will be placed near Saunders Street at the edge of the upper clay aquitard to a depth of approximately 60 feet (see Figure 7-3). At this pumping rate, it will require three years to collect one pore volume assuming a plume length of 1,500 feet. At a 70-percent contaminant recovery efficiency, it will take seven pore volumes and 21 years to reach TCLs. A total of 464 million gallons of groundwater will be extracted for treatment. At a 90-percent contaminant recovery efficiency, it will take 12 years to reach TCLs, and a total of 265 million gallons of groundwater will be extracted.

VOC INTERMEDIATE PLUME

Due to migration through the upper clay aquitard, significant VOC contamination reached the confined intermediate aquifer downgradient of the Kysor facility. Concentration levels were measured at 12 ppm during the 1987 sampling of groundwater at SLW-10D. The primary contaminants are TCE and t-1,2-DCE. The aquifer thickness in this area is approximately 120 feet and groundwater flow is to the northeast. The hydraulic conductivity of this aquifer is 4.0×10^{-2} cm/sec, and the horizontal hydraulic gradient is approximately 9.1×10^{-4} cm/cm.

Contamination in this intermediate aquifer, although less concentrated, is areally extensive requiring the development of a large capture zone. A pumping extraction system using three wells spaced approximately 795 feet apart was designed to capture the contaminated groundwater. The proposed locations of the wells are shown in Figure 7-2. To create the required capture zone width of approximately 5,248 feet, a pumping rate of 129 gpm from each well is required. The two southernmost pumping wells will be screened in the confined portion of the intermediate aquifer, with screens extending from approximately 60 to almost 200 feet below ground. The northernmost pumping well will be located where the intermediate aquifer is unconfined and will have a screened interval from 30 to 175 feet below ground. The plume length is estimated at 3,400 feet. At the 129-gpm pumping rate, it will require 19.4 years to extract one pore volume from the aquifer. Assuming a 90-percent contaminant recovery efficiency with each pore volume pumped, it will require 62 years to reach TCLs, and 13 billion gallons of groundwater will have been pumped for treatment.

NORTH PARK SUBDIVISION PLUMES

Two of the seven identified groundwater plumes in the Cadillac Industrial Park are located in the North Park Subdivision, a residential area north of Thirteenth Street, just outside Cadillac city limits. Both plumes are relatively small with weak concentrations. Based on hydrogeologic data, these plumes may be extensions of Four Star- and Joynts-related contamination.

The two plumes are at discrete depths, with the most downgradient plume being the deepest and more concentrated. Figure 7-2 shows the plumes as defined by the 1987 groundwater sampling. The deepest plume has a concentration in the range of 0.600 ppm in SLW-20. Screened auger sampling of the groundwater was

conducted to a depth of 180 feet below ground, at which point the augers could not be penetrated farther. These data indicate that the maximum contaminant concentration occurs at approximately 150 feet, but levels are still significant at the 180-foot depth. The shallower plume defined by SLW-4 and SLW-2 is approximately 55 feet deep with concentrations of about 0.090 ppm.

A one-well pumping system was designed to collect groundwater from the deeper plume. The well will be placed at a depth of 200 feet and screened the entire saturated thickness beginning at approximately 30 feet below ground. The pumping rate will be 56 gpm to create a capture zone width of approximately 656 feet. One pore volume will be collected in approximately three years and, assuming a 70-percent contaminant recovery efficiency, 12 years will be required for cleanup. A total of 360 million gallons of groundwater will be extracted for treatment.

The concentration of the shallower plume is approximately 0.090 ppm. At these low levels, it does not appear to be beneficial or cost-effective to extract this groundwater. With time and mixing with clean groundwater, these contaminant levels should approach nondetectable levels. Future groundwater monitoring will determine the effectiveness of in-situ dilution and the need for installing an extraction system for this plume.

HEXAVALENT-CHROMIUM PLUMES

Two hexavalent-chromium plumes were identified in the Cadillac Industrial Park: one in the shallow aquifer and one in the intermediate aquifer. The shallow aquifer plume was detected during just one time period in MW-5 located 500 feet downgradient of the Northernaire facility. The concentration of hexavalent chromium in this well was 1.6 ppm in February 1986, and had decreased to 0.5 ppm during the May/June 1987 sampling episode. Hexavalent chromium was not detected in subsequent sampling episodes, nor was it detected in prior episodes. This intermittent detection suggests a small, rapidly moving plume that should eventually reach the northeastward limit of the shallow aquifer while becoming diluted by groundwater mixing. Not knowing the exact position of this plume, development of an extraction system is not practical.

The intermediate aquifer hexavalent-chromium plume has been studied and sampled since the Northernaire groundwater investigation in 1985. The plume is only 10 to 20 feet thick, and is at a depth of approximately 160 feet situated on top of the lower clay aquitard. Concentrations in 1986 were in the 2.2-ppm range, and had decreased to 0.8 ppm during the 1987 sampling. The extent of the plume is well-defined and is located in the central portion of the industrial park. Plume length is approximately 1,000 feet elongated northwest-southeast in the direction of groundwater flow.

A one-well extraction system was designed to capture this deep hexavalent-chromium plume. The pumping well, located as shown in Figure 7-2, will be installed to a depth of 175 feet and will be screened from 145 to 175 feet. The short well screen will be used because of the plume thinness.

To create a capture zone width of approximately 1312 feet, a pumping rate of 32 gpm will be required. It is estimated that 67 million gallons of groundwater and four years will be needed to extract the hexavalent chromium at a contaminant recovery efficiency rate of 90 percent.

APPENDIX G
MONITORING WELL LOCATIONS
FOR THE EXTRACTION SYSTEM

